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**Buck**

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(54) **BIDIRECTIONAL MULTI-ENZYMATIC  
SCAFFOLDS FOR BIOSYNTHESIZING  
CANNABINOIDS**

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patent is extended or adjusted under 35  
U.S.C. 154(b) by 349 days.  
  
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None

See application file for complete search history.

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(57) **ABSTRACT**

This document relates to using bidirectional, multi-enzymatic scaffolds to biosynthesize cannabinoids in recombi-nant hosts.

**21 Claims, 55 Drawing Sheets**

**Specification includes a Sequence Listing.**

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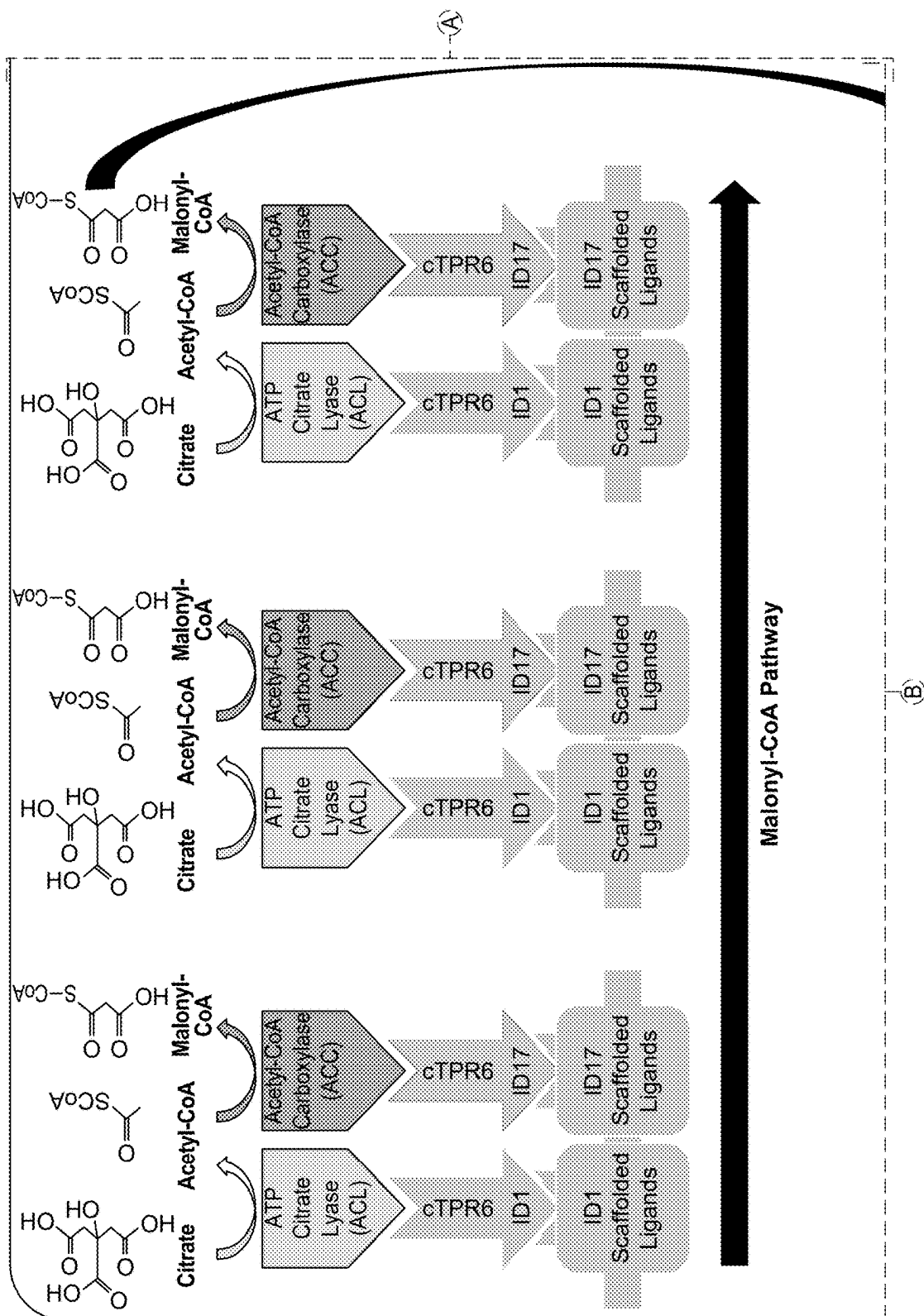


Figure 1A.

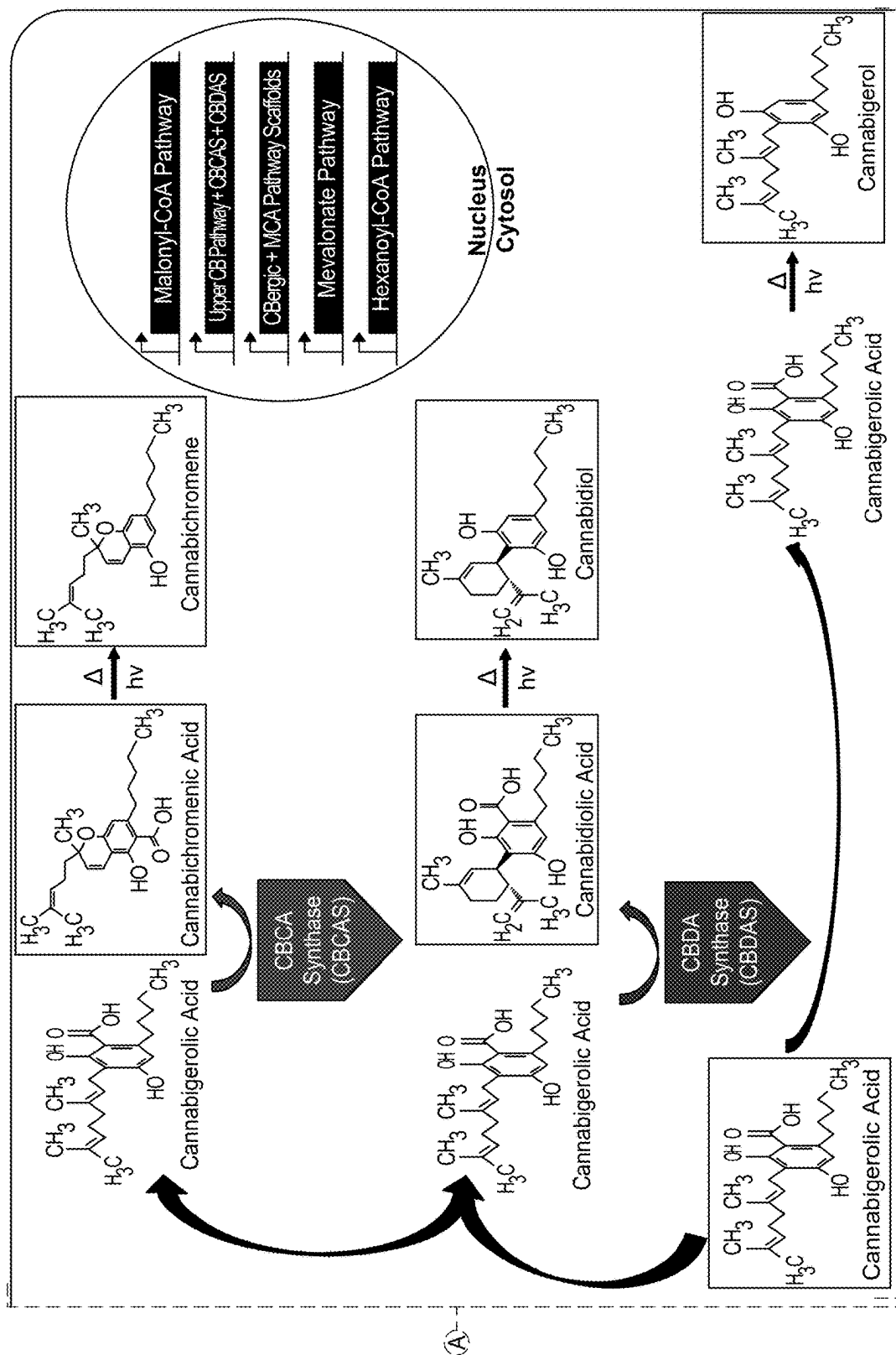


Figure 1A. (Cont'd)



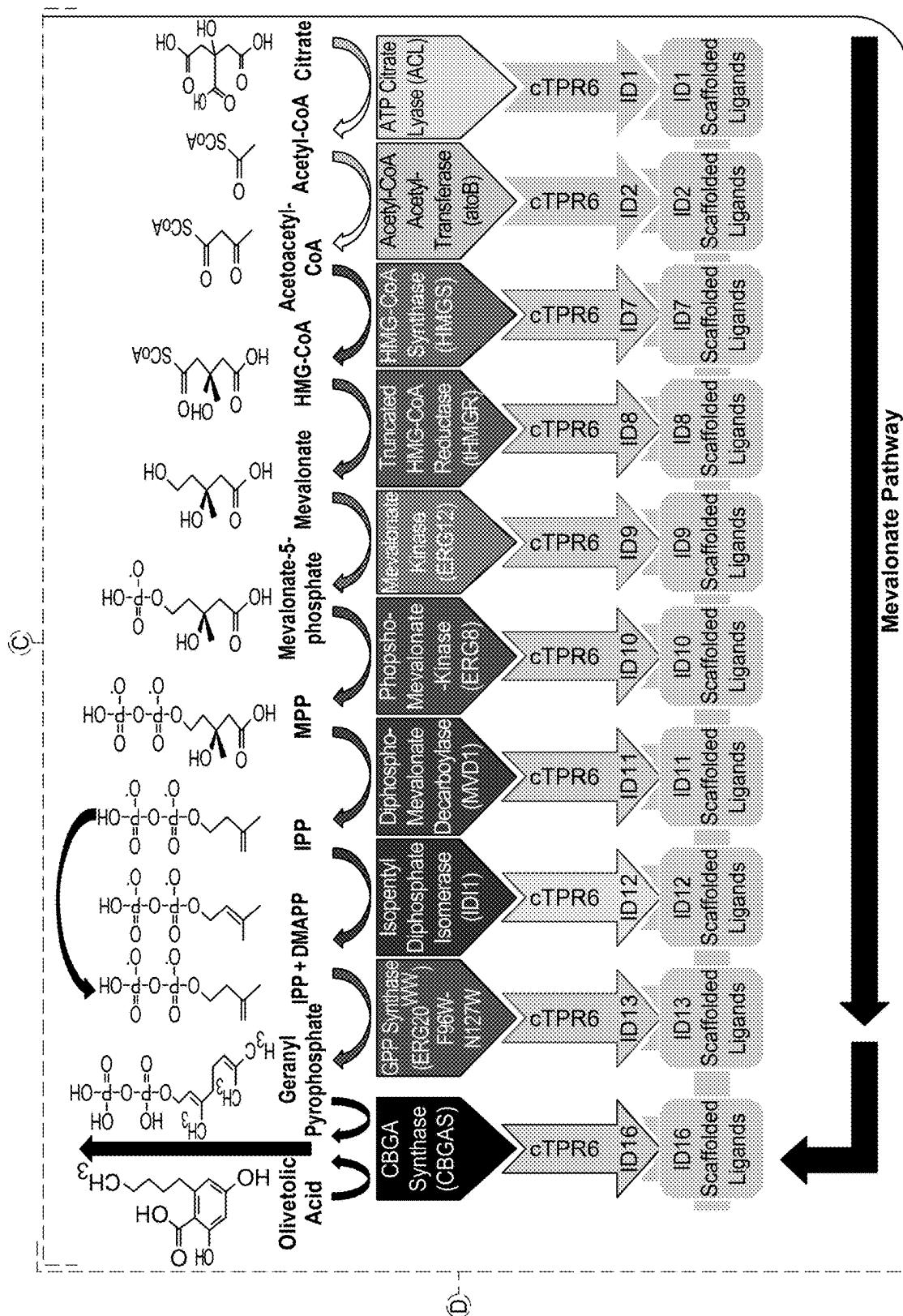


Figure 1A. (Cont'd)

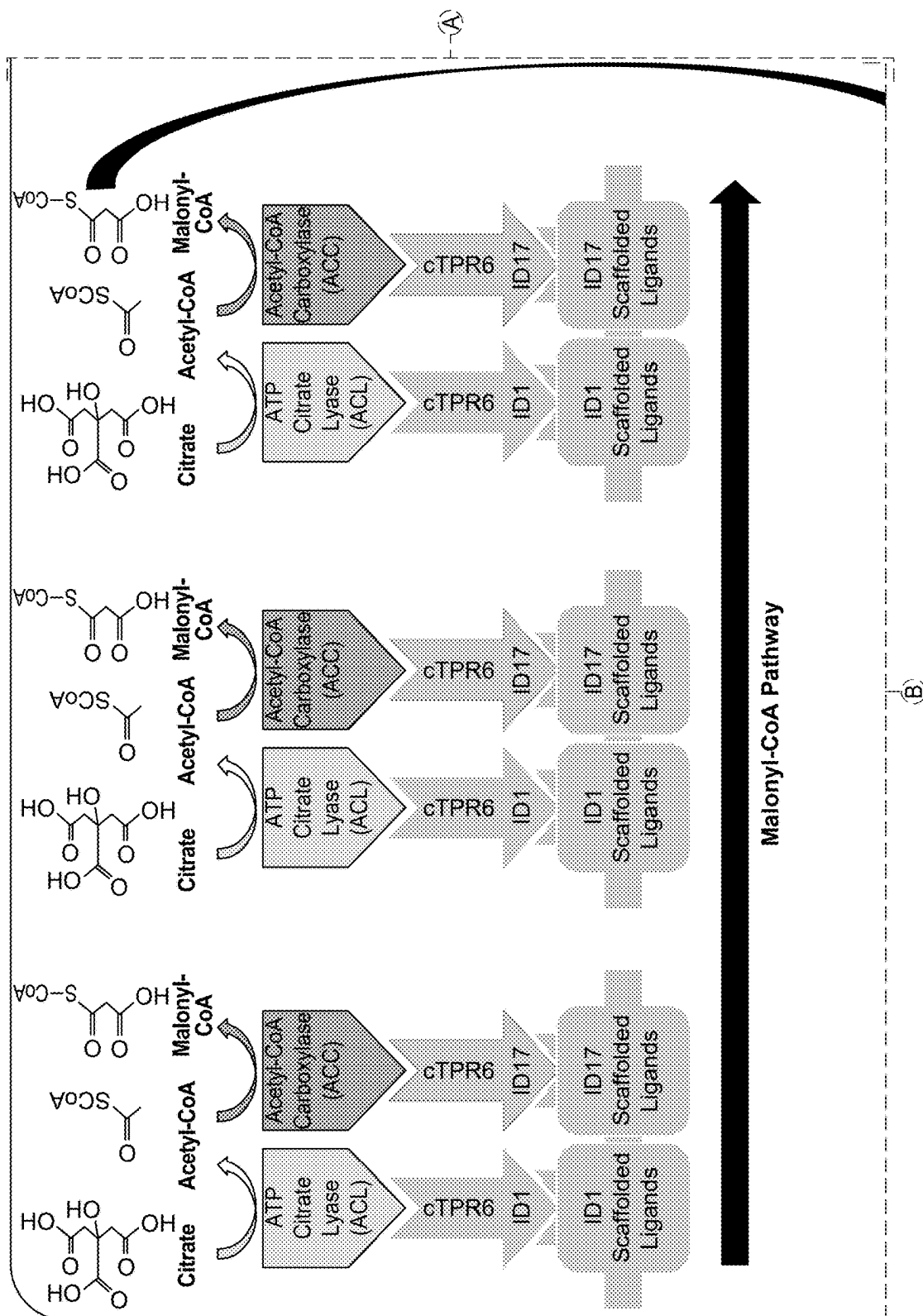


Figure 1B.

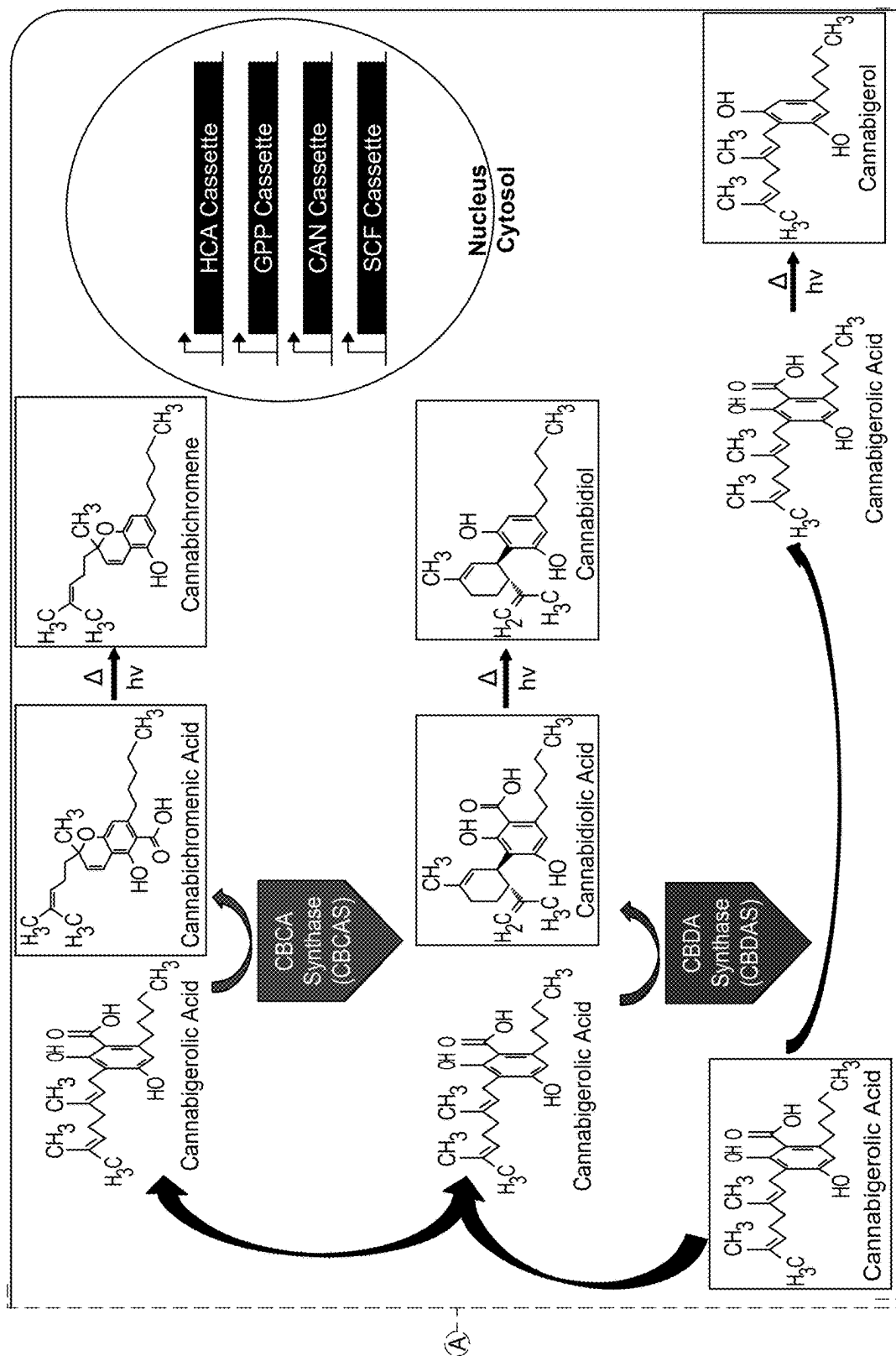


Figure 1B. (Cont'd)

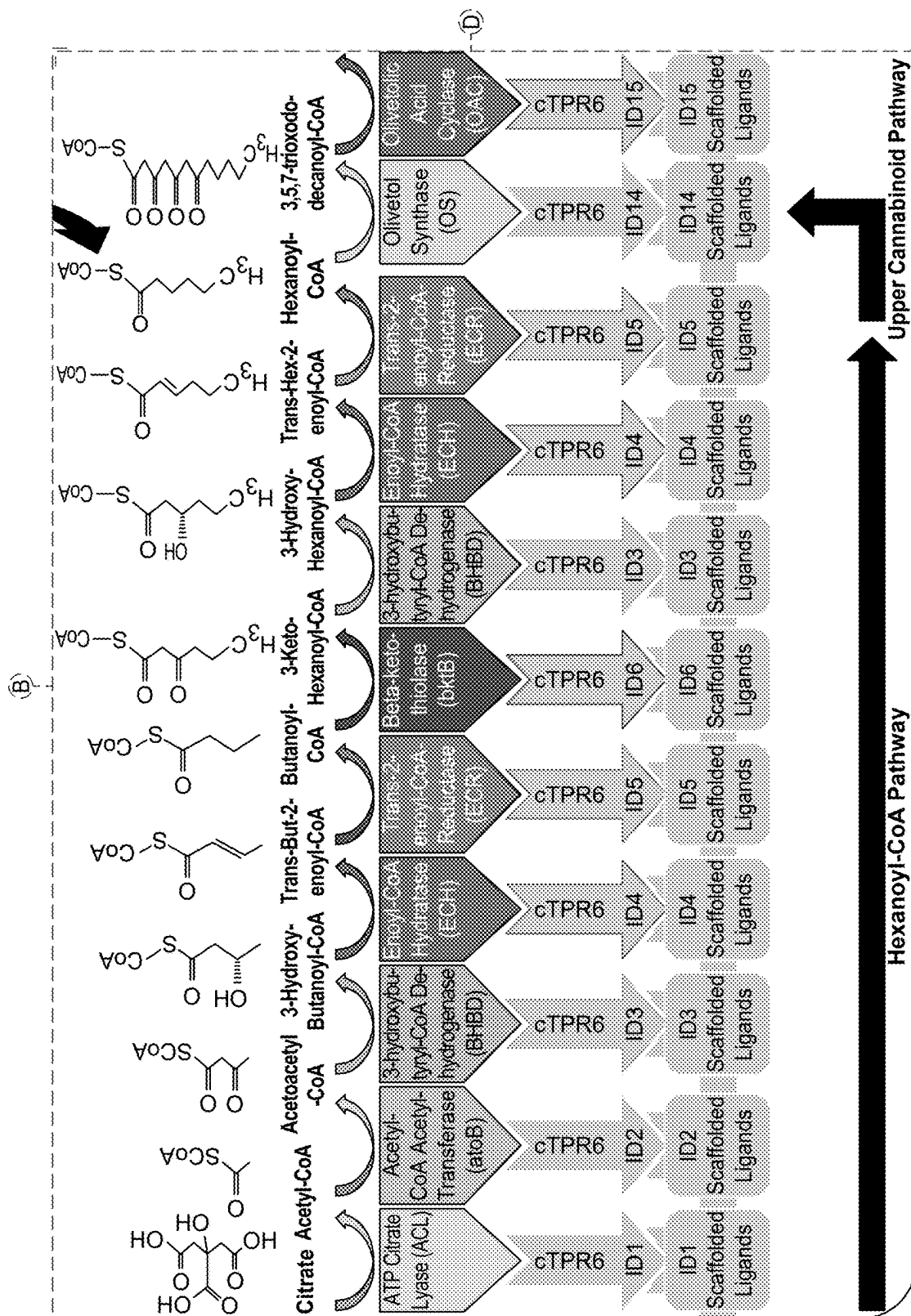


Figure 1B. (Cont'd)

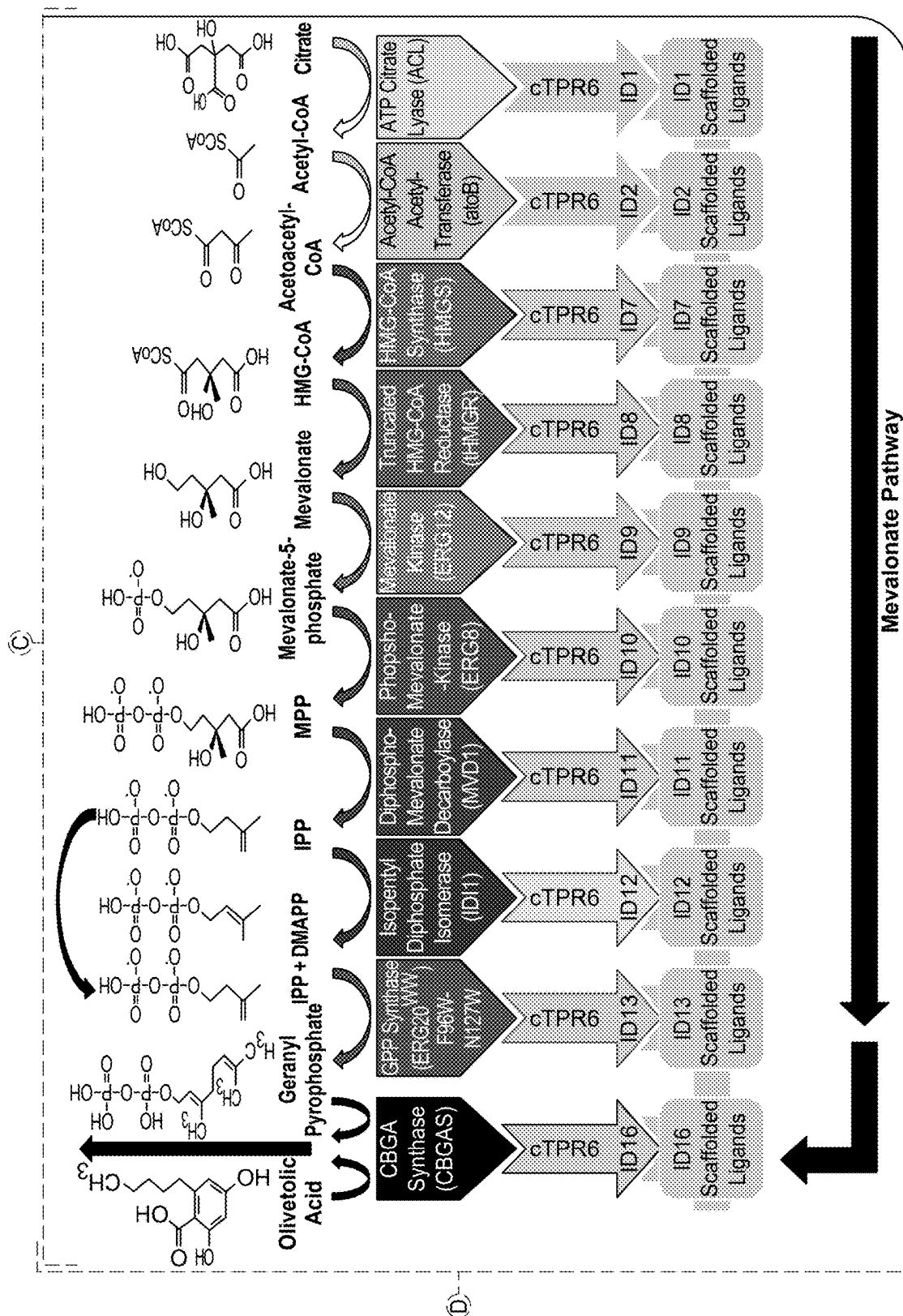


Figure 1B. (Cont'd)



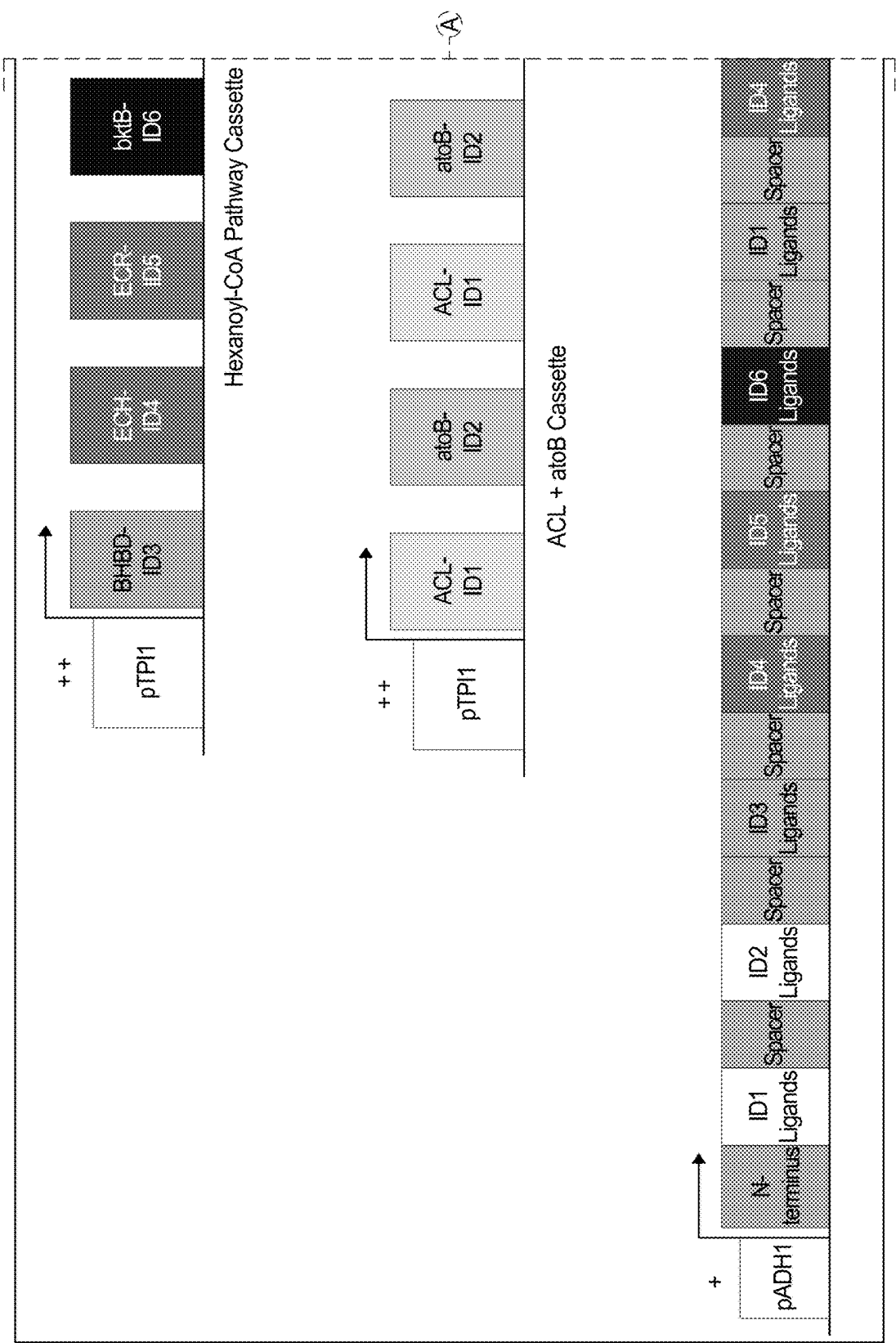


Figure 2A.

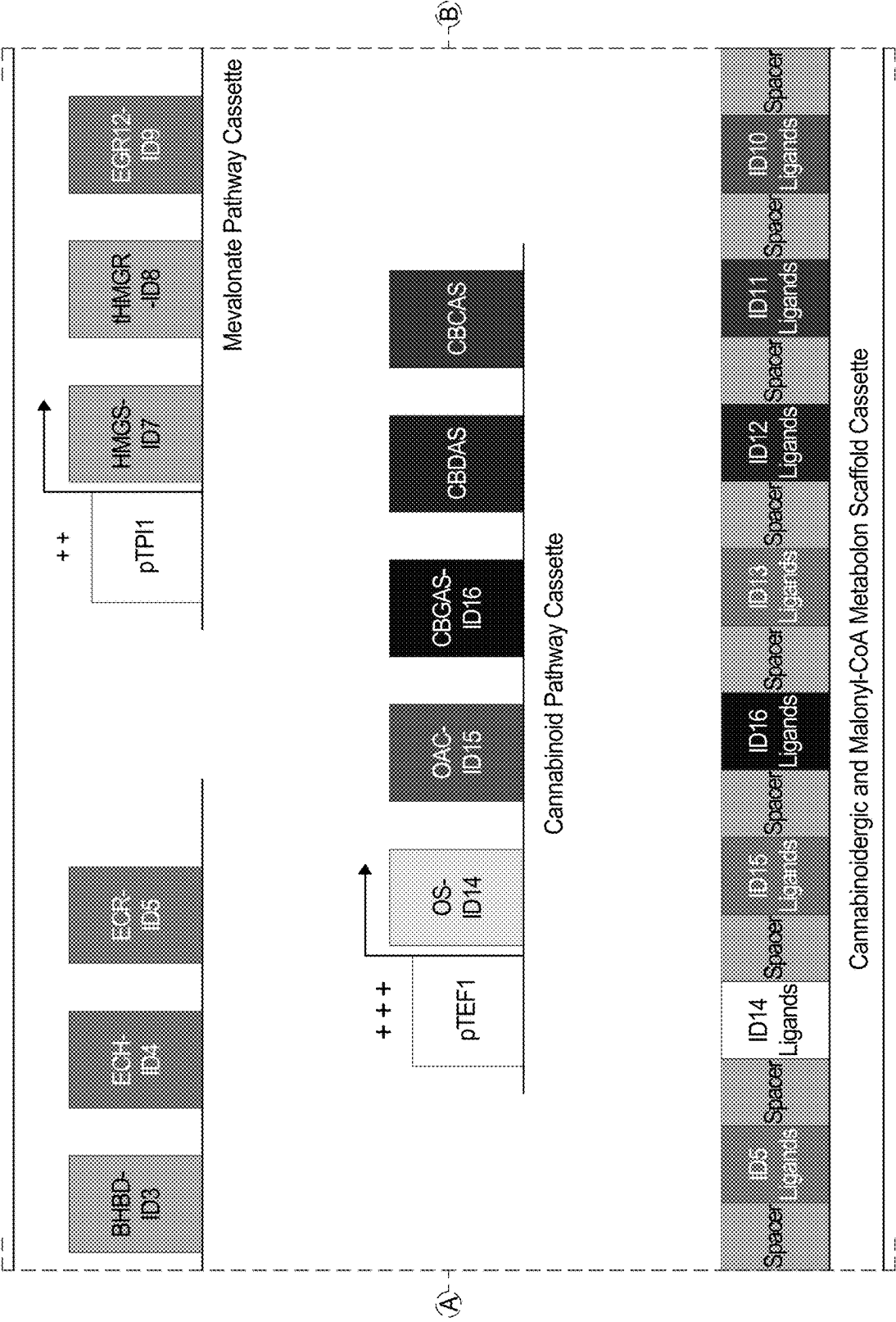


Figure 2A. (Cont'd)

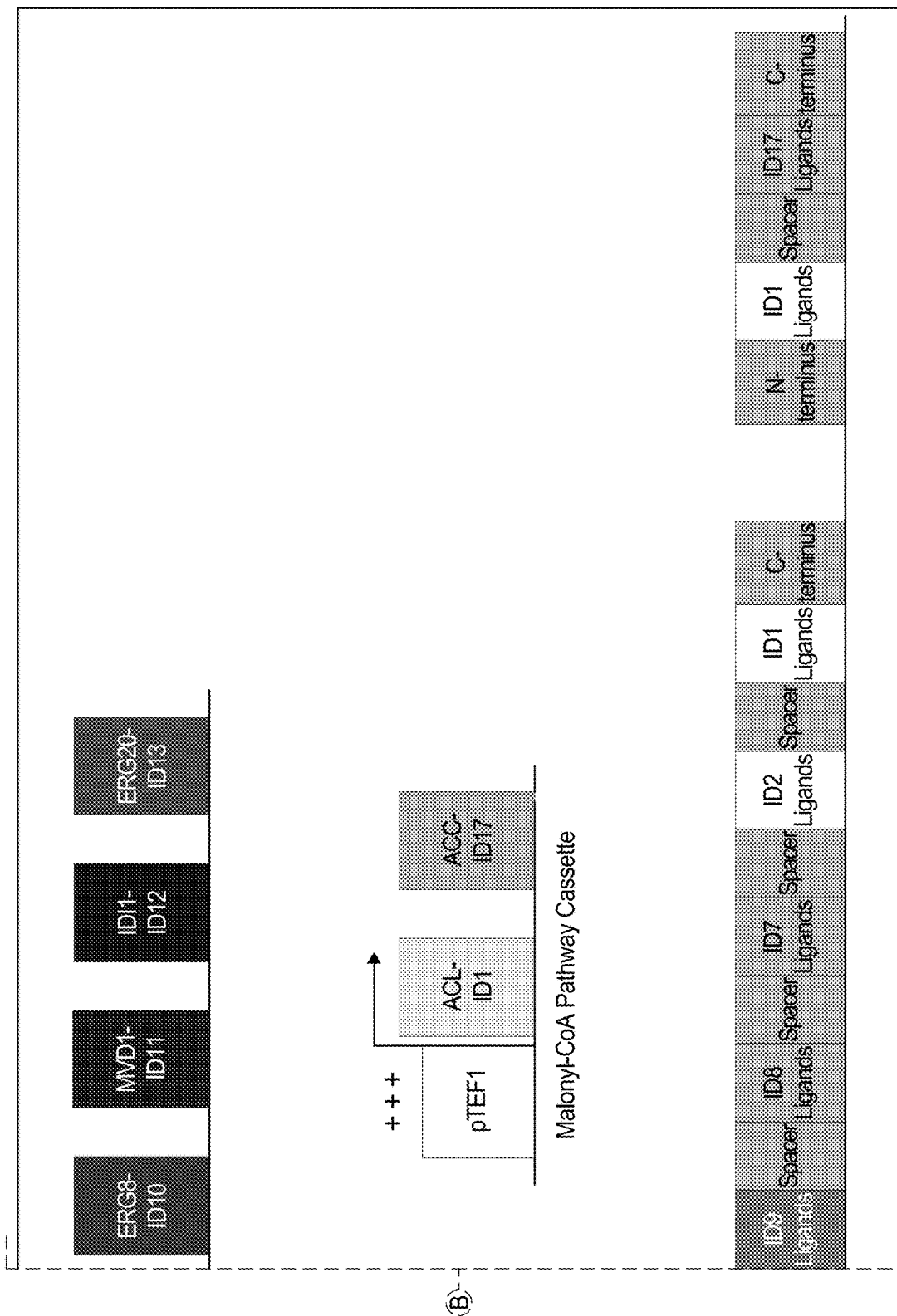


Figure 2A. (Cont'd)

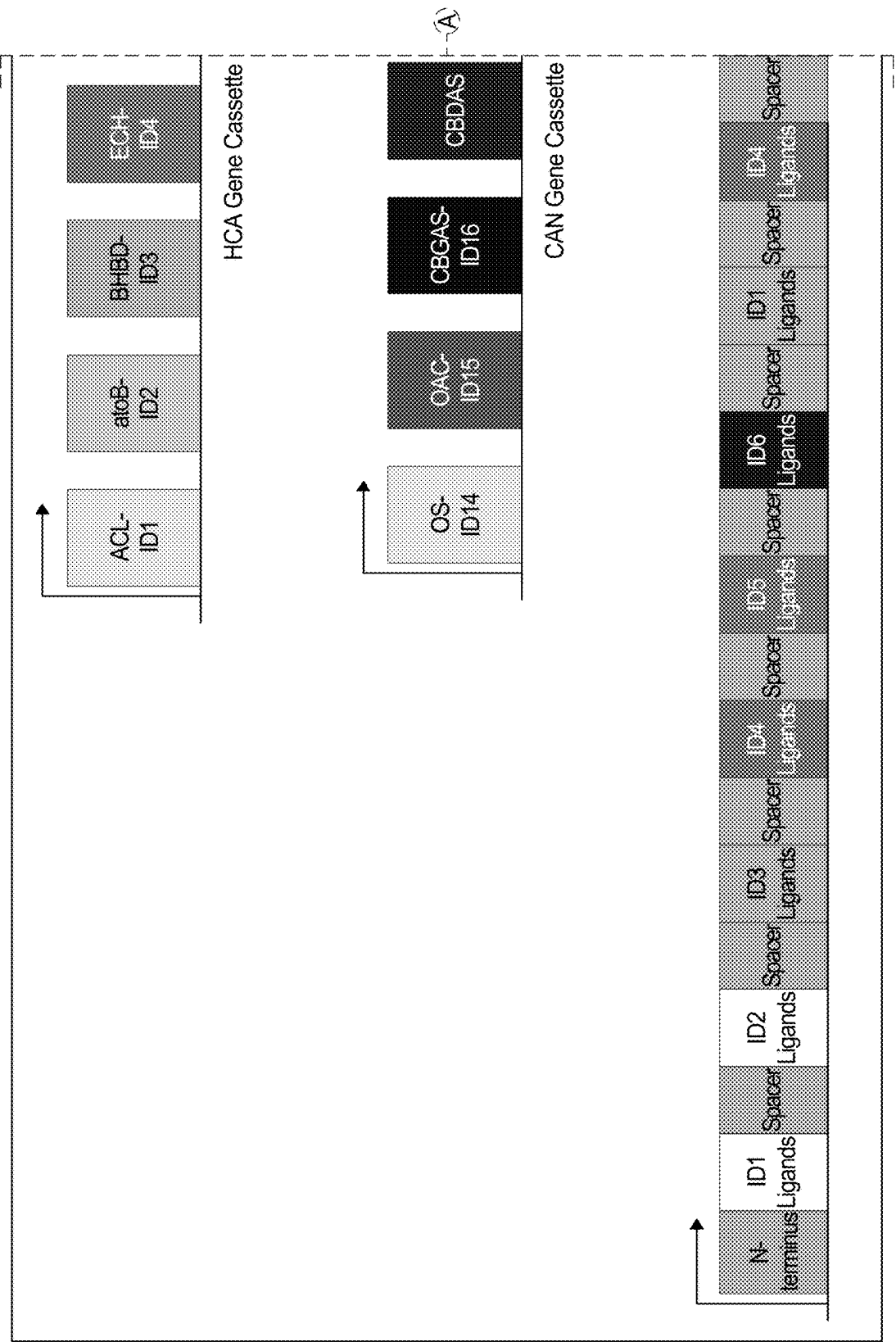


Figure 2B.

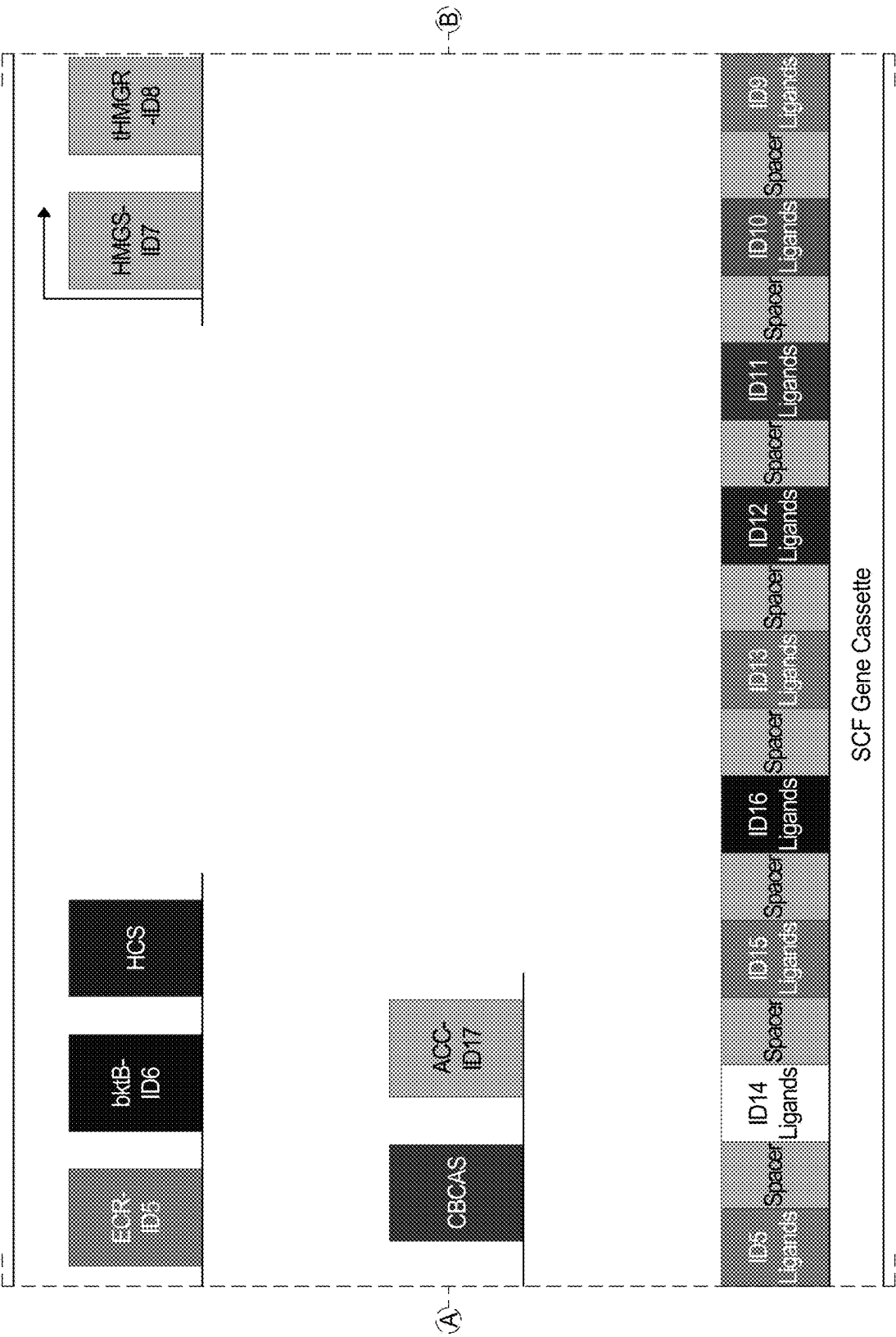


Figure 2B. (Cont'd)

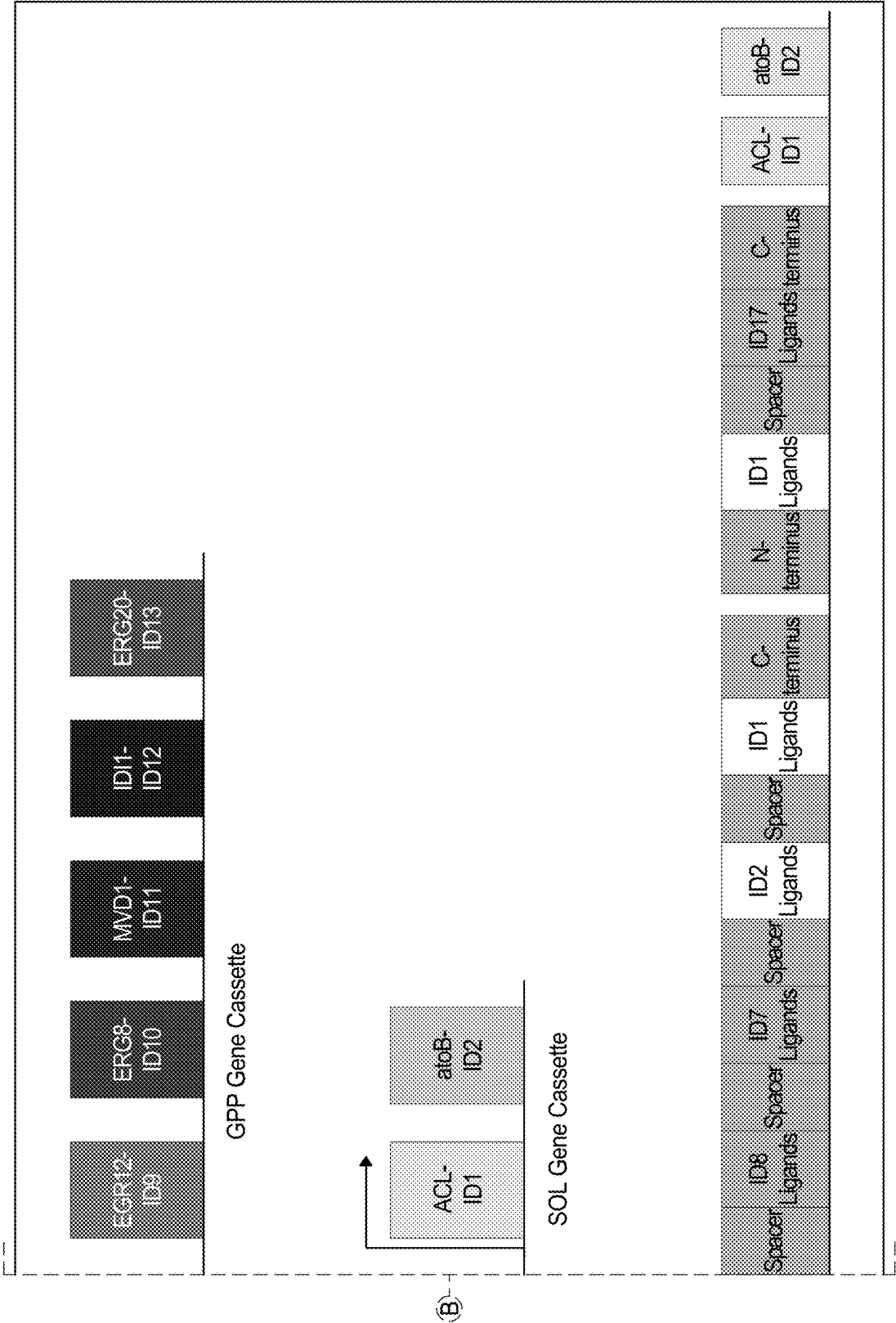


Figure 2B. (Cont'd)

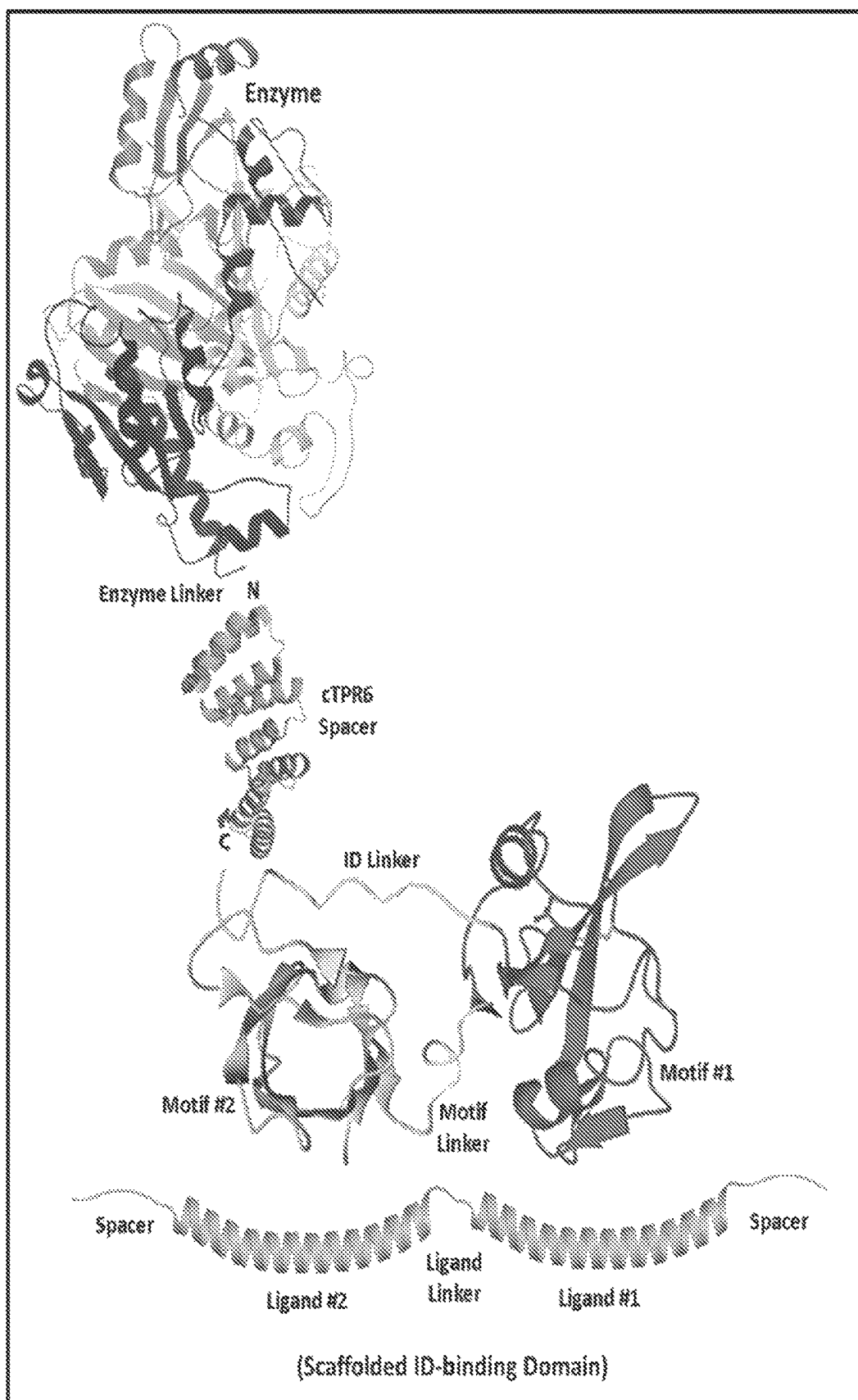


Figure 3.

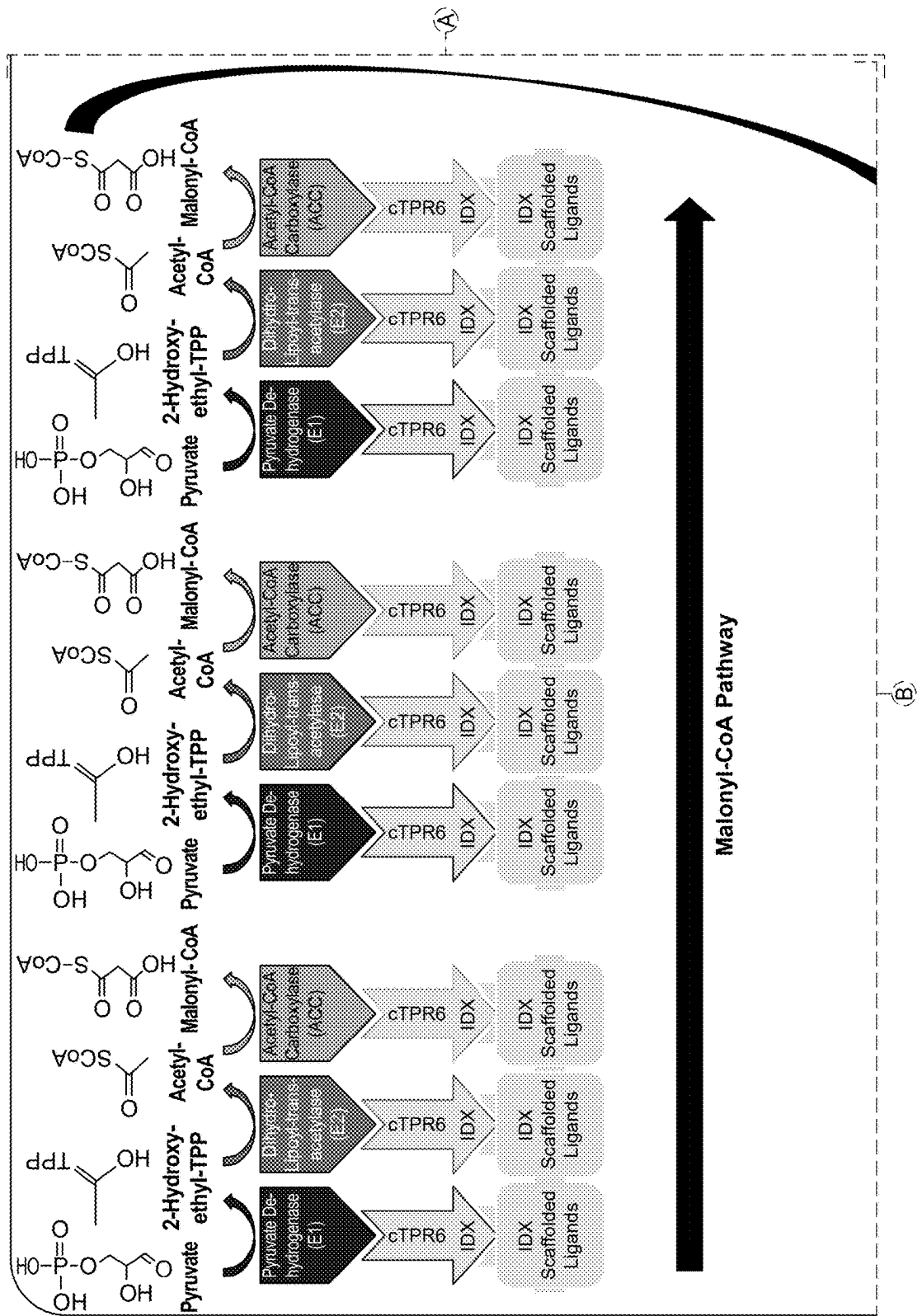


Figure 4.



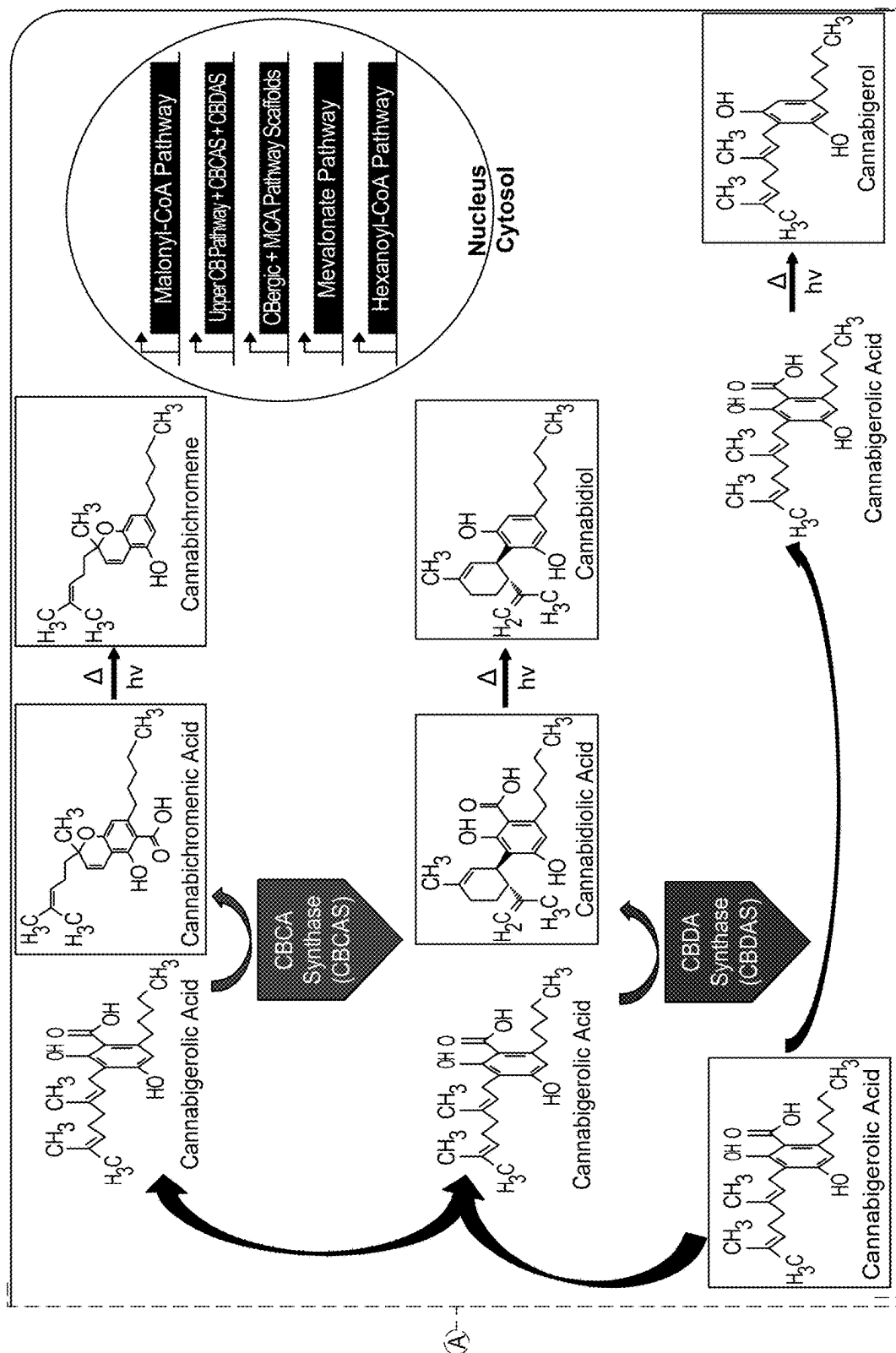


Figure 4. (Cont'd)

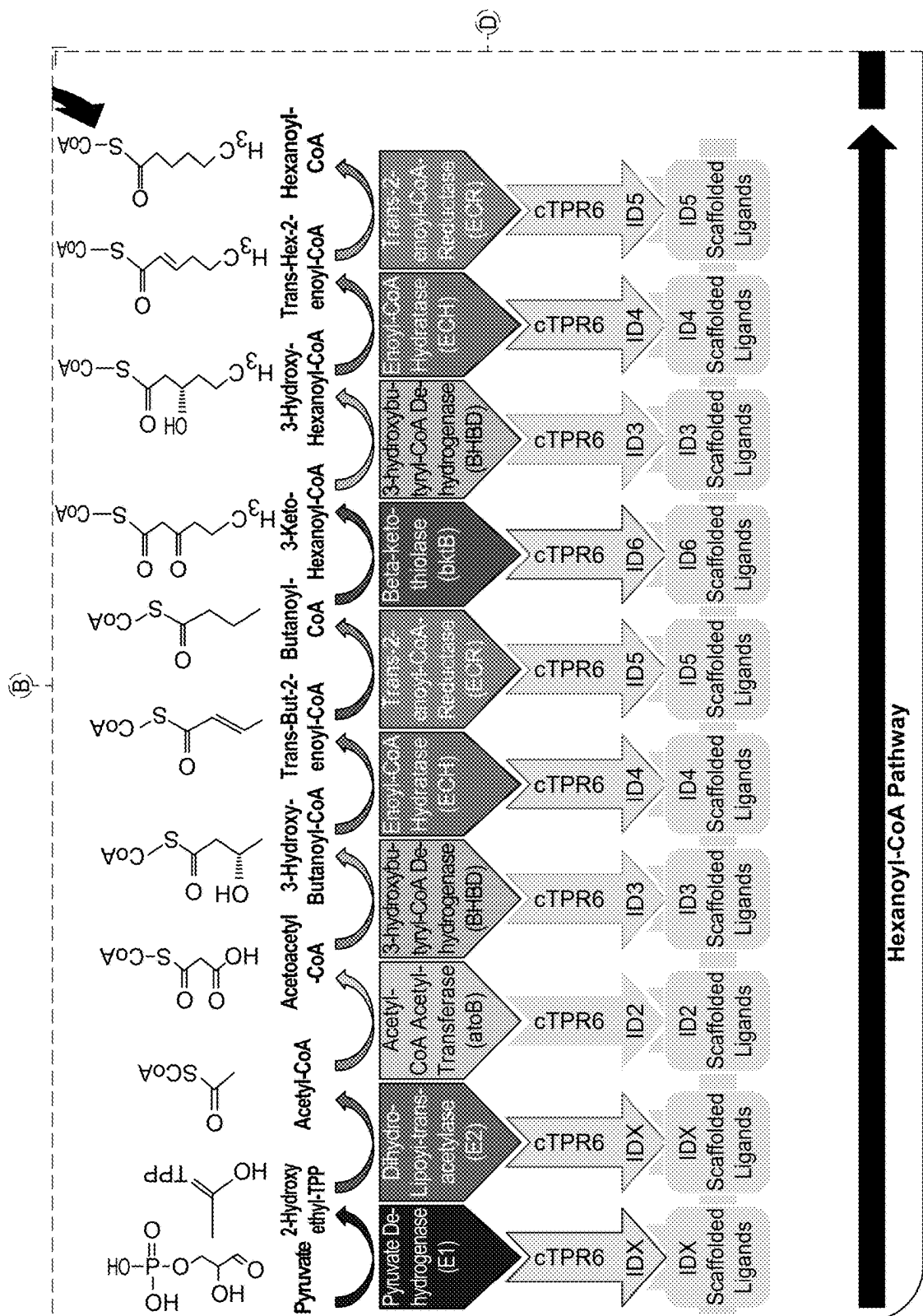
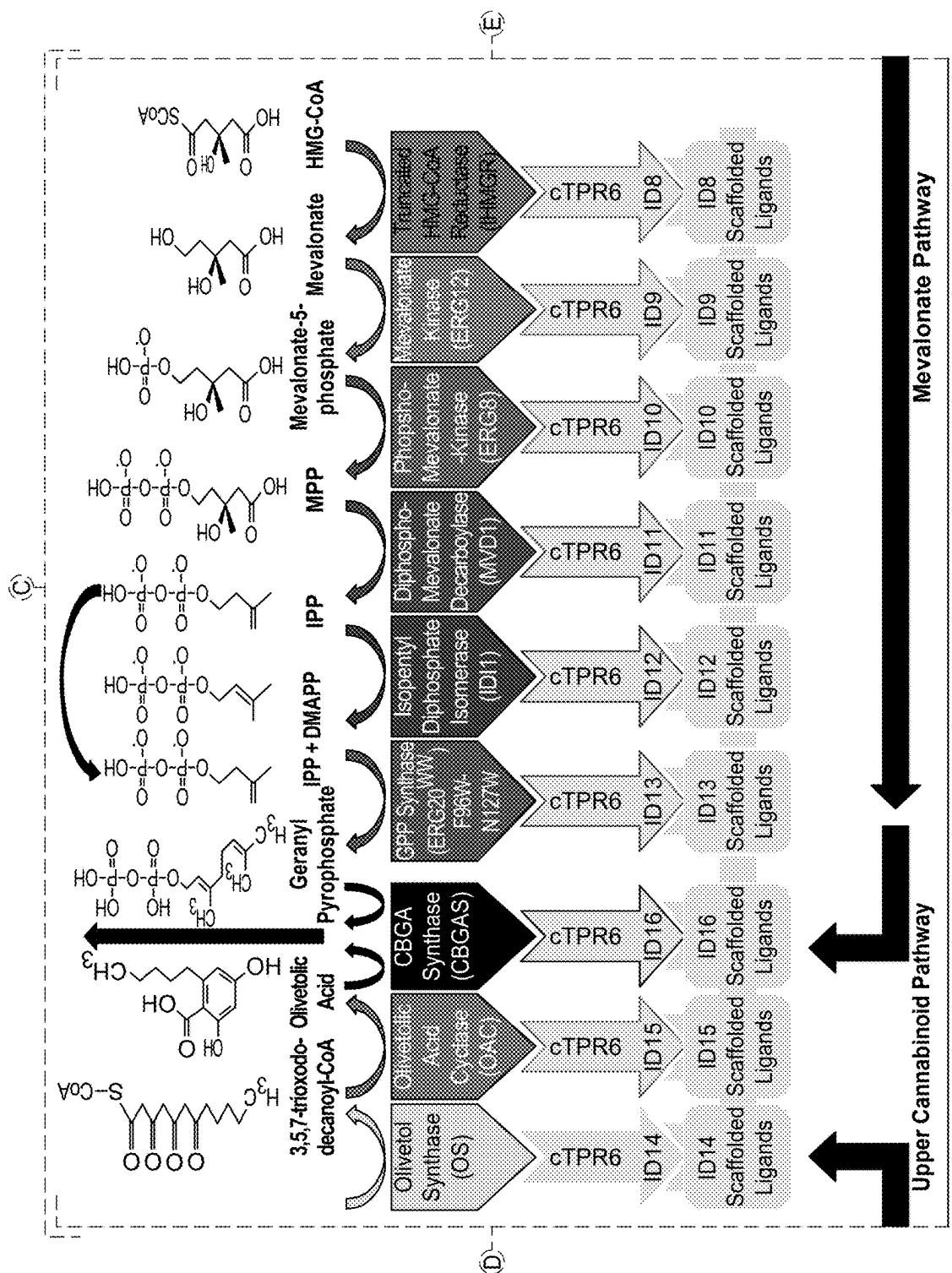


Figure 4. (Cont'd)



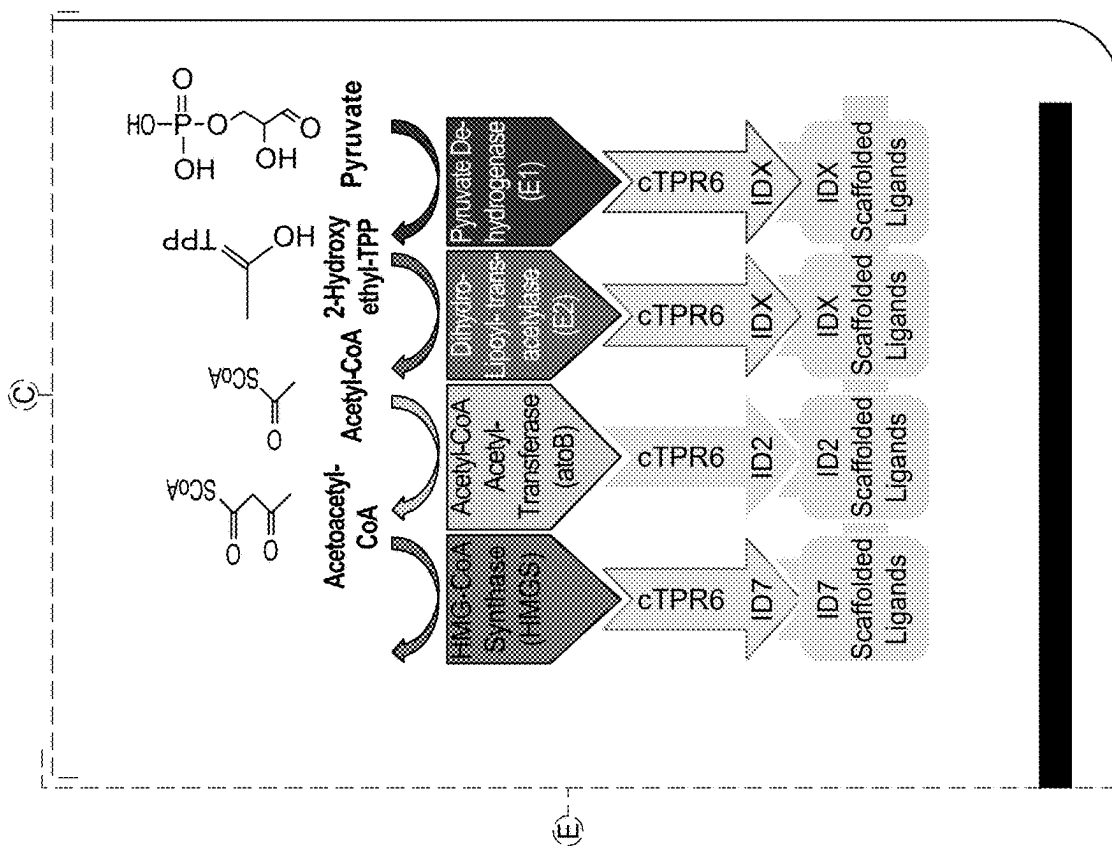


Figure 4. (Cont'd)

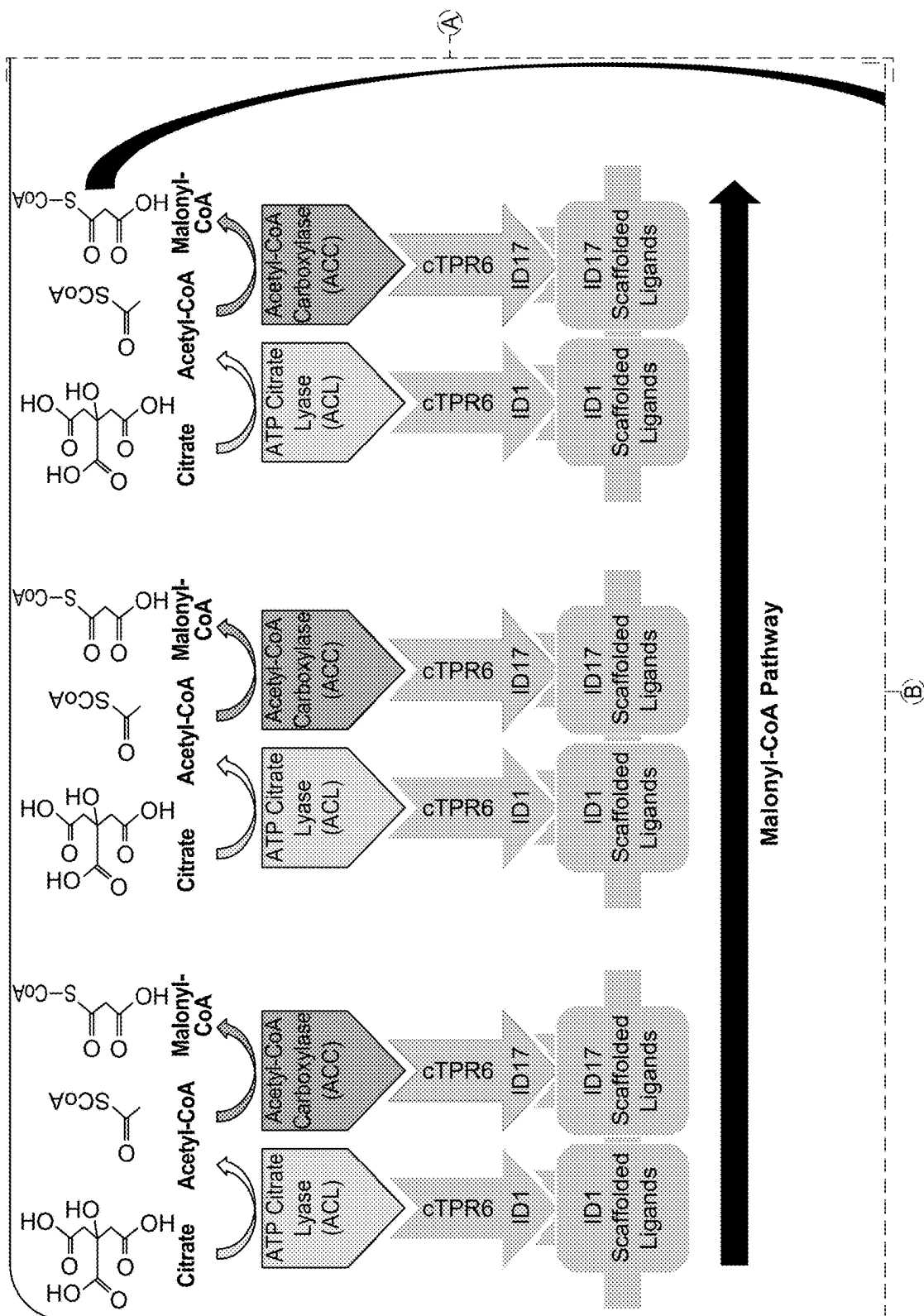


Figure 5.

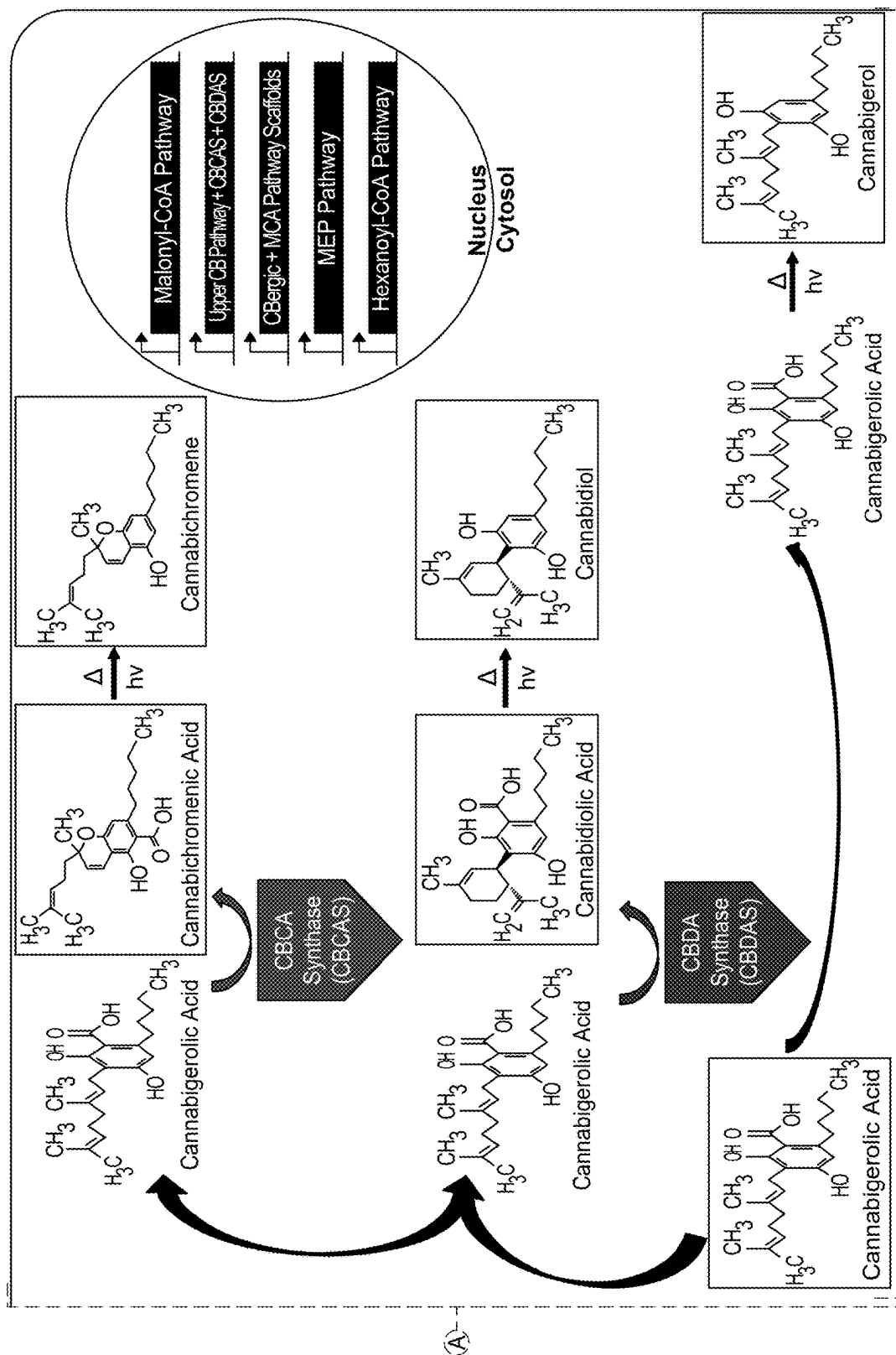


Figure 5. (Cont'd)

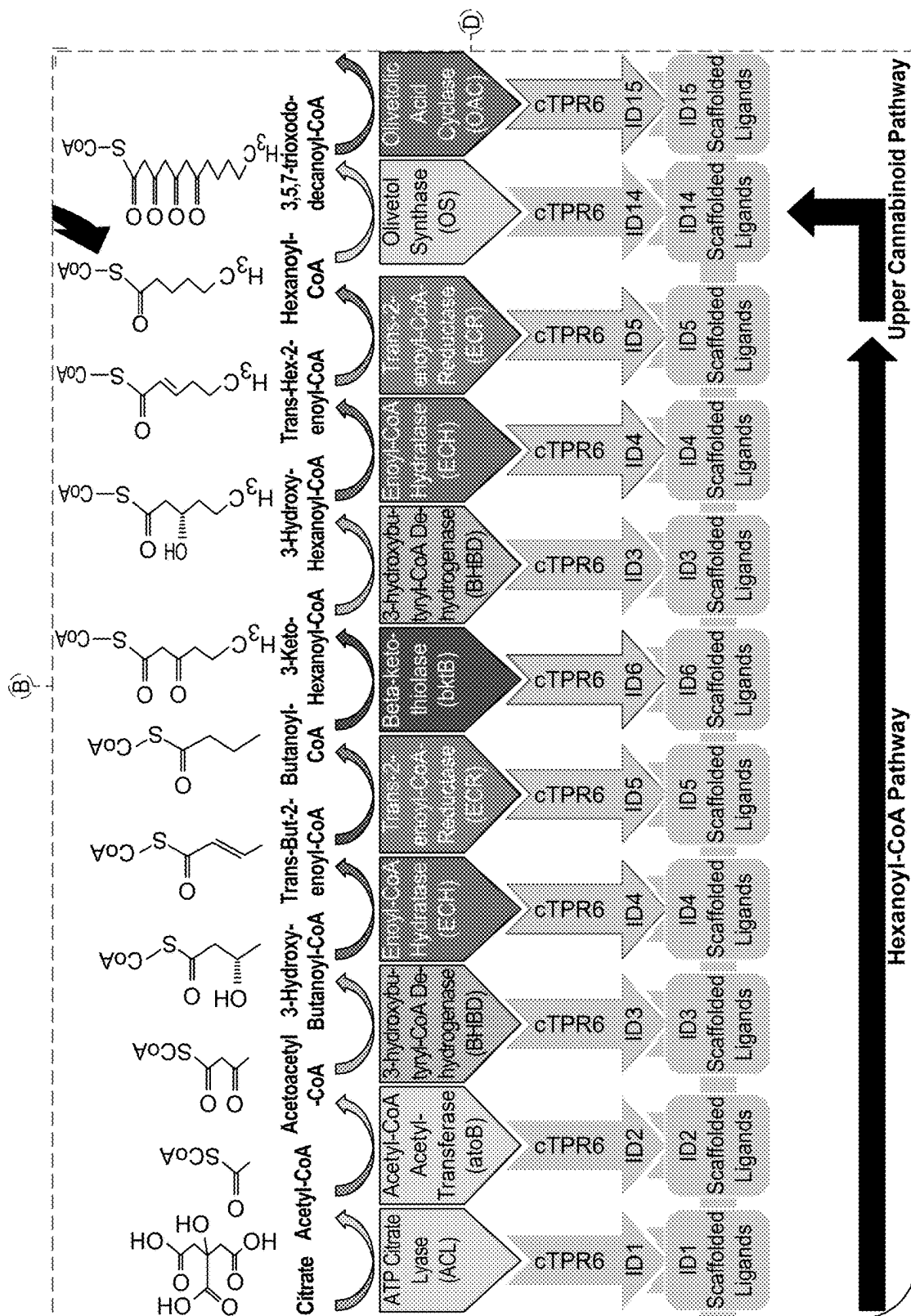


Figure 5. (Cont'd)

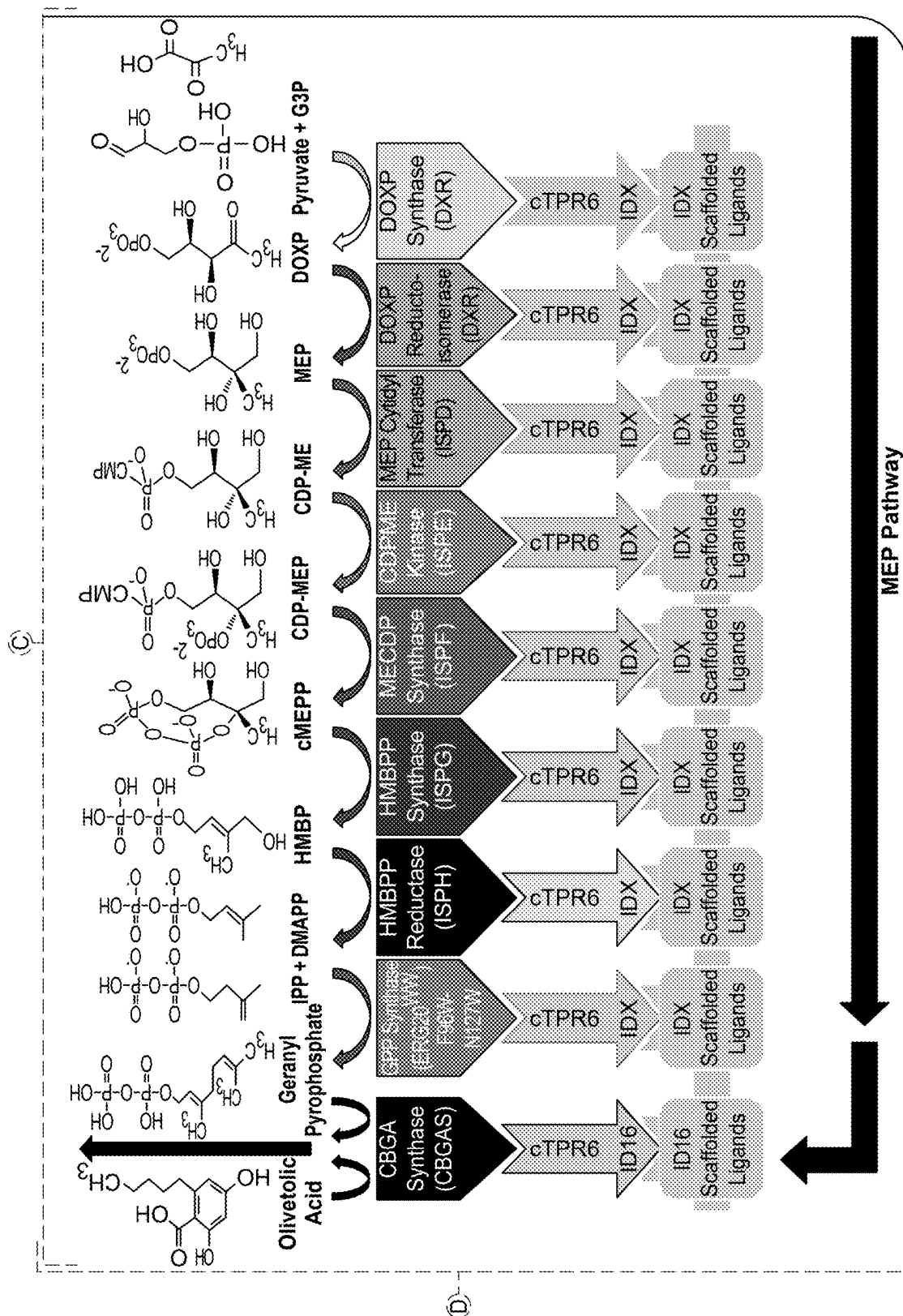


Figure 5. (Cont'd)



Figure 6A

**ATP Citrate Lyase**

MSAKAISEQTGKELLYKFICTTSAIQNRFKYARVTPD TDWARLLQDHPWLLSQNLVVKPDQLIKRRGKLGVLGVNLT  
DGVKSWLKPRLGQEATVGKATGFLKNFLIEPFVPHSQAEFFYVCIYATREGDYVLFHHEGGVDVGDVDAKAQKLLV  
GVDEKLPEDIKHLHVHAPEDKKEILASFISGLFNFYEDLYFTYLEINPLVVT KDGVYVLDLAAKV DATADYICKVKWG  
DIEFPPPPFGREAYPEEAYIADLDAKSGASLKLTLNPKGRIWTMVAGGGASVVYSDTICDLGGVNELANYGEYSGAP  
SEQQTYDYAKTILSLMTREKHPDGKILIGGSIANFTNVAATFKGIVRAIRDYQGPLKEHEVTIFVRRGGPNYQEGLRV  
MGEVGKTTGIPIHVFGTETHMTAIVGMALGHRPIPNQPPTAHTANFLLNASGSTSTPAPSRTASFSESRADEVAPAK  
KAKPAMPQDSVPSRSLQKGSTTLFSRHTKAIVWGMQTRAVQGM LDFDYVCSRDEPSVAAMVYPFTGDHKQKFY  
WGHKEILIPVFN MADAMRKHP EVDVLINFA SLRSAYDSTMETMNYAQIRTIAIAEGIPEALTRKLIKADQKGVTIIGP  
ATVGGIKPGCFKIGNTGGMLDNILASKLYRPGSVAYVSRSGGMSNELNNISRTTDGVYEGVAIGGDRYPGSTFMDH  
VLRYQDTPGVKMIVVLGEJGGTEYKICRGIKEGR LTKPIVCWCIGTCATMFSSEVQFGHAGACANQASETAVAKNQ  
ALKEAGVFVPRSFDELGEIISVYEDLVANGVIVPAQEVPPPTVPM DYSWARELGLIRKPASFMTSICDERGQELIYA  
GMPITEVFKEEMGIGGVLGLLWFQKRLPKYSCQFIEMCLMVTADHGPAVSGAHNTIICARAGKDLVSSLTSGLLTIGD  
RFGGALDAAAKMFSAFDSGIIPMEFVNKMKKEGKLIMGIGHRVKSINNPD MRVQILKDYVRQHFPATPLLDYALEVE  
KITTSKPNLILNVDGLIGVAFVDM LRNCGSFTREEADEYIDIGALNGIFVLGRSMGFIGHYLDQKRLKQGLYRHPWD  
DISYVLPEHMSM

**Acetyl-CoA Acetyltransferase (atoB)**

MKNCVIVSAVRTAIGSFNGSLASTSAIDL GATVIKAAIERAKIDSQHVDEVIMGNVLQAGLGQNPARQALLKSGLAETV  
CGFTVNKVCSSGLKSVLAAQAIQAGQAQSIVAGGMENMSLAPYLLDAKARSGYRLGDGQVYDVILRDGLMCATH  
GYHMGITAENVAKKEYGITREMQDELALHSQRKAAAIESGAFTAEIVPVNVVTRKKTFFVSQDEFPKANSTAEALGAL  
RPAFDKAGTVTAGNASGINDGAAALVIMEESAALAGLTPLARIKSYASGGVPPALMGMGMPVPATQKALQLAGLQLA  
DIDLIEANEAFAAQFLAVGKNLGF DSEKVVNVNGGAIALGHPIGASGARILVTL LHAMQARDKTLGLATLCIGGGQGIAM  
VIERLN

**3-Hydroxybutyryl-CoA Dehydrogenase**

MKKVCVIGAGTMGSGIAQFAAKGFEVVL RDIKDEFVDRGLDFINKNLSKLVKKGKIEEATKVEILTRISGTVDLNMAA  
DCDLVIEAAVERMDIKKQIFADLDNICKPETILASNTSSLSITEVASATKRDPKVIGMHFFNPAPVMKLV E VIRGIATSQE  
TFDAVKETSAIGKDPVEVAEAPGFV VNRILIPMINEAVGILAEGIASVEDIDKAMKLGANHPMGPLELGDFIGLDICLAI  
MDVLYSETGDSKYRPHTLLKKYVRAGWLGRKSGKGFDYSK

**Enoyl-CoA Hydratase**

MELNNVILEKEGKVAVVTINRPKALNALNSDTLKEMDYVIGEIENDSEVLAVILTGAGEKSFVAGADISEMKEMNTIEG  
RKFGILGNKVFRRLLELEKPVIAAVNGFALGGGCEIAMSCDIRIASSNARFGQPEVGLGITPGFGGTQRLSRLVGMGM  
AKQLIFTAQNIKADEALRIGLVNKVVEPEL MNTAKEIANKIVSNAPVAVKLSKQAINRGMQCDIDTALAFESEAFGECEF  
STEDQKDAMTAFIEKRKIEGFKNR

**Trans-Enoyl-CoA Reductase**

MIVKPMVRNNICLNAHPQGCKKGVEDQIEYTKKRITAEVKAGAKAPKNVLVLGCSNGYGLASRITAAFYGGAATIGVS  
FEKAGSETKYGTPGWYNNLAFDEAAKREGLYSVTIDGDAFSDEIKAQVIEEAKKKGIKFDLIVYSLASPVRTDPDTGIM  
HKSVLKPF GKTF TGKTVD PFTGELKEISAEPANDEEAAATVKVMGGEDWERWIKLSKEGLLEEGCITLAYSIGPEA  
TQALYRKGTIGKAKEHLEATAHRLNKENPSIRAFVSVNKG L VTRASAVIPVIPLYLASLFKVMKEKGNHEGCIEQITRLY  
AERLYRKDGTIPVDEENRIRIDDWELEEDVQKAVSALMEKVTGENAESLTDLAGYRHDFLASNGFDVEGINYEA EVE  
RFDRI

Figure 6A (continued)

**Beta-Ketothiolase (bktB)**

MTREVVVVSGVRTAIGTFGGSLKDVAPAEALGVVREALARAQVSGDDVGHVVFNGVIQTEPRDMYLGRVAAVNG  
GVTINAPALTVNRLCGSGLQAIVSAAQTILLGDTDAIGGGAESMSRAPYLAPAARWGARMGDAGLVDMMMLGALHD  
PFHRIHMGVTAENVAKEYDISRAQQDEAALESHRRASAAIKAGYFKDQIVPVVSKGRKGDVTFDTDEHVRHDATIDD  
MTKLRPVFKENGTVTAGNASGLNDAAAAMMERAEAEERRGLKPLARLVSYGHAGVDPKAMGIGPVPATKIALER  
AGLQVSDLDVIEANEAAQACAVTKALGLDPAKVNPNGSGISLGHPIGATGALITVKALHELNRVQGRYALVTMCIG  
GGQGIAAIFERI

**HMG-CoA Synthase**

MKLSTKLCWCGIKGRLRPQKQQQLHNTNLQMTELKKQKTAEQKTRPQNVGKGIQIYIPTQCVNQSELEKFDGVSQ  
GKYTIGLGQTNMSFVNDREDIYSMSLTVLSKLIKSYNIDTNKIGRLEVGTETLIDKSKSVKSVLMQLFGENTDVEGIDTL  
NACYGGTNALFNSLNWIESNAWDGRDAIVCGDIAIYDKGAARPTGGAGTVAMWIGPDAPIVFDVSRASYMEHAYD  
FYKPDFTSEYPYVDGHFSLTCYVKALDQVYKSYSKKAIKGLVSDPAGSDALNVLYFDYVNFHVPTCKLVTKSYGR  
LLYNDFRANPQLFPEVDAELATRDYDESLTDKNIEKTFVNVAKPFHKERVAQSLIVPTNTGNMYTASVYAAFASLLNY  
VGSDDLQGKRVGLFSYGSGLAASLYSCKIVGDVQHIKELDITNKLAKRITETPKDYEAAILRENAHLKKNFKPQGSIE  
HLQSGVYYLTNIDDKFRRSYDVKK

**Truncated HMG-CoA Reductase**

MVAVRRKALSILAEAPVLASDRLPYKNYDYDRVFGACCENVIGYMPLPVGVIGPLVIDGTSYHIPMATTEGCLVASAM  
RGCKAINAGGGATTVLTKDGMTRGPVVRFTLKRSGACKIWL DSEEGQNAIKKAFNSTSRFARLQHIQTCLAGDLLF  
MRFRTTTGDAMGMNMISKGVESLQKMVEEYGWEDMEVSVSGNYCTDKKPAAINWIEGRGKSVVAEATIPGDVV  
RKVLKSDVSALVELNIAKNLVGSAMAGSVGGFNAHAANLVAVFLALQDPAQNVESNCITLMKEVDGDLRISVSM  
PSIEVGTIGGGTVLEPQGAMLDLLGVRGPHATAPGTNARQLARIVACAVLAGELSLCAALAAGHLVQSHMTHNR

**Mevalonate Kinase**

MSLPFLTAPGKVIFGEHSAVYNKPAVAASVSALRTYLLISESSAPDTIELDFPDISFNHKWSINDFNAITEDQVNSQK  
LAKAQATDGLSQELVSLLDPLLAQLSESFHYHAAFCFLYMFVCLCPHAKNIKFSKSTLPIGAGLGSSASISVSLALA  
MAYLGGLIGSNDLEKLSENDKHIVNQWAFIGEKCIGHGTPSGIDNAVATYGNALLFEKDSHNGTINTNNFKFLDDFPAIP  
MILTYTRIPRSTKDLVARVRVLVTEKFPEVMKPILDAMGECALQGLEIMTKLSKCKGTDDAEVETNNELYEQLELIRIN  
HGLLVSIGVSHPGLELIKNLSDDLRIGSTKLTGAGGGGCSLTLLRRDITQEQIDSFKKKLQDDFSYETFETDLGGTGCC  
LLSAKNLNKDLKIKSLVFQLFENKTTTKQQIDDLLPGNTNLPWTS

**Phosphomevalonate Kinase**

MSELRAFSAPGKALLAGGYLVLDTKYEAFFVGLSARMHAVAHPYGSLLQGSDFEVVRVKSQKFDGEWLYHISPKSG  
FIPVSIIGSKNPFIEKVIANVFSYFKPNMDDYCNRNLFVIDIFSDDAYHSQEDSVTEHRGNRRLSFHSHRIEEVPKTGL  
GSSAGG<sub>[11]</sub>LVTLTALASFFVSDLENNVDKYREVIHNLQAQVHCAQAGKIGSGFDVAAAAYGSIRYRRFPALISNL  
PDIGSATYGSKLAHLVDEEDWNITIKSNHLPSTLWMDIKNGSETVKLVQKVKNWYDSHPESLKIYTELDHANS  
RFMDGLSKLDRLEHETHDDYSDQIFESLERNDCTQKYPEITEVRDAVATIRRSFRKITKESGADIEPPVQTSLLDDCCQ  
TLKGVLTCLIPAGGYDAIAVITKQDVLRAQTANDKRFKSVQWLDVTDQADWGVREKEDPETYLDK

Figure 6A (continued)

**Diphosphomevalonate Decarboxylase**

MTVYTASVTAPVNIATLKYGKRDTKLNLPTNSSISVTL SQDDLRLTSAATAPEFERDTLWLN GEPHSIDNERTQNC  
LRDLRQLRKEMESKDASLPTLSQWKLHIVSENNFPTAAGLASSAAGFAALVSAIAKLYQLPQSTSEISRIARKGSGSA  
CRSLFGGYVAVEMGKAEDGHDSMAVQIADSSDWPQMKACVLVSDIKKDVSSSTQGMQLTVATSELFKERIEHVVP  
KRFEVMRKAIVEKDFATFAKETMMDSNSFHATCLDSFPPIFYMNDTSKRIISWCHTINQFYGETIVAYTFDAGPNAVL  
YYLAENESKLF AFYIKLFGSVPGWDKKFTTEQLEAFNHQFESSNFTARELDLELQKDV ARVILTQVGS GPQETNESLI  
DAKTGLPKE

**Isopentenyl-Diphosphate Delta-Isomerase**

MTADNNSMPHGAVSSYAKLVQNQTPEDILEEFPEIIPLOQRPNTRSSETSND ESGETCFSGHDEEQIKLMNENCIVL  
DWDDNAIGAGTKKVCHLMENIEKGLLHRAFSVFIFNEQGELL LQQRATEKITFPDLWTNTCCSHPLCIDDELGLKGKL  
DDKIKGAIATAVRKLDHELGIPEDETKTRGKFHFLNRIHYMAPSN EPWGEHEIDYILFYKINAKENLTVNPNVNEVRDF  
KVVSPNDLKT MFADPSYKFTPWFKIICENYLFNWWEQ LDDLSEVENDRQIHRML

**Geranyl-Diphosphate Synthase (ERG20<sup>WW</sup>)**

MEAKIDELINNDPVWSSQNESLISKPYNHILLKPGKNFRLNLJVQINRVMNLPKDQLAIVSQIVELLHNSLLIDDIEDNA  
PLRRGQTTSHLIWGPSTINTANYMYFRAMQLVSQLTTKEPLYHWLITIFNEELINLHRGQGLDIYWRDFLPEIIPTE  
MYLNMVMNKTGGFLFRLTLRLMEALSPSSHGHSLVPFINLLGIYQIRDDYLNKDFQMSSEKGAEDITEGKLSFPIV  
HALNFTKTGQTEQHNEILRILLRTSDKDIKLLIQILEFDTNSLAYTKNFINQLVNMIKNDNENKYLPDLASHSDTATN  
LHDELLYIIDHSEL

**Olivetol Synthase**

MNHLRAEGPASVLAIGTANPENILLQDEFDPYYFRVTKSEHMTQLKEKFRKICDKSMIRKRNCF LNEEHLKONPRLVE  
HEMQTL DARQDMLVVEVPKLGKDACA KAIKEWGQPKSKITHLIFTSASTDMPGADYHCAKLLGLSPSVKRVMMYQ  
LGCYGGGTVLRIAKDIAENNKGARVLAVCCDIMACLFRGPSESDLELLVGQAIFGDGAAAVIVGAEPDES VGERPIFE  
LVSTGQTILPNSEG TIGGHIREAGLIFDLHKDVPMLISNNIEKCLIEAFTPIGISDWNSIFWITHPGGKAILDKVEEKLHLK  
SDKFVDSRHLVSEHGNMSSSTVLFVMD ELKRSL EEEGKSTTG DGF EWGVLF GFGPGLTVERVVVRSVPIKY

**Olivetolic Acid Cyclase**

MAVKHLIVLKFKDEITEAQKEEFFKTYVNLVNIIPAMKDVYWGKDVTQKNKEEGYTHIVEVTFESVETIQDYIIHPAHVG  
FGDVYRSFWEKLLIFDYTPRK

**CBGA Synthase**

MGLSSVCTFSFQTNHYHTLLNPHNNNP KTSLLCYRHPKTPIKYSYN NFPSKHCSTKSFHLQNKCS ELSIAKNSIRAAT  
TNQTEPPESDNH SVATKILNFGKACWK LQRPYTIIAFTSCACGLFGKELLHNTNLISWSLMFKAFFFLVAILCIASFTTTI  
NQIYDLHIDRINKPDLPLASGEISVNTAWIMSIAVALFGLIITIKMKG GPLYIFGYCFGIFGGIVYSVPPFRWKQNPSTAF L  
LNFLAHIIITNFTFYASRAALGLPFELRPSFTFLLAFMKSMG SALALIKDASDVEGDTKFGISTLASKYGS RNLTLCFSG  
IVLLSYVAAILAGIWPQAFNSNVMLLSHAILAFWLILQTRDFALTNYDPEAGR RFYEFMWKLYYAEYLVYVFI

Figure 6A (continued)

**Acetyl-CoA Carboxylase**

MSEESLFESSPOKMEYEITNYSERHTELPGHFIGLNTVDKLEESPLRDFVKSHGGHTVISKILIANNGIAAVKEIRSVRK  
WAYETFGDDRTVQFVAMATPEDLEANA EYIRMADQYIEVPGGTNNNNYANVDLIVDIAERADVDVWAGWGHASE  
NPLLPEKLSQSKRKVIFIGPPGNAMRSLGDKISSTIVAQSAKVPCIPWSGTGVDTVHVDEKTLGLVSVDDDIYQKGCCT  
SPEDGLQKAKRIGFPVMIKASEGGGGKIRQVEREEDFIALYHQAANEIPGSPIFIMKLAGRARHLEVQLLADQYGTNI  
SLFGRDCSVQRRHQKIEEAPVTIAKAETFHEMEKAAVRLGKLVGYVSAGTVEYLYSHDDGKFYFLELNPRLQVEHP  
TTEMVSGVNLPAQQLQIAMGIPMHRISDIRTLYGMPNHSASEIDFEFTQDATKKQRRPIPKGHCTACRITSEDNDG  
FKPSGGTLHELNFRSSSNVWGYFSVGNNGNIHSFSDSQGHIFAFGENRQASRKHMVVALKELSIRGDFRTTVEYLI  
KLELETEDFEDNTITTGWLDDLITHKMTAEKPDPTLAVICGAATKAFLASEEARHKYIESLQKGQVL SKDLLQTMFPVDF  
IHGKRYKFTVAKSGNDRYTLFINGSKCDIILRQLSDGGLLAIIGGKSHTIYWKEEVAATRLSVDSMTTLEVENDPQL  
RTPSPGKLVKFLVENGEHIKQPYAEIEVMKMQMPLVSQENGIVQLLKQPGSTIVAGDIMAIMTLDDPSKV KHALPFE  
GMLPDFGSPVIEGTKPAYKFKSLVSTLENILKGYDNQVIMNASLQQLEVLRLNPKLPYSEWKLHISALHSRLPAKLDEQ  
MEELVARSLRRGAVFPARQLSKLIDMAVNPEYNPDKLLGAVVEPLADIAHKYSNGLEAHEHSIFVHFLEEYEEVEKL  
FNGPNVREENIILKLRDENPKDLKVALTVLSHKSVAKNLILAILKHYQPLCKLSSKVS AIFSTPLQHIVELESKATAK  
VALQAREILIQGALPSVKERTEQIEHILKSSVVKVAYGSSNPKRSEPDNLKDLIDSNYVVFVLLQFLTHQDPVVTAA  
AAQVYIRRAYRAYTIGDIRVHEGVTPIVEWKFLPSAAFSTFPTVSKMGMNRAVSVDLSYVANSQSSPLREGILM  
AVDHLDDVDEILSQSLEVIPRHQSSSNGPAPDRSGSSASLSNVANVCVASTEGFESEEEILVRLREILDNLKQELINAS  
IRRTFMFGFKDGSYPKYTYFNGPNYNENETIRHIEPALAFQLELGRLSNFIKPIFTDNRNIHVYEA VSKTSPDKRFF  
TRGIIRTGHIRDDISIQEYL TSEANRLMSDILDNLEVTDTSNSDLNHIFINFI AVFDISPEDVEAAGGFLERFGKRLRLR  
VSSAEIRIIIKDPQTGAPVPLRALINNVSGYVIKTEMYTEVKNAGGEWVFKSLGKPGSMHLRPIATPYPVKEWLQPKRY  
KAHLMGTTYVYDFPELFRQASSSQWKNFSADVKLTDDFFISNELIEDENGELTEVEREPGANAIGMVAFKITVKTPEY  
PRGRQFVVVANDITFKIGSFQPEDEFFNKVTEYARKRIGPRIYLAANS GARIGMAEEIVPLFQVAVNDAANPDKGF  
QYLYLTSEGME TLKKFDKENS VLTERTVINGEERFVIKTIIGSEDLGVECLRGSLIAGATS RAYHDIFTITLVT CRSV  
GIGAYLVRLGQRAIQVEGQPIILTGAPAINKMLGREVYTSNLQLGGTQIMYNNGVSHLTAVDDLAGVEKIVEWMSYVP  
AKRNMPVPILET KDTWDRPVDFPTPTNDETYDVRWMI EGRETESGF EYGLFDKGSFFETLSGWAKGVVGRARLGGI  
PLGVIGVETRTVENLIPADPANPNSAETLIQEPGQVWHPNSAFKTAQAI NDFNNGEQLPMMILANWRGFSGGQORDM  
FNEVLKYGSFIVDALVDYKQPIIYIPTGELRGGSWVVVDPTINADQMEMYADV NARAGVLEPQGMVGIFRREKLL  
DTMNRLLDKYRELRSQLSNKS LAPEVHQISKQLADRERELPIYQGQISLQFADLH DRSRRMVAKGVISKELEWTEA  
RRFFFWRLRRRLNEEYLIKRLSHQVGEASRLKIRRSWYPASVDHEDDRQVATWIEENYKTLDDKLGKLGKLESFA  
QDLAKKIRSDHDNAIDGLSEVIKMLSTDDKEKLLKTLK

**CBDA Synthase**

MKCSTFSFWFVCKIIFFFSFNIQTSIANPRENFKCFSQYIPNNATNLKLVYTQNNPLYMSVLNSTIHNL RFTSDTTPK  
PLVIVTPSHVSHIQGTILCSKKVGLQIRTRSGGHDSEGMSYISQVPFVIVDLRNMRSIKIDVHSQTAWVEAGATLGEVY  
YWVNEKNENLSLAAGYCPTVCAGGHFGGGGYGPLMRNYGLAADNIIDAHLVNVHGKVLDRKSMGEDLFWALRGG  
GAESFGIIVAWKIRLVAVPKSTMFSVKKIMEIHELVLKLVNKWQNIAYKYDKDLLMTHFITR NITDNQGNKTAIHTYFS  
SVFLGGVDSLVDLMNKSFP ELGIKKTDCRQLSWIDTTIFYSGVVNYDTDNFNKEILLDRSAGQNGAFKIKLDYVKKPI  
ESVVFQILEKLYEEDIGAGMYALYPYGGIMDEISESAIPFPHRAGILYELWYICSWEKQEDNEKHLNWIRNIYNFMT  
VSKNPRLAYLNYRDLDIGINDPKPNPNYTQARIWGEKYFGKNFDRLVKVKTLVDPNNFFRNEQSIPPLPRHRH

**CBCA Synthase**

MNCSTFSFWFVCKIIFFFLSFNIQTSIANPQENFLKCFSEYIPNNPANPKFIYTQHDQLYMSVLNSTIQNL RFTSDTTPK  
PLVIVTPSNVSHIQASILCSKKVGLQIRTRSGGHDAEGLSYISQVPFAIVDLRNMHTVKVDIHSQTAWVEAGATLGEVY  
YWINEMNENFSFPGGYCPTVGVGGHFSGGGYGALMRNYGLAADNIIDAHLVNVGDKVLDRKSMGEDLFWAIRGGG  
GENFGIIAACKIKLVVPSKATIFSVKKNMEIHGLVKLFNKWQNIAYKYDKDMLTTHFRTRNITDNHGKNKTTVHGYF  
SSIFLGGVDSLVDLMNKSFP ELGIKKTDCRQLSWIDTTIFYSGVVNYNTANFKKEILLDRSAGKKTAFSIKLDYVKKLPI  
TAMVKILEKLYEEVEGVGMVLYPYGGIMDEISESAIPFPHRAGIMYELWYTATWEKQEDNEKHINWVRSVYNFTTP  
YVSQNPRLAYLNYRDLDLGKTNPESPNNYTQARIWGEKYFGKNFNRLVKVKTKADPNFFRNEQSIPPLPRRH

Figure 6A (continued)

**Hexanoyl-CoA Synthetase**

MGKNYKSLDSVVASDFIALGITSEVAETLHGRLAEIVCNYGAATPQTWINIANHILSPDLPFSLHQMLFYGCYKDFGPA  
PPAWIPDPEKVKSTNLGALLEKRGKEFLGVKYKDPISFSHFQEFVSRNPEVYWRVLMDEMKSFSKDPECILRRD  
DINNPGGSEWLPGGYLNSAKNCLNVNSNKKLNDTMIVWRDEGNDDLPLNKL TLDQLRKRVWLVG YALEEMGLEKG  
CAIAIDMPMHVDAVVIYLAIVLAGYVVVSIADSFSAPEISTRRLRSKAKAIFTQDHIIIRGKKRIPLYSRVVEAKSPMAIVIP  
CSGSNIGAELRDGDISWDYFLERAKEFKNCEFTAREQPVDAYTNILFSSGTTGEPKAIPWTQATPLKAAADGWSHLD  
IRKGDVIVWPTNLGMMMGPWL VYASLLNGASIALYNGSPLVSGFAKFVQDAKVTMLGVVPSIVRSWKSTNCVSGYD  
WSTIRCFSSSGEASNVD EYLWLMGRANYKPVIEMCGGTEIGGAFSAGSFLQAQSLSSFSSQCMGCTLYILDKNGY P  
MPKNKPGIGELALGPVMFGASKTLLNGNHHDVYFKGMPTLNGEVLR RHGDIFELTSNGYYHAHGRADDTMNIGGIKI  
SSIEIERVCNEVD DRVFETTAIGVPPLGGGPEQLVIFVFLKDSNDTTIDLNQLRLSFNLGLQKKLNPLFKVTRV VPLSSL  
PRTATNKIMRRVLRQQFSHF E

MKKVCVIGAGTMGSGIAQAAAKGFEVVLRLDIKDEFVDRGLDFINKNL SKLVKKGKIEEATKVEILTRISGTVDLNMAADCDL  
 VIEAAVERMDIKKQIFADLNDICKPETILASNTSSLSITEVASATKRPDKVIGMHFFNPAPVMKLVLEVIRGIATSQETFDVAVKET  
 SIAIGKDPVEVEAEAPGFVVNRILIPMINEAVGILAEGIASVEDIDKAMKLGANHPMGPLELGDFIGLDICLAIMDVLYSETGDSK  
 YRPHTLKKYVVRAGWLGRKSGKGFYDYSKLLSGGGSGGGSGGGSGAAEWYNLGNAYYKQGDYQKAIEYYQKALEL  
 DPNNAAEWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAAEWYNLGNAYYKQGDYQKAIEYQKALELDPNNLQAEAW  
 KNLGNAYYKQGDYQKAIEYYQKALELDPNNASAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAKAWYRRGNAYYKQ  
 GDYQKAIEDYQKALELDPNNRSRSGAGGGSGGGSGGGSGGASENLYFQGENLYFQGDSSESCWNCGRKASETCSGCNT  
 ARYCGSFCQHKDWEKHHHCIGQTLQAQGSAGSAAGSGSEFGSAEAAAEAAAKAGSAGSAAGSGSEFGSMAVSESOLK  
 KMVSKYKGRDLTVRETENVITIKDLKPKVLDSYVAGNDGSSRELMLNTGTPVYRGNTYNIPICLWLDDTYPNPPICFVKPT  
 SSMTIKTGHVDANGKGYLPYLHEWKHPQSDLLGLIQVMIVVFGDEPPVFSRP

Figure 6B (continued)

**Enoyl-CoA Hydratase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID4**

MELNNVILEKEGKVAVVTINRPKALNALNSDTLKEMDYVIGEIENDSEVLAVILTGAGEKSFVAGADISEMKEMNTIEG  
RKFGILGNKVFRRLLELEKPVIAAVNGFALGGGCEIAMSCDIRIASSNARFGQPEVGLGITPGFGGTQRLSRLVGMGM  
AKQLIFTAQNIKADEALRIGLVNKVVEPSELMNTAKEIANKIVSNAPVAVKLSKQAINRGMQCDIDTALAFESEAFGECE  
STEDQKDAMTAFIEKRKIEGFKNRKLSGGGGSGGGGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALEDPN  
NAEAWYNLGNAYYKQGDYQKAIEYYQKALEDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKALEDPNNLQAEAW  
KNLGNAYYKQGDYQKAIEYYQKALEDPNNASAWYNLGNAYYKQGDYQKAIEYYQKALEDPNNAKAWYRRGNAY  
YKQGDYQKAIEDYQKALEDPNNRSRSAGGGGSGGGGSGGGGASGPLGSPLTASMLASAPPQEQKQMLGERLFP  
LIQAMHPTLAGKITGMLLEIDNSELHMLESPESLRSKVDEAVAVLQAHQAKEAAQKAGSAGSAAGSGEFGSAEAAA  
KEAAAKAGSAGSAAGSGEFGSNTNMSVPTDGAVTTSQIPASEQETLVRPKPLLLKLLKSVGAQKDTYTMKEVLFYLG  
QYIMTKRLYDEKQQHIVYCSNDLLGDLFGVPSFSVKEHRKIYTMIRNLVV

**Trans-Enoyl-CoA Reductase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID5**

MIVKPMVRNNICLNAHPQGCKKGVEDQIEYTKKRITAEVKAGAKAPKNVLVLGCSNGYGLASRITAAGFYGAATIGVS  
FEKAGSETKYGTPGWYNNLAFDEAAKREGLYSVTIDGDAFSDEIKAQVIEEAKKKGIKFDLIVYSLASPVRTDPDTGIM  
HKSVLKPFGKFTFTGKTVDPFTGELKEISAEPADEEAAAATVKVMGGEDWERWIKQLSKEGLLEECCITLAYSIGPEA  
TQALYRKGTIGKAKEHLEATAHRLNKENPSIRAFVSVNKGVLTRASAVIPVIPLYLASLFKVMKEKGNHEGCIEQITRLY  
AERLYRKDGTIPVDEENRIRIDDWELEEDVQKAVSALMEKVTGENAESLTDLAGYRHDFLASNGFDVEGINYEAIVE  
RFDRIKLSGGGGSGGGGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALEDPNNAEAWYNLGNAYYKQGDY  
QKAIEYYQKALEDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKALEDPNNLQAEAWKNLGNAYYKQGDYQKAIEY  
YQKALEDPNNASAWYNLGNAYYKQGDYQKAIEYYQKALEDPNNAKAWYRRGNAYYKQGDYQKAIEDYQKALEL  
DPNNRSRSAGGGGSGGGGSGGGGASSYYHHHHHHLESTSLYKKAGSGSNLLATLRSTAAVLENENHVLEKEKEKL  
RKEKEQLLNKLEAYKGSAGSAAGSGEFGSAEAAAKEAAAKAGSAGSAAGSGEFGSSYYHHHHHHLESTSLYKKAG  
SGSKRIAYLRKKIAALKKDNANLEKDANLENEIERLIKEIKTLENEVASHEQ

**Beta-Ketothiolase (bktB) – Enzyme Linker – cTPR6 Spacer – ID Linker – ID6**

MTREVVVVSGVRTAIGTFGGSLKDVAPAEALGALVVREALARAQVSGDDVGHVVFGNVIQTEPRDMYLGRVAAVNG  
GVTINAPALTVNRLCGSGLQAIVSAAQTILLGDTDVAIGGGAESMSRAPYLAPAARWGARMGDAGLVDMMLGALHD  
PFHRIHMGVTAENVAKEYDISRAQQDEAALESHRRASAAIKAGYFKDQIVPVVSKGRKGDVTFDTHVHRHDATIDD  
MTKLRPVFKENGTVTAGNASGLNDAAA VMMERAEAEERRGLKPLARLVSYGHAGVDPKAMGIGPVPAKIALER  
AGLQVSDLDVIEANEAFAAQACAVTKALGLDPAKVNPNGSGISLGHPIGATGALITVKALHELNRVQGRYALVTMCIG  
GGQGI AIFERIKLSGGGGSGGGGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALEDPNNAEAWYNLGNAY  
YKQGDYQKAIEYYQKALEDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKALEDPNNLQAEAWKNLGNAYYKQGD  
YQKAIEYYQKALEDPNNASAWYNLGNAYYKQGDYQKAIEYYQKALEDPNNAKAWYRRGNAYYKQGDYQKAIED  
YQKALEDPNNRSRSAGGGGSGGGGSGGGGASDVMWEYKWENTGDAELYGPFTSAQMQTWVSEGYFPDGVYC  
RKLDPPGGQFYNSKRIDFDLYTGSAGSAAGSGEFGSAEAAAKEAAAKAGSAGSAAGSGEFGSESDSVEFNNAISYV  
NLIKTRFLDHPEIYRSFLEILHTYQKEQLHTKGRPFGRGMSEEEVFTEVANLFRGQEDLLSEFGQLPEAKR

Figure 6B (continued)

**HMG-CoA Synthase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID7**

MKLSTKLCWCGIKGRLRPQKQQQLHNTNLQMTLKKQKTAEQKTRPQNVGIKGIQIYIPTQCVNQSELEKFDGVSQ  
GKYTIGLGQTNMSFVNDREDIYSMSLTVLSKLIKSYNIDTNKIGRLEVGTETLIDKSKSVKSVLMQLFGENTDVEGIDTL  
NACYGGTNALFNSLNWIESNAWDGRDAIVCGDIAIDYDKGAARPTGGAGTVAMWIGPDAPIVFDSVRASYMEHAYD  
FYKPDFTSEYPYVDGHFSLTCYVKALDQVYKSYSKKAIKGLVSDPAGSDALNVLYFDYNVHFVPTCKLVTKSYGR  
LLYNDFRANPQLFPEVDAELATRDYDESLTDKNIEKTFVNVAKPFHKERVAQSLIVPTNTGNMYTASVYAAFASLLNY  
VGSDDLQGKRVGLFSYSGSLAASLYSCKIVGDVQHHIKELDITNKLAKRITETPKDYEAIELRENAHLKKNFKPQGSIE  
HLQSGVYYLTNIDDKFRRSYDVKKKLSGGGGSGGGGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALELDP  
NNAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKALELDPNNLQAEA  
WKNLGNAYYKQGDYQKAIEYYQKALELDPNNASAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAKAWYRRGN  
AYYKQGDYQKAIEDYQKALELDPNNRSRSAGGGGSGGGGSGGGGASLGPLPPGWVVRSTVSGRIYFVDHNNRTT  
QFTDPRHLHGSAGSAGSGEFGSAEAAAKEAAKAGSAGSAGSAGSGEFGSGAMGPLPPGWEKRTDSNGRVYFVNH  
NTRITQWEDPRS

**Truncated HMG-CoA Reductase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID8**

MVAVRRKALSILAEAPVLASDRLPYKNYDYDRVFGACCENVIGYMPLPVGVIGPLVIDGTSYHIPMATTEGCLVASAM  
RGCKAINAGGGATTVLTKDGMTRGPVVRFTLKRSGACKIWLDSSEGNNAIKKAFNSTSRFARLQHIQTCLAGDLLF  
MRFRTTTGDMGMNMISKGVEYSLKQMVVEEYGWEDMEVVSVSGNYCTDKKPAAINWIEGRGKSVVAEATIPGDVV  
RKVLKSDVSALVELNIAKNLVGSAMAGSVGGFNAAHANLVAVFLALGQDPAQNVESNCITLMKEVDGDLRISVSM  
PSIEVGTIGGGTVLEPQGAMLDLLGVRGPHATAPGTNARQLARIVACAVLAGELSLCAALAAGHLVQSHMTHNRKLS  
GGGGSGGGGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEYY  
QKALELDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKALELDPNNLQAEAWKNLGNAYYKQGDYQKAIEYYQKALE  
LDPNNASAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAKAWYRRGNAYYKQGDYQKAIEDYQKALELDPNNRS  
RSAGGGGSGGGGSGGGGASSYHHHHHHLESTSLYKKAGSEFFRRERNKMAAAKCRNRRRELTDTLQAEQDQLE  
DEKSALQTEIANLLKEKEKLEFILAAHRPACKIPDDLGFPEEMSLEGSAGSAGSAGSGEFGSAEAAAKEAAKAGSAGS  
AGSGEFGSSYHHHHHHLESTSLYKKAGSGSQKVESLKQKIEELKQRKAQLKNDIANLEKEIAYAET

**Mevalonate Kinase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID9**

MSLPFLTSAPGKVIFGEHSAVYNKPAVAASVSALRTYLLISESSAPDTIELDFPDISFNHKWSINDFNAITEDQVNSQK  
LAKAQATDGLSQELVSLDPLLAQLSESFHYHAAFCFLYMFVCLCPHAKNIKFLSKSTLPIGAGLGSSASISVSLALA  
MAYLGGLIGSNDLEKLESENKHIVNQWAFIGEKCIGHTPSGIDNAVATYGNALLFEKDSHNGTINTNNFKFLDDFPAIP  
MILTYTRIPRSTKDLVARVRVLVTEKFPEVMKPILDAMGECALQGLEIMTKLSKCKGTDDEAVETNNELYEQLLELIRIN  
HGLLVSIGVSHPGLELIKNSDDLRIIGSTKLTGAGGGGCSLTLLRRDITQEIQDSFKKKLQDDFSYETFETDLGGTGCC  
LLSAKNLNDKIKSLVQLFENKTTTKQIDDLLPGNTNLPWTSKLSGGGGSGGGGSGGGGSAEAWYNLGNAYY  
KQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQK  
AIEDYQKALELDPNNLQAEAWKNLGNAYYKQGDYQKAIEYYQKALELDPNNASAWYNLGNAYYKQGDYQKAIEYYQ  
KALELDPNNAKAWYRRGNAYYKQGDYQKAIEDYQKALELDPNNRSRSAGGGGSGGGGSGGGGASMEPAMEPET  
LEARINRATNPINKELDWASINGFCEQLNEDFEGPPLATRLAHKIQSPQEWAIQALTVLETCKMCKGKRHFHDEVG  
KFRFLNELIKVSPKYLGSRTSEKVKNKILELLYSWTVGLPEEVKIAEAYQMLKKQGIKSGSAGSAGSAGSGEFGSAEA  
AAKEAAKAGSAGSAGSGEFGSGAMGSMMAEAEESLESWLNKATNPNSNRQEDWEYIIGFCDQINKLEGPQIAV  
RLLAHKIQSPQEWALQALTVLEACMKNCGRRFHNEVGKFRFLNELIKVSPKYLGDVSEKVKTKVIELLYSWTMA  
LPEEAKIKDAYHMLKRQGIVQSDPPPIVDRTLIPSPPPRPKN



Figure 6B (continued)

**Phosphomevalonate Kinase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID10**

MSELRAFSAPGKALLAGGYLVLDTKYEAFFVGLSARMHAVAHPYGSLSQSDKFEVRVKSQKQFKDGEWLYHISPKSG  
FIPVSIIGGSKNPFIEKVIANVFSYFKPNMDDYCNRNLFVIDIFSDDAYHSQEDSVTEHRGNRRLSFHSRIEEVPKTGL  
GSSAGG<sub>[j2]</sub>LVTVLTTALASFFVSDLENNVDKYREVIHNLQAQVHCQAQKGISGFDVAAAAYGSIRYRRFPFALISNL  
PDIGSATYGSKLAHLVDEEDWNITIKSNHLPSTLWMDIKNGSETVKLVQKVKNWYDSDHMPESLKIYTELDHANS  
RFMDGLSKLDRLHETHDDYSDQIFESLERNDCTCQKYPEITEVRDAVATIRRSFRKITKESGADIEPPVQTSLLDDCQ  
TLKGVLTCLIPGAGGYDAIAVITKQDVLRAQTANDKRFSKVQWLDVTQADWGVREKDPETYLDDKLSGGGGSSGG  
GGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEYYQKALELDP  
NNAEAWYNLGNAYYKQGDYQKAIEDYQKALELDPNNLQAEAWKNLGNAYYKQGDYQKAIEYYQKALELDPNNA  
WYNLGNAYYKQGDYQKAIEYYQKALELDPNNAKAWYRRGNAYYKQGDYQKAIEDYQKALELDPNNRSRAGGGG  
SGGGGSGGGGASSYHHHHHHLESTSLYKKAGSGSQVEELKNKIAELENRNAVKNRV AHLKQEIAYLKDELAH  
EFEGSAGSAAGSGEFGSAEAAAKEAAAKAGSAGSAAGSGEFGSSYHHHHHHLESTSLYKKAGSGSFENVTHEFIL  
ATLENENAKLRRLEAKLERELARLRNEVAWL

**Diphosphomevalonate Decarboxylase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID11**

MTVYTASVTAPVNIATLKYWGKRDTKLNLPNTSSISVTLSQDDLRTLSAATAPEFERDTLWLNGEPHSIDNERTQNC  
LRDLRQLRKEMESKDASLPTLSQWKLHIVSENNFPTAAGLASSAAGFAALVSAIAKLYQLPQSTSEISRIARKGSGSA  
CRSLFGGYVAVWEMGKAEDGHDSMAVQIADSSDWPQMKACVLVSDIKKDVSSSQGMQLTVATSELFKERIEHVVP  
KRFEVMRKAIVEKDFATFAKETMDSNSFHATCLDSFPPIFYMNDTSKRIISWCHTINQFYGETIVAYTFDAGPNAVL  
YYLAENESKLFAYIYKLFGSVPGWDDKFTTEQLEAFNHQFESSNFTARELDLELQKDVARVILTQVSGSPQETNESLI  
DAKTGLPKEKLSGGGGSGGGGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYK  
QGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKALELDPNNLQAEAWKNLGNAYYKQGDYQ  
KAIEYYQKALELDPNNASAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAKAWYRRGNAYYKQGDYQKAIEDYQ  
KALELDPNNRSRAGGGGSGGGGSGGGGASAMADLEQKVLMEASTYDGVFIWKISDFPRKRQEA VAGRIPAIFS  
PAFYTSRYGYKMCRLIYLNQDGTGRGTHLSLFFVVMKGPNDALLRWPFNQKVTMLLDQNNREHVIDAFRPDVTSS  
SFQRPVNDMNIAAGCPLFCPVSKMEAKNSYVRDDAIFIKAIVDLTGLGSAGSAAGSGEFGSAEAAAKEAAAKAGSAG  
SAAGSGEFGSASIKLQSSDGEIFEVDVEIAKQSVTIKTMLDLGMDDEGDDDPVPLPNVNAAILKKVIQWCTHHKDDP  
PPPEDDENKEKRTDDIPVWDQEFKVDQGTFLFELILAANYLDIKGLLDVTCKTVANMIKGTPEEIRKTFNIKNDFTTE  
EEAQVRKENQWC

**Isopentenyl-Diphosphate Delta-Isomerase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID12**

MTADNNSMPHGAVSSYAKLVQNQTPEIDILEEFPEIPLQQRPNTRSSSETSNDSEGETCFSGHDEEQIKLMNENCIVL  
DWDDNAIGAGTKKVCHLMENIEKGLLHRAFSVFIFNEQGELLQQRATEKITFPDLWTNTCCSHPLCIDDELGLKGKL  
DDIKIGAITAAVRKLDHELPIPETKTRGKFHFLNRIHYMAPSNPWEHEIDYILFYKINAKENLTVNPNVNEVRDF  
KWWSPNDLKTMFADPSYKFTPWFKIICENYLFNWWQLDDLSEVENDRQIHRMLKLSGGGGSGGGGSGGGGSAE  
AWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGN  
AYYKQGDYQKAIEDYQKALELDPNNLQAEAWKNLGNAYYKQGDYQKAIEYYQKALELDPNNASAWYNLGNAYYKQ  
GDYQKAIEYYQKALELDPNNAKAWYRRGNAYYKQGDYQKAIEDYQKALELDPNNRSRAGGGGSGGGGSGGGGA  
SSYHHHHHHHLESTSLYKKAGSGSNTVKELKNYIQELEERNAELKNLKEHLKFAKAELEFELAAHKFEGSAGSAAGS  
GEFGSAEAAAKEAAAKAGSAGSAAGSGEFGSSYHHHHHHHLESTSLYKKAGSGSQKVAQLKNRVAYKLKENAKLE  
NIVARLENDNANLEKDIANLEKDIANLERDVAR

Figure 6B (continued)

**Geranyl-Diphosphate Synthase (ERG20<sup>WW</sup>) – Enzyme Linker – cTPR6 Spacer – ID Linker – ID13**

MEAKIDELINNDPVWSSQNESLISKPNHILLKPGKNFRLNLIVQINRVMNLPKDQLAIVSQIVELLHNSSLIDDIEDNA  
PLRRGQTTSHLIWGPSTINTANYMYFRAMQLVSQLTTKEPLYHWLITIFNEELINLHRGQGLDIYWRDFLEIIPTE  
MYLNMVMNKTGGLFRLTLRLMEALSPSSHGHSLVPFINLLGIYQIRDDYLNKDFQMSSEKGAEDITEGKLSFPIV  
HALNFTKTKGQTEQHNEILRILLRTSDKDIKLLIQILEFDTNSLAYTKNFQINLVNMIKNDNENKYLPLDASHSDTATN  
LHDELLYIIDHLSSELKLSGGGGSGGGGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLG  
NAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKALELDPNNLQAEAWKNLGNAYYK  
QGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAKAWYRRGNAYYKQGDYQK  
AIEDYQKALELDPNNRSRSAGGGSGGGGSGGGGASLCTMKKGPSGYGFNLHSDKSKPGQFIRSVDPDSPAES  
GLRAQDRIVEVNGVCMGKQHGDDVSAIRAGGDETLLVVDREGSAGSAAGSGEFGSAEAAAKEAAKAGSAGSA  
AGSGEFGSSSGAIYTVELKRYGGPLGITISGTEEPFDPIISSLTKGGLAERTGAIHIGDRILAINSSSLKGKPLSEAIHLL  
QMAGETVTLKIKKQTDAPASS

**Olivetol Synthase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID14**

MNHLRAEGPASVLAIGTANPENILLQDEFDYYFRVTKSEHMTQLKEKFRKICDKSMIRKRNCFLNEEHLKQNPRLVE  
HEMQTLARQDMLVVEVPKLGDACAKAIKEWGQPKSKITHLIFTSASTDMPGADYHCAKLLGLSPSVKRVMMYQ  
LGCYGGGTVLRIAKDIAENNKGARVLAVCCDIMACLFGRPSESLELLVGQAIFGDGAAAVIVGAEPDESVERPIFE  
LVSTGQTILPNSEGTTGGHIREAGLIFDLHKDVPMLISNIEKCLIEAFTPIGISDWSIFWITHPGGKAILDKVEEKLHLK  
SDKFVDSRHLVSEHGNMSSSTVLFVMDLKRKRSLEEGKSTTGDFEWGVLFGFGPGLTVERVVVRSVPKIKYKLSGG  
GGSGGGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEYYQKA  
LELDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKALELDPNNLQAEAWKNLGNAYYKQGDYQKAIEYYQKALELDP  
NNASAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAKAWYRRGNAYYKQGDYQKAIEDYQKALELDPNNRSRS  
GGGGSGGGSGGGGASGNNLETYEWYNKSISRDKAEKLLDGTGKEGAFMVRDSRTPGTYTVSVFTKAIISENPCIK  
HYHIKETNDSPKRYVYAEKYVFDSPILLIQYHQYNGGGLVTRLRYPVCSSAGSAGSAGSGEFGSAEAAAKEAAKAG  
SAGSAGSGEFGSGSHPWFFGKIPRAKAEMLSKQRHDGAFIRESSESAPGDFSLSVKFGNDVQHFVLRDAGAK  
YFLWVVKFNSLNLVDYHRSTSVSRNQIFLRDIEQVPQQPT

**Olivetolic Acid Cyclase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID15**

MAVKHLIVLKFDEITEAQKEEFFKTYVNLVNIIPAMKDVYWGKDVTKNKEEGYTHIVEVTFESVETIQDYIIHPAHVG  
FGDVYRSFWEKLLIFDYTPRKLSSGGGGSGGGGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNA  
EAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKALELDPNNLQAEAWKN  
LGNAYYKQGDYQKAIEYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAIEYYQKALELDPNNAKAWYRRGNAYYK  
QGDYQKAIEDYQKALELDPNNRSRSAGGGSGGGGSGGGGASGQDRSEATLIKRFKGEVRYKAKLIGIDEVSAA  
RGDKLCQDSMMKLKGVVAGARSKGEHKQKIFLTISFGGKIFDEKTGALQHHAHVHEISYIAKDITDHRAFGYVCGKE  
GNHRFVAIKTAQAAEPVILDRDLFQLIYELKQREELKAGSAGSAGSGEFGSAEAAAKEAAKAGSAGSAGSAGS  
EFGSGSHMGSQFWLTSQKTEASERCGLQGSYILRVEAKLTLLTLGAQSQILEPLLFWPYTLRRYGRDKVMFSFE  
AGRRCPSGPGTFTFQTSQGNDFQAVEAAIQQKKAQKVGQAQDILREHHHHHHH

Figure 6B (continued)

**CBGA Synthase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID16**

MGLSSVCTFSFQNTNYHTLLNPHNNNPKTSLLCYRHPKTPIKYSYNNFSPSKHCSTKSFHLQNKCESELSIAKNSIRAAT  
TNQTEPPESDNHVSATKILNFGKACWKLQRPYTIIFTSCACGLFGKELLHNTNLISWSLMFKAFFFLVAILCIASFTTTI  
NQIYDLHIDRINKPDLPLASGEISVNTAWIMSIIVALFGLIITIKMKGGPLYIFGYCFGIFGGIVYSVPPFRWKQNPSTAF  
LNFLAHIITNFTFYASRAALGLPFELRPSFTFLAFMKSMGSAALALIKDASDVEGDTKFGISTLASKYGSRNLTFCSG  
IVLLSYVAAILAGHWPOAFNSNVMLLSHAILAFWLILQTRDFALTNYDPEAGRFRFYEFMWKLYYAEYLVYVFIKLSGGG  
GSGGGGSGGGGSAEAWYNLGNAYYKQGDYQKAIEYYQKALEDPNNAEAWYNLGNAYYKQGDYQKAIEYYQKAL  
ELDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKALEDPNNLQAEAWKNLGNAYYKQGDYQKAIEYYQKALEDPN  
NASAWYNLGNAYYKQGDYQKAIEYYQKALEDPNNAKAWYRRGNAYYKQGDYQKAIEDYQKALEDPNNRSRSAG  
GGGSGGGGSGGGGSAEYVRALDFDNGNDEEDLPFKKGDIIRIDKPEEQWVNAEDSEGKRGMPVPYVEKYGS  
AGSAAGSGEFGSAEAAAKEAAKAGSAGSAAGSGEFGSLIKHMAEALDFTGNSKLELNFKAGDVIFLLSRINKDW  
LEGTVRGATGIFPLSFVKILK

**Acetyl-CoA Carboxylase – Enzyme Linker – cTPR6 Spacer – ID Linker – ID17**

MSEESLFESSPQKMEYEITNYSERHTELPGHFIGLNTVDKLEESPLRDFVKS HGGHTVISKILIANNGIAAVKEIRSVRK  
WAYETFGDDRTVQFVAMATPEDLEANA EYIRMADQYIEVPGGTNNNNYANVDLIVDIAERADVDVWAGWGHASE  
NPILLPEKLSQSKRKVIFIGPPGNAMRSLGDKISSTIVAQSAKVPCIPWSGTGVDTVHVDEKTGLVSVDDDIYQKGCCT  
SPEDGLQKAKRIGFPVMIKASEGGGKGIRQVEREEDFIALYHQAANEIPGSPIFIMKLARARHLEVQLLADQYGTNI  
SLFGRDCSVQRRHQKIIIEAPVTIAKAETFHEMEKAAVRLGKLVGYVSAGTVEYLYSHDDGKFYFLELNPRLQVEHP  
TTEMVSGVNLPAQAQLQIAMGIPMHRISDIRTLYGMNPHSAEIDFEFKTQDATKKQRRPIPKGHCTACRITSEDPNDG  
FKPSGGTLHELNFRSSSNVWGYFSVGNNGNIHSFSDSQFHIFAFGENRQASRKHMVVALKELSIRGDFRTTVEYLI  
KLETEDEFEDNTITTTGWLDDLITHKMTAEKPDPTLAVICGAATKAFLASEEARHKYIESLQKGQVLSKDLLQTMFPVDF  
IHGKRYKFTVAKSGNDRYTLFINGSKCDIILRQLSDGGLLIAGGKSHTIYWKEEVAATRLSVDSMTTLLEVENDPTQL  
RTPSPGKLVKFLVENGEHIKGGQPYAEIEVMKMQMPLVSEQENGIVQLLKQPGSTIVAGDIMAIMTLDDPSKV/KHALPFE  
GMLPDFGSPVIEGTPAYKFKSLVSTLENILKGYDNQVIMNASLQQLIEVLRNPKLPYSEWKLHISALHSRLPAKLEDEQ  
MEELVARSLRRGAVFPARQLSKLIDMAVKNPEYNPDKLLGAVVEPLADIAHKYSNGLEAHEHSIFVHFLEEYVEVEKL  
FNGPNVREENIILKLRDENPKDLKVALTVLSHKSVAKNLILAILKHYQPLCKLSSKVS AIFSTPLQHIVELESKATAK  
VALQAREILIQGALPSVKERTEQIEHILKSSVVKVAYGSSNPKRSEPDNLKDLIDSNYVVDVLLQFLTHQDPVVTAA  
AAQVYIRRAYRAYTIGDIRVHEGVTPIVWKFQLPSAAFSTFTVKS KMGMNRAVSVDLSYVANSQSSPLREGILM  
AVDHLDDVDEILSQSLEVIPRHQSSSNGPAPDRSGSSASLSNVANVCVASTEGFESEEEILVRLREILDNLKQELINAS  
IRRITFMFGFKDGSYPKYTTFNPNYENETIRHIEPALAFQLELGRLSNFIKPIFTDNRNIHVEAVSKTSPDKRFF  
TRGIHRTGHIRDDISIQEYLTSEANRLMSDILDNLEVTDTSNLDNLHIFINFIADVFDISPEDVEAAGGFLERFGKRLRLR  
VSSAEIRIIKDPQTGAPVPLRALINNVSGYVIKTEMYTEVKNAGGEWVFKSLGKPGSMHLRPIATPYPVKEWLQPKRY  
KAHLMGTTYVYDFPELFRQASSSQWKNFSADVKLTDDFFISNELIEDENGELTEVEREPGANAIGMVAFKITVKTPEY  
PRGRQFVVVANDITFKIGSFQPEDEFFNKVTEYARKRGIPRIYLAANS GARIGMAEEIVPLFQVAVNDAANPDKGF  
QYLYLTSEGME TLKKFDKENS VLTERTVINGEERFVIKTIIGSEDGLGVECLRGSGLIAGATSRAYHDIFTITLVTCRSV  
GIGAYLVRLGQRAIQVEGQPIILTGAPAINKMLGREVYTSNLQLGGTQIMYNNGVSHLTAVDDLAGVEKIVEWMYSYVP  
AKRNMPPVPILETKDTWDRPVDFPTNDETYDVRWMIEGRETESGFYGLFDKGSFFETLSGWAKGVVGRARLGGI  
PLGVIGVETRIVENLIPADPANPNSAETLIQEPGQVWHPNSAFKTAQAINDFNNGEQLPMMILANWRGFSGGQRDM  
FNEVLKYGSFIVDALVDYKQPIIYIPPTGELRGGSWVVVDPTINADQMEMYADV NARAGVLEPQGMVGIFRREKLL  
DTMNRLLDDKYRELSQLSNKSLAPEVHQISKQLADRERELPIYQGISLQFADLHDSRRMVAKGVISKELEWTEA  
RRFFFWRRLRRRLNEEYLIKRLSHQVGEASRLKIEARIRSWYPASVDHEDDRQVATWIEENYKTLDDKLGKLESFA  
QDLAKKIRSDHDNAIDGLSEVIKMLSTDDEKLLKTLKLSGGGSGGGGSGGGGSAEAWYNLGNAYYKQGDYQK  
AIEYYQKALEDPNNAEAWYNLGNAYYKQGDYQKAIEYYQKALEDPNNAEAWYNLGNAYYKQGDYQKAIEDYQKAL  
LEDPNNLQAEAWKNLGNAYYKQGDYQKAIEYYQKALEDPNNASAWYNLGNAYYKQGDYQKAIEYYQKALEDP  
NNAKAWYRRGNAYYKQGDYQKAIEDYQKALEDPNNRSRSAGGGGSGGGGSGGGGSGGSHMR LGAQSIQPTAN  
LDRTDDLVLVNMELVRVLELKNELAQLPPEGYVVVKNVGLTLRKLIGSVDDLPLSPSSSRTEIEGTQKLLNKDLA  
ELINKMR LAQQNAVTSLS ECKRQMLTASHTLAVDAKNLLDAVDQAKVLANLAHPPAEGSAGSAGSGEFGSAEAA  
AKEAAAKAGSAGSAAGSGEFGSGAMATPGSENVLPREPLIATAVKFLQNSRVRSPLATRR AFLKKKGLTDEEIDM  
AFQQSGTAAD EPPSSLW

Figure 6C

**Cannabinoidergic Metabolon Scaffold – (Myc)<sub>3</sub>**

MGSAGSAAGSGEFGSAGSAAGSGEFGSAGSAAGSGEFSYYHHHHHHLESTSLYKKAGSGSARNAYLRKKIA  
RLKKDNLQLERDEQNLEKIIANLRDEIARLENEVASHEQGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSG  
EFSYYHHHHHHLESTSLYKKAGSGSNLVAQLENEVASLENENETLKKKNLHKKDLIAYLEKEIANLRKKIEEGSA  
GSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSSYYHHHHHHLESTSLYKKAGSGSQ  
KVAELKNRVAVKLNREQLKNKVEELKNRNAYLKNELATLENEVARLENDVAEGSAGSAAGSGEFAEAAAKEA  
AAKAGSAGSAAGSGEFSYYHHHHHHLESTSLYKKAGSGSNEVTTLENDA AF IENENAYLEKEIARLRKEKAALR  
NRLAHKKGSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSRPPTISNPPPLISSAK  
HPSVGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFNFLQSRPEPTAPPEESFRSGGSAGSAAGSGE  
FGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSSKGTGLNPNKVVQEIAPGNGSAGSAAGSGEF  
AEAAAKEAAAKAGSAGSAAGSGEFGPDGTTFEHLWSSLEPDSTYGSAGSAAGSGEFGSAEAAAKEAAAKEA  
AAKEAAAKAGSAGSAAGSGEFGSSYYHHHHHHLESTSLYKKAGSGSKRIAYLRKKIAALKKDANLEKDIANLE  
NEIERLIKEIKTLENEVASHEQGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFSYYHHHHHHLESTSLY  
KKAGSGSNLLATLRSTAAVLENENHVLEKEKEKLKEKEQLLNKLEAYKGSAGSAAGSGEFGSAEAAAKEAAA  
KEAAAKEAAAKAGSAGSAAGSGEFGSPATSOHPPPPPGHRSQAPSHGSAGSAAGSGEFAEAAAKEAAAKAG  
SAGSAAGSGEFELNSLLILLEAAEYLERDRGSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSA  
AGSGEFGSRPPTISNPPPLISSAKHPSVGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFNFLQSRPEP  
TAPPEESFRSGGSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSSKGTGLNPNK  
KVVQEIAPGNGSAGSAAGSGEFAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGPDGTTFEHLWSSLEPDSTYGSAGS  
AAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSSYYHHHHHHLESTSLYKKAGSGSKRIA  
YLRKKIAALKKDANLEKDIANLENEIERLIKEIKTLENEVASHEQGSAGSAAGSGEFAEAAAKEAAAKAGSAGS  
AAGSGEFSYYHHHHHHLESTSLYKKAGSGSNLLATLRSTAAVLENENHVLEKEKEKLKEKEQLLNKLEAYKG  
SAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSALVDDAADYEPSPSNNEEALGSA  
GSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFRELFDDPSYVNVQNLDKARQGSAGSAAGSGEFGSAEAA  
AKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSKNTKSMNFDNPVYRKTTTEEGSAGSAAGSGEFAEAAAKE  
AAAKAGSAGSAAGSGEFGSRSLPSTWIENKLYGMSDPNWGSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAA  
KAGSAGSAAGSGEFGSVVDNSPPPALPPKKRQSAPSGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGE  
FTQRSKPQPAVPPRPSADLILGSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGST  
DEEREETEEEVYLLNSTTLGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFGDGNVSGTQRLDSATVRT  
YSCGSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSSYYHHHHHHLESTSLYKK  
AGSGSQKVAQLKNRVAYKLKENAKLENIVARLENDNANLEKDIANLEKDIANLERDVARGSAGSAAGSGEFAE  
AAAKEAAAKAGSAGSAAGSGEFSYYHHHHHHLESTSLYKKAGSGSNTVKELKNYIQELEERNAELKNLKEHLK  
FAKAELEFELAHHKFEGSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSHDDSLP  
HPQQATDDSGHESDGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFGSPNAGSVEQTPKKPGLRRR  
GSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSSYYHHHHHHLESTSLYKKAGS  
GSFENVTHEFILATLENENAKLRRLEAKLERELARLRNEVAWLGSAGSAAGSGEFAEAAAKEAAAKAGSAGSA  
AGSGEFSYYHHHHHHLESTSLYKKAGSGSQKVEELKNKIAELENRNNAVKKNRVAHLKQEIAYLKDELAHEFE  
GSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSVSSTKLVSFHDDSDDELLHIGS  
AGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFAATPISTFHDDSDDELLHVGSAAGSGEFGSAEAA  
AKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSSYYHHHHHHLESTSLYKKAGSGSQKVESLKQKIEELKQK  
AQLKNDIANLEKEIAYAETGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFSYYHHHHHHLESTSLYKK  
AGSEFFRRERNKMAAAKCRNRRELTDTLQAETDQLEDEKSALQTEIANLLKEKEKLEFILAAHRPACKIPDDL  
GFPEEMSLEGSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSFQMPADTPPPAY  
LPPEDPMTGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFERESNEEPPPPYEDPYWGNNGGSAGSA  
AGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSSYYHHHHHHLESTSLYKKAGSGSQKVA  
ELKNRVAVKLNREQLKNKVEELKNRNAYLKNELATLENEVARLENDVAEGSAGSAAGSGEFAEAAAKEAAAK

**Figure 6C (continued)**

AGSAGSAAGSGEFSYYHHHHHHLESTSLYKKAGSGSNEVTTLENDAAFIENENAYLEKEIARLRKEKAALRNRL  
AHKKSYYHHHHHHLESTSLYKKAGSGSARNAYLRKKIARLKKDNLQLERDEQNLEKIIANLRDEIARLENEVASH  
EQGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFSYYHHHHHHLESTSLYKKAGSGSNLVAQLENEV  
ASLENENETLKKKNLHKKDLIAYLEKEIANLRKKIEEGSAGSAAGSGEFGSAEAAAKEAAAKEAAAKEAAKAG  
SAGSAAGSGEFGSEQKLISEEDLEQKLISEEDLEQKLISEEDLGSAGSAAGSGEFGSAGSAAGSGEFGSAGSA  
AGSGEF

**Figure 6D****Malonyl-CoA Metabolon Scaffold – (FLAG)<sub>3</sub>**

MGSAGSAAGSGEFGSAGSAAGSGEFGSAGSAAGSGEFSYYHHHHHHLESTSLYKKAGSGSARNAYLRKKIA  
RLKKDNLQLERDEQNLEKIIANLRDEIARLENEVASHEQGSAGSAAGSGEFAEAAAKEAAAKAGSAGSAAGSG  
EFSYYHHHHHHLESTSLYKKAGSGSNLVAQLENEVASLENENETLKKKNLHKKDLIAYLEKEIANLRKKIEEGSA  
GSAAGSGEFGSAEAAAKEAAAKEAAAKEAAAKAGSAGSAAGSGEFGSSATRELDLMASLSDFKIQGGSAGS  
AAGSGEFAEAAAKEAAAKAGSAGSAAGSGEFDLALSENWAQEFLLAAGDAVDGSAGSAAGSGEFGSAEAAAK  
EAAAKEAAAKEAAAKAGSAGSAAGSGEFGSDYKDDDDKDYKDDDDKDYKDDDDKGSAGSAAGSGEFGSAG  
SAAGSGEFGSAGSAAGSGEF

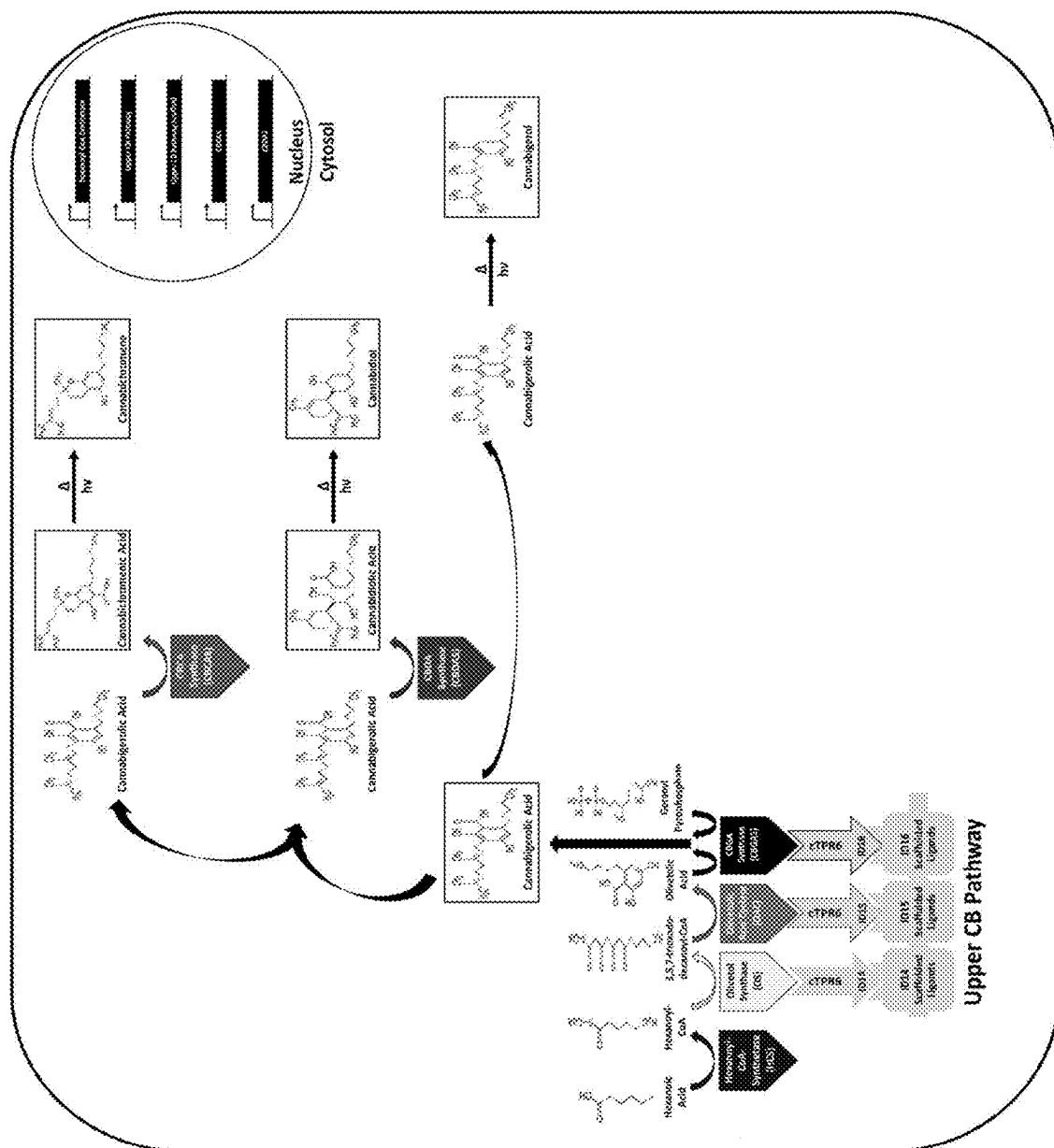


Figure 7.

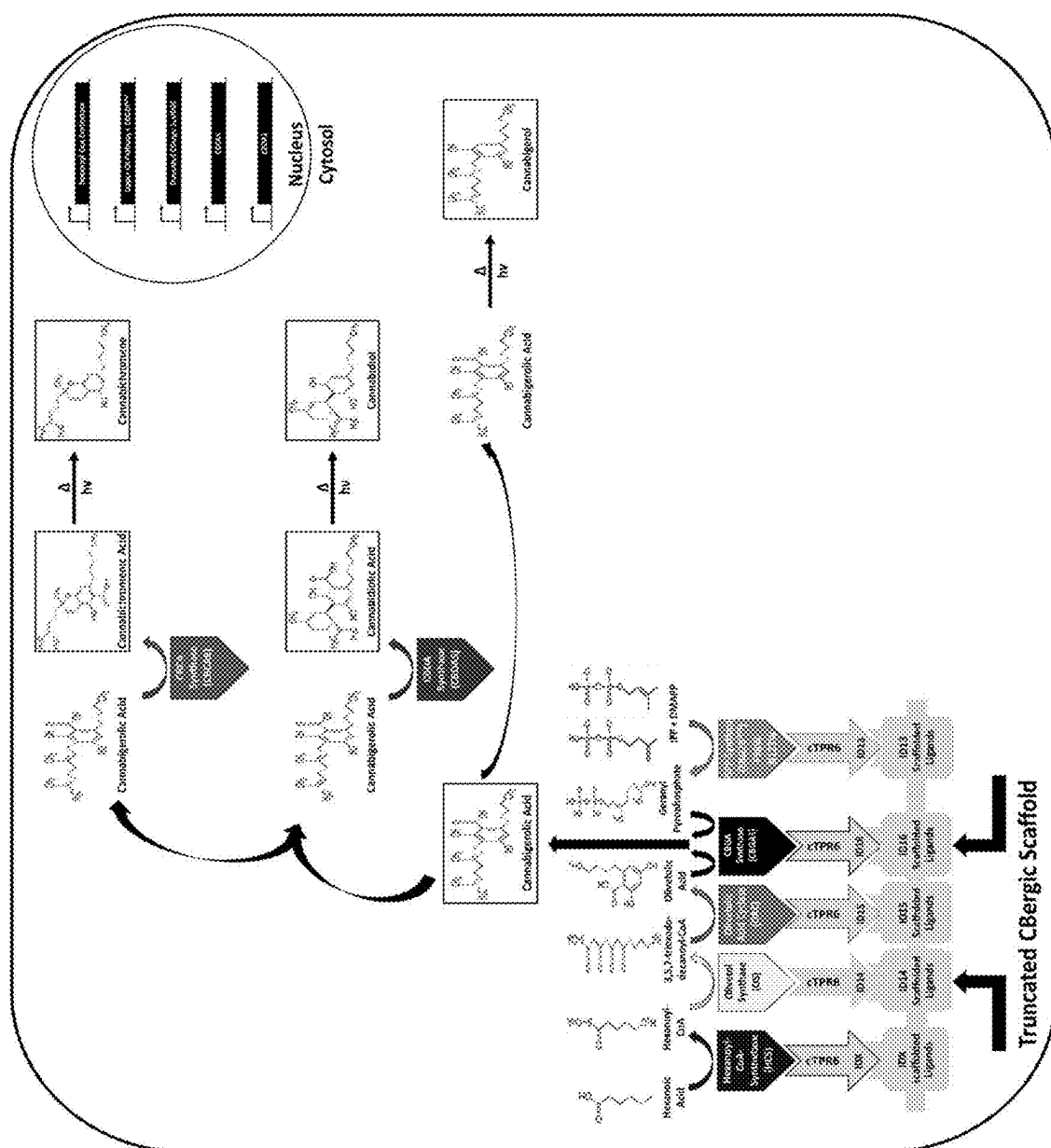


Figure 8.

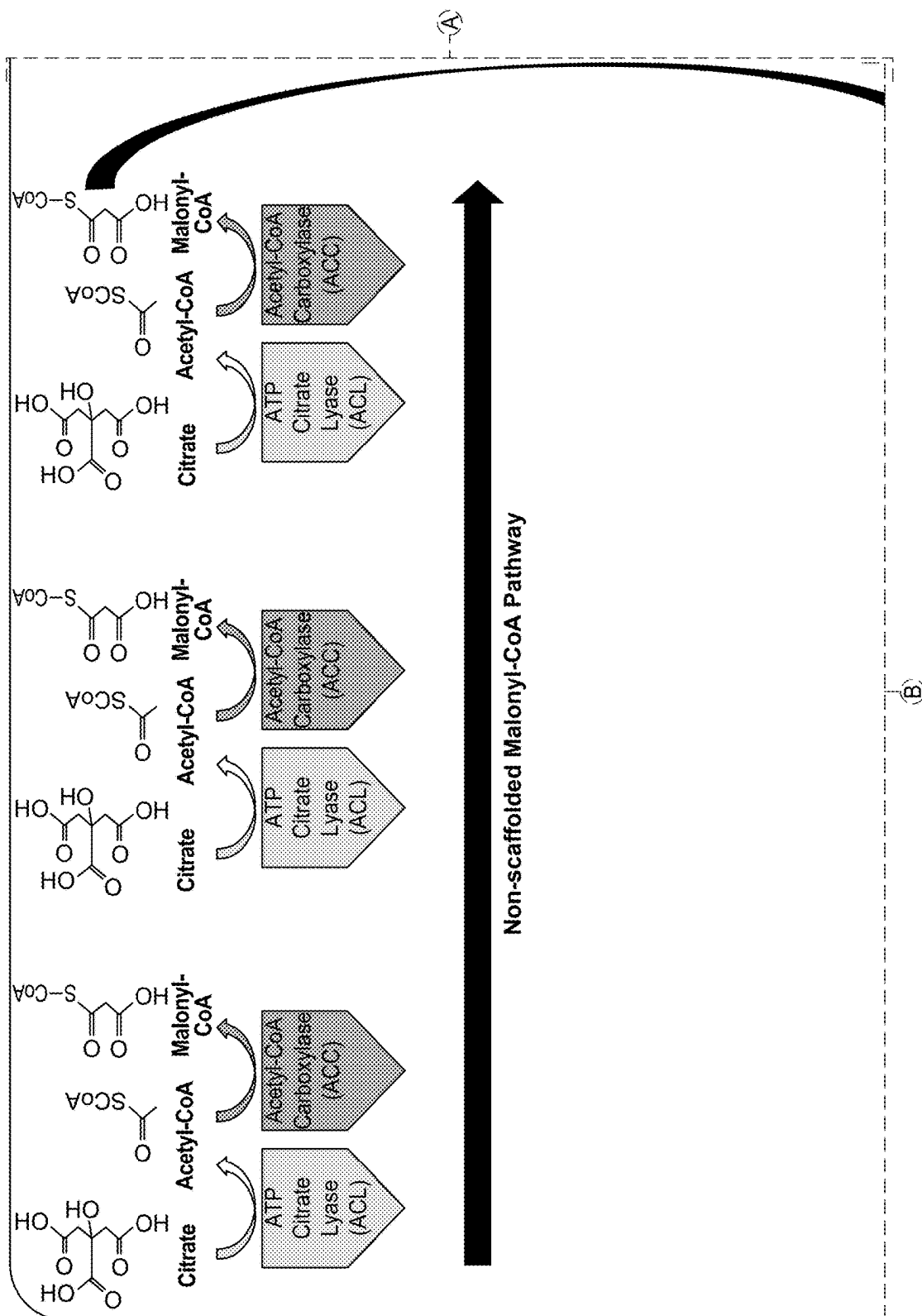


Figure 9.



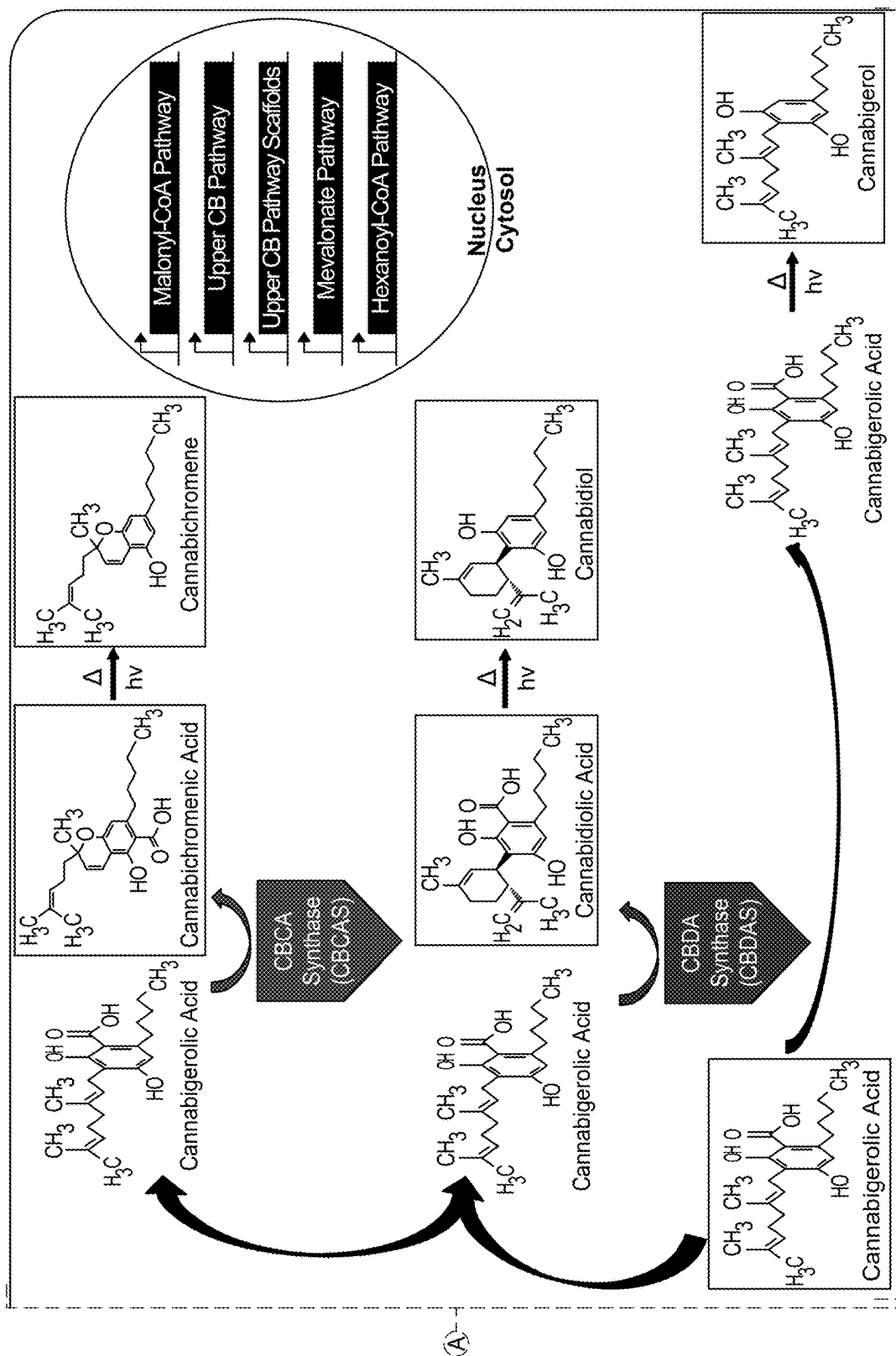


Figure 9. (Cont'd)

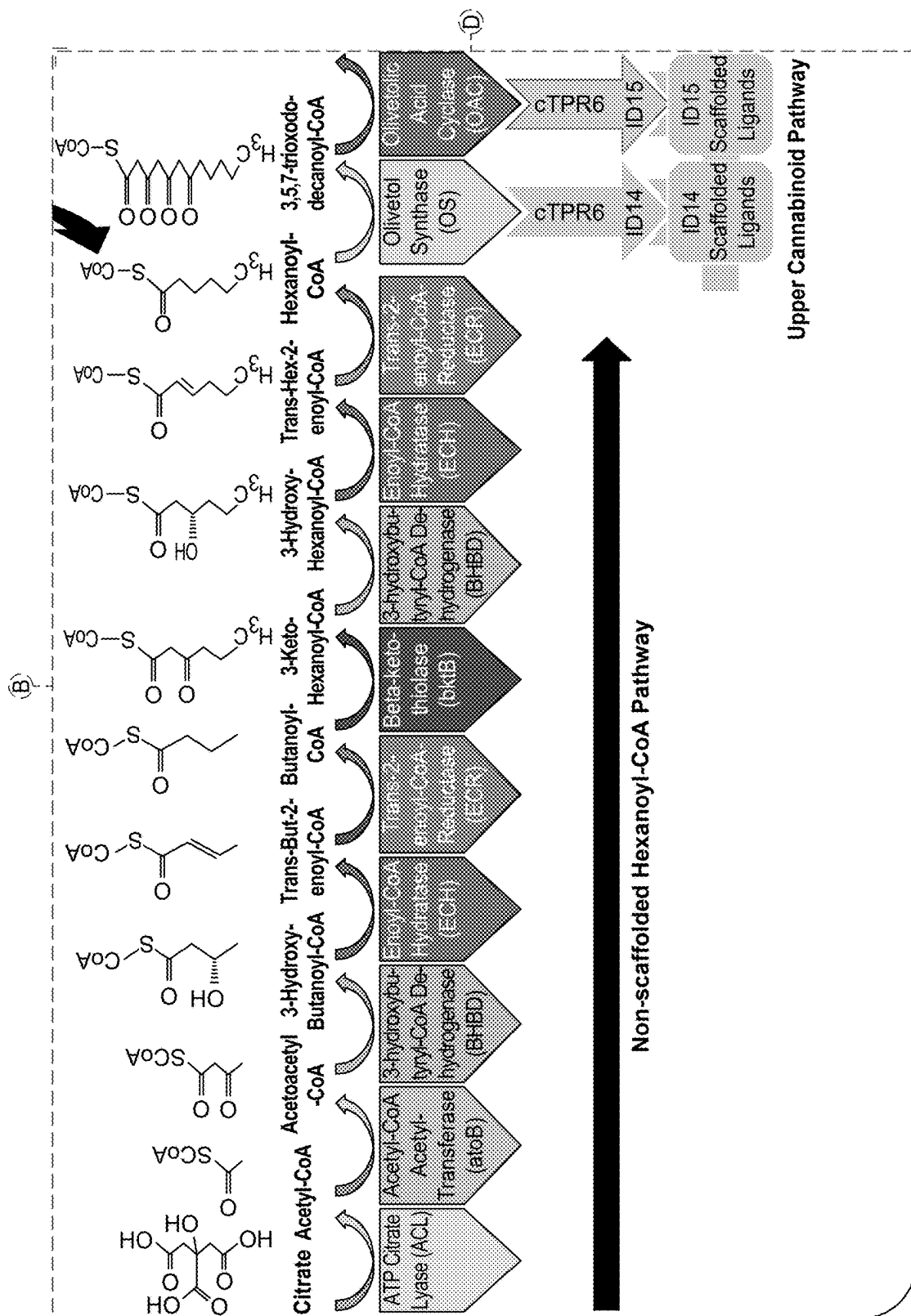


Figure 9. (Cont'd)

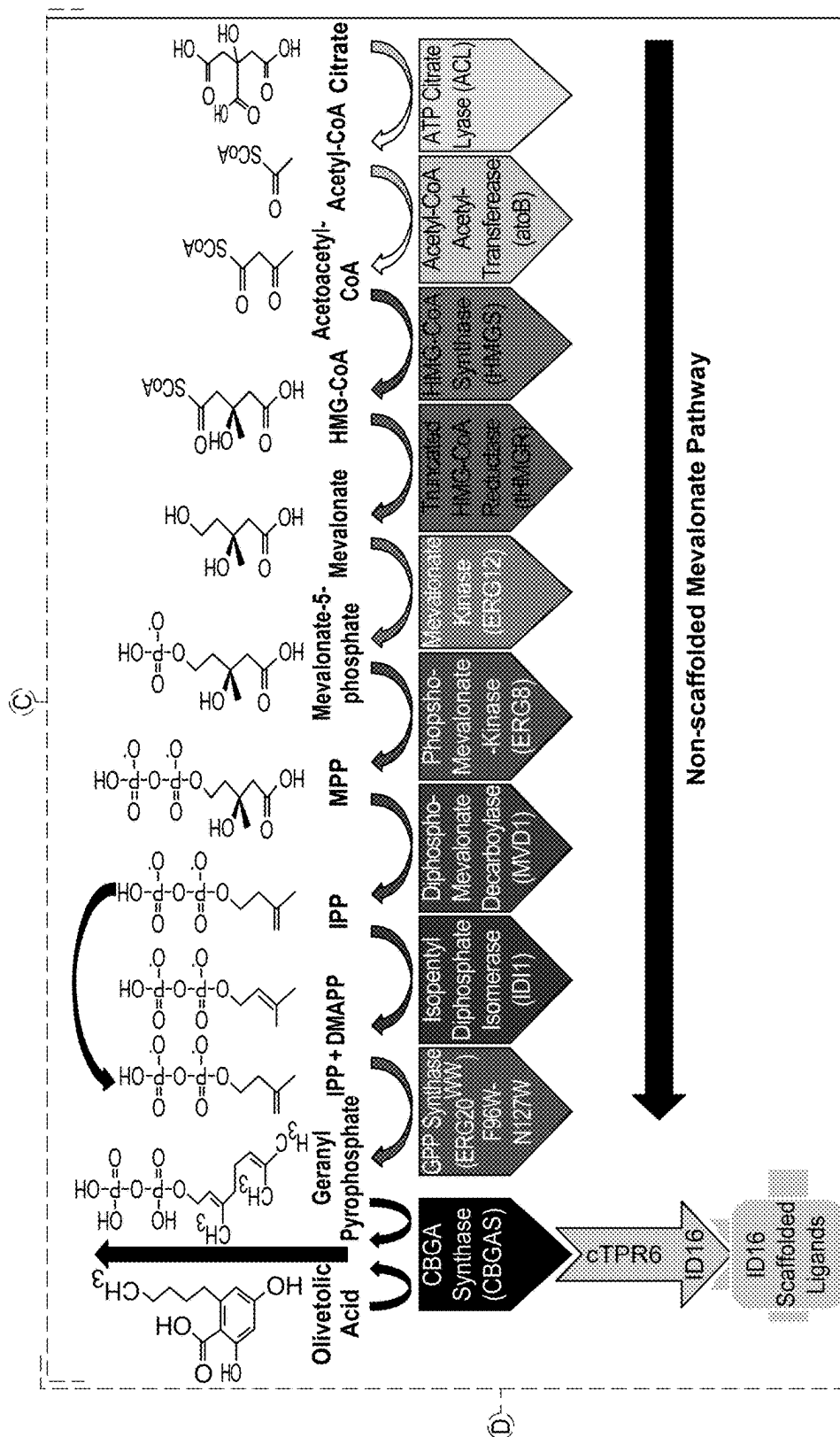


Figure 9. (Cont'd)

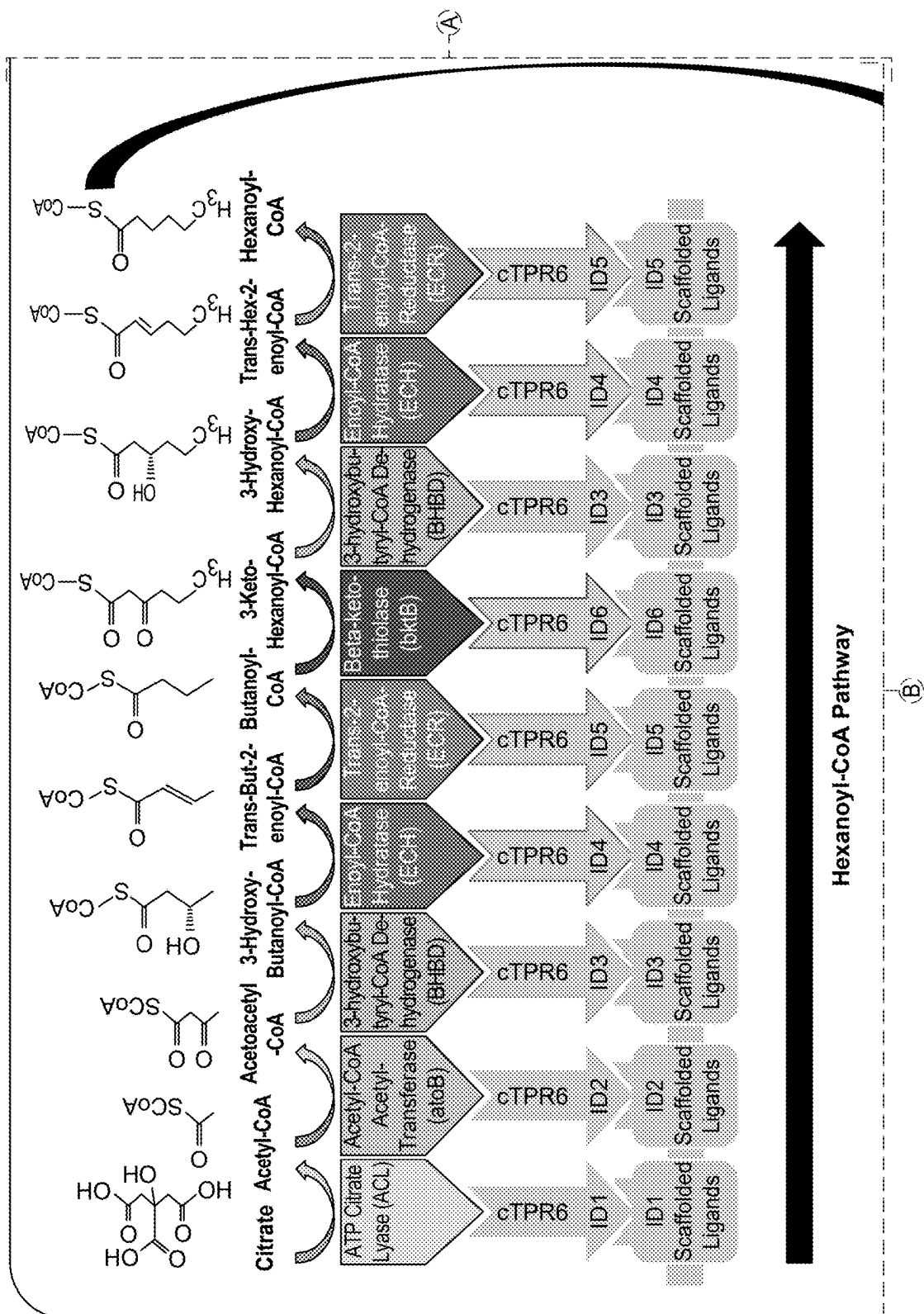


Figure 10.

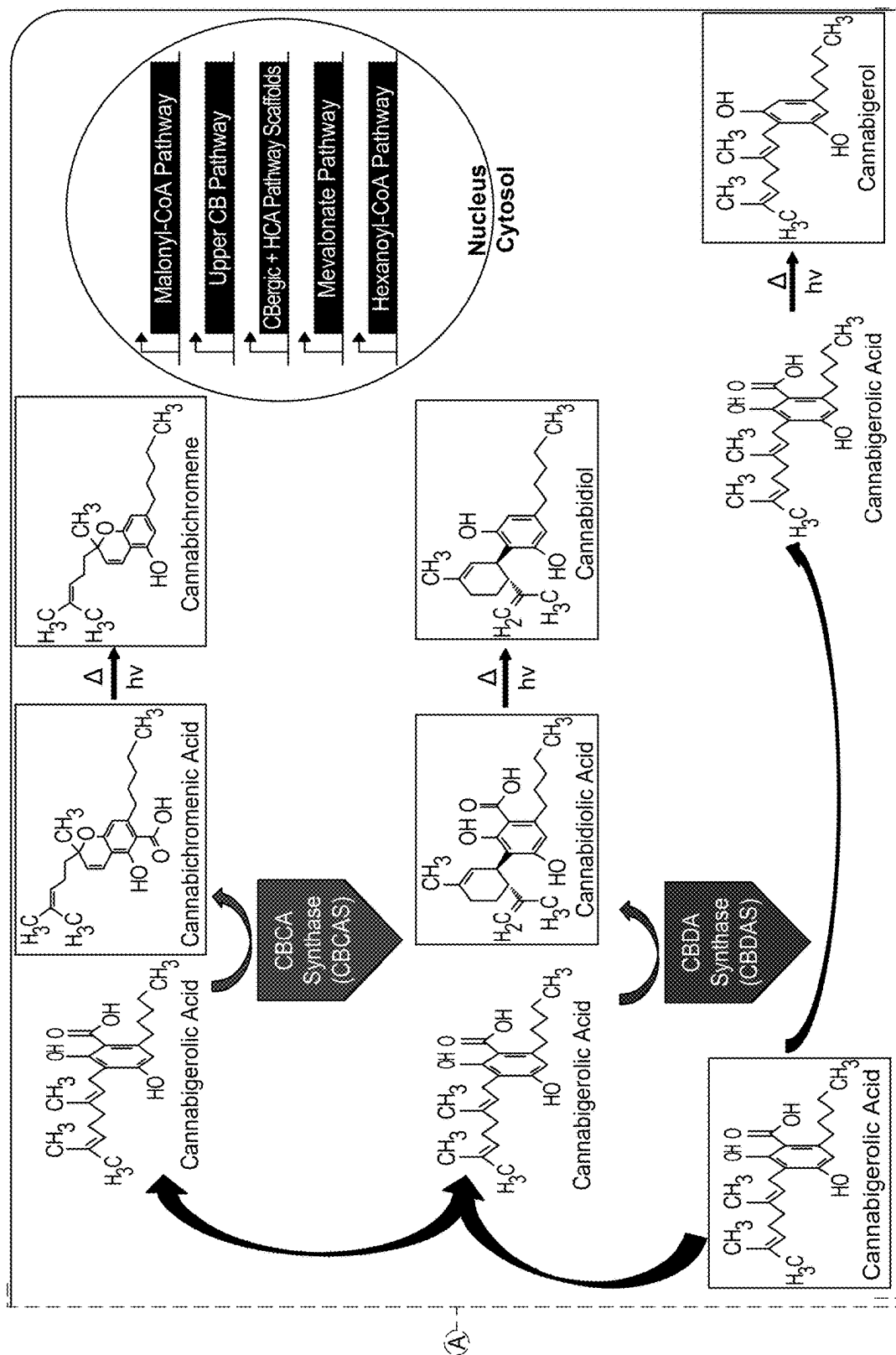
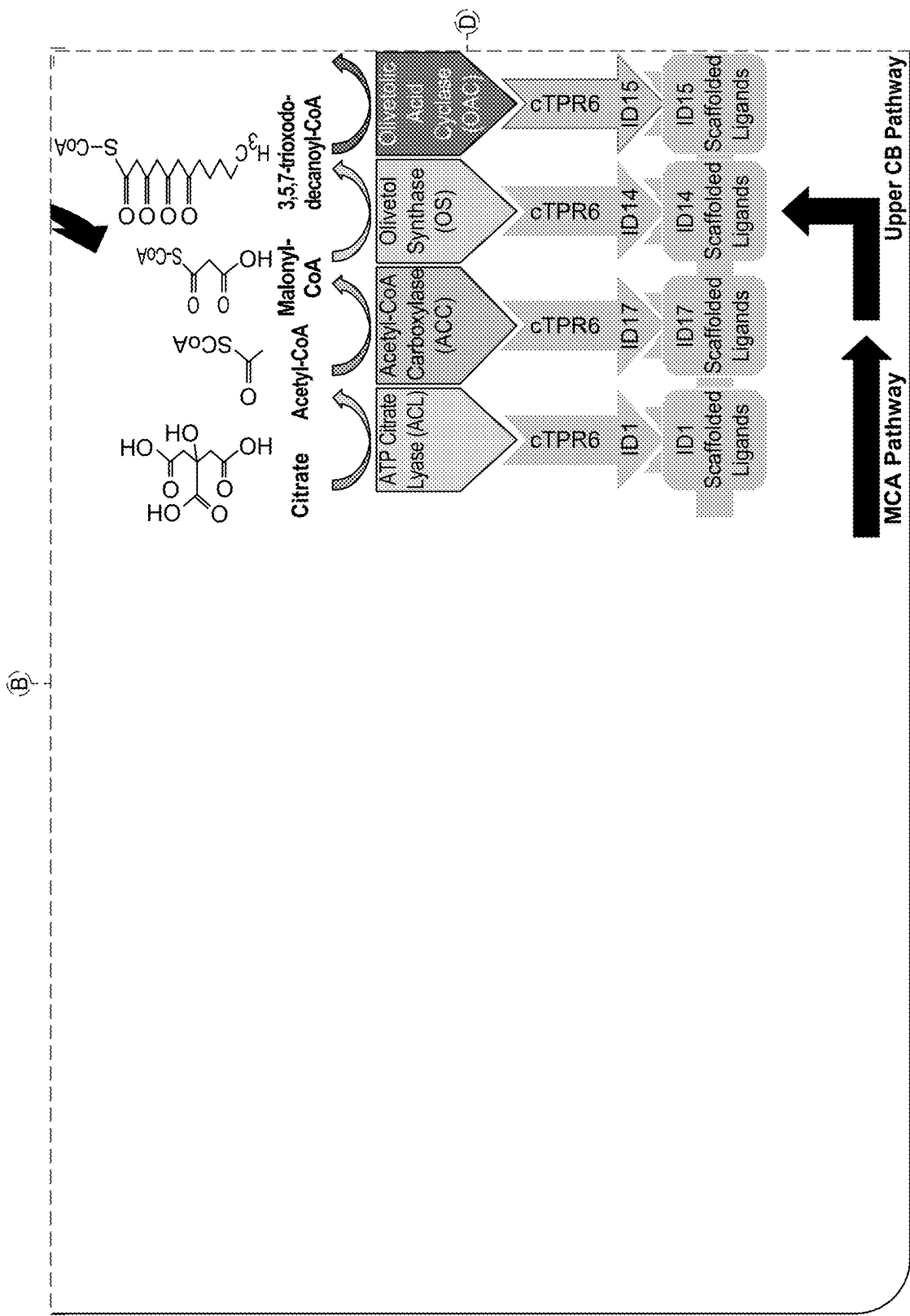


Figure 10. (Cont'd)



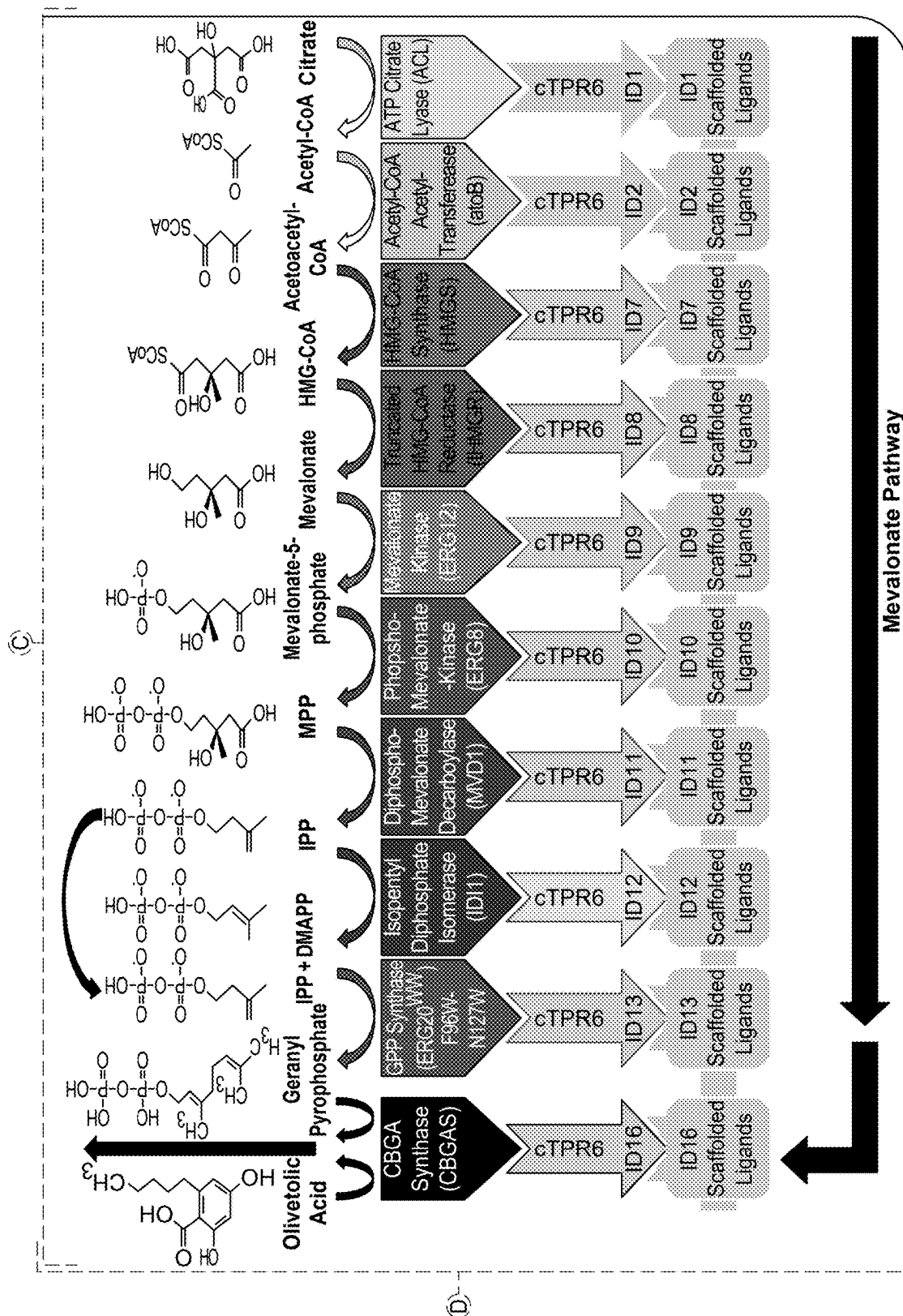


Figure 10. (Cont'd)

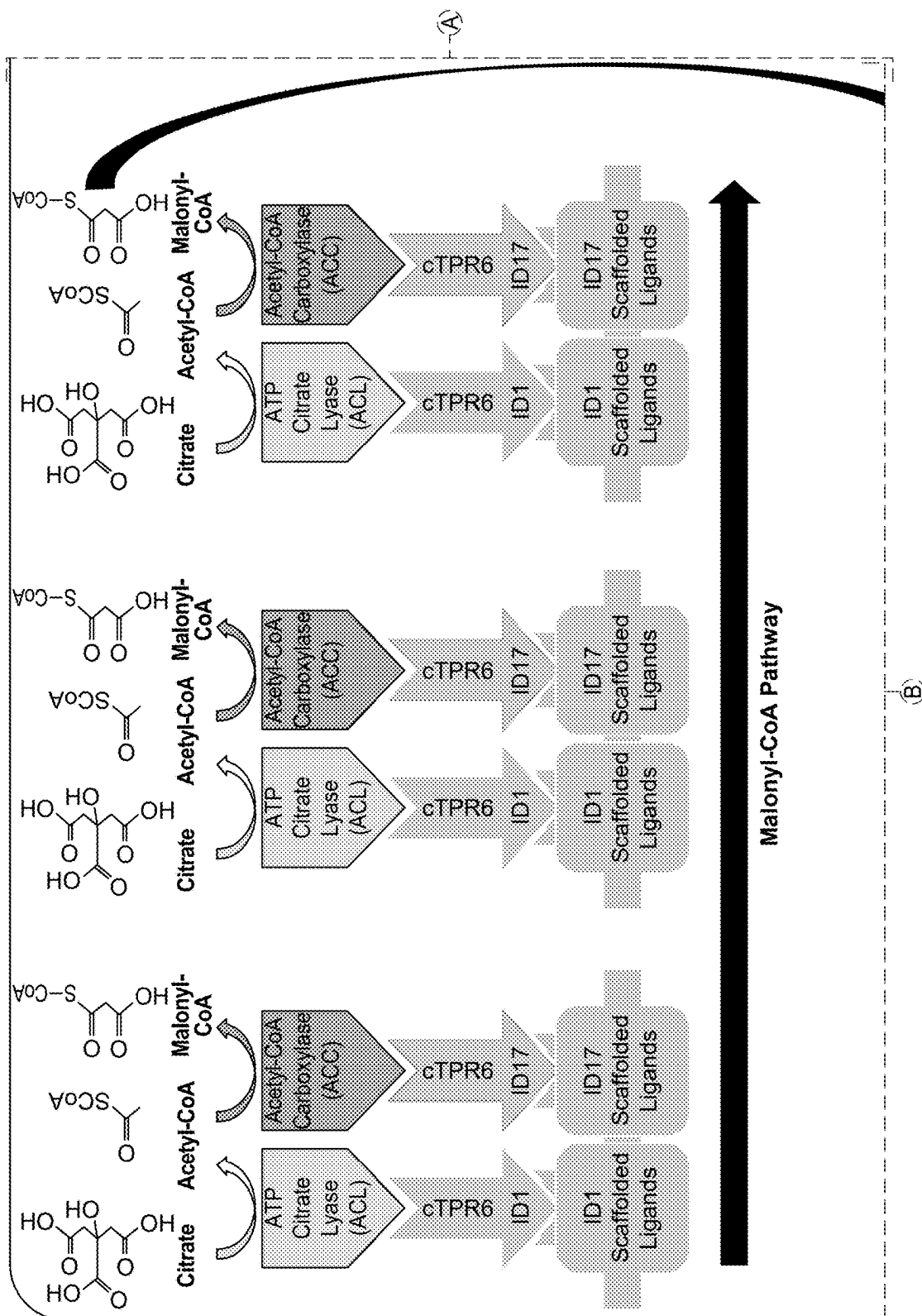


Figure 11.



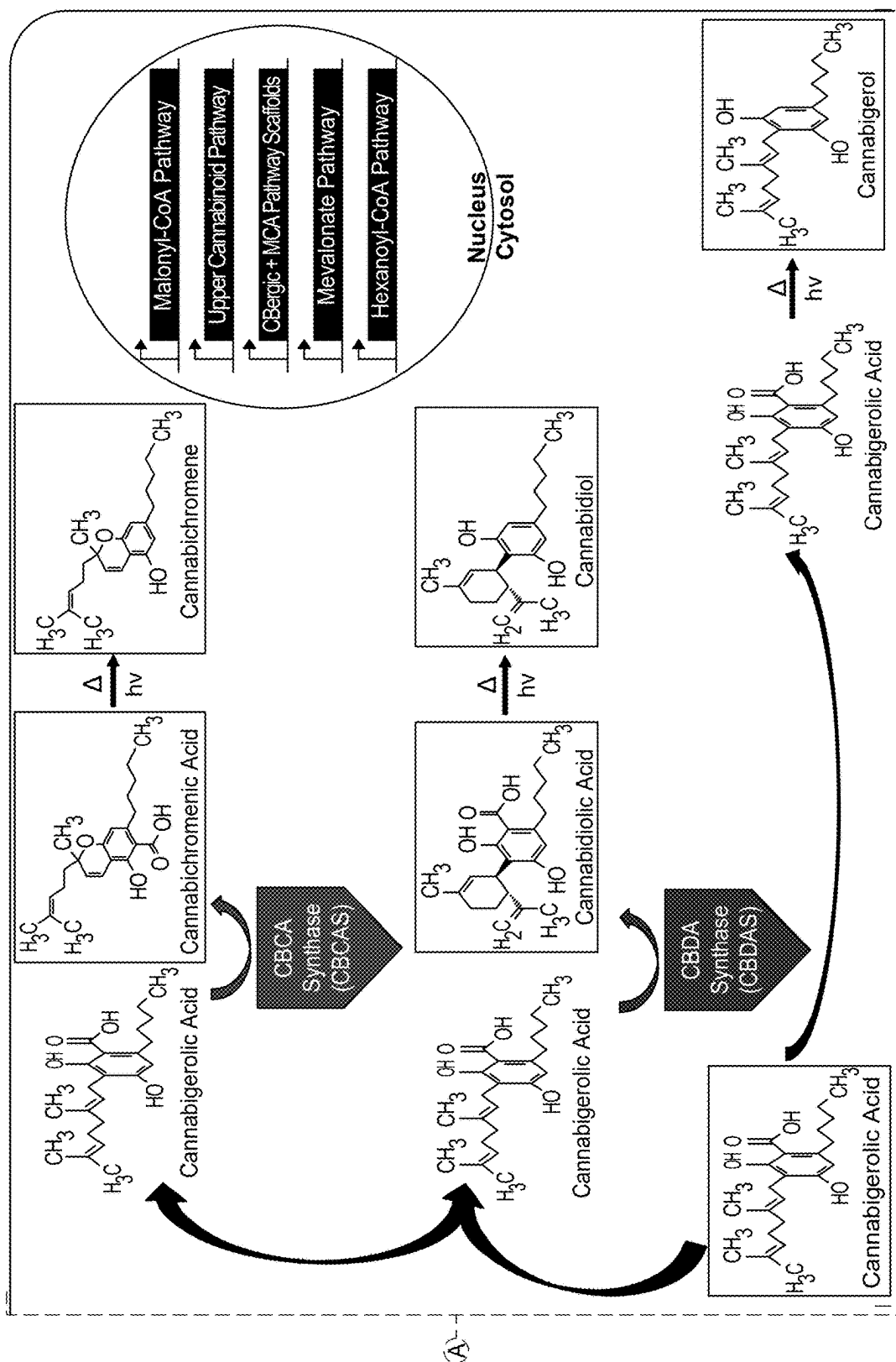


Figure 11. (Cont'd)

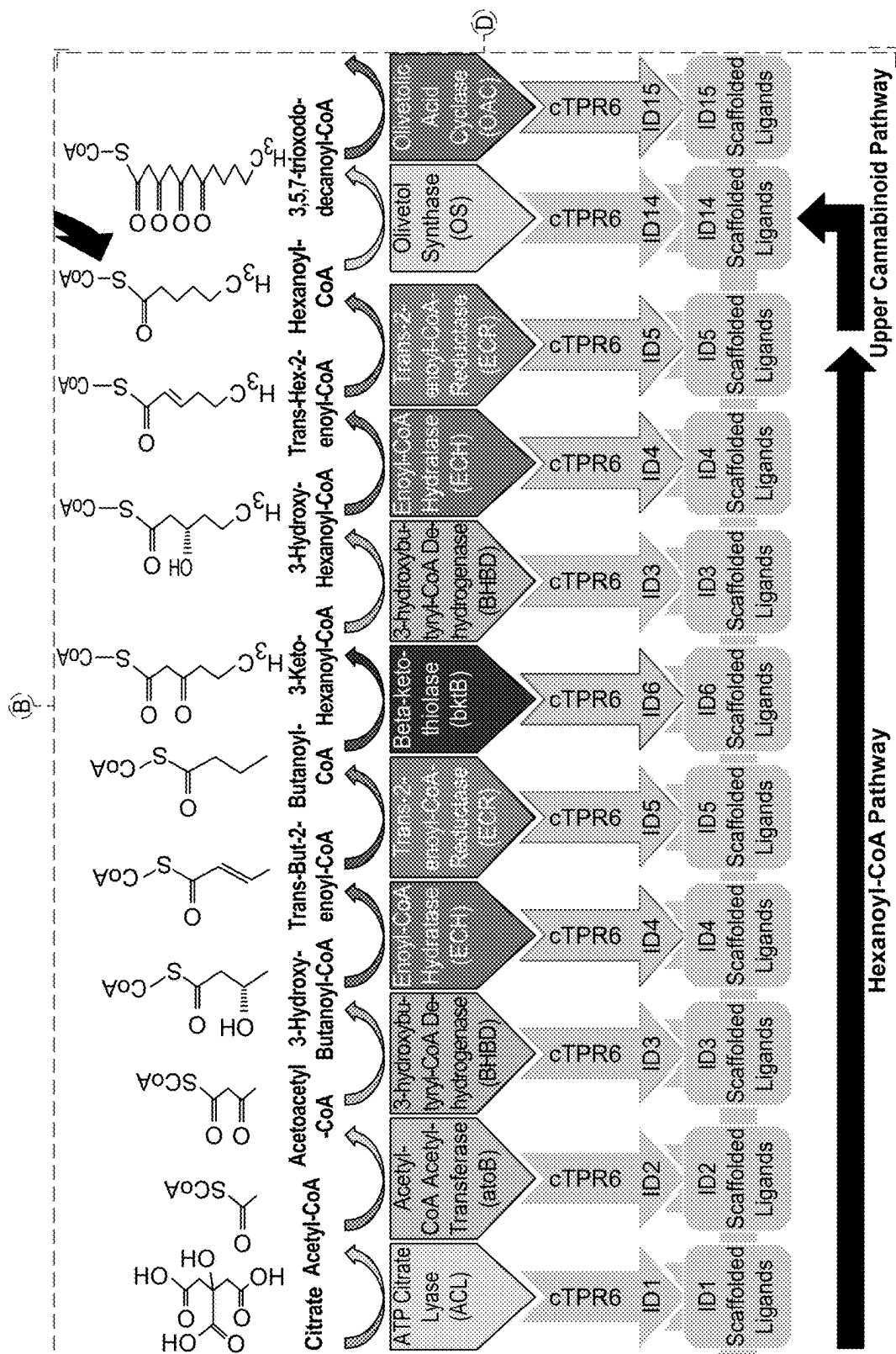


Figure 11. (Cont'd)

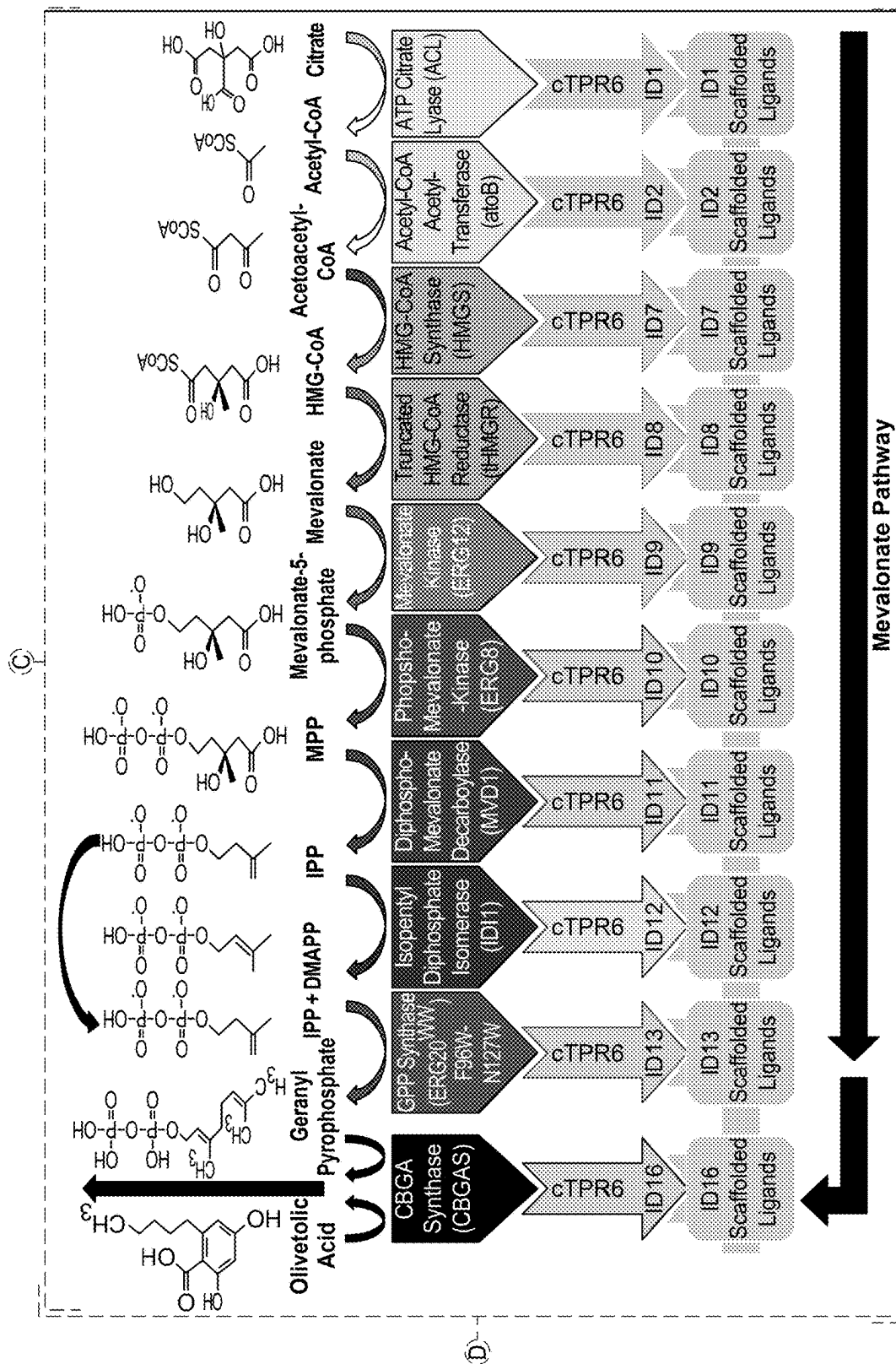


Figure 11. (Cont'd)

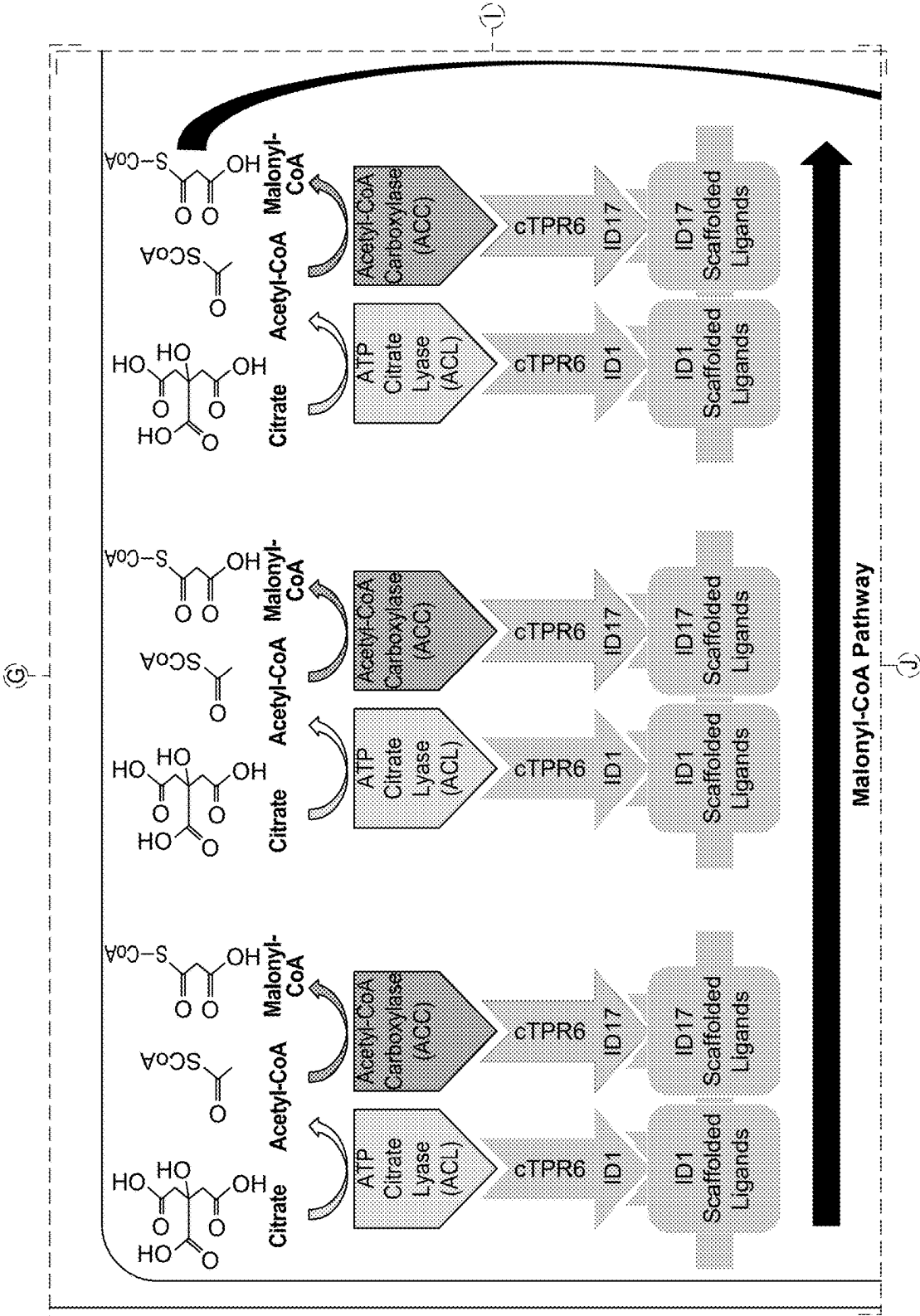


Figure 11. (Cont'd)

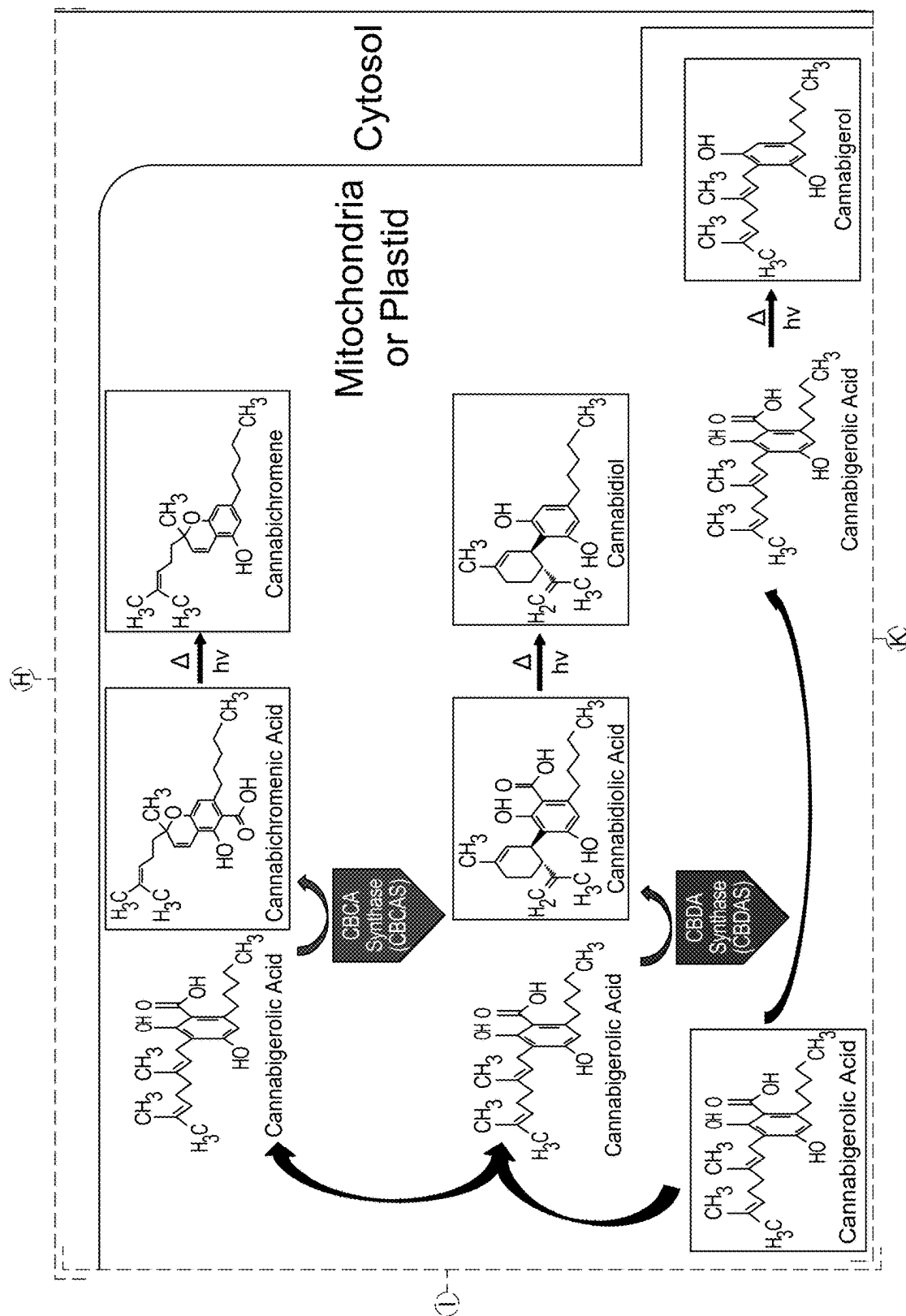


Figure 11. (Cont'd)

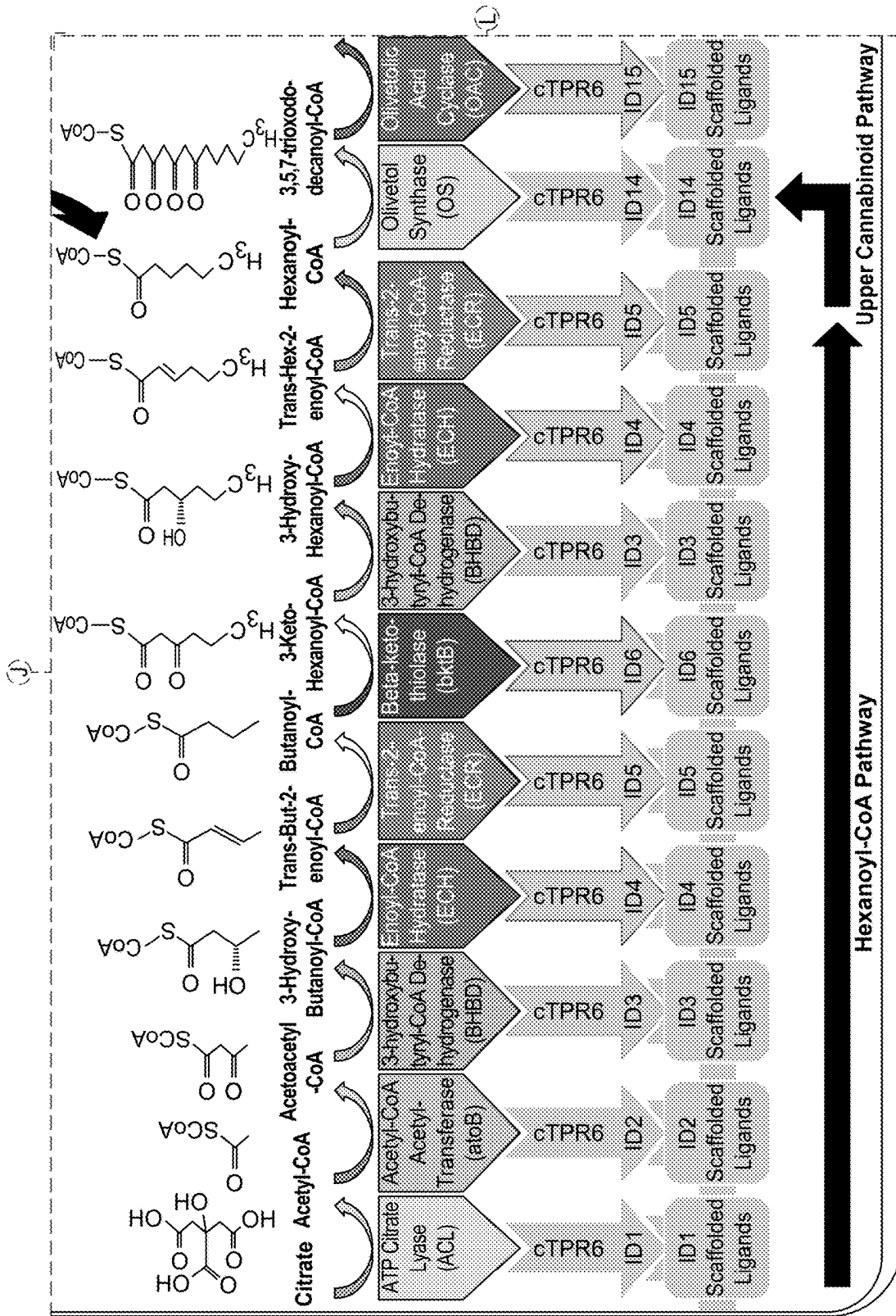


Figure 11. (Cont'd)

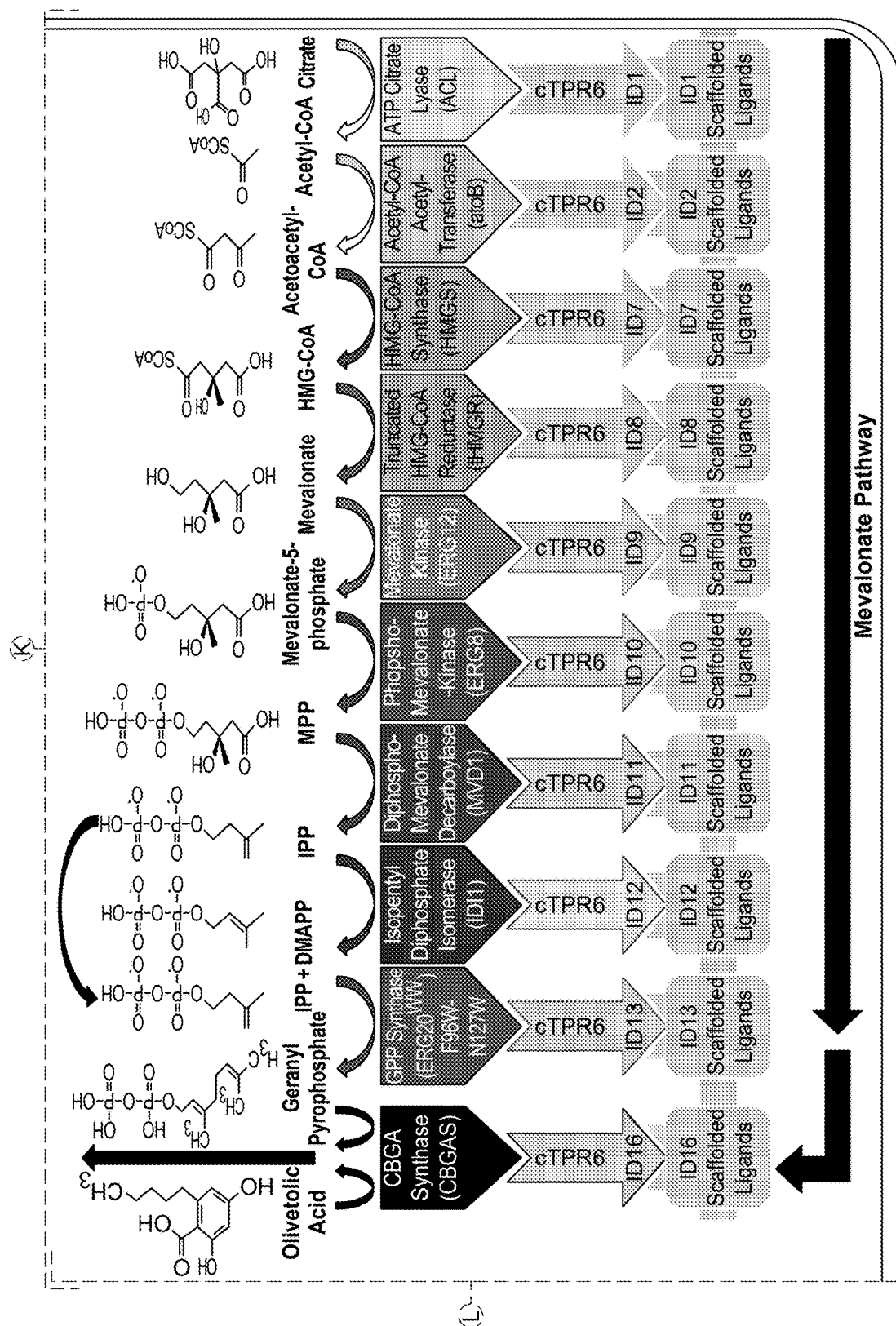


Figure 11. (Cont'd)

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# BIDIRECTIONAL MULTI-ENZYMATIC SCAFFOLDS FOR BIOSYNTHESIZING CANNABINOIDS

## CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/694,417, filed Nov. 25, 2019, which claims priority to U.S. Application Serial Nos. 62/836,265, filed on Apr. 19, 2019 and 62/771,839, filed on Nov. 27, 2018. The disclosures of the prior applications are considered part of (and are incorporated by reference in) the disclosure of this application.

## TECHNICAL FIELD

This document relates to methods and materials for biosynthesizing cannabinoids, and more particularly to using bidirectional multi-enzymatic scaffolds to biosynthesize cannabinoids.

## SEQUENCE LISTING

This application contains a Sequence Listing that has been submitted electronically as an XML file named “47300-0003002\_SL\_ST26.XML.” The XML file, created on May 12, 2023, is 487,591 bytes in size. The material in the XML file is hereby incorporated by reference in its entirety.

## BACKGROUND

The emerging therapeutic potential of cannabinoids warrants industrial-scale production to meet compounding future demands. Traditional cannabinoid production efforts rely on large-scale farming of *Cannabis sativa* L. However, agricultural cannabinoid production is problematic due to issues such as uncontrollable environmental factors and scaling limitations.

## SUMMARY

This document is based, at least in part, on the discovery that a bidirectional, multi-enzymatic scaffold can be engineered to allow high-throughput cannabinoid production in recombinant host cells. By controlling the localization, spatial orientation, and stoichiometry of enzymes catalyzing the biosynthesis of cannabinoids and cannabinoid precursors, the multi-enzymatic scaffolds described herein allow flux-optimized cannabinoid biosynthesis in genetically-engineered host cells.

In one aspect, this document features a host cell capable of producing one or more cannabinoids selected from the group consisting of cannabigerolic acid, cannabidiolic acid, and cannabichromenic acid. The host cell includes at least three different exogenous nucleic acids, wherein the first and the second exogenous nucleic acids each encode a plurality of engineered enzymes selected from the group consisting of acetyl-CoA acetyltransferase, a 3-hydroxybutyryl-CoA dehydrogenase, an enoyl-CoA hydratase, a beto-ketothiolase, a trans-enoyl-CoA reductase, an HMG-CoA synthetase, an HMG-CoA reductase, a mevalonate kinase, a phosphomevalonate kinase, a diphosphomevalonate decarboxylase, an isopentenyl-diphosphate delta isomerase, a geranyl-diphosphate synthase, an olivetol synthase, an olivetolic acid cyclase, and a CBGA synthase; wherein each of the engineered enzymes includes a heterologous interac-

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tion domain, wherein the heterologous interaction domain comprises a first and a second peptide motif, and wherein each heterologous interaction domain is different from each other; and wherein the third exogenous nucleic acid encodes a polypeptide scaffold comprising a plurality of peptide ligands, wherein each peptide ligand comprises an amino acid sequence that can bind to the first or the second peptide motif of one of the heterologous interaction domains. The plurality of engineered enzymes further can include an ATP citrate lyase and an acetyl-CoA carboxylase. The host cell further can include an exogenous nucleic acid encoding a cannabidiolic acid synthase (CBDAS) and a cannabichromenic acid synthase (CBCAS). The host cell can include an exogenous CBDAS. The host cell can include an exogenous CBCAS. The host cell can include an exogenous CBDAS and an exogenous CBCAS. The host cell can include an exogenous hexanoyl-CoA synthetase. The host cell can include at least four different exogenous nucleic acids, wherein the first, second, and fourth nucleic acids each encode a plurality of the engineered enzymes. The host cell can include at least five different exogenous nucleic acids, wherein the first, second, fourth, and fifth nucleic acid each encode a plurality of the engineered enzymes. The host cell can include at least six different exogenous nucleic acids, wherein the first, second, fourth, fifth, and sixth nucleic acids each encode a plurality of the engineered enzymes. Each exogenous nucleic acid can include a constitutive promoter operably linked to the sequence encoding the engineered enzyme or polypeptide scaffold or an inducible promoter operably linked to the sequence encoding the engineered enzyme or polypeptide scaffold. In some embodiments, the promoter is a GAL1-10 promoter. In some embodiments, a constitutive promoter used to express the polypeptide scaffold has weaker constitutive activity level than a constitutive promoter used to express the engineered enzymes. In some embodiments, a constitutive promoter is used to express the engineered enzymes and an inducible promoter is used to express the polypeptide scaffold. In some embodiments, an inducible promoter is used to express the engineered enzymes and a constitutive promoter is used to express the polypeptide scaffold.

Any of the host cells can be bacterial, yeast, algae, or plant cells. A bacterial cell can be selected from the group consisting of *Escherichia coli*, *Bacillus*, *Brevibacterium*, *Streptomyces*, and *Pseudomonas* cells. A yeast cell can be selected from the group consisting of *Pichia pastoris*, *Saccharomyces cerevisiae*, *Yarrowia lipolytica*, *Kluyveromyces marxianus*, and *Komagataella phaffii* cells. An algae cell can be *Dunaliella* sp., *Chlorella variabilis*, *Euglena mutabilis*, or *Chlamydomonas reinhardtii* cells. A plant cell can be a *Cannabis* or tobacco cell.

In some embodiments, each of the engineered enzymes is of the formula: enzyme—linker<sub>1</sub>—spacer—linker<sub>2</sub>—motif<sub>1</sub>—linker<sub>3</sub>—motif<sub>2</sub>, where linkers 1, 2, and 3 can be the same or different, motif 1 and motif 2 can be the same or different, and where motif 1 and motif 2 form the heterologous interaction domain. A scaffold polypeptide can be of the formula: N-terminus—[Ligand 1—linker—Ligand 2—Spacer]<sub>n</sub>—(optionally-tagged)C-terminus, where n is the number of heterologous interaction domains, and where ligand 1 and ligand 2 bind motif 1 and motif 2, respectively, of the heterologous interaction domain. The scaffold polypeptide can be tagged with a MYC tag, FLAG tag, or HA tag. The host cell further can include a nucleic acid encoding a second polypeptide scaffold comprising a plurality of peptide ligands, wherein each peptide ligand comprises an amino acid sequence that can bind to a different motif of the



heterologous interaction domain. The linker can have a flexible GS-rich sequence flanking a rigid  $\alpha$ -helical moiety. The spacer can be the cTPR6 spacer.

This document also features a method of producing one or more cannabinoids selected from the group consisting of cannabigerolic acid, cannabidiolic acid, and cannabichromenic acid. The method can include culturing any of the host cells described herein under conditions wherein the host cell produces the one or more cannabinoids. The host cells can be cultured in a culture medium supplemented with citrate, glucose, hexanoic acid, and/or other carbon source, and/or in a culture medium supplemented with malonyl-CoA. The method further can include extracting the one or more cannabinoids from the host cells.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention pertains. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, suitable methods and materials are described below. All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

Other features and advantages of the invention will be apparent from the following detailed description, and from the claims.

#### DESCRIPTION OF DRAWINGS

FIG. 1A is a schematic of one representative embodiment of a multi-enzymatic cannabinoidergic scaffold within a cell. The multi-enzymatic scaffold includes enzymes of the hexanoyl-CoA pathway, enzymes of the upper cannabinoid pathway, and enzymes of the mevalonate pathway. The schematic also depicts a second scaffold according to one embodiment containing enzymes of the malonyl-CoA pathway and depicts a non-scaffolded cannabidiolic acid synthase (CBDAS) and a non-scaffolded cannabichromenic acid synthase (CBCAS). ID refers to enzyme-linked interaction domain; cTPR6 refers to a spacer sequence; scaffolded ligands refer to the tandem peptide ligands that form the scaffold-binding sites specific for each enzyme-linked ID. The target products cannabigerolic acid (CBGA), cannabigerol (CBG), cannabidiolic acid (CBDA), cannabidiol (CBD), cannabichromenic acid (CBCA), and cannabichromene (CBC), are boxed for emphasis. CBG can be produced by decarboxylation of CBGA, CBD can be produced by decarboxylation of CBDA, and CBC can be produced by decarboxylation of CBCA. For each decarboxylation, the 'Δ' symbols represent heat and the 'hv' symbols represent light.

FIG. 1B is a schematic of one representative embodiment of a bidirectional, multi-enzymatic scaffold within a cell (e.g., a yeast cell). The multi-enzymatic scaffold (referred to as SCF gene cassette in the nucleus) includes enzymes of the hexanoyl-CoA pathway (referred to as HCA cassette in nucleus), enzymes of the upper cannabinoid pathway (referred to as CAN cassette in nucleus), and enzymes of the mevalonate pathway (referred to as GPP cassette in nucleus). The schematic also depicts a second scaffold according to one embodiment containing enzymes of the malonyl-CoA pathway and depicts a non-scaffolded CBDAS and a non-scaffolded CBCAS. ID refers to enzyme-linked interaction domain; cTPR6 refers to a spacer

sequence; scaffolded ligands refer to the tandem peptide ligands that form the scaffold-binding sites specific for each enzyme-linked ID. The target products CBGA, CBG, CBDA, CBD, CBCA, and CBC are boxed for emphasis. CBG can be produced by decarboxylation of CBGA, CBD can be produced by decarboxylation of CBDA, and CBC can be produced by decarboxylation of CBCA. For each decarboxylation, the 'Δ' symbols represent heat and the 'hv' symbols represent light.

FIG. 2A is a schematic of gene cassettes according to one embodiment for the engineering of cannabinoidergic cells.

FIG. 2B is a schematic of gene cassettes used in Examples 2-4 for biosynthesizing cannabinoids in yeast.

FIG. 3 is an example of an enzyme-scaffold complex.

FIG. 4 is a schematic of one representative embodiment of a multi-enzymatic cannabinoidergic scaffold within a cell. The multi-enzymatic scaffold includes enzymes of the hexanoyl-CoA pathway, enzymes of the upper cannabinoid pathway, and enzymes of the mevalonate pathway. The schematic also depicts a second scaffold according to one embodiment containing enzymes of the malonyl-CoA pathway and depicts a non-scaffolded CBDAS and a non-scaffolded CBCAS. Pyruvate dehydrogenase (E1) and dihydrolipoyl transacetylase (E2) are substituted for ATP citrate lyase in both of the depicted scaffolds. ID refers to enzyme-linked interaction domain; cTPR6 refers to a spacer sequence; scaffolded ligands refer to the tandem peptide ligands that form the scaffold-binding sites specific for each enzyme-linked ID. The target products CBGA, CBG, CBDA, CBD, CBCA, and CBC are boxed for emphasis. CBG can be produced by decarboxylation of CBGA, CBD can be produced by decarboxylation of CBDA, and CBC can be produced by decarboxylation of CBCA. For each decarboxylation, the 'Δ' symbols represent heat and the 'hv' symbols represent light.

FIG. 5 is a schematic of one representative embodiment of a multi-enzymatic cannabinoidergic scaffold within a cell. The multi-enzymatic scaffold includes enzymes of the hexanoyl-CoA pathway, enzymes of the upper cannabinoid pathway, and enzymes of the MEP (2-C-methylerythritol 4-phosphate) pathway. The schematic also depicts a second scaffold according to one embodiment containing enzymes of the malonyl-CoA pathway and depicts a non-scaffolded CBDAS and a non-scaffolded CBCAS. ID refers to enzyme-linked interaction domain; cTPR6 refers to a spacer sequence; scaffolded ligands refer to the tandem peptide ligands that form the scaffold-binding sites specific for each enzyme-linked ID. The target products CBGA, CBG, CBDA, CBD, CBCA, and CBC are boxed for emphasis. CBG can be produced by decarboxylation of CBGA, CBD can be produced by decarboxylation of CBDA, and CBC can be produced by decarboxylation of CBCA. For each decarboxylation, the 'Δ' symbols represent heat and the 'hv' symbols represent light.

FIG. 6A contains the amino acid sequence of each of the following enzymes: an ATP citrate lyase (SEQ ID NO:83), acetyl-CoA acetyltransferase (atoB) (SEQ ID NO:84), a 3-hydroxybutyryl-CoA dehydrogenase (SEQ ID NO:85), an enoyl-CoA hydratase (SEQ ID NO:86), a trans-enoyl-CoA reductase (SEQ ID NO:88), a beta-ketothiolase (bktB) (SEQ ID NO:87), an HMG-CoA synthase (SEQ ID NO:90), a truncated HMG-CoA reductase (SEQ ID NO:91), a mevalonate kinase (SEQ ID NO:92), a phosphomevalonate kinase (SEQ ID NO:93), a diphosphomevalonate decarboxylase (SEQ ID NO:94), an isopentenyl-diphosphate delta isomerase (SEQ ID NO:95), a mutant geranyl-diphosphate synthase (ERG20<sup>HW</sup>) (SEQ ID NO:96), an olivetol synthase

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(SEQ ID NO:98), an olivetolic acid cyclase (SEQ ID NO:99), a CBGA synthase (SEQ ID NO:100), an acetyl-CoA carboxylase (SEQ ID NO:97), a CBDA synthase (SEQ ID NO:101), a CBCA synthase (SEQ ID NO:102), and a hexanoyl-CoA synthetase (SEQ ID NO:89).

FIG. 6B contains the amino acid sequence of engineered enzymes of the formula Enzyme-Enzyme Linker-cTPR6 Spacer-ID Linker-ID Motif #1-ID Motif Linker-ID Motif #2, where the linkers (enzyme linker, ID linker, and ID motif linker) can be the same or different, and ID motif #1 and ID motif #2 can be the same or different. The amino acid sequence of the following engineered enzymes are provided: ATP citrate lyase (ID1) (SEQ ID NO:103), an acetyl-CoA acetyltransferase (atoB) (ID2) (SEQ ID NO:104), a 3-hydroxybutyryl-CoA dehydrogenase (ID3) (SEQ ID NO:105), an enoyl-CoA hydratase (ID4) (SEQ ID NO:106), a trans-enoyl-CoA reductase (ID5) (SEQ ID NO:107), a beto-ketothiolase (bktB) (ID6) (SEQ ID NO:108), an HMG-CoA synthase (ID7) (SEQ ID NO:109), a truncated HMG-CoA reductase (ID8) (SEQ ID NO:110), a mevalonate kinase (ID9) (SEQ ID NO:111), a phosphomevalonate kinase (ID10) (SEQ ID NO:112), a diphosphomevalonate decarboxylase (ID11) (SEQ ID NO:113), an isopentenyl-diphosphate delta isomerase (ID12) (SEQ ID NO:114), a mutant geranyl-diphosphate synthase (ERG20<sup>WW</sup>) (ID13) (SEQ ID NO:115), an olivetol synthase (ID14) (SEQ ID NO:116), an olivetolic acid cyclase (ID15) (SEQ ID NO:117), a CBGA synthase (ID16) (SEQ ID NO:118), and an acetyl-CoA carboxylase (ID17) (SEQ ID NO:211).

FIG. 6C contains the amino acid sequence of a polypeptide scaffold of the formula: N-terminus-[Ligand #1-ID Motif #1 Ligand-Linker-ID Motif #2 Ligand-Scaffolded ID-binding Site Spacer]<sub>n</sub>-(Myc)<sup>3</sup>-tagged C-terminus, where n is 16 and the ID motif ligands correspond to the motifs for IDs 1-16 as shown in Table 2. See SEQ ID NO:119.

FIG. 6D contains the amino acid sequence of a polypeptide scaffold of the formula: N-terminus-[Ligand #1-ID Motif #1 Ligand-Linker-ID Motif #2 Ligand-Scaffolded ID-binding Site Spacer]<sub>n</sub>-(FLAG)<sup>3</sup>-tagged C-terminus, where n is 2 and the ID motif ligands correspond to the motifs for IDs 1 and 17 as shown in Table 2. See SEQ ID NO:120.

FIG. 7 is a schematic of one representative embodiment of a scaffold with the minimal requirements for cannabinoid acid synthesis. The scaffold contains enzymes of the upper cannabinoid pathway. In this embodiment, a non-scaffolded hexanoyl-CoA synthetase (HCS), a non-scaffolded CBDAS, and a non-scaffolded CBCAS also are used. ID refers to enzyme-linked interaction domain; cTPR6 refers to a spacer sequence; scaffolded ligands refer to the tandem peptide ligands that form the scaffold-binding sites specific for each enzyme-linked ID. The target products CBGA, CBG, CBDA, CBD, CBCA, and CBC are boxed for emphasis. CBG can be produced by decarboxylation of CBGA, CBD can be produced by decarboxylation of CBDA, and CBC can be produced by decarboxylation of CBCA. For each decarboxylation, the 'Δ' symbols represent heat and the 'hv' symbols represent light.

FIG. 8 is a schematic of one representative embodiment of a bi-directional scaffold containing a HCS on the N-terminus of the scaffold, a geranyl pyrophosphate synthase (GPPS) on the C-terminus of the scaffold, and the enzymes of the upper cannabinoid pathway between the HCS and GPPS. In this embodiment, a non-scaffolded CBDAS and a non-scaffolded CBCAS also can be used. ID refers to enzyme-linked interaction domain; cTPR6 refers to a spacer

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sequence; scaffolded ligands refer to the tandem peptide ligands that form the scaffold-binding sites specific for each enzyme-linked ID. The target products CBGA, CBG, CBDA, CBD, CBCA, and CBC are boxed for emphasis.

CBG can be produced by decarboxylation of CBGA, CBD can be produced by decarboxylation of CBDA, and CBC can be produced by decarboxylation of CBCA. For each decarboxylation, the 'Δ' symbols represent heat and the 'hv' symbols represent light.

FIG. 9 is a schematic of one representative embodiment of a unidirectional scaffold containing enzymes of the upper cannabinoid pathway, shown with soluble enzymes from the precursor pathways (hexanoyl-CoA pathway, mevalonate pathway, and malonyl-CoA pathway), and soluble CBDAS and CBCAS. ID refers to enzyme-linked interaction domain; cTPR6 refers to a spacer sequence; scaffolded ligands refer to the tandem peptide ligands that form the scaffold-binding sites specific for each enzyme-linked ID. The target products CBGA, CBG, CBDA, CBD, CBCA, and CBC are boxed for emphasis. CBG can be produced by decarboxylation of CBGA, CBD can be produced by decarboxylation of CBDA, and CBC can be produced by decarboxylation of CBCA. For each decarboxylation, the 'Δ' symbols represent heat and the 'hv' symbols represent light.

FIG. 10 is a schematic of one representative embodiment of a multi-enzymatic cannabinoidergic scaffold within a cell. The multi-enzymatic scaffold includes enzymes of the malonyl-CoA (MCA) pathway, enzymes of the upper cannabinoid pathway, and enzymes of the mevalonate pathway. The schematic also depicts a separate scaffold according to one embodiment containing enzymes of the hexanoyl-CoA pathway and depicts a non-scaffolded CBDAS and a non-scaffolded CBCAS. ID refers to enzyme-linked interaction domain; cTPR6 refers to a spacer sequence; scaffolded ligands refer to the tandem peptide ligands that form the scaffold-binding sites specific for each enzyme-linked ID. The target products CBGA, CBG, CBDA, CBD, CBCA, and CBC are boxed for emphasis. CBG can be produced by decarboxylation of CBGA, CBD can be produced by decarboxylation of CBDA, and CBC can be produced by decarboxylation of CBCA. For each decarboxylation, the 'Δ' symbols represent heat and the 'hv' symbols represent light.

FIG. 11 is a schematic of one representative embodiment of a multi-enzymatic cannabinoidergic scaffold within dual compartments of a cell, the cytosol and mitochondria/plastid.

FIG. 12A contains the nucleotide sequences encoding each of the following: an ATP citrate lyase (SEQ ID NO:121), an acetyl-CoA acetyltransferase (atoB) (SEQ ID NO:122), a 3-hydroxybutyryl-CoA dehydrogenase (SEQ ID NO:123), an enoyl-CoA hydratase (SEQ ID NO:124), a trans-enoyl-CoA reductase (SEQ ID NO:125), a beto-ketothiolase (bktB) (SEQ ID NO:126), an HMG-CoA synthase (SEQ ID NO:127), a truncated HMG-CoA reductase (SEQ ID NO:128), a mevalonate kinase (SEQ ID NO:129), a phosphomevalonate kinase (SEQ ID NO:130), a diphosphomevalonate decarboxylase (SEQ ID NO:131), an isopentenyl-diphosphate delta isomerase (SEQ ID NO:132), a geranyl-diphosphate synthase (ERG20<sup>WW</sup>) (SEQ ID NO:133), an olivetol synthase (SEQ ID NO:134), an olivetolic acid cyclase (SEQ ID NO:135), a CBGA synthase (SEQ ID NO:136), an acetyl-CoA carboxylase (SEQ ID NO:137), a CBDA synthase (SEQ ID NO:138), a CBCA synthase (SEQ ID NO:139), and a hexanoyl-CoA synthetase (SEQ ID NO:140).

FIG. 12B contains the nucleotide sequences encoding engineered enzymes of the formula: Enzyme-Enzyme

Linker—cTPR6 Spacer— ID Linker— ID Motif #1— ID Motif Linker— ID Motif #2, where the Enzyme Linker, ID Linker, and ID Motif Linker can be the same or different, and where ID Motif #1 and ID Motif #2 can be the same or different. The nucleotide sequences encoding the following engineered enzymes are provided: ATP citrate lyase (ID1) (SEQ ID NO:141), an acetyl-CoA acetyltransferase (atoB) (ID2) (SEQ ID NO:142), a 3-hydroxybutyryl-CoA dehydrogenase (ID3) (SEQ ID NO:143), an enoyl-CoA hydratase (ID4) (SEQ ID NO:144), a trans-enoyl-CoA reductase (ID5) (SEQ ID NO:145), a bktB (ID6) (SEQ ID NO:146), an HMG-CoA synthase (ID7) (SEQ ID NO:147), a truncated HMG-CoA reductase (ID8) (SEQ ID NO:148), a mevalonate kinase (ID9) (SEQ ID NO:149), a phosphomevalonate kinase (ID10) (SEQ ID NO:150), a diphosphomevalonate decarboxylase (ID11) (SEQ ID NO:151), an isopentenyl-diphosphate delta isomerase (ID12) (SEQ ID NO:152), a mutant geranyl-diphosphate synthase (ERG20<sup>trw</sup>) (ID13) (SEQ ID NO:153), an olivetol synthase (ID14) (SEQ ID NO:154), an olivetolic acid cyclase (ID15) (SEQ ID NO:155), a CBGA synthase (ID16) (SEQ ID NO:156), and an acetyl-CoA carboxylase (ID17) (SEQ ID NO:157).

FIG. 12C contains the nucleotide sequence (SEQ ID NO:158) encoding a scaffold polypeptide that contains the peptide ligands corresponding to IDs 1-16 as shown in Table 2 and a triplicate myc tag on the C-terminus.

FIG. 12D contains the nucleic acid sequence (SEQ ID NO:159) encoding a scaffold polypeptide that contains the peptide ligands corresponding to IDs 1 and 17, and a triplicate FLAG tag on the C-terminus.

FIG. 13A contains the amino acid sequence of scaffold-binding engineered enzymes and a soluble hexanoyl-CoA synthetase (HCS) (SEQ ID NO:209) encoded by the HCA gene cassette. The scaffold-binding engineered enzymes are ATP Citrate Lyase (ACL) (ACL— Enzyme Linker—cTPR6 Spacer— ID Linker— ID1) (SEQ ID NO:160); Acetyl-CoA Acetyltransferase (atoB) (atoB— Enzyme Linker—cTPR6 Spacer— ID Linker— ID2) (SEQ ID NO:161); 3-Hydroxybutyryl-CoA Dehydrogenase (BHBD) (BHBD— Enzyme Linker—cTPR6 Spacer— ID Linker— ID3) (SEQ ID NO:162); Enoyl-CoA Hydratase (ECH) (ECH— Enzyme Linker—cTPR6 Spacer— ID Linker— ID4) (SEQ ID NO:163); Trans-Enoyl-CoA Reductase (ECR) (ECR— Enzyme Linker—cTPR6 Spacer— ID Linker— ID5) (SEQ ID NO:164); and Beta-Ketothiolase (bktB) (bktB— Enzyme Linker—cTPR6 Spacer— ID Linker— ID6) (SEQ ID NO:165).

FIG. 13B contains the amino acid sequences of scaffold-binding engineered enzymes encoded by the GPP gene cassette. The scaffold-binding engineered enzymes are HMG-CoA Synthase (HMGS) (HMGS— Enzyme Linker—cTPR6 Spacer— ID Linker— ID7) (SEQ ID NO:166); truncated HMG-CoA Reductase (tHMGR) (tHMGR— Enzyme Linker—cTPR6 Spacer— ID Linker— ID8) (SEQ ID NO:167); Mevalonate Kinase (ERG12) (ERG12— Enzyme Linker—cTPR6 Spacer— ID Linker— ID9) (SEQ ID NO:168); Phosphomevalonate Kinase (ERG8) (ERG8— Enzyme Linker—cTPR6 Spacer— ID Linker— ID10) (SEQ ID NO:169); Diphosphomevalonate Decarboxylase (MVD1) (MVD1— Enzyme Linker—cTPR6 Spacer— ID Linker— ID11) (SEQ ID NO:170); Isopentenyl-Diphosphate Delta-Isomerase (IDI1) (IDI1— Enzyme Linker—cTPR6 Spacer— ID Linker— ID12) (SEQ ID NO:171); and Geranyl-Diphosphate Synthase (ERG20<sup>trw</sup>) (ERG20<sup>trw</sup>— Enzyme Linker—cTPR6 Spacer— ID Linker— ID13) (SEQ ID NO:172).

FIG. 13C contains the amino acid sequences of scaffold-binding engineered enzymes, a soluble CBDA synthase (SEQ ID NO:173), and a soluble CBCA synthase (SEQ ID NO:174) encoded by the CAN gene cassette. The scaffold-binding engineered enzymes are Olivetol Synthase (OS) (OS— Enzyme Linker—cTPR6 Spacer— ID Linker— ID14) (SEQ ID NO:175); Olivetolic Acid Cyclase (OAC) (OAC— Enzyme Linker—cTPR6 Spacer— ID Linker— ID15) (SEQ ID NO:176); CBGA Synthase (CBGAS— Enzyme Linker—cTPR6 Spacer— ID Linker— ID16) (SEQ ID NO:177); and Acetyl-CoA Carboxylase (ACC) (ACC— Enzyme Linker—cTPR6 Spacer— ID Linker— ID17) (SEQ ID NO:178).

FIG. 13D contains the amino acid sequences of the Cannabinoidergic Metabolon Scaffold (CBSCFLD)— (Myc)3 (SEQ ID NO:179) and the Malonyl-CoA Metabolon Scaffold (MCASCFLD)— (FLAG)3 (SEQ ID NO:180).

FIG. 14A contains codon-optimized nucleotide sequences (SEQ ID NOs:181-187) encoding the enzymes of FIG. 13A.

FIG. 14B contains the codon-optimized nucleotide sequences (SEQ ID NOs:188-194) encoding the enzymes of FIG. 13B.

FIG. 14C contains the codon-optimized nucleotide sequences (SEQ ID NOs:195-200) encoding the enzymes of FIG. 13C.

FIG. 14D contains the codon-optimized nucleotide sequences (SEQ ID NO:201 and SEQ ID NO:202) encoding the scaffolds of FIG. 13D.

FIG. 15A contains the nucleotide sequence of the HCA gene cassette (SEQ ID NO:203).

FIG. 15B contains the nucleotide sequence of the GPP gene cassette (SEQ ID NO:204).

FIG. 15C contains the nucleotide sequence of the CAN gene cassette (SEQ ID NO:205).

FIG. 15D contains the nucleotide sequence of the SCF gene cassette (SEQ ID NO:206).

FIG. 15E contains the nucleotide sequence of the SOL gene cassette (SEQ ID NO:207).

FIG. 16 is a map of the pCCI-Brick plasmid construct.

FIG. 17 is a map of a pESC-TRP (“vHCA”) vector construct. In this map, the vector contains a TRP gene allowing selection in tryptophan deficient media. Similar vectors also were made in which the TRP gene was replaced with a LEU gene allowing selection in leucine deficient media, a HIS3 gene allowing selection in histidine deficient media, or a URA3 gene allowing selection in uracil deficient media.

FIG. 18 is a graph of the proliferation curves for yCBSCF and yCBSOL cultures. Line plots depicting cell proliferation curves were fitted via nonlinear regression of cell density measurements (OD<sub>600nm</sub>) recorded in 12-hour intervals over a 48-hour incubation period for yCBSCF and yCBSOL cultures. Initial cell densities for all cultures were standardized to OD<sub>600nm</sub>=0.3. For all measures, n=3 biological replicates for yCBSCF and yCBSOL cultures. Floating data points depict means with 95% confidence intervals. Dotted lines represent 95% confidence intervals for regression curve fits.

FIGS. 19A-19E show a comparison of cannabinoid and precursor titers for scaffolded and soluble cannabinoid biosynthesis. Representative mass spectra of target analytes isolated from (FIG. 19A) yCBSOL and (FIG. 19B) yCBSCF cultures incubated for 48 hours in basal culture media. Bar plots depicting (FIG. 19C) Total (aggregate) cannabinoid (CBGA+CBDA+CBCA+CBG+CBD+CBC) titers, (FIG. 19D) cannabinoid precursor (OVA) titers and summated parent and decarboxylation derivative (CBGA+CBG,

CBDA+CBD, and CBCA+CBC) cannabinoid titers, and (FIG. 19E) separated parent (COO(H)) cannabinoid (CBGA, CBDA, and CBCA) and decarboxylation derivative (ACOOH) cannabinoid (CBG, CBD, and CBC) titers for 48-hour yCBSOL (left) and yCBSCF (right) cultures grown in basal culture media. For all measures,  $n=3$  biological replicates for yCBSCF and yCBSOL cultures. CB, cannabinoid; Cannabigerolic acid, CBGA; cannabigerol, CBG; cannabidiolic acid, CBDA; cannabidiol, CBD; cannabichromenic acid, CBCA; cannabichromene, CBC; olivetolic acid, OVA. Floating asterisks indicate statistically significant (determined by Bonferroni's multiple comparisons post-hoc test;  $\alpha=0.05$ ) between-strain differences for yCBSCF versus yCBSOL cultures. Bar plots depict means with 95% confidence intervals. \* $p<0.05$ ; \*\* $p<0.01$ ; \*\*\* $p<0.001$ ; \*\*\*\* $p<0.0001$ .

FIG. 20 is a bar plot of the impact of citrate and hexanoate supplementation on scaffolded and soluble cannabinoid biosynthesis. Total cannabinoid (CBGA+CBDA+CBCA+CBG+CBD+CBC) titers are shown for yCBSOL and yCBSCF cultures incubated for 48 hours in basal, hexanoate (300 mg/L)-supplemented, and buffered (pH 6.0) citrate (300 mg/L)-supplemented culture media. Floating asterisks indicate statistically significant (determined by Bonferroni's multiple comparisons post-hoc test;  $\alpha=0.05$ ) between-strain differences for yCBSCF versus yCBSOL cultures. Lines with asterisks indicate statistically significant (determined by Bonferroni's multiple comparisons post-hoc test;  $\alpha=0.05$ ) within-strain differences for basal media total cannabinoid titers versus citrate-supplemented media total cannabinoid titers for yCBSCF cultures. Bar plots depict means with 95% confidence intervals. \* $p<0.05$ ; \*\* $p<0.01$ ; \*\*\* $p<0.001$ ; \*\*\*\* $p<0.0001$ .

FIGS. 21A and 21B show concentration-response parameterization of scaffolded and soluble cannabinoid biosynthesis from citrate. In FIG. 21A, line plots are shown depicting eight-point concentration ([citrate])—response (total cannabinoid titers) curves fitted via asymmetric sigmoidal (five-parameter) logistic regression and in FIG. 21B, bar graphs are shown depicting concentration-response parameter estimates ( $CB_{Max}$ , the estimated maximum total cannabinoid titers and citrate  $EC_{50}$ , the estimated citrate concentration yielding half-maximal total cannabinoid titers) for 48-hour yCBSCF and yCBSOL cultures incubated for 48 hours in culture media supplemented with 0, 10, 30, 100, 300, 1000, 3000, or 10000 mg/L buffered (pH 6.0) citrate. For all measures,  $n=3$  biological replicates for yCBSCF and yCBSOL cultures. Floating asterisks indicate statistically significant (determined by Bonferroni's multiple comparisons post-hoc test;  $\alpha=0.05$ ) between-strain differences for yCBSCF versus yCBSOL cultures. Floating data points and bar plots depict means with 95% confidence intervals. Dotted lines represent 95% confidence intervals for regression curve fits \* $p<0.05$ ; \*\* $p<0.01$ ; \*\*\* $p<0.001$ ; \*\*\*\* $p<0.0001$ .

#### DETAILED DESCRIPTION

This document provides methods and materials for producing cannabinoids in host cells or in vitro using a bidirectional, multi-enzymatic scaffold, which can control the localization and stoichiometry of enzymes catalyzing the biosynthesis of cannabinoids and cannabinoid precursors. As described herein, one or more cannabinoids including cannabigerolic acid (CBGA), cannabidiolic acid (CBDA), cannabichromenic acid (CBCA), and tetrahydrocannabinolic acid, can be produced using a bidirectional, multi-enzymatic scaffold and one or more soluble cannabinoid

synthesis enzymes, and the conjugate bases, cannabigerolate, cannabidiolate, cannabichromenate, and tetrahydrocannabinolate, respectively, and decarboxylation products, cannabigerol (CBG), cannabidiol (CBD), cannabichromene (CBC), and tetrahydrocannabinol, respectively, of these cannabinoids also can be produced, as can the tetrahydrocannabinolic acid oxidation product cannabinolic acid and its decarboxylation product cannabinol. The bidirectional, multi-enzymatic scaffold described herein results in significant increases in cannabinoid production in recombinant hosts, including total cannabinoid, CBGA, CBG, CBDA, CBD, CBCA, CBC, and olivetolic acid precursor production, as compared with cannabinoid production in recombinant hosts using the same enzymes that are not bound to a scaffold. As used herein, enzymes that are not bound to a scaffold are referred to as soluble or non-scaffolded. While one particular form of a cannabinoid or other compound may be referenced herein, it is understood that any of its neutral or ionized forms, including any salt forms thereof or decarboxylation derivatives thereof (e.g., produced in the presence of heat and light), are included unless otherwise indicated. It is understood by those skilled in the art that the specific form will depend on factors such as pH and carboxylation status.

In general, enzymes described herein, which can be co-localized on one or more scaffolds and used for producing cannabinoids or cannabinoid precursors, are engineered to contain an interaction domain (ID), which can be separated from the enzyme by an amino acid spacer sequence at the N- or C-terminus of the enzyme. The ID can be composed of two or more scaffold-binding motifs. The engineered enzymes also can include one or more linkers between the enzyme, spacer, and/or ID. The engineered enzymes can bind to a scaffold, which is a polypeptide that contains unique ID-binding domains, i.e., tandem peptide ligands, as shown in FIG. 1A and FIG. 1B, such that the enzymes are co-localized to the scaffold. In other words, each enzyme can be engineered to contain a protein-protein interaction domain that is specific for ligand or ligands (binding site) on the scaffold such that the enzyme can be localized to a discrete location along the scaffold via non-covalent interactions. In some cases, the engineered enzymes can be chimeric enzymes. The scaffolded ligands can be separated using amino acid linkers or spacers. See, for example, Horn and Sticht, *Frontiers in Bioengineering and Biotechnology*, 2015, volume 3, article 191; Whitaker and Dueber, *Methods in Enzymology*, Chapter 19, "Metabolic Pathway Flux Enhancement by Synthetic Protein Scaffolding," Volume 497, 2011, for descriptions of IDs, binding domains, linkers and spacers. IDs also can be referred to as adaptor domains.

Typically, each interaction domain consists of two tandem scaffold-binding motifs that continue/extend from the C-terminus of the engineered enzyme and that can bind to their corresponding scaffolded peptide ligands, which are constructed in tandem along the scaffold. Dual-binding of enzymes to the scaffold ensures fixed spatial orientation, increases binding specificity for each ID-scaffold interaction, and better tethers each enzyme to the scaffold, all of which can improve pathway flux by enabling substrate channeling through each enzymatic step in the scaffolded biosynthetic pathways.

In some embodiments, there are more than two, e.g., three, four, five, six, seven, eight, nine, or ten, or more molecules of each enzyme localized to the scaffold. In addition, the ratio of any given enzyme in a biosynthetic pathway to any other enzyme in the biosynthetic pathway

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can be varied. For example, the ratio of one engineered enzyme in a pathway to a second engineered enzyme in the same pathway can be varied, e.g., from about 1:5 to about 5:1, e.g., from about 1:5 to about 2:5, from about 2:5 to about 3:5, from about 3:5 to about 5:5, from about 5:5 to about 5:3, from about 5:3 to about 5:2, or from about 5:2 to about 5:1.

The peptide ligands are typically short peptide sequences, ranging in length from 3 to 50 amino acid residues. For example, a peptide ligand can be 3-10, 7-15, 10-20, 15-25, 20-30, 25-35, 30-40, 35-45, or 40-50 amino acids in length. There is a database of over 200 different motifs available on the web at [elm.eu.org](http://elm.eu.org) that can be used as described herein. See, for example, Dinkel et al., *Nucleic Acids Res.* 2014; 42(Database issue): D259—D266.

An ID can be a peptide sequence ranging in length 3 to 200 amino acid residues. For example, the ID can be 3-10, 7-15, 10-20, 15-25, 20-30, 25-35, 30-40, 35-45, 40-50, 45-55, 50-60, 65-75, 70-80, 85-95, 90-100, 100-110, 105-115, 110-120, 115-125, 120-130, 125-135, 130-140, 135-145, 140-150, 135-145, 140-150, 145-155, 150-160, 165-175, 170-180, 175-185, 180-190, 185-195, or 190-200 amino acids in length. For example, an ID can be a SH2 domain, a SH3 domain, a PDZ domain, a GTPase binding domain (GBD), a leucine zipper domain, a PTB domain, an FHA domain, a WW domain, a 14-3-3 domain, a death domain, a caspase recruitment domain, a bromodomain, a chromatin organization modifier, a shadow chromo domain, an F-box domain, a HECT domain, a RING finger domain, a sterile alpha motif domain, a glycine-tyrosine-phenylalanine domain, a SNAP domain, a VHS domain, an ANK repeat, an armadillo repeat, a WD40 repeat, an MH2 domain, a calponin homology domain, a Dbl homology domain, a gelsolin homology domain, a PB1 domain, a SOCS box, an RGS domain, a Toll/IL-1 receptor domain, a tetratricopeptide repeat, a TRAF domain, a Bcl-2 homology domain, a coiled-coil domain, a bZIP domain, a fibronectin receptor domain, a FNDC domain, a SAMD domain, a WBP domain, and/or a SASH domain. See, e.g., U.S. Pat. No. 9,856,460 for a list of domains that can be used as an ID as described herein.

For example, an ID can be a “Src homology2” (SH2) or a “Src homology3” (SH3) domain. SH2 domains are highly conserved structures of approximately 100 amino acid residues that comprise two  $\alpha$ -helices and seven  $\beta$ -strands. The SH2 domain can have a promiscuous or strict specificity for a 3-5 amino acid motif flanking a phosphorylated tyrosine. See, Horn and Sticht, 2015, *supra*. For example, a SH2 domain that can be used as an ID as described herein can be residues 5-122 of a mouse Ctl0 regulator of kinase adaptor (Crk) protein having GenBank Accession No. AAH31149.

SH3 domains are small modules of approximately 60 residues that bind proline-rich ligands, which bind to the domain surface at three shallow grooves formed by conserved aromatic residues and exhibit two different binding orientations. See, Horn and Sticht, 2015, *supra*. In some embodiments, the proline-rich ligand can have a core PXXP motif flanked by a positively charged residue. Class I PZP domains recognize ligands conforming to the consensus +XXPXXP (where + is either Arg or Lys), while Class II domains recognize PXXPX+ motifs and bind to ligands in the opposite orientation. See, Teyra, et al., *FEBS Lett.*, 2012 586(17):2631-7. Individual SH3 domains do not measurably interact with other SH3 domain family ligands within an organism, minimizing cross-talk and increasing the number of domain/ligand pairs available for simultaneous use. See, Whitaker and Dueber, 2011, *supra*. For example, a SH3

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domain that can be used as an ID as described herein can be residues 134-190 of a mouse Crk protein having GenBank Accession No. AAH31149 and its peptide ligand can be

(SEQ ID NO: 1)

PPPALPPKRRR.

For example, an ID can be a PDZ (PSD-95/Discs-large/ZO1) domain. PDZ domains are approximately 100 amino acid residues in length and target specific motifs at the C-terminus of the binding partner. The peptide ligand adopts a  $\beta$ -strand and extends an existing  $\beta$ -sheet within the PDZ domain upon binding. At least four different classes of ligands are known for PDZ domains exhibiting a distinct binding specificity. See, Horn and Sticht, 2015, *supra*. For example, grouped PDZ domains into two main specificity classes based on distinct ligand signatures: Class I PDZ domains recognize a (X[T/S]X $\phi$ COOH) motif, Class II PDZ domains recognize a (X $\phi$ X $\phi$ INCOOH) motif, and Class III PDZ domains recognize a X[ED]X $\phi$ COOH motif, where X is any residue and  $\phi$  is a hydrophobic amino acid. See, Teyra, et al., 2012, *supra*. PDZ and SH3 domains are found throughout eukaryotic and eubacterial genomes. For example, a PDZ domain that can be used as an ID as described herein can be residues 77-171 of a mouse  $\alpha$ -syn-trophin protein having GenBank Accession No. EDL06069 and the peptide ligand can be

(SEQ ID NO: 208)

GVKESLV.

For example, an ID can be a GBD domain from a protein such as the Wiskott-Aldrich syndrome-like protein (N-WASP). Isolated GBD domains do not adopt a single, discrete structure under physiological conditions but rather exhibit multiple, loosely packed conformations in solution. The corresponding peptide ligand has been deduced from the autoinhibited form of the GBD. See, Horn and Sticht, 2015, *supra*. For example, a GBD domain that can be used as an ID described herein can include residues 196 to 274 of a rat N-WASP protein having GenBank Accession No. BAA21534, and its peptide ligand, which can be LVGALMHVMQKRSRAIHSSDEGEDQAGDEDED (SEQ ID NO:2), can be used as a peptide ligand as described herein.

For example, an ID can have a leucine zipper or synthetic coiled-coil domain. A leucine zipper domain can include multiple interspersed leucine residues approximately seven amino acid residues apart. Havranek, and Harbury ((2003), *Nat. Struct. Biol.* 10, 45-52) identified new pairs of homodimers or heterodimers by altering residues between leucine zipper pairs based on computational prediction. Reinke, et al. ((2010). *J. Am. Chem. Soc.* 132, 6025-6031) identified three pairs of synthetic coiled coils that do not exhibit measurable self-association. See, Whitaker and Dueber, 2011, *supra*. One example of an ID that can be used as described herein can be ITIRAAFLKENTALRTEIAEL-EKEVGRCNIVSKYETRYGPL (SEQ ID NO:3), and its peptide ligand for use as described herein can be

(SEQ ID NO: 4)

LEIRAAFLKENTALRTRAAELRKRVRGNIVSKYETRYGPL.

For example, an ID can be a dockerin polypeptide, which can localize to a specific cohesion polypeptide on a scaffold described herein. Cohesion-dockerin pairs are particularly

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useful for ex vivo applications as binding is calcium dependent. See, Whitaker and Dueber, 2011, *supra*.

Combinations of IDs that have high affinity for their peptide ligands and high specificity, i.e., minimal cross-reactivity, can be used as described herein to allow for binding of multiple, different enzymes to a scaffold provided herein. For example, at least three different enzymes can be localized on a scaffold. In some embodiments, at least four different enzymes can be localized on a scaffold. In some embodiments, at least five different enzymes can be localized on a scaffold. In some embodiments, at least six different enzymes can be localized on a scaffold. In some embodiments, at least seven different enzymes can be localized on a scaffold. In some embodiments, at least eight different enzymes can be localized on a scaffold. In some embodiments, at least nine different enzymes can be localized on a scaffold. In some embodiments, at least ten different enzymes can be localized on a scaffold. In some embodiments, at least eleven different enzymes can be localized on a scaffold. In some embodiments, at least twelve different enzymes can be localized on a scaffold. In

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some embodiments, at least fifteen different enzymes can be localized on a scaffold. In some embodiments, at least seventeen different enzymes can be localized on a scaffold. In some embodiments, at least eighteen different enzymes can be localized on a scaffold. In some embodiments, at least twenty different enzymes can be localized on a scaffold. In some embodiments, at least twenty-one different enzymes can be localized on a scaffold.

Table 1 provide exemplary combinations of heterologous IDs, i.e., IDs that are different from each other, that can be used in seventeen different engineered enzymes and Table 2 provides the corresponding exemplary combinations of peptide ligands that can be used to localize the seventeen different enzymes to one or more scaffolds. In the embodiments shown in Tables 1 and 2, each ID is composed of two tandem peptide motifs as are the corresponding peptide ligands, which interact with the tandem peptide motifs. It will be appreciated that any one of the enzymes listed in Tables 1 and 2 can be used in combination with any of the listed combinations of IDs and corresponding peptide ligands.

TABLE 1

| Interaction Domain Motif Sequences in Engineered Enzymes |      |             |                                                                                                                                       |             |                                                                                                                                                                                         |
|----------------------------------------------------------|------|-------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enzyme                                                   | ID # | ID Motif #1 | ID Motif #1 Amino Acid Sequence                                                                                                       | ID Motif #2 | ID Motif #2 Amino Acid Sequence                                                                                                                                                         |
| ATP Citrate Lyase                                        | 1    | SYNZIP1     | SYHHHHHHLESTSLYKKAGSG<br>SNLVAQLENEVASLENENETLK<br>KKNLHKDLIAYLEKEIANLRK<br>KIEE ((SEQ ID NO: 5))                                     | SYNZIP2     | SYHHHHHHLESTSLYKKAGSGS<br>ARNAYLRKKIARLKKDNQLERD<br>EQNLEKIIANLRDEIARLENEVASH<br>EQ (SEQ ID NO: 6)                                                                                      |
| Acetyl-CoA<br>Acetyltransferase<br>(atoB)                | 2    | SYNZIP3     | SYHHHHHHLESTSLYKKAGSG<br>SNEVTTLENDAAFIENENAYLE<br>KEIARLRKEKAALRNRLAHKK<br>(SEQ ID NO: 7)                                            | SYNZIP4     | SYHHHHHHLESTSLYKKAGSGS<br>QKVAELKNRVAVKLNRNEQLKNK<br>VEELKNRNAYLKNELATLENEVA<br>RLENDVAE (SEQ ID NO: 8)                                                                                 |
| 3-hydroxybutyryl-<br>CoA Dehydrogenase                   | 3    | MYND        | ENLYFQGENLYFQGDSSSESCWN<br>CGRKASETCSCGNTARYCGSFC<br>QHKDWEKHHHICGQTLQAQQ<br>(SEQ ID NO: 9)                                           | UEV         | MAVSESQKKMVSQYKYRDLTVR<br>ETVNVITLYKDLKPVLDSSYFNDG<br>SSRELMLNTGTIPVPYRGNTYNIPI<br>CLWLLDTYPYNPPICFVKPTSSMTI<br>KTGKHVDANGKIYLPYLHEWKHP<br>QSDLLGLIQVMIVVFGDEPPVFSRP<br>(SEQ ID NO: 10) |
| Enoyl-CoA<br>Hydratase                                   | 4    | PABP        | GPLGSPLTASMLASAPPQEQKQ<br>MLGERLFPPLIQAMHPTLAGKITG<br>MLLEIDNSELLHMLESPESLRSK<br>VDEAVAVLQAHAQKEAAQKA<br>(SEQ ID NO: 11)              | MDM2        | NTNMSVPTDGAVTTSQIPASEQET<br>LVRPKPLLLKLLKSVGAQKDTYT<br>MKEVLFYLGQYIMTKRLYDEKQQ<br>HIVYCSNDLLGDLFGVPSPFSVKEH<br>RKIYTMIRNLVV<br>(SEQ ID NO: 12)                                          |
| Trans-Enoyl-CoA<br>Reductase                             | 5    | SYNZIP10    | SYHHHHHHLESTSLYKKAGSG<br>SNLLATLRSTAAVLENENHVLE<br>KEKEKLKEKEQLLNKLEAYK<br>(SEQ ID NO: 13)                                            | SYNZIP22    | SYHHHHHHLESTSLYKKAGSGS<br>KRIAYLRKKIAALKKDNANLEKDI<br>ANLENEIERLIKEIKTLENEVASHE<br>Q (SEQ ID NO: 14)                                                                                    |
| Beta-ketothiolase<br>(bktB)                              | 6    | GYF         | DVMWEYKVENTGDAELYGPFT<br>SAQMQTWVSEGYFPDGVYCRK<br>LDPPGGQFYNSKRIDFDLYT<br>(SEQ ID NO: 15)                                             | PAH         | ESDSVEFNNAISYVNKIKTRFLDHP<br>EIYRSFLEILHTYQKEQLHTKGRPF<br>RGMSEEEVFTEVANLFRGQEDLLS<br>EFGQFLPEAKR (SEQ ID NO: 16)                                                                       |
| HMG-CoA Synthase                                         | 7    | WW1A        | LGPLPPGWEVRSTVSGRIYFVD<br>HNNRTTQFTDPRHL (SEQ ID<br>NO: 17)                                                                           | WW1B        | GAMGPLPPGWEKRTDSNGRVYFV<br>NHNTRITQWEDPRS (SEQ ID<br>NO: 18)                                                                                                                            |
| HMG-CoA<br>Reductase                                     | 8    | FOS         | SYHHHHHHLESTSLYKKAGSE<br>FFRRERNKMAAAKCRNRREL<br>DTLQAETDQLEDEKSALQTEIA<br>NLLKEKEKLEFILAAHRPACKIP<br>DDLGFPEEMSLE (SEQ ID<br>NO: 19) | SYNZIP9     | SYHHHHHHLESTSLYKKAGSGS<br>QKVESLKQKIEELKQKQAQLKNDI<br>ANLEKEIAYAET<br>(SEQ ID NO: 20)                                                                                                   |

TABLE 1-continued

| Interaction Domain Motif Sequences in Engineered Enzymes |      |             |                                                                                                                                                                                                                         |             |                                                                                                                                                                                                                          |
|----------------------------------------------------------|------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enzyme                                                   | ID # | ID Motif #1 | ID Motif #1<br>Amino Acid Sequence                                                                                                                                                                                      | ID Motif #2 | ID Motif #2<br>Amino Acid Sequence                                                                                                                                                                                       |
| Mevalonate Kinase                                        | 9    | VHS1        | MEPAMEPETLEARINRATNPLN<br>KELDWASINGFCEQLNEDFEGP<br>PLATRLLAHKIQSPQEWELQAL<br>TVLETCMKSCGKRPHDEVGKFR<br>PLNELIKVVSPKYLGSRTSEKVK<br>NKILELLYSWTVGLPEEVKIAEA<br>YQMLKKQGIVKS (SEQ ID<br>NO: 21)                           | VHS2        | GAMGSMAEABEGESLESWLNKATN<br>PSNRQEDWEYIIGFCDQINKELEGP<br>QIAVRLLAHKIQSPQEWELQALT<br>VLEACMKNCGRFRPHNEVGKFRFL<br>NELIKVVSPKYLGDVSEKVKTKV<br>IELLYSWTMALPEEAKIKDAYHML<br>KRQGIVQSDPPPIPVDRTLIPSPPPRP<br>KN (SEQ ID NO: 22) |
| Phosphomevalonate Kinase                                 | 10   | SYNZIP13    | SYHHHHHHLESTSLYKKAGSG<br>SQKVEELNKNIAELENRNAVKK<br>NRVAHLKQEIAYLKDELAHEF<br>E (SEQ ID NO: 23)                                                                                                                           | SYNZIP15    | SYHHHHHHLESTSLYKKAGSGSF<br>ENVTHEFILATLENENAKLRRLEA<br>KLRELARLRNEVAWL (SEQ ID<br>NO: 24)                                                                                                                                |
| Diphospho-<br>mevalonate<br>Decarboxylase                | 11   | MATH        | AMADLEQKVLEMEASTYDGVFI<br>WKISDFPRKRQEAIVAGRIPAIFS<br>PAFYTSRYGYMKCLRIYLNQDG<br>TGRGTHLSLFFVVMKGNPDALL<br>RWPFNQKVTMLLDQNNREHV<br>IDAFRPDVTSSSFQRPVNDMNIA<br>SGCPLFCPVSKMEAKNSYVRDD<br>AIFIKAIVDLTGL (SEQ ID<br>NO: 25) | SKP1        | ASIKLQSSDGEIFEVDVEIAKQSVTI<br>KTMLEDLGMDDEGDDDPVPLPNV<br>NAAILKKVIQWCTHHKDDPPPPED<br>DENKEKRTDDIPVWDQEFKLKVQD<br>GTLFELILAANYLDIKGLLDVTCKT<br>VANMIKGKTPPEIRKTFNIKNDPTE<br>EEEEQVRKENQWC (SEQ ID<br>NO: 26)              |
| Isopentenyl-<br>Diphosphate Delta-<br>Isomerase          | 12   | SYNZIP5     | SYHHHHHHLESTSLYKKAGSG<br>SNTVKELKNYIQELEEERNAELK<br>NLKEHLKFAKAELEFELAAHKF<br>E (SEQ ID NO: 27)                                                                                                                         | SYNZIP6     | SYHHHHHHLESTSLYKKAGSGS<br>QKVAQLKNRVAYKLKENAKLENI<br>VARLENDNANLEKDIANLEKDIAN<br>LERDVAR (SEQ ID NO: 28)                                                                                                                 |
| Geranyl-Diphosphate<br>Synthase                          | 13   | PDZ1        | LCTMKKGPSGYGNLHSDKSKP<br>GQFIRSVDPDSPAASGLRAQDR<br>IVEVNGVCMGKQHGDVVSATR<br>AGGDETKLLVVDRE (SEQ ID<br>NO: 29)                                                                                                           | PDZ2        | SSGALIYTVELKRYGGPLGITISGTE<br>EPFDPPIISSLTGKGLAERTGAIHIG<br>DRILAINSSSLKGLPLSEAIHLLQM<br>AGETVTLKIKKQTDAPASS (SEQ<br>ID NO: 30)                                                                                          |
| Olivetol Synthase                                        | 14   | SH2A        | GNNLETYEWYNSISRDKAEL<br>LLDTGKEGAFMVRDRTPGTYT<br>VSVFTKAIISENPCIKHYHIKET<br>NDSPKRYVVAEKYVFDISPLLIQ<br>YHQYNGGGLVTRLRYPVCG<br>(SEQ ID NO: 31)                                                                           | SH2B        | GSHPWFFGKIPRAKAEEMLSKQRH<br>DGAFLIRESEAPGDFSLSVKFGND<br>VQHFVLRDVGAGKYFLVVVKFNS<br>LNELVDYHRSTSVSRNQQIFLDIE<br>QVPQQPT (SEQ ID NO: 32)                                                                                   |
| Olivetolic Acid<br>Cyclase                               | 15   | PTB1        | GQDRSEATLIKRFKGEVRYKA<br>KLIGIDEVSAARGDKLCQDSMM<br>KLKGVVAGARSKGEHKQKIFLT<br>ISFGGIKIFDEKTGALQHHAHVH<br>EISYIAKDITDHRAFGYVCGKEG<br>NHRFVAIKTAQAEPVILDLRDL<br>FQLIYELKQREELEKKA (SEQ<br>ID NO: 33)                       | PTB2        | GSHMGSQFWVTSQKTEASERCGL<br>QGSYILRVEAEKLTLLTLGAQSQIL<br>EPLLFWPYTLRLRYGRDKVMFSFE<br>AGRRCPSPGPGTFTFQTSQGNDFQ<br>AVEAAIQQQAQKQVGAQDILR<br>LEHHHHHH (SEQ ID NO: 210)                                                       |
| CBGA Synthase                                            | 16   | SH3A        | AEYVRALPDFNGNDEEDLPFKK<br>GDILIRDKPEEQWNAEDSEG<br>KRGMI PVVVEKY (SEQ ID<br>NO: 34)                                                                                                                                      | SH3B        | LIKHMRAEALFDFTGNSKLELNFK<br>AGDVIFLLSRINKDWLEGTVRGAT<br>GIFPLSFVKILK<br>(SEQ ID NO: 35)                                                                                                                                  |
| Acetyl-CoA<br>Carboxylase                                | 17   | FAT         | GSHMRLGAQSIQPTANLDRD<br>LVVNLNVMELVRAVLELKNELA<br>QLPPEGVYVVVKNVGLTLRKL<br>GSVDDLLPSLSSSRTEIEGTQK<br>LLNKDLAELINKMRLAQONAVTS<br>LSECKRQMLTASHTLAVDAKN<br>LLDAVDQAKVLANLAHPAE<br>(SEQ ID NO: 36)                         | PEX         | GAMATPGSENVLPREPLIATAVKF<br>LQNSRVRSPLATRRFLKKKGLT<br>DEEIDMAFQQSGTADEPSSLW<br>(SEQ ID NO: 37)                                                                                                                           |

TABLE 2

| Tandem Peptide Ligand Sequences in Scaffold     |      |             |                                                                                                             |             |                                                                                                                                    |
|-------------------------------------------------|------|-------------|-------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------|
| Enzyme                                          | ID # | ID Motif #1 | ID Motif #1 Scaffolded Ligand Amino Acid Sequence                                                           | ID Motif #2 | ID Motif #2 Scaffolded Ligand Amino Acid Sequence                                                                                  |
| ATP Citrate Lyase                               | 1    | SYNZIP1     | SYHHHHHHLESTSLYKKAGS<br>GSARNAYLRKKIARLKKDNLQ<br>LERDEQNLEKIIANLRDEIARLE<br>NEVASHEQ (SEQ ID NO: 6)         | SYNZIP2     | SYHHHHHHLESTSLYKKAGSGS<br>NLVAQLENEVASLENEENETLKKK<br>NLHKKDLIAYLEKEIANLRKKIEE<br>(SEQ ID NO: 5)                                   |
| Acetyl-CoA<br>Acetyltransferase<br>(atoB)       | 2    | SYNZIP3     | SYHHHHHHLESTSLYKKAGS<br>GSQKVAELKNRVAVKLNRENEQ<br>LKNKVEELKNRNAYLKNELAT<br>LENEVARLENDVAE (SEQ ID<br>NO: 8) | SYNZIP4     | SYHHHHHHLESTSLYKKAGSGS<br>NEVTTLENDAAFIENENAYLEKEI<br>ARLRKEKAALRNRLAHKK (SEQ<br>ID NO: 7)                                         |
| 3-hydroxybutyryl-<br>CoA Dehydrogenase          | 3    | MYND        | RPPTISNPPPLISSAKHPSV<br>(SEQ ID NO: 38)                                                                     | UEV         | NFLQSRPEPTAPPEESFRSG (SEQ<br>ID NO: 39)                                                                                            |
| Enoyl-CoA Hydratase                             | 4    | PABP        | SKGTGLNPNKAVQEIAPGN<br>(SEQ ID NO: 40)                                                                      | MDM2        | PDGGTTFEHLWSSLEPDSTY (SEQ<br>ID NO: 41)                                                                                            |
| Trans-Enoyl-CoA<br>Reductase                    | 5    | SYNZIP10    | SYHHHHHHLESTSLYKKAGS<br>GSKRIAYLRKKIAALKKDNAN<br>LEKDIANLENEIERLIKEIKTLE<br>NEVASHEQ (SEQ ID NO: 14)        | SYNZIP22    | SYHHHHHHLESTSLYKKAGSGS<br>NLLATLRSTAAVLENEHVLEKEK<br>EKLKKEKEQLLNKLEAYK (SEQ ID<br>NO: 13)                                         |
| Beta-Ketothiolase<br>(bktB)                     | 6    | GYF         | PATSQHPPPPQGHRSQAPSH<br>(SEQ ID NO: 42)                                                                     | PAH         | ELNSLLILLEAAEYLERRDR (SEQ<br>ID NO: 43)                                                                                            |
| HMG-CoA Synthase                                | 7    | WW1A        | FQMPADTPPPAYLPEDPMT<br>(SEQ ID NO: 44)                                                                      | WW1B        | ERESNEEPPPPYEDPYWNGN (SEQ<br>ID NO: 45)                                                                                            |
| HMG-CoA Reductase                               | 8    | FOS         | SYHHHHHHLESTSLYKKAGS<br>GSQKVESLKQKIEELKQKQAQL<br>KNDIANLEKEIAYAET (SEQ<br>ID NO: 20)                       | SYNZIP9     | SYHHHHHHLESTSLYKKAGSEFF<br>RRERNKMAAAKCRNRRELDTL<br>QAETDQLEDEKSALQTEIANLLKE<br>KEKLEFILAAHRPACKIPDDLGFPE<br>EMSLE (SEQ ID NO: 19) |
| Mevalonate Kinase                               | 9    | VHS1        | VSSTKLVSFHDSDDEDLLHI<br>(SEQ ID NO: 46)                                                                     | VHS2        | AAATPISTFHDSDDEDLLHV (SEQ<br>ID NO: 47)                                                                                            |
| Phosphomevalonate<br>Kinase                     | 10   | SYNZIP13    | SYHHHHHHLESTSLYKKAGS<br>GSFENVTHEFILATLENENAKL<br>RRLEAKLERELARLRNEVAWL<br>(SEQ ID NO: 24)                  | SYNZIP15    | SYHHHHHHLESTSLYKKAGSGS<br>QKVEELKNKIAELENRNNAVKKNR<br>VAHLKQETIAYLKDELAHEFE<br>(SEQ ID NO: 23)                                     |
| Diphosphomevalonate<br>Decarboxylase            | 11   | MATH        | HDDSLPHQPQATDDSGHESD<br>(SEQ ID NO: 48)                                                                     | SKP1        | GSPNAGSVEQTPKKPGLRRR (SEQ<br>ID NO: 49)                                                                                            |
| Isopentenyl-<br>Diphosphate Delta-<br>Isomerase | 12   | SYNZIP5     | SYHHHHHHLESTSLYKKAGS<br>GSQKVAQLKNRVAYKLKENA<br>KLENIVARLENDNANLEKDIAN<br>LEKDIANLERDVAR (SEQ ID<br>NO: 28) | SYNZIP6     | SYHHHHHHLESTSLYKKAGSGS<br>NTVKELKNYIQELEERNAEKLNK<br>EHLKFAKAELEFELAAHKFE (SEQ<br>ID NO: 27)                                       |
| Geranyl-Diphosphate<br>Synthase                 | 13   | PDZ1        | TDEEREETEEVYLLNSTTL<br>(SEQ ID NO: 50)                                                                      | PDZ2        | DGNVSGTQRLDSATVRTYSY (SEQ<br>ID NO: 51)                                                                                            |
| Olivetol Synthase                               | 14   | SH2A        | ALVDDAADYEPSPSNNEEAL<br>(SEQ ID NO: 52)                                                                     | SH2B        | RELFDPPSYVNVQNLDKARQ (SEQ<br>ID NO: 53)                                                                                            |
| Olivetolic Acid<br>Cyclase                      | 15   | PTB1        | KNTKSMNFDNPVYRKTTEE<br>(SEQ ID NO: 54)                                                                      | PTB2        | RSLPSTWIENKLYGMSDPNW (SEQ<br>ID NO: 55)                                                                                            |
| CBGA Synthase                                   | 16   | SH3A        | VVDNSPPPALPPKRQSAPS<br>(SEQ ID NO: 56)                                                                      | SH3B        | TORSKPQPAVPPRPSADLIL (SEQ<br>ID NO: 57)                                                                                            |
| Acetyl-CoA<br>Carboxylase                       | 17   | FAT         | SATRELDLMASLSDFKIQG<br>(SEQ ID NO: 58)                                                                      | PEX         | DLALSENWAQEFLLAAGDAVD<br>(SEQ ID NO: 59)                                                                                           |

The spacers or linkers connecting an enzyme and ID, as well as a binding domain on a scaffold, can be peptide sequences ranging in length from 6 to 250 amino acid residues. The term “spacer” typically refers to a longer and more structurally-rigid peptide sequence and the term “linker” typically refers to a shorter and more structurally-

flexible peptide sequence. In embodiments in which both terms are used, linker typically refers to a sequence that is about 3 to about 50 amino acids in length and spacer typically refers to a sequence that is longer (e.g., about 36 to about 250 amino acids in length). For example, a linker can be 6-15, 10-20, 15-25, 20-30, 25-35, 30-40, 35-45, or 40-50



amino acids in length. A spacer can be, for example, 36-40, 40-50, 45-55, 50-60, 55-65, 60-70, 65-75, 70-80, 75-85, 90-100, 95-105, 100-110, 105-115, 110-120, 115-125, 120-130, 125-135, 130-140, 135-145, 140-150, 145-155, 150-160, 165-175, 170-180, 175-185, 180-190, 185-195, 190-200, 195-205, 200-210, 205-215, 210-220, 215-225, 220-230, 225-235, 230-240, 235-245, or 240-250 amino acids in length. See, for example, Chen, et al., *Adv Drug Deliv Rev* 2013 65(10): 1357-1369. In either case, the linker/spacer can be a series of small and/or hydrophilic and/or other amino acid residues that can adapt flexible and/or rigid structures. For example, the linker can be a series of glycine residues, a series of alanine residues, a series of serine residues, or a series of alternating glycine and serine (or threonine) residues such as (G-S)<sub>8</sub> (SEQ ID NO:60), (G-S)<sub>10</sub> (SEQ ID NO:61), or (G-S)<sub>15</sub> (SEQ ID NO:62), or contain mainly glycine residues such as (GGGGS)<sub>3</sub> (SEQ ID NO:63) or (GGGGS)<sub>4</sub> (SEQ ID NO:64), or contain any other series of canonical or non-canonical amino acid residues or combinations thereof. In some embodiments, a linker can include glutamic acid, alanine, and lysine residues such as (EAAAK)<sub>2</sub> (SEQ ID NO:65), (EAAAK)<sub>3</sub> (SEQ ID NO:66), or (EAAAK)<sub>4</sub> (SEQ ID NO:67). See, Horn and Sticht, 2015, *supra*. In some embodiments, a linker can be a combination of glycine, alanine, proline and methionine residues, such as AAAGGM (SEQ ID NO:68), AAAGGMPPAAAGGM (SEQ ID NO:69), AAAGGM (SEQ ID NO:70), or PPAAAGGMM (SEQ ID NO:71). See, e.g., U.S. Pat. No. 9,856,460.

Based on amino acid composition, linkers or spacers can be either structured or intrinsically unstructured. For example, in some embodiments, a spacer can have a sequence that adopts a more structurally-rigid  $\alpha$ -helical conformation and a linker can have a GS-rich peptide sequence that is more structurally-flexible. For example, in some embodiments, a linker can include flexible GS-rich sequences flanking one or more rigid  $\alpha$ -helical moieties, e.g., GS-rich sequences flanking duplicate, triplicate, or quadruplicate  $\alpha$ -helical moieties. For example, in some embodiments, a linker or spacer can have the sequence GSAGSAAGSGEF (SEQ ID NO:72), KLSGGGGSGGGSGGGGS (SEQ ID NO:73), GSAGSAAGSGEFSAEAAAKEAAK-AGSAGSAAGSGEF (SEQ ID NO:74), GSAGSAAGSGEFAEAAAKEAAK-AGSAGSAAGSGEF (SEQ ID NO:75), or GSAGSAAGSGEF-SAEAAAKEAAAKEAAAKEAAK-AGSAGSAAGSGEF (SEQ ID NO:76).

In some embodiments, the ligands on the scaffold can be separated by linkers that are 20-50 amino acid residues in length (e.g., 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50 amino acid residues in length). In some embodiments, the IDs engineered at the C-terminus or N-terminus of each scaffolded enzyme can contain a linker (e.g., a flexible linker) of 15 to (e.g., 20) amino acid residues in length flanking a spacer of 15 to 50 (e.g. 36) amino acid residues. In some embodiments, the ID can be separated from the enzyme by a spacer sequence such as the cTPR6 spacer, which includes sextuplicate rigid  $\alpha$ -helical moieties and can have the sequence:

(SEQ ID NO: 77)

AEAWYNLGNAYYKQGDYQKAEIYYQKALELDPNNAEAWYNLGNAYYKQ

5 GDYQKAEIYYQKALELDPNNAEAWYNLGNAYYKQGDYQKAEIYYQKAL

ELDPNNLQAEAWKNLGNAYYKQGDYQKAEIYYQKALELDPNNASAWYN

LGNAYYKQGDYQKAEIYYQKALELDPNNAKAWYRRGNAYYKQGDYQKA

10 IEDYQKALELDPNNRSRA.

In some embodiments, the engineered enzyme can be of a formula: enzyme-linker<sub>1</sub>-spacer-linker<sub>2</sub>-motif<sub>1</sub>-linker<sub>3</sub>-motif<sub>2</sub>, where linkers 1, 2, and 3 can be the same or different, and motif 1 and motif 2 can be the same or different. In some embodiments, linker 1 can be referred to as the enzyme linker, i.e., it connects the enzyme to the spacer such as cTPR6 spacer, and can include flexible GS-rich moieties flanking a rigid  $\alpha$ -helical moiety such as KLSGGGGSGGGSGGGGS (SEQ ID NO:73). In some embodiments, linker 2 can be referred to as the ID linker and can include, for example, flexible GS-rich moieties flanking a rigid  $\alpha$ -helical moiety such as GGGSGGGSGGGSGGGAS (SEQ ID NO:78). In some embodiments, linker 3 can be referred to as the motif linker and can include flexible GS-rich moieties flanking a rigid  $\alpha$ -helical moiety such as

(SEQ ID NO: 74)

GSAGSAAGSGEFSAEAAAKEAAKAGSAGSAAGSGEF.

30 Table 1 provides non-limiting examples of motifs 1 and motifs 2, which are used together to form heterologous IDs. FIG. 3 contains a schematic of an exemplary engineered enzyme of this formula complexed with a scaffold. FIG. 6B and FIGS. 13A-C contain the amino acid sequence of an ATP citrate lyase, atoB, a 3-hydroxybutyryl-CoA dehydrogenase, an enoyl-CoA hydratase, a trans-enoyl-CoA reductase, a beto-ketothiolase (bktB), an HMG-CoA synthase, a truncated HMG-CoA reductase, a mevalonate kinase, a phosphomevalonate kinase, a diphosphomevalonate decarboxylase, an isopentenyl-diphosphate delta isomerase, a geranyl-diphosphate synthase (ERG20<sup>WW</sup>), an olivetol synthase, an olivetolic acid cyclase, a CBGA synthase, and an acetyl-CoA carboxylase according to this formula. In some embodiments, linkers 1 and 2 can be (G<sub>4</sub>S)<sub>3</sub>, the spacer can be the cTPR6 sequence, and linker 3 can be (GS)<sub>8</sub>.

In some embodiments, a scaffold can be of a formula: N-terminus-[Ligand #1-linker-Ligand #2-Spacer]<sub>n</sub>-(optionally-tagged)C-terminus, where n is the number of interaction domains. The linker can be referred to as a scaffolded ligand linker and can be used to connect and separate paired motif-binding ligands that recruit/localize each enzyme to its scaffold-binding site. Such a linker can include flexible GS-rich moieties flanking a rigid  $\alpha$ -helical moiety and have a sequence such as GSAGSAAGSGEFAEAAAKEAAK-AGSAGSAAGSGEF (SEQ ID NO:75). The spacer can be referred to as a scaffolded ID-binding site spacer and can be used to connect and separate the scaffold-binding sites (composed of the paired motif binding ligands) for each enzyme. Such a spacer can include flexible GS-rich moieties flanking a rigid  $\alpha$ -helical moiety and have a sequence such as GSAGSAAGSGEF-SAEAAAKEAAAKEAAAKEAAK-AGSAGSAAGSGEF (SEQ ID NO:76). The N-terminus can include a flexible GS-rich sequence to help stabilize and solubilize the scaffold. For example, the N-terminus can have the sequence

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GSAGSAAGSGEFGSAGSAAGSGEFGSAGSAAGSGEF (SEQ ID NO:79). The C-terminus can include a flexible GS rich sequence flanking a rigid  $\alpha$ -helical moiety to stabilize and solubilize the scaffold and can be optionally tagged (e.g., with a MYC tag, a FLAG tag, or other tag described below) to ease purification or detection of the scaffold. For example, a C-terminal sequence with a triplicate MYC tag can have the sequence GSAGSAAGSGEFG-SAEAAAKEAAAKEAAAKEAAAK-AGSAGSAAGSGEFGSEQK LISEEDLEQKLISEED-LEQKLISEEDLGSAGSAAGSGEFGSAGSAAGSGEFGSAGSAAGSGEF (SEQ ID NO:80). For example, a C-terminal sequence with a triplicate FLAG tag can have the sequence GSAGSAAGSGEFG-SAEAAAKEAAAKEAAAKEAAAK-AGSAGSAAGSGEFGSDYK DDDDKDYKDDDDKDYKDDDDKGSAGSAAGSGEF GSAGSAAGSGEFGSAGSAA GSGEF (SEQ ID NO:81). FIG. 6C and FIG. 13D each contain an example of a scaffold polypeptide of this formula that contains the peptide ligands corresponding to IDS 1-16 as shown in Table 2, and a triplicate MYC tag on the C-terminus. For example, FIG. 13D contains an example of a scaffold polypeptide (see SCF gene cassette of FIG. 2B) containing a triplicate MYC tag. FIG. 6D and FIG. 13D each contain an example of a scaffold polypeptide that contains the peptide ligands corresponding to IDs 1 and 17 as shown in Table 2 and a triplicate FLAG tag on the C-terminus. Accordingly, the amino acid sequence of a scaffold can depend on the sequence of the peptide ligands that can bind to the selected ID motif of the enzymes.

In some embodiments, any one of the enzymes can be engineered to include an N-terminal or C-terminal linker motif that allows covalent (isopeptide) bonding to the scaffold. See, for example, the SpyTag and SpyCatcher system described by Zakeri, et al., *Proc. Natl. Acad. Sci.*, 2012 109 (12) E690-E697.

In some embodiments involving multi-enzymatic scaffolds described herein, the first engineered enzyme of a biosynthetic pathway can produce a first product that can be a substrate for the second engineered enzyme of the biosynthetic pathway, the second engineered enzyme of the biosynthetic pathway can produce a second product that can be a substrate for the third engineered enzyme of the biosynthetic pathway, and so forth. In some cases, the second engineered enzyme can be immobilized on the scaffold such that it is positioned adjacent to or very close to the first engineered enzyme. The third engineered enzyme can be immobilized on the scaffold such that it is positioned adjacent or very close the second engineered enzyme. In this way, the effective concentration of the first product can be high, and the second engineered enzyme can act efficiently on the first product, the third engineered enzyme can act efficiently on the second product, and so forth.

As shown in FIGS. 1A and 1B, one example of a multi-enzymatic scaffold contains enzymes of the hexanoyl-CoA pathway on the N-terminus of the scaffold, enzymes of the mevalonate pathway on the C-terminus of the scaffold, and enzymes of the upper cannabinoid pathway in between. Within any of the pathways, the enzymes can be from a single source, i.e., from one species or genera, or can be from multiple sources, i.e., different species or genera. Nucleic acids encoding the enzymes described herein have been identified from various organisms and are readily available in publicly available databases such as GenBank or EMBL (see below).

A fully-assembled multi-enzymatic scaffold provided herein can adopt stoichiometry and a spatial arrangement

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that can help maximize pathway flux and minimize accumulation of pathway intermediates and by-products. Such scaffolds can facilitate substrate channeling both within and between cannabinoid and cannabinoid precursor pathways. Specifically, this scaffolding system can facilitate unidirectional flux through each of the primary cannabinoid precursor pathways, and converging near the midpoint of the scaffold. The hexanoyl-CoA/olivetolic acid (OVA) pathway can begin at the N-terminus of the scaffold, and the mevalonate or MEP pathway can begin at the C-terminus of the scaffold. The enzyme catalyzing the rate-limiting/committed step in cannabinoid biosynthesis, a CBGA synthase, can be localized at the intersection of these precursor pathways near the scaffold midpoint.

By this design, the two primary precursors for cannabinoid biosynthesis, hexanoyl-CoA/olivetolic acid and geranyl pyrophosphate, can be bi-directionally delivered to a CBGA synthase at this intersection. The CBGA synthase can catalyze biosynthesis of CBGA, the primary cannabinoid from which all other cannabinoids are derived. Substrate channeling within and between the scaffolded pathways can accelerate the kinetics of the composite pathway in accordance with the law of mass action.

In the embodiment shown in FIGS. 1A and 1B, the N-terminal hexanoyl-CoA pathway can include an ATP citrate lyase (ACL) (also can be referred to as an ATP citrate synthase), an acetyl-CoA acetyltransferase (atoB), two 3-hydroxy-acyl-CoA dehydrogenases (BHBDs), two enoyl-CoA hydratases (ECHs), a beta-ketothiolase (bktB), and two trans-2-enoyl-CoA-reductases (ECRs).

In the hexanoyl-CoA pathway shown in FIGS. 1A and 1B, citrate, from cellular metabolism and/or supplemented in the growth medium, can be used as a substrate for ACL-catalyzed acetyl-CoA synthesis. ACL is classified under EC 2.3.3.8. Acetyl-CoA can be used as a substrate for atoB-catalyzed acetoacetyl-CoA synthesis. atoB is classified under EC 2.3.1.9. Acetoacetyl-CoA can serve as the substrate for BHBD-catalyzed 3-hydroxybutanoyl-CoA synthesis. BHBD is classified under EC 1.1.1.157. 3-hydroxybutanoyl-CoA can serve as the substrate for ECH-catalyzed trans-but-2-enoyl-CoA synthesis. ECH is classified under EC 4.2.1.17. Trans-but-2-enoyl-CoA can serve as the substrate for ECR-catalyzed butanoyl-CoA synthesis. ECR is classified under EC 1.3.8.1. Butanoyl-CoA can serve as the substrate for bktB-catalyzed 3-keto-hexanoyl-CoA synthesis. bktB is classified under EC 2.3.1.9. The bktB catalyzing the production of 3-ketohexanoyl CoA from butanoyl-CoA can be the same as, or different from, the atoB used to catalyze the production of acetoacetyl-CoA from acetyl-CoA. 3-ketohexanoyl-CoA is the substrate for BHBD-catalyzed 3-hydroxyhexanoyl-CoA synthesis. BHBD is classified under EC 1.1.1.157. The BHBD catalyzing the production of 3-hydroxyhexanoyl-CoA can be the same as, or different from, the BHBD used to catalyze the production of 3-hydroxybutanoyl-CoA. 3-hydroxyhexanoyl-CoA can be the substrate for ECH-catalyzed trans-hex-2-enoyl-CoA synthesis. ECH is classified under 4.2.1.17. The ECH catalyzing the production of trans-hex-2-enoyl-CoA can be the same as, or different from, the ECH used to catalyze the production of trans-but-2-enoyl-CoA. Trans-hex-2-enoyl-CoA can be the substrate for ECR-catalyzed hexanoyl-CoA synthesis. ECR is classified under EC 1.3.1.38 or EC 1.3.1.44. The ECR catalyzing the production of hexanoyl-CoA can be the same as, or different from, the ECR used to catalyze the production of butanoyl-CoA.

In some embodiments, a hexanoyl-CoA synthetase (HCS) enzyme can be substituted for the scaffolded enzymes of the

hexanoyl-CoA pathway or can be included in a soluble form in addition to the scaffolded enzymes of the hexanoyl-CoA pathway, and in some embodiments, hexanoic acid can be added to the growth media as a substrate for HCS-catalyzed hexanoyl-CoA production. The HCS can be included on the scaffold, N-terminal to the upper cannabinoid pathway in FIGS. 1A and 1B, and/or it can be non-scaffolded (soluble).

In the embodiment shown in FIGS. 1A and 1B, the C-terminal mevalonate pathway can include an ACL, an atoB, a hydroxymethylglutaryl-CoA, an HMG-CoA synthase (HMGS), an HMG-CoA reductase (HMGR), a mevalonate kinase (ERG12), a phosphomevalonate kinase (ERGS), a diphospho mevalonate decarboxylase (MVD1), an isopentyl diphosphate isomerase (IDI1), and a mutant GPP synthase (mGPPS). In the mevalonate pathway shown in FIGS. 1A and 1B, citrate from cellular metabolism and/or supplemented in the growth medium, can be used as a substrate for ACL-catalyzed acetyl-CoA synthesis. ACL is classified under EC 2.3.3. Acetyl-CoA can be used as a substrate for bktB-catalyzed acetoacetyl-CoA synthesis. bktB is classified under EC 2.3.1.9. Acetoacetyl-CoA can be the substrate for HMGS-catalyzed HMG-CoA synthesis. HMG-CoA can be the substrate for HMGR catalyzed mevalonate synthesis. HMGR is classified under EC 1.1.1.88 or 1.1.1.34. Mevalonate can be the substrate for mevalonate kinase-catalyzed mevalonate-5 phosphate synthesis. Mevalonate kinase is classified under EC 2.7.1.36. Mevalonate-5-phosphate can be the substrate for phosphomevalonate kinase-catalyzed mevalonate pyrophosphate synthesis. Phosphomevalonate kinase is classified under EC 2.7.4.2. Mevalonate pyrophosphate can be the substrate for diphosphomevalonate decarboxylase-catalyzed isopentyl pyrophosphate synthesis. Diphosphomevalonate decarboxylase is classified under EC 4.1.1.33. Isopentyl pyrophosphate can be the substrate for isopentyl diphosphate isomerase-catalyzed dimethylallyl pyrophosphate synthesis. Isopentyl diphosphate isomerase is classified under EC 5.3.3.2. Dimethylallyl pyrophosphate can be the substrate for geranyl pyrophosphate synthase (GPPS)-catalyzed geranyl pyrophosphate synthesis. GPPS is classified under EC 2.5.1.1.

As acetyl-CoA can be the initial substrate for the hexanoyl-CoA, mevalonate/geranyl pyrophosphate, and malonyl-CoA cannabinoid precursor biosynthetic pathways, the inclusion of ACL at both the N-terminus and C-terminus of the multi-enzymatic scaffold in FIGS. 1A and 1B can directly couple the scaffolded pathways to cellular metabolism via ACL-catalyzed production of acetyl-CoA from citric acid cycle-derived citrate. The citrate also can be supplemented into the culture medium (e.g., as buffered citrate). In some embodiments, the ACL enzyme is included only at the N-terminus of the scaffold. In some embodiments, the ACL enzyme is included only at the C-terminus of the scaffold. In some embodiments, the ACL enzyme is included in soluble form.

In some embodiments, the 2-C-methylerythritol 4-phosphate (MEP) pathway, which also can produce geranyl pyrophosphate, can be substituted for the scaffolded mevalonate pathway at the C-terminus of the scaffold or can be included in a soluble form in addition to the scaffolded mevalonate pathway. For example, as shown in FIG. 5, the C-terminus of the scaffold can include a 1-deoxy-D-xylulose-5-phosphate (DOXP) synthase, a DOXP reductoisomerase, a MEP cytidyl transferase, a 4-diphosphocytidyl-2-C-methylerythritol (CDPME) kinase, a 2-C-methyl-D-erythritol 2,4-cyclodiphosphate (MECDP) synthase, a 4-hydroxy-3-methyl-but-2-enyl pyrophosphate (HIVIBPP) synthase, a HMBPP reductase, and a GPPS. Pyruvate and

glyceraldehyde-3-phosphate (G3P) can be used as substrates for DOXP-synthase-catalyzed DOXP synthesis. DOXP is classified under EC 2.2.1.7. DOXP can be the substrate for DOXP reductoisomerase (DXR)-catalyzed MEP synthesis. DXR is classified under EC 1.1.1.267. MEP can be the substrate for 2-C-methyl-D-erythritol 4-phosphate cytidyltransferase (ISPD)-catalyzed 4-diphosphocytidyl-2-C-methylerythritol (CDP-ME) synthesis. ISPD is classified under EC 2.7.7.60. CDP-ME can be the substrate for 4-diphosphocytidyl-2-C-methyl-D-erythritol kinase (ISPE)-catalyzed 4-diphosphocytidyl-2-C-methyl-D-erythritol 2-phosphate (CDP-MEP) synthesis. ISPE is classified under EC 2.7.1.148. CDP-MEP can be the substrate for 2-C-methyl-D-erythritol 2,4-cyclodiphosphate synthase (ISPF)-catalyzed 2-C-methyl-D-erythritol 2,4-cyclodiphosphate (cMEPP) synthesis. ISPF is classified under EC 4.6.1.12. cMEPP can be the substrate for HMB-PP synthase (ISPG)-catalyzed (E)-4-Hydroxy-3-methyl-but-2-enyl pyrophosphate (HMBPP) synthesis. ISPG is classified under EC 1.17.7.1. HMBPP can be the substrate for 4-hydroxy-3-methylbut-2-enyl diphosphate reductase (ISPH)-catalyzed isopentenyl pyrophosphate (IPP) and dimethylallyl pyrophosphate (DMAPP) synthesis. ISPH is classified under EC 1.17.1.2. IPP and DMAPP can be substrates for GPPS-catalyzed geranyl pyrophosphate synthesis. GPPS is classified under EC 2.5.1.1.

In some embodiments, the mevalonate pathway can be substituted for the scaffolded MEP pathway at the C-terminus of the scaffold or can be included in a soluble form in addition to the scaffolded MEP pathway.

In the embodiment shown in FIG. 1A and FIG. 1B, a second multi-enzymatic scaffold can be co-expressed to enhance cytosolic titers of malonyl-CoA, another secondary substrate which can be used in cannabinoid biosynthesis. Such a scaffold can include an ATP citrate lyase (ACL) and acetyl-CoA carboxylase (ACC) in tandem. In some embodiments, the ACL and ACC are paired in duplicate or triplicate along the scaffold. If the ACL and ACC are paired in duplicate or triplicate, the two or three ACLs on the scaffold can be the same or different, and the two or three ACCs can be the same or different. In any of the embodiments, malonyl-CoA can be supplemented into the growth media instead of, or in addition to, being supplied by a scaffolded malonyl-CoA pathway.

In any of the embodiments in which an ACL enzyme is used, a pyruvate dehydrogenase (E1) and a dihydrolipoyl transacetylase (E2) can be substituted for the ACL. For example, as shown in FIG. 4, a pyruvate dehydrogenase (E1) and a dihydrolipoyl transacetylase (E2) can be substituted upstream of scaffolded mevalonate, hexanoyl-CoA, and malonyl-CoA pathways. Using both a pyruvate dehydrogenase (E1) and a dihydrolipoyl transacetylase can allow acetyl-CoA to be produced using pyruvate rather than citrate as the primary substrate. In such embodiments, pyruvate also can be supplemented in the growth media. Pyruvate dehydrogenases and dihydrolipoyl transacetylases are constituents of the multi-enzyme pyruvate dehydrogenase complex that catalyze acetyl-CoA production from pyruvate. E1 and E2 are found in bacteria and eukaryotes.

As shown in FIG. 1A and FIG. 1B, the co-scaffolded upper cannabinoid pathway can include an olivetol synthase (OS), an olivetolic acid cyclase (OAC), and an aromatic prenyl-transferase (APT) such as a CBGA synthase (CBGAS). The upper cannabinoid pathway can begin using hexanoyl-CoA and three malonyl CoAs as the substrate for olivetol synthase-catalyzed 3,5,7-trioxododecanoyl-CoA synthesis. Olivetol synthase is classified under EC 2.3.1.206.

3,5,7-trioxododecanoyl-CoA can be used as a substrate for OAC-catalyzed olivetolic acid synthesis. OAC is classified under EC 4.4.1.26.

At the flux intersection of the converging N-terminal hexanoyl-CoA/upper cannabinoid and C-terminal mevalonate/MEP pathways (near the scaffold midpoint), an APT such as CBGAS can use olivetolic acid from the hexanoyl-CoA/upper cannabinoid pathways and geranyl pyrophosphate from the mevalonate or MEP pathway as substrates for cannabigerolate synthesis. A suitable APT is classified under EC 2.5.1.102.

In some embodiments, enzymes in the upper cannabinoid pathway can be scaffolded with a hexanoyl-CoA synthetase (HCS) to biosynthesize cannabigerolate. In some embodiments, a soluble HCS can be used with scaffolded enzymes of the upper cannabinoid pathway to biosynthesize cannabigerolate as shown in FIG. 7. Suitable enzymes for the upper cannabinoid pathway are described above.

In some embodiments, a minimal bidirectional scaffold, such as the one depicted in FIG. 8, can be used in which HCS is on the N-terminus of the scaffold, a GPPS is on the C-terminus of the scaffold, and enzymes in the upper cannabinoid pathway are scaffolded between the HCS and GPPS.

In some embodiments, such as the embodiment shown in FIG. 9, the enzymes in the upper cannabinoid pathway can be scaffolded, while the enzymes in the hexanoyl-CoA pathway, enzymes in the mevalonate pathway, and enzymes in the malonyl-CoA pathway can be soluble. In some embodiments, the enzymes in the upper cannabinoid pathway can be scaffolded, while the enzymes in the hexanoyl-CoA pathway, enzymes in the MEP pathway, and enzymes in the malonyl-CoA pathway can be soluble. In such embodiments, HCS can be substituted for the soluble forms of the enzymes of the hexanoyl-CoA pathway. Suitable enzymes for each of these pathways are described above.

In some embodiments, the enzymes in the upper cannabinoid pathway can be scaffolded, while a hexanoyl-CoA synthase, enzymes in the mevalonate or MEP pathway, and enzymes in the malonyl-CoA pathway can be soluble. Suitable enzymes for each of these pathways are described above.

In some embodiments, a HCS can be scaffolded N-terminally relative to the scaffolded enzymes in the upper cannabinoid pathway, while enzymes in the mevalonate or MEP pathway, and enzymes in the malonyl-CoA pathway can be soluble. Suitable enzymes for each of these pathways are described above.

In some embodiments, the enzymes in the upper cannabinoid pathway can be scaffolded, while the enzymes in the hexanoyl-CoA pathway or a hexanoyl-CoA synthase and enzymes in the mevalonate or MEP pathways can be soluble. In some embodiments, the enzymes in the hexanoyl-CoA pathway or a hexanoyl-CoA synthase can be scaffolded N-terminal to the enzymes in the upper cannabinoid pathway, and enzymes in the mevalonate or MEP pathways can be soluble. In such embodiments, malonyl-CoA can be supplemented. Suitable enzymes for each of these pathways are described above.

In some embodiments, such as the embodiment shown in FIG. 10, a bi-directional scaffold can include enzymes of the malonyl-CoA (MCA) pathway on the N-terminus of the scaffold, enzymes of the mevalonate pathway on the C-terminus of the scaffold, and enzymes in the upper cannabinoid pathway in between. In some embodiments, a bi-directional scaffold can include enzymes of the malonyl-CoA pathway on the N-terminus of the scaffold, enzymes of the MEP

pathway on the C-terminus of the scaffold, and enzymes in the upper cannabinoid pathway in between. In such embodiments, enzymes of the hexanoyl-CoA pathway can be on a separate scaffold or can be soluble. In some embodiments, HCS can be substituted for scaffolded or soluble enzymes of the hexanoyl-CoA pathway.

In some embodiments, each of the pathways are on separate scaffolds. For example, in one embodiment, enzymes of the upper cannabinoid pathway can be on one scaffold, enzymes of the mevalonate or MEP pathway can be localized on one scaffold, enzymes of the hexanoyl-CoA pathway can be localized on one scaffold, and enzymes of the malonyl-CoA pathway can be localized on another scaffold.

Cannabigerolic acid biosynthesized in any of the embodiments described herein can be isolated and/or can be used as a substrate for synthesis of other secondary and tertiary cannabinoids using downstream cannabinoid synthases. In order to generate a more diverse profile of cannabinoids, the downstream cannabinoid synthases typically are not scaffolded, as scaffolding would favor production of the terminal cannabinoid. In some embodiments, however, one or more of the downstream cannabinoid synthases can be included on a scaffold described herein.

For example, one or more of cannabidiolic acid synthase (CBDAS), cannabichromenic acid synthase (CBCAS), tetrahydrocannabinolic acid synthase (THCAS), or other cannabinoid synthases can be used to produce additional cannabigerolate-derived cannabinoids. For example, a CBDAS; a CBCAS; a THCAS; a CBDAS and a CBCAS; a CBDAS and a THCAS; a CBCAS and a THCAS; or a CBDAS, CBCAS, and THCAS can be used to produce additional cannabigerolate-derived cannabinoids such as one or more of cannabidiolic acid, cannabichromenic acid, and delta-9 tetrahydrocannabinolic acid. CBDAS is classified under EC 1.21.3.8 and can catalyze the synthesis of cannabidiolic acid from cannabigerolic acid. CBCAS is classified under EC 1.3.3—and can catalyze the synthesis of cannabichromenic acid from cannabigerolic acid. THCAS is classified under EC 1.21.3.7 and can catalyze the synthesis of delta-9 tetrahydrocannabinolic acid from cannabigerolic acid.

#### Host Cells for Producing Cannabinoids

Cannabinoids can be produced in host cells or in vitro using a multi-enzymatic scaffold as described herein. Suitable host cells include any microorganism, eukaryotic or prokaryotic, such as bacteria (e.g., *Escherichia coli*, *Bacillus*, *Brevibacterium*, *Streptomyces*, or *Pseudomonas*), yeast (e.g., *Pichia pastoris*, *Saccharomyces cerevisiae*, *Yarrowia lipolytica*, *Kluyveromyces marxianus*, or *Komagataella phaffii*) and other fungi (e.g., *Neurospora crassa*), and green algae (e.g., *Dunaliella* sp., *Chlorella variabilis*, *Euglena mutabilis*, or *Chlamydomonas reinhardtii*), as well as plant cells (e.g., tobacco, *Cannabis*, or other photosynthetic plant cells) that can be maintained in culture or, in the case of plant cells such as those from tobacco or *cannabis* plants, can be engineered in culture and cultivated as intact transgenic plants. Such host cells or plant may or may not naturally produce cannabinoids.

A host cell can be modified to contain one or more exogenous nucleic acids that encode a scaffold as described herein and one or more exogenous nucleic acids that encode the engineered enzymes. The term “nucleic acid” as used herein encompasses both RNA and DNA, including cDNA, genomic DNA, and synthetic (e.g., chemically synthesized) DNA. The nucleic acid can be double-stranded or single-

stranded. Where single-stranded, the nucleic acid can be the sense strand or the antisense strand. In addition, nucleic acid can be circular or linear.

The term “exogenous” as used herein with reference to nucleic acid and a particular host cell refers to any nucleic acid that does not originate from that particular host cell as found in nature. Thus, non-naturally-occurring nucleic acid is considered to be exogenous to a host cell once introduced into the host cell. It is important to note that non-naturally-occurring nucleic acid can contain nucleic acid sequences or fragments of nucleic acid sequences that are found in nature provided the nucleic acid as a whole does not exist in nature. For example, a nucleic acid molecule containing a genomic DNA sequence within an expression vector is non-naturally-occurring nucleic acid, and thus is exogenous to a host cell once introduced into the host cell, since that nucleic acid molecule as a whole (genomic DNA plus vector DNA) does not exist in nature. Thus, any vector, autonomously replicating plasmid, or virus (e.g., retrovirus, adenovirus, or herpes virus) that as a whole does not exist in nature is considered to be non-naturally-occurring nucleic acid. It follows that genomic DNA fragments produced by PCR or restriction endonuclease treatment as well as cDNAs are considered to be non-naturally-occurring nucleic acid since they exist as separate molecules not found in nature. It also follows that any nucleic acid containing a promoter sequence and polypeptide-encoding sequence (e.g., cDNA or genomic DNA) in an arrangement not found in nature is non-naturally-occurring nucleic acid.

A nucleic acid that is naturally-occurring can be exogenous to a particular cell. For example, an entire chromosome isolated from a cell of organism X is an exogenous nucleic acid with respect to a cell of organism Y once that chromosome is introduced into Y's cell.

It is noted that a host cell can be given an exogenous nucleic acid molecule that encodes a polypeptide having an enzymatic activity that catalyzes the production of a compound not normally produced by that host cell. Alternatively, or additionally, a host cell can be given an exogenous nucleic acid molecule that encodes a polypeptide having an enzymatic activity that catalyzes the production of a compound that is normally produced by that host cell. In this case, the recombinant host cell can produce more of the compound, or can produce the compound more efficiently, than a similar host cell not having the genetic modification.

An enzyme having a particular enzymatic activity can be a polypeptide that is either naturally-occurring or non-naturally-occurring. A naturally-occurring polypeptide is any polypeptide having an amino acid sequence as found in nature, including wild-type and polymorphic polypeptides. Such naturally-occurring polypeptides can be obtained from any species including, without limitation, animal (e.g., mammalian), plant, fungal, and bacterial species. A non-naturally-occurring polypeptide is any polypeptide having an amino acid sequence that is not found in nature. Thus, a non-naturally-occurring polypeptide can be a mutated version of a naturally-occurring polypeptide, or an engineered polypeptide such as the engineered enzymes described herein that contain IDs. For example, a non-naturally-occurring polypeptide having geranyl pyrophosphate synthase activity can be a mutated version of a naturally-occurring polypeptide having geranyl pyrophosphate synthase activity. For example, the GPPS encoded by Erg20 may include a substitution of a tryptophan for phenylalanine at position 96 and a substitution of a tryptophan for asparagine at position 127 (referred to as Erg20<sup>rw</sup>). Erg20<sup>rw</sup> favors production of geranyl pyrophosphate over farnesyl

pyrophosphate. See, Jiang, et al., *Metab Eng.* 2017, 41:57-66. For example, a truncated HMGR (tHMGR) such as an N-terminally truncated HMGR that includes the catalytic domain but not the transmembrane or regulatory domains of HMGR can be used. For example, the HMGR from *A. thaliana* (GenBank Accession No. J04537) or a HMGR from *S. cerevisiae* (which contains only residues 646-1025) can be truncated to remove the transmembrane and/or regulatory domains and used in a scaffold described herein to remove a bottleneck in the mevalonate pathway. HMGR catalyzes the rate-limiting step in the mevalonate pathway (see, e.g., Song et al., 2017, *Scientific reports*, doi:10.1038/s41598-017-15005-4). For example, the nucleic acid encoding an atoB from *S. cerevisiae* can be modified to contain a synthetic 5' UTR (such as the synthetic 5' UTR sequence: 5'-cggcaccctcaaacagaaggaatataaa-3' (SEQ ID NO:82)) and can be used in the scaffold as it alters atoB expression to facilitate flux-rebalancing in favor of production of acetoacetyl-CoA over the reverse reaction product butyryl-CoA (see Kim et al., 2018, *Bioresour Technol.* doi: 10.1016/j.biortech. 2017.10.014). A polypeptide can be mutated by, for example, sequence additions, deletions, substitutions, or combinations thereof.

Any of the enzymes described herein that can be used to produce one or more cannabinoids can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of the corresponding wild-type enzyme. It will be appreciated that the sequence identity can be determined on the basis of the mature enzyme (e.g., with any signal sequence removed).

For example, an ACL can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *Homo sapiens* ACL (see SEQ ID NO:83, FIG. 6A), or an ACL from *Rattus norvegicus*, *Mus musculus*, or *Ciona intestinalis*, e.g., GenBank Accession Nos. AAA74463, AAK56081, and BAB00624, respectively.

For example, an acetyl-CoA acetyltransferase (atoB) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of an *Escherichia coli* atoB (see SEQ ID NO:84, FIG. 6A), or an atoB from *Cupriavidus necator*, *Clostridium acetobutylicum*, or *Arabidopsis thaliana*, e.g., GenBank Accession Nos. CAJ92573, AAK80816, and AAM67058, respectively. In some embodiments, a malonyl-CoA acyl carrier protein transacylase from *Saccharomyces cerevisiae*, *Homo sapiens*, *Serratia plymuthica*, or *Dickeya paradisiaca* can be substituted for atoB, e.g., GenBank Accession Nos. DAA10992, AAH30985, AG055277, and ACS85236, respectively.

For example, a 3-hydroxy-butyryl-CoA dehydrogenase (BHBD) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *Clostridium acetobutylicum* BHBD (see SEQ ID NO:85, FIG. 6A), or a BHBD from *Escherichia coli*, *Treponema denticola*, or *Arabidopsis thaliana*, e.g., GenBank Accession Nos. AIZ91493, AAS11105, and AAN17431, respectively.

For example, an enoyl-CoA hydratase (ECH) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *Clostridium acetobutylicum* ECH (see SEQ ID NO:86, FIG. 6A), or an ECH from *Acinetobacter oleivorans*, *Cupriavidus necator*, or *Acinetobacter baumannii*, e.g., GenBank Accession Nos. ADI91469, CAJ91294, and ACJ57023, respectively.

For example, a beta-ketothiolase (bktB) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *Cupriavidus necator* bktB (see SEQ ID NO:87, FIG. 6A), or a bktB from *Escherichia coli*, *Lactobacillus casei*, or *Clostridium acetobutylicum*, e.g., GenBank Accession Nos. ALI39443, CAQ67083, and AAK80816, respectively.

For example, a trans-2-enoyl-CoA-reductase (ECR) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *Treponema denticola* ECR (see SEQ ID NO:88, FIG. 6A), or an ECR from *Cupriavidus necator*, *Saccharomyces cerevisiae*, or *Klebsiella michiganensis*, e.g., GenBank Accession Nos. AAP86010, DAA07148, and AIE72439, respectively.

For example, a hexanoyl-CoA synthetase (HCS), which is a type of acyl-activating enzyme (AAE), can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *C. sativa* AAE1 (see SEQ ID NO:89, FIG. 6A, GenBank Accession No. AFD33345) or *C. sativa* AAE3 (GenBank Accession No. AFD33347). The *C. sativa* AAE1 and AAE3 each can use hexanoate as a substrate. See, Stout, et al., *Plant* 1, 71(3): 353-365 (2012). In some embodiments, the AAE encoded by CsAAE1 can be used. See, GenBank Accession No. JN717233 for the coding sequence. In some embodiments, the AAE encoded by CsAAE3 can be used. See, GenBank Accession No. JN717233 for the coding sequence. In some embodiments, both CsAAE1 and CsAAE3 can be used.

For example, an HMG-CoA synthase (HMGS) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *S. cerevisiae* HMGS (see SEQ ID NO:90, FIG. 6A), or an HMGS from *Arabidopsis thaliana*, *Lactobacillus casei*, or *Homo sapiens*, e.g., GenBank Accession Nos. AEE83052, CAQ67081, and AAA62411, respectively.

For example, an HMG-CoA reductase (HMGR), N-terminally truncated or canonical, can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *S. cerevisiae* HMGS (see SEQ ID NO:91, FIG. 6A), or an HMGR from *Arabidopsis thaliana*, *Lactobacillus casei*, or *Homo sapiens*, e.g., GenBank Accession Nos. AEE35849, CAQ67082, and AAA52679, respectively.

For example, a mevalonate kinase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *S. cerevisiae* mevalonate kinase (see SEQ ID NO:92, FIG. 6A), or a mevalonate kinase from *Arabidopsis thaliana*, *Lactobacillus casei*, or *Homo sapiens*, e.g., GenBank Accession Nos. AAD31719, CAQ66794, and AAF82407, respectively.

For example, a phosphomevalonate kinase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *S. cerevisiae* phosphomevalonate kinase (see SEQ ID NO:93, FIG. 6A), or a mevalonate kinase from *Scheffersomyces stipitis*, *Lactobacillus casei*, or *Homo sapiens*, e.g., GenBank Accession Nos. EAZ63544, CAQ66339, and AAH06089, respectively.

For example, a diphosphomevalonate decarboxylase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *S. cerevisiae* diphosphomevalonate decarboxylase (see SEQ ID NO:94, FIG. 6A), or a diphosphomevalonate decarboxylase from *Arabidopsis thaliana*,

*Lactobacillus casei*, or *Homo sapiens*, e.g., GenBank Accession Nos. AAC67348, CAQ66795, and AAC50440, respectively.

For example, an isopentyl diphosphate isomerase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *S. cerevisiae* isopentyl diphosphate isomerase (see SEQ ID NO:95, FIG. 6A), or an isopentyl diphosphate isomerase from *Arabidopsis thaliana*, *Lactobacillus casei*, or *Homo sapiens*, e.g., GenBank Accession Nos. AAC49920, CAQ66796, and AAP35407, respectively.

For example, a geranyl pyrophosphate synthase (GPPS) (also known as a geranyl-diphosphate synthase) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of the *S. cerevisiae* GPS or a GPPS from *Acinetobacter baumannii*, *Lactobacillus casei*, or *Homo sapiens*, e.g., GenBank Accession Nos. ACJ56139, CAQ66932, and AAH10004, respectively. In some embodiments, a mutant GPPS can be used. For example, the GPPS encoded by Erg20 may include a substitution of a tryptophan for phenylalanine at position 96 and a substitution of a tryptophan for asparagine at position 127 (referred to as Erg20<sup>FW</sup>) (see SEQ ID NO:96, FIG. 6A). Erg20<sup>FW</sup> favors production of geranyl pyrophosphate over farnesyl pyrophosphate. See, Jiang, et al., *Metab Eng.* 2017 41:57-66. In some cases, substituting a glutamic acid for lysine at position 179 of Erg20 (Erg20<sup>K179E</sup>) can be used to produce a GPPS that favors production of geranyl pyrophosphate. See, WO2016010827A1.

For example, a DOXP synthase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of an *Escherichia coli*, *Clostridium acetobutylicum*, *Treponema denticola*, or *Arabidopsis thaliana* DOXP synthase, e.g., GenBank Accession Nos. CDH63925, AAK80036, AAS12424, and ANM65835, respectively.

For example, a DOXP reductoisomerase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of an *Escherichia coli*, *Clostridium acetobutylicum*, *Treponema denticola*, or *Arabidopsis thaliana* DOXP reductoisomerase, e.g., GenBank Accession Nos. CDH63708, AAK79760, AAS12860, and AAM61343, respectively.

For example, a MEP cytidyl transferase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of an *Escherichia coli*, *Clostridium acetobutylicum*, *Treponema denticola*, or *Arabidopsis thaliana* MEP cytidyl transferase, e.g., GenBank Accession Nos. CDH66380, AAK81121, AAS12810, and BAB21592, respectively.

For example, a CDPME kinase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of an *Escherichia coli*, *Clostridium acetobutylicum*, *Treponema denticola*, or *Arabidopsis thaliana* CDPME kinase, e.g., GenBank Accession Nos. CDH64802, AAK80844, AAS11855, and AEC07908, respectively.

For example, a MECDP synthase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of an *Escherichia coli*, *Nicotiana tabacum*, *Treponema denticola*, or *Acinetobacter baumannii* MECDP synthase, e.g., GenBank Accession Nos. CDH66379, AHM22925, AAS12811, and ACJ59227, respectively.

For example, an HMBPP synthase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%,

97%, 98%, 99%, or 100%) to the amino acid sequence of an *Escherichia coli*, *Acinetobacter baumannii*, *Treponema denticola*, or *Arabidopsis thaliana* HMBPP synthase, e.g., GenBank Accession Nos. AAN81487, ACJ58210, AAS11783, and AED97354, respectively.

For example, an HMBPP reductase can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of an *Escherichia coli*, *Acinetobacter baumannii*, *Treponema denticola*, or *Arabidopsis thaliana* HMBPP reductase, e.g., GenBank Accession Nos. CDH63564, ACJ57384, AAS11585, and AEE86362, respectively.

For example, an acetyl-CoA carboxylase (ACC) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *S. cerevisiae* acetyl-CoA carboxylase (see SEQ ID NO:97, FIG. 6A), or an acetyl-CoA carboxylase from *Homo sapiens*, *Treponema denticola*, or *Cupriavidus necator*, e.g., GenBank Accession Nos. AAP94122, AAS11086, and CAQ67359, respectively.

For example, a pyruvate dehydrogenase (E1) and dihydrolipoyl transacetylase (E2) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a *Saccharomyces cerevisiae*, *Escherichia coli*, *Clostridium acetobutylicum*, or *Cupriavidus necator* E1 and E2, e.g., GenBank Accession Nos. DAA07337, AMC97367, CAQ66617, and CAJ92510 for E1, and DAA10474, AUG14916, CAQ66619, and CAJ92511 for E2, respectively.

For example, an olivetol synthase (OS) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of an OS from *C. sativa* set forth in SEQ ID NO:98 (FIG. 6A) or the OS from *C. sativa* having GenBank Accession No. BAG14339. See, for example, Taura, et al., *FEBS Letters* 583 (2009) 2061-2066.

For example, an olivetolic acid cyclase (OAC) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of an OAC from *C. sativa* set forth in SEQ ID NO:99 (FIG. 6A) or the OAC from *C. sativa* having GenBank Accession No. AFN42527. See, for example, Gagne, et al., *Proc. Natl. Acad. Sci. USA*, 2012 109 (31) 12811-12816.

For example, a CBGAS can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of an aromatic prenyl-transferase (APT) from *Cannabis sativa* such as the CBGAS set forth in SEQ ID NO:100 (FIG. 6A). See, for example, U.S. Patent Publication No. 20120144523A1 and U.S. Pat. No. 8,884,100B2. In some embodiments, a soluble APT from *Streptomyces* (e.g., NphB) can be used. See, for example, Carvalho et al., *FEMS Yeast Research*, 17, 2017, fox037.

For example, a cannabidiolic acid synthase (CBDAS) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a CBDAS from *C. sativa* set forth in SEQ ID NO:101 (FIG. 6A) or the amino acid sequence of a CBDAS from *C. sativa* having GenBank Accession No. BAF65033. See, for example, Taura, et al., *FEBS Lett.* 581 (16), 2929-2934 (2007).

For example, a cannabichromenic acid synthase (CBCAS) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a CBCAS from *C. sativa* set

forth in SEQ ID NO:102 (FIG. 6A) or the amino acid sequence of a CBCAS from *C. sativa* as set forth in SEQ ID NO:2 of WO 2015/196275 A1. SEQ ID NO:2 of WO 2015/196275 A1 includes an N-terminal 28 amino acid signal peptide. All or a portion of the signal peptide can be removed from the sequence. The CBDAS from *C. indica* or *C. ruderalis* also can be used. In some embodiments, an *Escherichia coli* or yeast optimized nucleic acid sequence encoding a *C. sativa* CBCAS as set forth in SEQ ID NOs: 8 and 9, respectively, of WO 2015/196275 A1 can be used.

For example, a tetrahydrocannabinolic acid synthase (THCAS) can have at least 70% sequence identity (e.g., at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%) to the amino acid sequence of a THCAS from *C. sativa* having GenBank Accession No. BAC41356. See, for example, Sirikantaramas, et al., *J. Biol. Chem.* 279 (38), 39767-39774 (2004).

The percent identity (homology) between two amino acid sequences can be determined as follows. First, the amino acid sequences are aligned using the BLAST 2 Sequences (B12seq) program from the stand-alone version of BLASTZ containing BLASTP version 2.0.14. This stand-alone version of BLASTZ can be obtained from Fish & Richardson's web site (e.g., [www.fr.com/blast/](http://www.fr.com/blast/)) or the U.S. government's National Center for Biotechnology Information web site ([www.ncbi.nlm.nih.gov](http://www.ncbi.nlm.nih.gov)). Instructions explaining how to use the B12seq program can be found in the readme file accompanying BLASTZ. B12seq performs a comparison between two amino acid sequences using the BLASTP algorithm. To compare two amino acid sequences, the options of B12seq are set as follows: -i is set to a file containing the first amino acid sequence to be compared (e.g., C:\seq1.txt); -j is set to a file containing the second amino acid sequence to be compared (e.g., C:\seq2.txt); -p is set to blastp; -o is set to any desired file name (e.g., C:\output.txt); and all other options are left at their default setting. For example, the following command can be used to generate an output file containing a comparison between two amino acid sequences: C:\B12seq-i c:\seq1.txt-j c:\seq2.txt-p blastp-o c:\output.txt. If the two compared sequences share homology (identity), then the designated output file will present those regions of homology as aligned sequences. If the two compared sequences do not share homology (identity), then the designated output file will not present aligned sequences. Similar procedures can be following for nucleic acid sequences except that blastn is used.

Once aligned, the number of matches is determined by counting the number of positions where an identical amino acid residue is presented in both sequences. The percent identity (homology) is determined by dividing the number of matches by the length of the full-length polypeptide amino acid sequence followed by multiplying the resulting value by 100. It is noted that the percent identity (homology) value is rounded to the nearest tenth. For example, 78.11, 78.12, 78.13, and 78.14 is rounded down to 78.1, while 78.15, 78.16, 78.17, 78.18, and 78.19 is rounded up to 78.2. It also is noted that the length value will always be an integer.

It will be appreciated that a number of nucleic acids can encode a polypeptide having a particular amino acid sequence. The degeneracy of the genetic code is well known to the art; i.e., for many amino acids, there is more than one nucleotide triplet that serves as the codon for the amino acid. For example, codons in the coding sequence for a given enzyme can be modified such that optimal expression in a particular species (e.g., bacteria or fungus) can be attained, using appropriate codon bias tables for that species. For example, the nucleotide sequences set forth in FIG. 12A are



the nucleic acid sequences encoding an ATP citrate lyase, an atoB, a 3-hydroxybutyryl-CoA dehydrogenase, an enoyl-CoA hydratase, a beto-ketothiolase (bktB), a trans-enoyl-CoA reductase, an HMG-CoA synthase, an HMG-CoA reductase, a mevalonate kinase, a phosphomevalonate kinase, a diphosphomevalonate decarboxylase, an isopentenyl-diphosphate delta isomerase, a geranyl-diphosphate synthase (ERG20<sup>WW</sup>), an olivetol synthase, an olivetolic acid cyclase, a CBGA synthase, a CBDA synthase, a CBCA synthase, an acetyl-CoA carboxylase, and a hexanoyl-CoA synthetase. The nucleic acid sequences for the ATP citrate lyase, atoB, 3-hydroxybutyryl-CoA dehydrogenase, enoyl-CoA hydratase, trans-enoyl-CoA reductase, bktB, olivetol synthase, olivetolic acid cyclase, CBGA synthase, CBDA synthase, and CBCA synthase have been codon optimized for expression in yeast. FIGS. 14A-14C contain codon optimized (for expression in yeast) nucleic acid sequences encoding the engineered enzymes of FIGS. 13A-13C.

In addition to sequence similarity, it will be appreciated that enzymes and scaffolds with structural and/or functional similarity to the enzymes and scaffolds described herein are also encompassed within the scope of the document.

This document provides recombinant host cells that can be used to produce one or more cannabinoids as described herein. For example, an individual host cell can contain exogenous nucleic acid such that the scaffold polypeptide and each of the enzymes to be immobilized on the scaffold are expressed. It is important to note that such host cells can contain any number and/or combination of exogenous nucleic acid molecules. For example, a particular host cell can contain an exogenous nucleic acid encoding the scaffold, and additional exogenous nucleic acids encoding the enzymes of the malonyl-CoA pathway, enzymes of the hexanoyl-CoA pathway or encoding a HCS, and enzymes of the mevalonate or MEP pathway. A single exogenous nucleic acid can encode one enzyme or more than one enzyme (e.g., one or more copies of from one to ten (or more) enzymes, from one to eight, from one to seven, from one to six, from one to five, from one to four, or from two to three enzymes). Thus, the number of different exogenous nucleic acids needed to produce the engineered enzymes to be localized on the scaffold will depend on the design of the scaffold and/or the particular embodiment. FIG. 2A and FIG. 2B each provide a non-limiting schematic of suitable gene cassettes for expressing the scaffolds and enzymes. FIG. 12C provides the nucleic acid sequence encoding a scaffold polypeptide containing the peptide ligands corresponding to IDs 1-16 as shown in Table 2 and a triplicate MYC tag. See also FIG. 14D for the codon-optimized nucleic acid sequence encoding the scaffold polypeptide of FIG. 13D. FIG. 12D provides the nucleic acid sequence encoding a scaffold polypeptide that contains the peptide ligands corresponding to IDs 1 and 17, and a triplicate FLAG tag. See also FIG. 14D.

In some embodiments, multiple nucleic acids encoding polypeptides (e.g., the nucleic acids of a gene cassette such as in FIG. 2A or FIG. 2B) can be linked together using a nucleic acid sequence encoding a self-cleaving peptide. During translation of the transcripts, the growing polypeptide can be cleaved at the 2A peptide with translation continuing through to the next polypeptide. When designing a vector to express the polypeptides as a polycistronic unit, the nucleic acid encoding the polypeptides and the self-cleaving peptide (e.g., a 2A peptide) can be designed such that they are in translational frame with each other. Examples of 2A peptides that can be used as described herein include, without limitation, a 2A peptide of foot-and-

mouth disease virus (FMDV), a 2A peptide of equine rhinitis A virus (ERAVO), a 2A peptide of Thossea asigna virus (TaV), or a 2A peptide of porcine teschovirus-1 (PTV-1) or porcine teschovirus-2 (PTV-2). The 2A peptides from PTV-1 and PTV-2 are referred to as P2A peptides. See, e.g., SEQ ID NO:212 for a codon-optimized nucleotide sequence (for *S. cerevisiae*) encoding a P2A peptide.

Further, the cells described herein can contain a single copy or multiple copies (e.g., about 5, 10, 20, 35, 50, 75, 100 or 150 copies), of a particular exogenous nucleic acid molecule. Again, the cells described herein can contain more than one particular exogenous nucleic acid molecule and/or copies thereof. For example, a particular cell can contain about 50 copies of exogenous nucleic acid molecule X as well as about 75 copies of exogenous nucleic acid molecule Y.

Any method can be used to introduce an exogenous nucleic acid molecule into a host cell. In fact, many methods for introducing nucleic acid into host cells such as bacteria and yeast are well known to those skilled in the art. For example, heat shock, lipofection, electroporation, nucleofection, conjugation, fusion of protoplasts, and biolistic delivery are common methods for introducing nucleic acid into bacteria and yeast cells. See, e.g., Ito et al., *J. Bacteriol.* 153:163-168 (1983); Durrens et al., *Curr. Genet.* 18:7-12 (1990); and Becker and Guarente, *Methods in Enzymology* 194:182-187 (1991).

An exogenous nucleic acid molecule contained within a particular host cell can be maintained within that host cell in any form. For example, exogenous nucleic acid molecules can be integrated into the genome of the microorganism or maintained in an episomal state. In other words, a microorganism can be a stable or transient transformant. Again, a microorganism described herein can contain a single copy, or multiple copies (e.g., about 5, 10, 20, 35, 50, 75, 100 or 150 copies), of a particular exogenous nucleic acid molecule as described herein.

Suitable nucleic acid constructs for expressing the engineered enzymes and scaffolds include, for example, CRISPR plasmids, baculovirus vectors, bacteriophage vectors, plasmids, phagemids, cosmids, fosmids, bacterial artificial chromosomes, viral vectors (for example, viral vectors based on vaccinia virus, poliovirus, adenovirus, adeno-associated virus, SV40, herpes simplex virus, and the like), P1-based artificial chromosomes, yeast plasmids, yeast artificial chromosomes, and other vectors. Typically such constructs include a regulatory element that promotes the expression of a nucleic acid sequence that encodes a polypeptide. Typically, regulatory elements are DNA sequences that regulate the expression of other DNA sequences at the level of transcription. Thus, regulatory elements include, without limitation, promoters, enhancers, and the like. Any type of promoter can be used to express an amino acid sequence from an exogenous nucleic acid molecule. Examples of promoters include, without limitation, constitutive promoters, tissue-specific promoters, and inducible or repressible promoters that are responsive or unresponsive to a particular stimulus (e.g., light, oxygen, chemical concentration, sound, and the like).

In some embodiments, endogenous yeast promoters with varying constitutive activity levels can be used to express the engineered enzymes and/or scaffolds. To maintain an excess of enzymes relative to scaffold molecules, the scaffolds can be expressed under control of the weakest promoter. For example, one or more of the following yeast promoters can be used: the promoter from the gene encoding transcriptional elongation factor EF-1 $\alpha$  (pTEF1), the pro-



motor from the gene encoding phosphoglycerate kinase (PGK1), the promoter from the gene encoding triose phosphate isomerase (pTPI1), the promoter from the gene encoding a hexose transporter (pHXT7), HXT7, the promoter from the gene encoding pyruvate kinase 1 (pPYK1), the promoter from the gene encoding alcohol dehydrogenase 1 (pADH1), or the promoter from the gene encoding triphosphate dehydrogenase (pTDH3). For example, in the embodiment shown in FIG. 2A, the pTPI1 promoter can be used to express enzymes of the upper hexanoyl-CoA (HCA), enzymes of the lower HCA pathway, enzymes of the upper mevalonate (MVA) pathway, enzymes of the lower MVA pathway, and enzymes of the lower cannabinoid (CB) pathway, while the pTEF1 promoter can be used to express enzymes of the upper CB pathway, the *atoB* enzyme, and the enzymes of the malonyl-CoA pathway, and the pADH1 promoter can be used to express the scaffold. Of these promoters, the pADH1 promoter has the weakest activity (+ in FIG. 2A), the pTEF1 promoter has the strongest activity (+++ in FIG. 2A), and the activity of the pTPI1 promoter is between the other two (++ in FIG. 2A). In some embodiments, the Gal 1-10 promoter (e.g., from *S. cerevisiae*) can be used. See, e.g., FIG. 17.

A nucleic acid construct also can include a selectable marker, e.g., for an antibiotic such as neomycin resistance, ampicillin resistance, tetracycline resistance, chloramphenicol resistance, or kanamycin resistance). In some embodiments, a nutritional marker gene that confers prototrophy for an essential nutrient such as tryptophan (TRP1), uracil (URA3), histidine (HIS3), leucine (LEU2), lysine (LYS2), or methionine can be included on a nucleic acid construct. See, e.g., FIG. 17. As shown in Example 3, four different auxotrophic markers were used to sequentially select for transformed cells containing the desired combinations of nucleic acids encoding the enzymes and scaffold. For example, yeast cells transformed with a vector containing a TRP gene and the nucleic acids encoding enzymes of the hexanoyl-CoA pathway were grown in tryptophan deficient media. The transformed cells that grew in the tryptophan deficient media were selected and further transformed with a vector containing a LEU gene and nucleic acid encoding enzymes of the mevalonate pathway. The resulting transformed cells were grown on media lacking tryptophan and leucine, and the cells that grew in the media lacking tryptophan and leucine were transformed with a vector containing a HIS gene and nucleic acids encoding enzymes of the upper cannabinoid pathway. The resulting transformed cells were grown on media lacking tryptophan, leucine, and histidine, and the cells that grew in the media lacking tryptophan, leucine, and histidine were transformed with a vector containing a URA3 gene and a nucleic acid encoding a scaffold. The resulting transformed cells were grown on media lacking tryptophan, leucine, histidine, and uracil. Cells that grew in media lacking tryptophan, leucine, histidine, and uracil contained the desired combination of enzymes and scaffold as shown in FIG. 1B.

In some embodiments, the encoded enzymes (e.g., one or more enzymes from the cannabinoid biosynthesis pathway, mevalonate pathway, MEP pathway, hexanoyl-CoA pathway, or a hexanoyl-CoA synthetase) and/or the scaffold can include a targeting sequence that can be used to direct the enzymes or scaffold to one of several different intracellular compartments, including, for example, the endoplasmic reticulum (ER), mitochondria, plastids (such as chloroplasts), the vacuole, the Golgi apparatus, or protein storage vesicles (PSV). For example, a mitochondrial or plastidial targeting sequence can be used to facilitate mitochondrial or

plastidial compartmentalization of cannabinoid/cannabinoid precursor biosynthesis such that the encoded enzymes and scaffold are expressed in the mitochondria or plastids of the host cell.

In some embodiments, cannabinoid/cannabinoid precursor biosynthesis can be performed in two compartments by co-expressing one or more engineered enzymes and a scaffold in both the cytosolic compartment and either the plastids or mitochondria of the host cell. See, for example, FIG. 11. It will be appreciated that while FIG. 11 depicts a scaffold containing enzymes of the hexanoyl-CoA pathway, enzymes of the upper cannabinoid pathway, and enzymes of the mevalonate pathway, dual-compartment engineering can be performed with any of the scaffolds and enzymes described herein. For example, dual-compartment engineering can be performed in two compartments by co-expressing a scaffold and enzymes of the hexanoyl-CoA pathway, enzymes of the upper cannabinoid pathway, and enzymes of the MEP pathway in both the cytosolic compartment and either the plastids or mitochondria of the host cell. Dual-compartment engineering also can be achieved by engineering separate haploid yeast strains for cytosolic and mitochondrial/plastidial cannabinoid biosynthesis, and then mating these two haploid strains to produce a diploid lineage that is heterozygous for cytosolic and mitochondrial/plastidial cannabinoid biosynthesis.

In some embodiments, the engineered enzymes and/or scaffolds also contain a tag that can be used for purification of the recombinant protein (e.g., c-myc, FLAG, polyhistidine (e.g., hexahistidine), hemagglutinin (HA), glutathione-S-transferase (GST), or maltose binding protein (MBP)) or as a detectable marker (e.g., luciferase, green fluorescent protein (GFP), or chloramphenicol acetyl transferase (CAT)). For example, in the embodiment shown in FIG. 6C and FIG. 6D, a scaffold can include a myc tag (e.g., (Myc)<sub>3</sub> tag) or a FLAG tag (FLAG)<sub>3</sub> tag at the C-terminus.

In some embodiments, a host cell can be engineered to increase acetyl-CoA availability for cannabinoid and cannabinoid precursor biosynthesis. For example, the mitochondrial enzyme isocitrate dehydrogenase-1 (IDH1) can be placed under transient micro-RNA-mediated inducible repression. Since mitochondrial IDH1 is primarily responsible for depletion of the cellular citrate pool, micro-RNA-mediated repression of IDH1 can increase the availability and cytosolic shuttling of citrate for production of acetyl-CoA by ATP citrate lyase. The resulting increase in acetyl-CoA bioavailability can further enhance downstream hexanoyl-CoA and geranyl pyrophosphate titers by improving initial substrate availability for the hexanoyl-CoA and mevalonate pathways. The combinatorial metabolic engineering of acetyl-CoA can mitigate issues related to the siphoning of acetyl-CoA away from the endogenous metabolism of the host cells.

In some embodiments, one or more conventional and/or contemporary gene editing techniques can be used to produce recombinant hosts. For example, clustered, regularly interspaced, short palindromic repeat (CRISPR) technology can be used to modify expression of an endogenous nucleic acid. The CRISPR/Cas system includes components of a prokaryotic adaptive immune system that is functionally analogous to eukaryotic RNA interference, using RNA base pairing to direct DNA or RNA cleavage. The Cas9 protein functions as an endonuclease, and CRISPR RNA (crRNA) and trans-activating RNA (tracrRNA) sequences complex with the Cas9 enzyme and direct it to a target DNA sequence (Makarova et al., *Nat Rev Microbiol* 9(6):467-477, 2011). The modification of a single targeting RNA can be sufficient

to alter the nucleotide target of a Cas protein. In some cases, crRNA and tracrRNA can be engineered as a single crRNA/tracrRNA hybrid (also referred to as a “guide RNA” or “gRNA”) to direct Cas9 cleavage activity (Jinek et al., *Science*, 337(6096):816-821, 2012). The CRISPR/Cas system can be used in a variety of prokaryotic and eukaryotic organisms (see, e.g., Jiang et al., *Nat Biotechnol*, 31(3):233-239, 2013; Dicarlo et al., *Nucleic Acids Res*, doi:10.1093/nar/gkt135, 2013; Cong et al., *Science*, 339(6121):819-823, 2013; Mali et al., *Science*, 339(6121):823-826, 2013; Cho et al., *Nat Biotechnol*, 31(3):230-232, 2013; and Hwang et al., *Nat Biotechnol*, 31(3):227-229, 2013).

Another gene-editing technique can include a sequence-specific nuclease created by fusing transcription activator-like effectors (TALEs) to, for example, the catalytic domain of the FokI endonuclease. Both native and custom TALE-nuclease (“TALEN”) fusions direct DNA double-strand breaks to specific, targeted sites. See, for example, Christian, et al., *Genetics* 186: 757-761 (2010) and U.S. Patent Publication No. 20110145940.

Other suitable gene insertion techniques include the use of retroviral vectors and biolistic particle gene delivery systems (colloquially known as “gene guns”).

Methods of identifying and/or selecting host cells that contain exogenous nucleic acid or a modified endogenous nucleic acid are well known to those skilled in the art. Such methods include, without limitation, the introduction and expression of a negative selection marker such as an antibiotic resistance gene, PCR, and nucleic acid hybridization techniques such as Northern and Southern analyses. In some cases, immunohistochemistry and biochemical techniques can be used to determine if a microorganism contains a particular nucleic acid by detecting the expression of the encoded enzymatic polypeptide encoded by that particular nucleic acid molecule. For example, an antibody having specificity for an encoded enzyme can be used to determine whether or not a particular cell contains that encoded enzyme. Further, biochemical techniques can be used to determine if a cell contains a particular nucleic acid molecule encoding an enzymatic polypeptide by detecting an organic product produced as a result of the expression of the enzymatic polypeptide.

This document also provides isolated nucleic acids molecules. The term “isolated” as used herein with reference to nucleic acid refers to a naturally-occurring nucleic acid that is not immediately contiguous with both of the sequences with which it is immediately contiguous (one on the 5' end and one on the 3' end) in the naturally-occurring genome of the organism from which it is derived. For example, an isolated nucleic acid can be, without limitation, a recombinant DNA molecule of any length, provided one of the nucleic acid sequences normally found immediately flanking that recombinant DNA molecule in a naturally-occurring genome is removed or absent. Thus, an isolated nucleic acid includes, without limitation, a recombinant DNA that exists as a separate molecule (e.g., a cDNA or a genomic DNA fragment produced by PCR or restriction endonuclease treatment) independent of other sequences as well as recombinant DNA that is incorporated into a vector, an autonomously replicating plasmid, a virus (e.g., a retrovirus, adenovirus, or herpes virus), or into the genomic DNA of a prokaryote or eukaryote. In addition, an isolated nucleic acid can include a recombinant DNA molecule that is part of a hybrid or fusion nucleic acid sequence.

The term “isolated” as used herein with reference to nucleic acid also includes any non-naturally-occurring nucleic acid since non-naturally-occurring nucleic acid

sequences are not found in nature and do not have immediately contiguous sequences in a naturally-occurring genome. For example, non-naturally-occurring nucleic acid such as an engineered nucleic acid is considered to be isolated nucleic acid. Engineered nucleic acid can be made using common molecular cloning or chemical nucleic acid synthesis techniques. Isolated non-naturally-occurring nucleic acid can be independent of other sequences, or incorporated into a vector, an autonomously replicating plasmid, a virus (e.g., a retrovirus, adenovirus, or herpes virus), or the genomic DNA of a prokaryote or eukaryote. In addition, a non-naturally-occurring nucleic acid can include a nucleic acid molecule that is part of a hybrid or fusion nucleic acid sequence.

It will be apparent to those of skill in the art that a nucleic acid existing among hundreds to millions of other nucleic acid molecules within, for example, cDNA or genomic libraries, or gel slices containing a genomic DNA restriction digest is not to be considered an isolated nucleic acid.

In some embodiments, the production of one or more cannabinoids can be performed in vitro using the scaffold and immobilized enzymes described herein, using a lysate (e.g., a buffered cell lysate) from a recombinant host cell as a source of the scaffold and enzymes, using a plurality of lysates from different host cells as the source of the scaffold and enzymes, or using an acellular reaction buffer such as a synthetic reaction buffer. For example, following co-immunoprecipitation of C-terminal Myc/Flag-tagged enzyme-bound scaffolds, scaffold-enzyme complexes can be maintained in a citrate-supplemented and/or glucose-supplemented (or other carbon source-supplemented) reaction buffer which allows in-vitro scaffolded cannabinoid biosynthesis.

Producing Cannabinoids Using a Recombinant Host

Typically, one or more cannabinoids can be produced by providing a recombinant host such as a recombinant microorganism and culturing the microorganism with a culture medium. In general, the culture media and/or culture conditions can be such that the microorganisms grow to an adequate density and produce cannabinoids efficiently. For example, the microorganisms can be subjected to aerobic batch fermentation. In some embodiments, one or more precursors (e.g., citrate, glucose, hexanoic acid, and/or other carbon source and/or malonyl-CoA) are supplemented in the culture medium. In some embodiments, about 30 mg/L to about 10,000 mg/L (e.g., about 100 mg/L to about 5,000 mg/L, about 200 mg/L to about 4,000 mg/L, about 300 mg/L to about 3,000 mg/L, or about 350 mg/L to about 1,000 mg/L) of buffered citrate, pH 6.0 can be added to the culture medium.

For large-scale production processes, any method can be used such as those described elsewhere (Manual of Industrial Microbiology and Biotechnology, 2nd Edition, Editors: A. L. Demain and J. E. Davies, ASM Press; and Principles of Fermentation Technology, P. F. Stanbury and A. Whitaker, Pergamon). Briefly, a large vessel (e.g., a 100 gallon, 200 gallon, 500 gallon, or higher volume vessel) containing an appropriate culture medium is inoculated with a particular microorganism. After inoculation, the microorganism is incubated to allow biomass to be produced. Once a desired biomass or cellular confluency is attained, a portion or all of the broth containing the microorganisms can be transferred to a second vessel. This second vessel can be any size. For example, the second vessel can be larger, smaller, or the same size as the first vessel. Typically, the second vessel is larger than the first such that additional culture medium can be added to the broth from the first vessel. In addition, the

culture medium within this second vessel can be the same as, or different from, that used in the first vessel. This system can expand to include an array consisting of any number of individual vessels.

Once transferred, the microorganisms can be incubated to allow for the production of one or more cannabinoids. Once produced, any method can be used to isolate cannabinoids. For example, common separation techniques can be used to remove the biomass from the broth, and common isolation procedures (e.g., extraction such as non-polar extraction with hexane followed by ethyl-acetate), high-performance liquid chromatography (e.g., HPLC with a diode array detector (HPLC-DAD)), gas chromatography-flame ionization detection (GC-FID), or ion-exchange procedures) can be used to obtain the cannabinoids from the biomass.

A host cell described herein can produce one or more cannabinoids at a concentration of at least about 10 mg per L (e.g., at least about 15 mg/L, 25 mg/L, 50 mg/L, 75 mg/L, 100 mg/L, 150 mg/L, 200 mg/L, 250 mg/L or more). For example, in some embodiments, total cannabinoids (total of CBG, CBGA, CBD, CBDA, CBC, and CBCA) can be produced at a concentration of at least about 10 mg/L, 15 mg/L, 20 mg/L, 40 mg/L, 60 mg/L, 80 mg/L, or 100 mg/L or more. For example, in some embodiments, total cannabinoids (total of CBG, CBGA, CBD, CBDA, CBC, and CBCA) can be produced at a concentration from about 10 mg/L to about 500 mg/L (e.g., 20 mg/L to 450 mg/L, 40 mg/L to 380 mg/L, 60 mg/L to 280 mg/L, 80 mg/L to 250 mg/L, 100 mg/L to 150 mg/L, 120 mg/L to 400 mg/L, 140 mg/L to 300 mg/L, 160 mg/L to 250 mg/L, 180 mg/L to 200 mg/L, 200 mg/L to 175 mg/L, 220 mg/L to 400 mg/L, 240 mg/L to 300 mg/L, 260 mg/L to 250 mg/L, or 280 mg/L to 150 mg/L). In some embodiments, one or more individual cannabinoids (e.g., one or more of CBG, CBGA, CBD, CBDA, CBC, and CBCA) can be produced at concentrations of at least about 1 mg/L, 2 mg/L, 5 mg/L, 10 mg/L, 15 mg/L, 20 mg/L, 25 mg/L, 30 mg/L, 35 mg/L, 40 mg/L, 45 mg/L, 50 mg/L, 55 mg/L, 60 mg/L, 65 mg/L, 70 mg/L, 75 mg/L, 80 mg/L, 85 mg/L, 90 mg/L, 95 mg/L, 100 mg/L or more. For example, in some embodiments, one or more individual cannabinoids can be produced at a concentration from about 1 mg/L to about 100 mg/L (e.g., 2 to 90 mg/L, 2 to 80 mg/L, 2 to 70 mg/L, 2 to 60 mg/L, 2 to 50 mg/L, 2 to 40 mg/L, 2 to 30 mg/L, 2 to 20 mg/L, 2 to 15 mg/L, 3 to 90 mg/L, 3 to 80 mg/L, 3 to 70 mg/L, 3 to 60 mg/L, 3 to 50 mg/L, 3 to 40 mg/L, 3 to 30 mg/L, 3 to 20 mg/L, 3 to 15 mg/L, 4 to 90 mg/L, 4 to 80 mg/L, 4 to 70 mg/L, 4 to 60 mg/L, 4 to 50 mg/L, 4 to 40 mg/L, 4 to 30 mg/L, 4 to 20 mg/L, or 4 to 15 mg/L).

The invention will be further described in the following examples, which do not limit the scope of the invention described in the claims.

## EXAMPLES

### Example 1—General Methods

#### Enzymatic Constructs

Each enzyme construct is designed to include an interaction domain (ID) which is comprised of two tandem N-terminal or C-terminal ligand-binding motifs which are separated from the given enzyme and from one another by an amino acid sequence containing flexible GS-rich linkers flanking a rigid  $\alpha$ -helical spacer sequence. The motifs comprising the ID of each enzyme specifically bind tandem peptide ligands which form ID-binding sites at discrete locations along a synthetic intracellular polypeptide scaffold.

fold. Expression of each enzyme is controlled by a constitutive or inducible promoter. The nucleic acid encoding the enzyme can be codon optimized, e.g., for expression in yeast.

#### Scaffolding Constructs

ID-binding sites containing tandem peptide ligands that are specific for the tandem scaffold-binding motifs, which comprise the ID of each enzyme, are inserted at discrete positions along an intracellular polypeptide scaffold.

The tandem ligands which comprise each scaffolded ID-binding site are separated from one another by a 36 amino acid residue sequence containing flexible GS-rich linkers flanking a rigid  $\alpha$ -helical spacer sequence, while the scaffolded ID-binding sites themselves are separated from one another by a 50 amino acid residue sequence (or any other number of amino acid residues) containing flexible GS-rich linkers flanking a rigid  $\alpha$ -helical spacer sequence. Specifically, the scaffold binding sites for each enzyme in the hexanoyl-CoA pathway are positioned (in order of catalysis) proximally to ATP citrate lyase and acetyl-CoA acetyltransferase at the N-terminus of the primary scaffold. Scaffold binding sites for each enzyme in the upper cannabinoid pathway are positioned proximally to (immediately downstream of) the binding sites for the hexanoyl-CoA pathway enzymes. The scaffold binding sites for each enzyme in the mevalonate (or MEP) pathway are positioned (in order of catalysis) proximally to ATP citrate lyase and acetyl-CoA acetyltransferase at the C-terminus of the primary scaffold. The enzyme catalyzing the rate-limiting/committed step in cannabinoid biosynthesis (CBGA synthase, the final enzymatic step in the upper cannabinoid pathway) is located at the intersection of the converging cannabinoid precursor pathways near the scaffold midpoint. Assessment of Cannabinoidergic Potential by Transient Transfection

Competent yeast and/or green algae cells are transiently transfected with plasmids encoding various permutations of the scaffold and enzymes. To establish baseline cannabinoidergic capacity, cells first undergo transient transfection with the enzymes required for cannabinoid biosynthesis (but not the scaffolds), and biosynthesized cannabinoids are extracted, isolated, and quantified as described below (see "Cannabinoid Extraction, Isolation, and Analytical Characterization"). To measure the improvement in cannabinoidergic capacity conferred by multi-enzymatic scaffolding, a subset of the aforementioned cells is co-transfected with plasmids encoding one or more of the multi-enzymatic scaffolds described herein, and biosynthesized cannabinoids are extracted, isolated, and quantified. The presence of the plasmid DNA is confirmed by PCR, functional gene expression is confirmed by qRT-PCR, protein/polypeptide production is confirmed by Western blotting, and scaffolding of each enzyme is confirmed by co-immunoprecipitation of C-terminal myc/flag-tagged scaffolds followed by Western blot analysis of each co-immunoprecipitated enzyme.

#### Engineering of Stable Cannabinoidergic Cell Lines

The constructs can be integrated into the genome of host cells such as yeast, green algae, or other suitable hosts via stable transfection. Gene integration is confirmed by PCR, functional gene expression is confirmed by qRT-PCR, and protein/polypeptide production is confirmed by Western blotting. Gene expression/protein synthesis is confirmed by comparing both qRT-PCR and Western blot results among samples with and without genetic engineering. To assess the improvement in cannabinoidergic capacity conferred by multi-enzymatic scaffolding for stably engineered cannabinoidergic cell lines, cannabinoid biosynthesis will be com-

pared among cells that are stimulated for enzyme but not scaffold expression and cells that are stimulated for enzyme and scaffold expression.

#### Validation of Multi-enzymatic Scaffolding

To verify successful multi-enzymatic scaffolding in both transiently transfected and stably engineered cells, a myc-tag (or other immunoprecipitable tag) is inserted at the N-terminal or C-terminal of the polypeptide scaffold(s). Scaffolded enzymes are selectively co-immunoprecipitated by affinity chromatography using anti-myc affinity beads. Western blots are performed to detect and quantify each co-immunoprecipitated enzyme.

#### Aerobic Fed-batch Fermentation

Stably engineered cannabinoidergic yeast, green algae, or other host cells are grown in bioreactors (or any other vessel) via aerobic batch fermentation (or any other culture technique).

#### Cannabinoid Extraction, Isolation, and Analytical Characterization

Following sufficient elicitation of cannabinoid biosynthesis, engineered yeast/green algae cells are pelleted by centrifugation and washed with TBS. The supernatant (liquid culture media) is decanted and collected. Following washing with TBS, pelleted cells are resuspended in NaOH adjusted ethanol and lysed by iterative freeze-thawing and ultrasonication. Biosynthesized cannabinoid fermentates are then harvested from both lysates and supernatants via triplicate nonpolar extractions using hexane followed by ethyl-acetate. The resulting organic fractions are pooled and roto-evaporated. High-performance liquid chromatography with a diode array detector (HPLC-DAD) or gas chromatography-flame ionization detection (GC-FID) is then applied for quantitative and qualitative measurement of biosynthesized cannabinoids.

In the following examples, each 48-hour culture was lysed/homogenized by ultrasonication. Ultrasonicated samples were then subjected to triplicate liquid-liquid extractions with ethyl acetate (one volumetric equivalent of ethyl acetate per extraction). Following separation, the ethyl acetate fractions collected from each sample were pooled, and the pooled samples were centrifugally filtered. Ethyl acetate was then removed from each sample in a vacuum oven, and the residual samples were resuspended in 10 mL methanol for analytical characterization. Analytical characterization of all samples was conducted by a licensed, independent, third-party analytical testing facility (Precision Plant Molecules, Denver, CO). HPLC-DAD was utilized for quantitative and qualitative measurement of each parent and derivative cannabinoid as well as the cannabinoid precursor OVA.

#### Example 2—Synthetic Gene Cassette Assembly/Synthesis, Plasmid Preparation, and Polycistronic Vector Construction

Five synthetic gene cassettes (entitled HCA, GPP, CAN, SCF, and SOL) were constructed for biosynthesizing cannabinoids in heterologous cells or acellular reaction buffers. See, FIG. 2B. The cassettes collectively encode all scaffold-binding engineered enzymes and the polypeptide scaffolds to which the engineered enzymes can bind.

The HCA gene cassette encoded scaffold-binding engineered enzymes for scaffolded hexanoyl-CoA biosynthesis, namely ACL, atoB, BHBD, ECH, ECR, and bktB, and encoded a soluble HCS for additional hexanoyl-CoA production from hexanoate-supplemented culture media or acellular reaction buffer. See, FIG. 13A. The GPP gene

cassette encoded scaffold-binding engineered enzymes for scaffolded geranyl pyrophosphate (GPP) biosynthesis, namely HMGS, tHMGR, ERG12, ERGS, MVD1, IDI1, and ERG20<sup>HW</sup>. See, FIG. 13B. The CAN gene cassette encoded scaffold-binding engineered enzymes for scaffolded OAC, malonyl-CoA, and CBGA biosynthesis, namely OS and OAC, ACC, and CBGAS, respectively, as well all enzymes for soluble (non-scaffolded) CBDA and CBCA biosynthesis, namely CBDAS and CBCAS, respectively. See, FIG. 13C. The SCF gene cassette encoded the polypeptide scaffolds for bidirectional scaffolded cannabinoid biosynthesis and scaffolded malonyl-CoA biosynthesis, namely the cannabinoidergic metabolon scaffold (CBSCF) and the malonyl-CoA metabolon scaffold (MCASCF), respectively, as well as additional copies of both ACL and atoB to enhance acetyl-CoA biosynthesis from supplemental and/or endogenous citrate and acetoacetyl-CoA biosynthesis from acetyl-CoA, respectively. See, FIG. 13D. The SOL gene cassette lacked the polypeptide scaffolds for bidirectional scaffolded cannabinoid biosynthesis and scaffolded malonyl-CoA biosynthesis (i.e., it was used for soluble cannabinoid biosynthesis) but, analogous to the SCF gene cassette, encoded additional copies of ACL and atoB to enhance acetyl-CoA biosynthesis from supplemental and/or endogenous citrate and acetoacetyl-CoA biosynthesis from acetyl-CoA. See FIG. 13A for the amino acids sequences of the ACL and atoB engineered enzymes.

Gene cassettes were assembled/synthesized using self-cleaving 2A peptides (P2As) to link multiple codon-optimized (for *S. cerevisiae*) gene sequences assigned to each cassette. To improve P2A cleavage, a GSG linker (comprised of a single serine residue flanked by single glycine residues) was inserted at the interface between each constituent gene sequence and the P2A linker sequence to which it was fused (of the format: gene cassette sequence 1—SG—P2A linker—gene cassette sequence 2—GSG—P2A linker—gene cassette sequence 3—GSG—P2A linker—) and so forth. See, FIGS. 14A-14D for codon-optimized nucleic acid sequences encoding the engineered enzymes and scaffolds. Following assembly, each synthetic gene cassette was inserted into a pCCI-Brick plasmid, resulting in plasmids entitled pHCA, pGPP, pCAN, pSCF, and pSOL as described in Table 3. See, FIGS. 15A-15E for the complete gene cassette inserted into the plasmids. Each of these plasmids then were used to amplify each synthetic gene cassette via standard plasmid prep. Plasmid DNA encoding each complete synthetic gene cassette was cloned into the SpeI/XhoI cloning site of polycistronic yeast auxotrophic selection vectors, resulting in vectors entitled vHCA, vGPP, vCAN, vSCF, and vSOL as described in Table 3, to allow iterative antibiotic/auxotrophic selection of only those cells that were transformants of one or more such polycistronic vector(s).

TABLE 3

| HCA Gene Cassette |                   |                  |              |                 |
|-------------------|-------------------|------------------|--------------|-----------------|
| Gene ID           | Cassette Position | pCCI-Brick #1 ID | Yeast Vector | Yeast Vector ID |
| ACL               | 1                 | pHCA             | pESC-TRP     | vHCA            |
| atoB              | 2                 |                  |              |                 |
| BHBD              | 3                 |                  |              |                 |
| ECH               | 4                 |                  |              |                 |
| ECR               | 5                 |                  |              |                 |

TABLE 3-continued

|                      |                   |                  |              |                 |
|----------------------|-------------------|------------------|--------------|-----------------|
| bktB                 | 6                 |                  |              |                 |
| HCS                  | 7                 |                  |              |                 |
| MVA Gene Cassette    |                   |                  |              |                 |
| Gene ID              | Cassette Position | pCCI-Brick #2 ID | Yeast Vector | Yeast Vector ID |
| HMGS                 | 1                 | pGPP             | pESC-LEU     | vGPP            |
| tHMGR                | 2                 |                  |              |                 |
| ERG12                | 3                 |                  |              |                 |
| ERG8                 | 4                 |                  |              |                 |
| MVD1                 | 5                 |                  |              |                 |
| IDI1                 | 6                 |                  |              |                 |
| ERG20 <sup>HW</sup>  | 7                 |                  |              |                 |
| CAN Gene Cassette    |                   |                  |              |                 |
| Gene ID              | Cassette Position | pCCI-Brick #3 ID | Yeast Vector | Yeast Vector ID |
| OS                   | 1                 | pCAN             | pESC-HIS     | vCAN            |
| OAC                  | 2                 |                  |              |                 |
| CBGAS                | 3                 |                  |              |                 |
| CBDAS                | 4                 |                  |              |                 |
| CBCAS                | 5                 |                  |              |                 |
| ACC                  | 6                 |                  |              |                 |
| SCFLD Gene Cassette  |                   |                  |              |                 |
| Gene ID              | Cassette Position | pCCI-Brick #4 ID | Yeast Vector | Yeast Vector ID |
| CBSCF                | 1                 | pSCF             | pESC-URA #1  | vSCF            |
| MCASCF               | 2                 |                  |              |                 |
| ACL                  | 3                 |                  |              |                 |
| atoB                 | 4                 |                  |              |                 |
| NSCFLD Gene Cassette |                   |                  |              |                 |
| Gene ID              | Cassette Position | pCCI-Brick #5 ID | Yeast Vector | Yeast Vector ID |
| ACL                  | 1                 | pSOL             | pESC-URA #2  | vSOL            |
| atoB                 | 2                 |                  |              |                 |

The genes assigned to each synthetic gene cassette as well as the plasmids and vectors into which each synthetic gene cassette was inserted are listed in Table 3, the amino acid sequences encoded by each synthetic gene cassette are provided in FIGS. 13A-13D, the codon-optimized nucleotide sequence fragments comprising each synthetic gene cassette are detailed in FIGS. 14A-14D, the complete nucleotide sequences of each fully-assembled synthetic gene cassette (the complete insert sequences for each plasmid and expression vector) are provided in FIGS. 15A-15E, a general map of pCCI-Brick plasmids is shown in FIG. 16, and a general map of a polycistronic yeast auxotrophic selection vector is shown in FIG. 17.

#### Example 3—Engineering of Cannabinoidergic Cells

To engineer a novel heterologous pathway for the biosynthesis of cannabinoids from citrate, and to evaluate the impacts of bidirectional multi-enzymatic scaffolding thereon, competent *S. cerevisiae* cells were sequentially/iteratively transformed with, and auxotrophically selected for, expression of vHCA, vGPP, vCAN, and either vSCF (for scaffolded cannabinoid biosynthesis) or vSOL (for non-scaffolded/soluble cannabinoid biosynthesis) constructs.

All vector transformation and auxotrophic selection procedures were conducted as follows. An aliquot of an overnight *S. cerevisiae* culture was inoculated into 100 mL YPD

media (10 g/L yeast nitrogen base, 20 g/L peptone, and 20 g/L D-(+)-glucose) to OD<sub>600nm</sub>=0.3 (stationary phase) and grown to OD<sub>600nm</sub>=1.6 in an orbital shaker at 30° C. and 225 RPM. Cells then were harvested by centrifugation at 3000×g for 3 minutes followed by aspiration of media. The harvested cell pellet was next washed 2× with 50 mL chilled nuclease-free water and 1× with 50 mL chilled electroporation buffer (1M sorbitol/1 mM CaCl<sub>2</sub>). Washed cells were conditioned by incubation for 30 minutes in 20 mL 0.1M LiAc/10 mM DTT in an orbital shaker at 30° C. and 225 RPM, harvested, washed 1× with 50 mL electroporation buffer, harvested, and resuspended in 100 µL electroporation buffer. The resuspended cells were transformed with a quantity of vector containing 3 µg of the target DNA insert (calculated using the vector-insert ratio for each vector) by electroporation at 2.5 kV and 25 g. To the electroporated cell suspension was then added 8 mL of YPD media containing 1M sorbitol, and the resulting suspension was incubated for one hour in an orbital shaker at 30° C. and 225 RPM. To isolate target transformants by auxotrophic selection, cells were harvested, resuspended in the appropriate yeast nitrogen base (YNB) dropout (selection) media as subsequently described for each iterative transformation step, transferred to a baffled culture flask, and incubated overnight in an orbital shaker at 30° C. and 225 RPM. The transformation and selection protocols were utilized sequentially for each assigned vector.

Applying the aforementioned approach, an initial culture of electrocompetent *S. cerevisiae* cells was first transformed with vHCA, which encodes scaffold-binding engineered enzymes required for biosynthesis of HCA from citrate. Cells transformed with vHCA (designated yHCA) were selected for by resuspension and incubation in tryptophan-deficient YNB media. Selected yHCA cells (i.e., cells that grew in tryptophan-deficient YNB media) were next transformed with vGPP, which encodes scaffold-binding engineered enzymes required for biosynthesis of GPP from citrate. Cells co-transformed with vHCA and vGPP (designated yHCAGPP) were selected for by resuspension and incubation in tryptophan- and leucine-deficient YNB media. Selected yHCAGPP cells (i.e., cells that grew in tryptophan- and leucine-deficient YNB media) were then transformed with vCAN, which encodes scaffold-binding engineered enzymes required for biosynthesis of malonyl-CoA from citrate, olivetol from HCA and malonyl-CoA, OVA (olivetol acid) from olivetol, and CBGA from OVA and GPP as well as soluble enzymes required for biosynthesis of CBDA and CBCA from CBGA). Cells co-transformed with vHCA, vGPP, and vCAN (designated yCB<sub>Parent</sub>) were selected for by resuspension and incubation in tryptophan-, leucine-, and histidine-deficient YNB media.

The yCB<sub>Parent</sub> culture containing cells that grew in tryptophan-, leucine-, and histidine-deficient YNB media then was split into two separate cultures. The first of the split yCB<sub>Parent</sub> cultures was transformed with vSCF, which encodes CBSCF (cannabinoidergic metabolon scaffold) and MCASCF (malonyl-CoA metabolon scaffold) as well as additional copies of ACL and atoB. Cells co-transformed with vHCA, vGPP, vCAN, and vSCF (designated yCB<sub>SCF</sub>) were selected for by resuspension and incubation in tryptophan-, leucine-, histidine-, and uracil-deficient YNB media. The second of the split yCB<sub>Parent</sub> cultures was transformed with vSOL, which encodes additional copies of ACL and atoB but lacks both CB SCF and MCASCF. Cells co-transformed with vHCA, vGPP, vCAN, and vSOL (desig-

nated yCB<sub>SOL</sub>) were also selected for by resuspension and incubation in tryptophan-, leucine-, histidine-, and uracil-deficient YNB media.

To quantify the improvement in cannabinoidergic capacity conferred by multi-enzymatic scaffolding, cannabinoid titers were compared between triplicate yCB<sub>SOL</sub> and yCB<sub>SCF</sub> cultures grown in 100 mL YPD media for 48 hours at 30° C. and 400 RPM in an incubator-shaker. To compare the proliferation rates of yCB<sub>SOL</sub> and yCB<sub>SCF</sub>, each culture was initially diluted to OD<sub>600nm</sub>=0.3, and OD<sub>600 nm</sub> measurements were recorded in 12-hour intervals thereafter. Proliferation curves are depicted in FIG. 18. The extra sum-of-squares F-test indicated that the proliferation curves of yCB<sub>SCF</sub> and yCBSOL cultures did not significantly differ for any parameter over the 48-hour incubation period, indicating that scaffolding does not impact cellular proliferation.

Total cannabinoid titers, parent (carboxylated) cannabinoid (CBGA, CBDA, and CBCA) titers, derivative (decarboxylated) cannabinoid (CBG, CBD, and CBC) titers, and cannabinoid precursor (OVA) titers were measured. As shown in FIGS. 19A-19E, mixed ANOVA detected main effects of strain ( $F_{1,4}=943.8$ ;  $p<0.0001$ ) and analyte (cannabinoid and cannabinoid precursor) titers ( $F_{10,40}=216.4$ ;  $p<0.0001$ ) and a significant strain x analyte interaction ( $F_{10,40}=131.4$ ;  $p<0.0001$ ). Relative to yCBSOL cultures, yCBSCF cultures exhibited increased total cannabinoid ( $p<0.0001$ ), OVA precursor ( $p<0.0001$ ), CBG(A) ( $p<0.0001$ ), CBD(A) ( $p<0.0001$ ), CBC(A) ( $p<0.0001$ ), CBGA ( $p<0.0001$ ), CBDA ( $p<0.0001$ ), CBCA ( $p<0.0001$ ), CBG ( $p<0.0001$ ), CBD ( $p<0.01$ ), and CBC ( $p<0.001$ ) titers.

#### Example 4—Impacts of Citrate and Hexanoate Supplementation on Scaffolded and Soluble Cannabinoid Biosynthesis

To evaluate the impacts of culture media supplementation with citrate and hexanoate precursors, cannabinoid titers were compared between triplicate yCB<sub>SOL</sub> and yCB<sub>SCF</sub> cultures grown in 100 mL YPD media containing 300 mg/L of either buffered citrate (pH 6.0) or hexanoate for 48 hours at 30° C. and 400 RPM in an orbital shaker. All cultures were initially diluted to OD<sub>600nm</sub>=0.3. Cannabinoid titers for cultures grown in YPD media, citrate-supplemented YPD media, and hexanoate-supplemented YPD media were assessed and analyzed by ANOVA. As shown in FIG. 20, mixed ANOVA detected main effects of strain ( $F_{1,4}=457.5$ ;

$p<0.0001$ ) and culture media supplementation ( $F_{2,8}=312.5$ ;  $p<0.0001$ ) and a significant strain x culture media supplementation interaction ( $F_{2,8}=289.6$ ;  $p<0.0001$ ). Compared to basal media cultures, yCBSCF but not yCBSOL cultures exhibited increased total cannabinoid titers when cultured in media supplemented with 300 mg/L citrate ( $p<0.0001$ ). Neither yCBSCF nor yCBSOL cultures differed in total cannabinoid titers relative to basal media when cultured in media supplemented with 300 mg/L hexanoate. For all measures,  $n=3$  biological replicates for yCBSCF and yCBSOL cultures. Moreover, relative to yCBSOL cultures, yCBSCF cultures exhibited increased total cannabinoid titers when cultured in basal media ( $p<0.0001$ , data also reported in FIG. 19) as well as media supplemented with 300 mg/L citrate ( $p<0.0001$ ) and hexanoate ( $p<0.0001$ ).

To delineate concentration-response relationships for the supplementation of culture media with citrate, cannabinoid titers were compared between triplicate yCB<sub>SOL</sub> and yCB<sub>SCF</sub> cultures grown in 100 mL YPD media containing 0, 10, 30, 100, 300, 1000, 3000, and 10000 mg/L buffered citrate (pH 6.0) for 48 hours at 30° C. and 400 RPM in an orbital shaker. All cultures were initially diluted to OD<sub>600nm</sub>=0.3. Following quantification, asymmetric sigmoidal (five-parameter) logistic regressions were computed to fit concentration-response curves, from which were derived estimates of the maximal cannabinoid titer (CB<sub>Max</sub>) and citrate EC<sub>50</sub> for cannabinoid biosynthesis in yCB<sub>SOL</sub> and yCB<sub>SCF</sub> cultures. Concentration-response curves, CB<sub>Max</sub> estimates, and citrate EC<sub>50</sub> estimates are depicted in FIGS. 21A and 21B. Mixed ANOVA detected main effects of strain ( $F_{1,8}=69.9$ ;  $p<0.0001$ ) and parameter ( $F_{1,8}=66.7$ ;  $p<0.0001$ ) and a significant strain x parameter interaction ( $F_{1,8}=5.3$ ;  $p<0.05$ ) for concentration-response parameter estimates (CB<sub>Max</sub> and citrate EC<sub>50</sub>). Compared to yCB<sub>SOL</sub> cultures, yCB<sub>SCF</sub> cultures exhibited markedly increased CB<sub>Max</sub> ( $p<0.0001$ ) and citrate EC<sub>50</sub> ( $p<0.001$ ) estimates.

#### OTHER EMBODIMENTS

It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

#### SEQUENCE LISTING

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                     organism = synthetic construct

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IANLRKKIEE 70

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IARLENEVAS HEQ 73

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                 organism = synthetic construct
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LAHKK 65

SEQ ID NO: 8      moltype = AA length = 77
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                 organism = synthetic construct
SEQUENCE: 8
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LATLENEVAR LENDVAE 77

SEQ ID NO: 9      moltype = AA length = 64
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                 organism = synthetic construct
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QAQQ 64

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                 organism = synthetic construct
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VPYRGNTYNI PICLWLLDTY PYNPPICFVK PTSSMTIKTG KHVDANGKIY LPYLHEWKHP 120
QSDLLGLIQV MIVVFGDEPP VFSRP 145

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SPESLRSKVD EAVAVLQAHQ AKAAQA 88

SEQ ID NO: 12     moltype = AA length = 107
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TKRLYDEKQQ HIVYCSNDLL GDLEFGVPSFS VKEHRKIYTM IYRNLVV 107

SEQ ID NO: 13         moltype = AA length = 65
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                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 13
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LEAYK 65

SEQ ID NO: 14         moltype = AA length = 73
FEATURE              Location/Qualifiers
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                      organism = synthetic construct

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IKTLENEVAS HEQ 73

SEQ ID NO: 15         moltype = AA length = 62
FEATURE              Location/Qualifiers
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                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 15
DVMWEYKWEN TGDAELYGPF TSAQMQTWVS EGYFPDGVYC RKLDPGGQF YNSKRIDFDL 60
YT 62

SEQ ID NO: 16         moltype = AA length = 85
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                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 16
ESDSVEFNNA ISYVNKIKTR FLDHPEIYRS FLEILHTYQK EQLHTKGRPF RGMSEEEVFT 60
EVANLFRGQE DLLSEFGQFL PEAKR 85

SEQ ID NO: 17         moltype = AA length = 36
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                      organism = synthetic construct

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SEQ ID NO: 18         moltype = AA length = 37
FEATURE              Location/Qualifiers
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                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 18
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SEQ ID NO: 19         moltype = AA length = 101
FEATURE              Location/Qualifiers
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                      organism = synthetic construct

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LQTEIANLLK EKEKLEFILA AHRPACKIPD DLGPFPEMSL E 101

SEQ ID NO: 20         moltype = AA length = 59
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                      mol_type = protein
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SEQ ID NO: 21         moltype = AA length = 147
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mol_type = protein
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MEPAMEPETL EARINRATNP LNKELDWASI NGFCEQLNED FEGPPLATRL LAHKIQSPQE 60
WEAIQALTVL ETCMKSCGKR FHDEVGKFRF LNELIKVVSP KYLGSRTSEK VKNKILELLY 120
SWTVGLPEEV KIAEAYQLMK KQGIVKS 147

SEQ ID NO: 22      moltype = AA length = 171
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                  organism = synthetic construct

SEQUENCE: 22
GAMGSMABAE GESLESWLNK ATNPSNRQED WEYIIGFCDQ INKELEGPOI AVRLLAHKIQ 60
SPQEWELQA LTVLEACMKR CGRRFHNEVG KFRFLNELIK VVSPKYLGR VSEKVKTKVI 120
ELLYSWTMAL PEEAKIKDAY HMLKRQGIVQ SDPPIPVDRT LIPSPPPRPK N 171

SEQ ID NO: 23      moltype = AA length = 67
FEATURE           Location/Qualifiers
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SEQUENCE: 23
SYHHHHHHHL ESTSLYKAG SGSQKVEELK NKIAELENRN AVKKNRVAHL KQEIAYLKDE 60
LAAHEFE 67

SEQ ID NO: 24      moltype = AA length = 64
FEATURE           Location/Qualifiers
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                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 24
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VAWL 64

SEQ ID NO: 25      moltype = AA length = 168
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AMADLEQKVL EMEASTYDGV FIWKISDFPR KRQEAAGRI PAIFSPAFYT SRYGYKMLR 60
IYLNGDGTGR GTHLSLFFVV MKGPNDAALL WPFNQKVTLM LLDQNNREHV IDAFRPDVT 120
SSFQRPVNDM NIASGCPLFC PVSMEAKNS YVRDDAIFIK AIVDLTGL 168

SEQ ID NO: 26      moltype = AA length = 159
FEATURE           Location/Qualifiers
source            1..159
                  mol_type = protein
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SEQUENCE: 26
ASIKLQSSDG EIFEVDVEIA KQSVTIKTM EDLGMDDGDD DDPVPLPNVN AAILKKVIQW 60
CTHHKDDPPP PEDDENKEKR TDDIPVWDQE FLKVDQGTLF ELILAANYLD IKGLLDVTCK 120
TVANMIKGKT PEEIRKTFNI KNDFTEEEE QVRKENQWC 159

SEQ ID NO: 27      moltype = AA length = 67
FEATURE           Location/Qualifiers
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                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 27
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LAAHKFE 67

SEQ ID NO: 28      moltype = AA length = 77
FEATURE           Location/Qualifiers
source            1..77
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 28
SYHHHHHHHL ESTSLYKAG SGSQKVAQLK NRVAYKLKEN AKLENIVARL ENDNANLEKD 60
IANLEKDIAN LERDVAR 77

SEQ ID NO: 29      moltype = AA length = 81
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LCTMKKGPSG YGFNLHSDKS KPGQFIRSVD PDSPAASGL RAQDRIVEVN GVCMEGKQHG 60  
 DVVSAIRAGG DETKLLVVDR E 81

SEQ ID NO: 30 moltype = AA length = 97  
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 organism = synthetic construct

SEQUENCE: 30  
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 SSLKGGKPLSE AIHLLQMAE TVTLKIKKQT DAQPASS 97

SEQ ID NO: 31 moltype = AA length = 109  
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 mol\_type = protein  
 organism = synthetic construct

SEQUENCE: 31  
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 HYHIKETNDS PKRYVVAEKY VFDSIPLLIQ YHQYNGGGLV TRLRYPVCG 109

SEQ ID NO: 32 moltype = AA length = 104  
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 organism = synthetic construct

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 GKYFLWVVKF NSLNELVDYH RSTSVSRNQQ IFLRDIEQVP QQPT 104

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 organism = synthetic construct

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 QKIPLTISFG GIKIFDEKTG ALQHHHAVHE ISYIAKDITD HRAFGYVCGK EGNHRFVAIK 120  
 TAQAAPVIL DLRDLFQLIY ELKQRELEK KA 152

SEQ ID NO: 34 moltype = AA length = 57  
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 mol\_type = protein  
 organism = synthetic construct

SEQUENCE: 34  
 AEYVRALPDF NGNDEEDLPF KKGDLIRIRD KPBEQWNAE DSEGKRGMIP VPYVEKY 57

SEQ ID NO: 35 moltype = AA length = 60  
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 source 1..60  
 mol\_type = protein  
 organism = synthetic construct

SEQUENCE: 35  
 LIKHMRAEAL PDFTGNSKLE LNFKAGDVIF LLSRINKDWL EGTVRGATGI FPLSFVKILK 60

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 organism = synthetic construct

SEQUENCE: 36  
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 LRKLIGSVDD LLPSLPSSSR TEIEGTQKLL NKDLAELINK MRLAQQNAV T SLSECKRQM 120  
 LTASHTLAVD AKNLLDAVDQ AKVLANLAHP PAE 153

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 TAADEPSSLW 70

SEQ ID NO: 38 moltype = AA length = 20  
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| SEQUENCE: 46          |                                |    |
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| SEQ ID NO: 47         | moltype = AA length = 20       |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
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|                       | organism = synthetic construct |    |
| SEQUENCE: 47          |                                |    |
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| FEATURE               | Location/Qualifiers            |    |
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| FEATURE               | Location/Qualifiers            |    |
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|                       | organism = synthetic construct |    |
| SEQUENCE: 49          |                                |    |
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| FEATURE               | Location/Qualifiers            |    |
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|                       | mol_type = protein             |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 50          |                                |    |
| TDEEREETEE EVYLLNSTTL |                                | 20 |
| SEQ ID NO: 51         | moltype = AA length = 20       |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = protein             |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 51          |                                |    |
| DGNVSGTQRL DSATVRTYSC |                                | 20 |
| SEQ ID NO: 52         | moltype = AA length = 20       |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = protein             |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 52          |                                |    |
| ALVDDAADYE PPPSNNEAL  |                                | 20 |
| SEQ ID NO: 53         | moltype = AA length = 20       |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = protein             |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 53          |                                |    |
| RELFDGPSYV NVQNLDKARQ |                                | 20 |
| SEQ ID NO: 54         | moltype = AA length = 20       |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = protein             |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 54          |                                |    |
| KNTKSMNFDN PVYRKTTEE  |                                | 20 |
| SEQ ID NO: 55         | moltype = AA length = 20       |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = protein             |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 55          |                                |    |
| RSLPSTWIEN KLYGMSDPNW |                                | 20 |
| SEQ ID NO: 56         | moltype = AA length = 20       |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = protein             |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 56          |                                |    |
| VVDNSPPPAL PPKKRQSAPS |                                | 20 |
| SEQ ID NO: 57         | moltype = AA length = 20       |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = protein             |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 57          |                                |    |
| TQRSKQPAPV PPRPSADLIL |                                | 20 |
| SEQ ID NO: 58         | moltype = AA length = 20       |    |
| FEATURE               | Location/Qualifiers            |    |

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|                                                  |                                                                                                                  |    |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----|
| source                                           | 1..20<br>mol_type = protein<br>organism = synthetic construct                                                    |    |
| SEQUENCE: 58<br>SATRELDLM ASLSDFKIQQ             |                                                                                                                  | 20 |
| SEQ ID NO: 59<br>FEATURE<br>source               | moltype = AA length = 20<br>Location/Qualifiers<br>1..20<br>mol_type = protein<br>organism = synthetic construct |    |
| SEQUENCE: 59<br>DLALSENWAQ EFLAAGDAVD            |                                                                                                                  | 20 |
| SEQ ID NO: 60<br>FEATURE<br>source               | moltype = AA length = 16<br>Location/Qualifiers<br>1..16<br>mol_type = protein<br>organism = synthetic construct |    |
| SEQUENCE: 60<br>GSGSGSGSGS GSGSGS                |                                                                                                                  | 16 |
| SEQ ID NO: 61<br>FEATURE<br>source               | moltype = AA length = 20<br>Location/Qualifiers<br>1..20<br>mol_type = protein<br>organism = synthetic construct |    |
| SEQUENCE: 61<br>GSGSGSGSGS GSGSGSGSGS            |                                                                                                                  | 20 |
| SEQ ID NO: 62<br>FEATURE<br>source               | moltype = AA length = 30<br>Location/Qualifiers<br>1..30<br>mol_type = protein<br>organism = synthetic construct |    |
| SEQUENCE: 62<br>GSGSGSGSGS GSGSGSGSGS GSGSGSGSGS |                                                                                                                  | 30 |
| SEQ ID NO: 63<br>FEATURE<br>source               | moltype = AA length = 15<br>Location/Qualifiers<br>1..15<br>mol_type = protein<br>organism = synthetic construct |    |
| SEQUENCE: 63<br>GGGSGGGGS GGGGS                  |                                                                                                                  | 15 |
| SEQ ID NO: 64<br>FEATURE<br>source               | moltype = AA length = 20<br>Location/Qualifiers<br>1..20<br>mol_type = protein<br>organism = synthetic construct |    |
| SEQUENCE: 64<br>GGGSGGGGS GGGSGGGGS              |                                                                                                                  | 20 |
| SEQ ID NO: 65<br>FEATURE<br>source               | moltype = AA length = 10<br>Location/Qualifiers<br>1..10<br>mol_type = protein<br>organism = synthetic construct |    |
| SEQUENCE: 65<br>EAAAKEAAAK                       |                                                                                                                  | 10 |
| SEQ ID NO: 66<br>FEATURE<br>source               | moltype = AA length = 15<br>Location/Qualifiers<br>1..15<br>mol_type = protein<br>organism = synthetic construct |    |
| SEQUENCE: 66<br>EAAAKEAAAK EAAAK                 |                                                                                                                  | 15 |
| SEQ ID NO: 67<br>FEATURE<br>source               | moltype = AA length = 20<br>Location/Qualifiers<br>1..20<br>mol_type = protein<br>organism = synthetic construct |    |
| SEQUENCE: 67<br>EAAAKEAAAK EAAAKEAAAK            |                                                                                                                  | 20 |
| SEQ ID NO: 68                                    | moltype = AA length = 6                                                                                          |    |

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|                                                                   |                                |     |
|-------------------------------------------------------------------|--------------------------------|-----|
| FEATURE                                                           | Location/Qualifiers            |     |
| source                                                            | 1..6                           |     |
|                                                                   | mol_type = protein             |     |
|                                                                   | organism = synthetic construct |     |
| SEQUENCE: 68                                                      |                                |     |
| AAAGGM                                                            |                                | 6   |
| SEQ ID NO: 69                                                     | moltype = AA length = 14       |     |
| FEATURE                                                           | Location/Qualifiers            |     |
| source                                                            | 1..14                          |     |
|                                                                   | mol_type = protein             |     |
|                                                                   | organism = synthetic construct |     |
| SEQUENCE: 69                                                      |                                |     |
| AAAGGMPPAA AGGM                                                   |                                | 14  |
| SEQ ID NO: 70                                                     | moltype = AA length = 6        |     |
| FEATURE                                                           | Location/Qualifiers            |     |
| source                                                            | 1..6                           |     |
|                                                                   | mol_type = protein             |     |
|                                                                   | organism = synthetic construct |     |
| SEQUENCE: 70                                                      |                                |     |
| AAAGGM                                                            |                                | 6   |
| SEQ ID NO: 71                                                     | moltype = AA length = 9        |     |
| FEATURE                                                           | Location/Qualifiers            |     |
| source                                                            | 1..9                           |     |
|                                                                   | mol_type = protein             |     |
|                                                                   | organism = synthetic construct |     |
| SEQUENCE: 71                                                      |                                |     |
| PPAAAGGMM                                                         |                                | 9   |
| SEQ ID NO: 72                                                     | moltype = AA length = 12       |     |
| FEATURE                                                           | Location/Qualifiers            |     |
| source                                                            | 1..12                          |     |
|                                                                   | mol_type = protein             |     |
|                                                                   | organism = synthetic construct |     |
| SEQUENCE: 72                                                      |                                |     |
| GSAGSAAGSG EF                                                     |                                | 12  |
| SEQ ID NO: 73                                                     | moltype = AA length = 18       |     |
| FEATURE                                                           | Location/Qualifiers            |     |
| source                                                            | 1..18                          |     |
|                                                                   | mol_type = protein             |     |
|                                                                   | organism = synthetic construct |     |
| SEQUENCE: 73                                                      |                                |     |
| KLSGGGGSGG GSGGGGS                                                |                                | 18  |
| SEQ ID NO: 74                                                     | moltype = AA length = 40       |     |
| FEATURE                                                           | Location/Qualifiers            |     |
| source                                                            | 1..40                          |     |
|                                                                   | mol_type = protein             |     |
|                                                                   | organism = synthetic construct |     |
| SEQUENCE: 74                                                      |                                |     |
| GSAGSAAGSG EFGSAEAAK EAAAKAGSAG SAAGSGEFGS                        |                                | 40  |
| SEQ ID NO: 75                                                     | moltype = AA length = 36       |     |
| FEATURE                                                           | Location/Qualifiers            |     |
| source                                                            | 1..36                          |     |
|                                                                   | mol_type = protein             |     |
|                                                                   | organism = synthetic construct |     |
| SEQUENCE: 75                                                      |                                |     |
| GSAGSAAGSG EFAEAAAKEA AAKAGSAGSA AGSGEF                           |                                | 36  |
| SEQ ID NO: 76                                                     | moltype = AA length = 50       |     |
| FEATURE                                                           | Location/Qualifiers            |     |
| source                                                            | 1..50                          |     |
|                                                                   | mol_type = protein             |     |
|                                                                   | organism = synthetic construct |     |
| SEQUENCE: 76                                                      |                                |     |
| GSAGSAAGSG EFGSAEAAK EAAAKEAAK EAAAKAGSAG SAAGSGEFGS              |                                | 50  |
| SEQ ID NO: 77                                                     | moltype = AA length = 211      |     |
| FEATURE                                                           | Location/Qualifiers            |     |
| source                                                            | 1..211                         |     |
|                                                                   | mol_type = protein             |     |
|                                                                   | organism = synthetic construct |     |
| SEQUENCE: 77                                                      |                                |     |
| AEAWNLGNA YYKQGDYQKA IEYYQKALEL DPNNAEAWYN LGNAYYKQGD YQKAIEYYQK  |                                | 60  |
| ALELDPNNAE AWYNLGNAYY KQGDYQKAIE DYQKALELDP NNLQAEAWKN LGNAYYKQGD |                                | 120 |

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YQKAIEYYQK ALELDPNNAS AWYNLGNAYY KQGDYQKAIE YYQKALELDP NNAKAWYRRG 180  
 NAYYKQGDYQ KAIEDYQKAL ELDPNRNRSS A 211

SEQ ID NO: 78                   moltype = AA   length = 16  
 FEATURE                      Location/Qualifiers  
 source                       1..16  
                              mol\_type = protein  
                              organism = synthetic construct

SEQUENCE: 78  
 GGGSGGGGS GGGGAS 16

SEQ ID NO: 79                   moltype = AA   length = 36  
 FEATURE                      Location/Qualifiers  
 source                       1..36  
                              mol\_type = protein  
                              organism = synthetic construct

SEQUENCE: 79  
 GSAGSAAGSG EFGSAGSAAG SGEFGSAGSA AGSGEF 36

SEQ ID NO: 80                   moltype = AA   length = 116  
 FEATURE                      Location/Qualifiers  
 source                       1..116  
                              mol\_type = protein  
                              organism = synthetic construct

SEQUENCE: 80  
 GSAGSAAGSG EFGSAEAAAK EAAAKEAAAK EAAAKAGSAG SAAGSGEFGS EQKLISEEDL 60  
 EQKLISEEDL EQKLISEEDL GSAGSAAGSG EFGSAGSAAG SGEFGSAGSA AGSGEF 116

SEQ ID NO: 81                   moltype = AA   length = 110  
 FEATURE                      Location/Qualifiers  
 source                       1..110  
                              mol\_type = protein  
                              organism = synthetic construct

SEQUENCE: 81  
 GSAGSAAGSG EFGSAEAAAK EAAAKEAAAK EAAAKAGSAG SAAGSGEFGS DYKDDDDKDY 60  
 KDDDDKDYD DDDKGSAGSA AGSGEFGSAG SAAGSGEFGS AGSAAGSGEF 110

SEQ ID NO: 82                   moltype = DNA   length = 30  
 FEATURE                      Location/Qualifiers  
 source                       1..30  
                              mol\_type = other DNA  
                              organism = synthetic construct

SEQUENCE: 82  
 cggcaccctt acaaacagaa ggaatataaa 30

SEQ ID NO: 83                   moltype = AA   length = 1101  
 FEATURE                      Location/Qualifiers  
 source                       1..1101  
                              mol\_type = protein  
                              organism = Homo sapiens

SEQUENCE: 83  
 MSAKAISEQT GKELLYKFIC TTSAIQNRFK YARVTPDTDW ARLLQDHPWL LSQNLVVKPD 60  
 QLIKRRGKLG LVGVNLTLTG VKSWLKPRLG QEATVGKATG FLKNFLIEPF VPHSQAEFFY 120  
 VCIYATREGD YVLFHHEGGV DVGVDVDAKA KLLVGVDKEL NPEDIKKHLL VHAPEDKKEI 180  
 LASFISGLFN FYEDLYFTYL EINPLVVTGD GYVLDLAAK VDATADYICK VKWGDIEFPP 240  
 PFGREAYPEE AYIADLDKAS GASLKLTLN PKGRIWTMVA GGGASVVYSD TICDLGGVNE 300  
 LANYGEYSAG PSEQQTYDYA KTLISLMTRE KHPDGKILII GGSIANFTNV AATFKGIVRA 360  
 IRDYQGPLEKE HEVTIFVRRG GPNYQGLRV MGEVGKTTGI PIHVFGTETH MTAIVGMALG 420  
 HRPINQPPPT AAHTANFLN ASGSTSTPAP SRTASFSESR ADEVAPAKKA KPAMPQDSVP 480  
 SPRSLQGGST TLFSRHTKAI VWGMQTRAVQ GMLDPDYVCS RDEPSVAAMV YPFTGDHKQK 540  
 FYWGHKEILI PVFKNMADAM RKHPEVDVLI NFASLRSAYD STMETMNYAQ IRTIAIIAEG 600  
 IPEALTRKLI KKADQKGVTI IGPATVGGIK PGCCKIGNTG GMLDNILASK LYRPGSVAYV 660  
 SRSGGMSNEL NNIISRTTDG VYEGVAIGGD RYPGSTFMDH VLRVQDTPGV KMIVVLGEIG 720  
 GTEEYKICRG IKEGRLTKPI VCWCIGTCAT MFSSEVQFGH AGACANQASE TAVAKNQALK 780  
 EAGVFVPRSF DELGEIIQSV YEDLVANGVI VPAQEVPPPT VPMDYWARE LGLIRKPASF 840  
 MTSICDERGQ ELIYAGMPIT EVFKEEMGIG GVLGLLWFQK RLPKYSCQFI EMCLMVTADH 900  
 GPAVSGAHNT IICARAGKDL VSSLTSGLLT IGDRFGGALD AAAKMFSAKAF DSGIIPMEFV 960  
 NMMKKEGLI MGIGHRVKSI NNPDMRVQIL KDYVRQHPPA TPLLDYALEV EKITTSKKPN 1020  
 LILNVDGLIG VAFVDMLENC GSPTREEADE YIDIGALNGI FVLGRSMGFI GHYLDQKRLK 1080  
 QGLYRHPWDD ISYVLPEHMS M 1101

SEQ ID NO: 84                   moltype = AA   length = 394  
 FEATURE                      Location/Qualifiers  
 source                       1..394  
                              mol\_type = protein  
                              organism = Escherichia coli

SEQUENCE: 84  
 MKNCVIVSAV RTAIGSFNGS LASTSAIDLG ATVIKAAIER AKIDSQHVDE VIMGNVLQAG 60  
 LGQNPARQAL LKSGLAETVC GFTVNVKCGS GLKSVALAAQ AIQAGQAQSI VAGGMENMSL 120

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|            |            |            |            |            |             |     |
|------------|------------|------------|------------|------------|-------------|-----|
| APYLLDAKAR | SGYRLDGGQV | YDVILRDGLM | CATHGYHMG  | TAENVAKEYG | ITREMQDELA  | 180 |
| LHSQRKAAAA | IESGAFTAEI | VPVNVVTRKK | TFVFSQDEFP | KANSTAEALG | ALRPAPFDKAG | 240 |
| TVTAGNASGI | NDGAAALVIM | EESAALAAGL | TPLARIKSYA | SGGVPPALMG | MGPVPATQKA  | 300 |
| LQLAGLQLAD | IDLIEANEAF | AAQFLAVGKN | LGFDSEKVN  | NGGAIALGHP | IGASGARILV  | 360 |
| TLLHAMQARD | KTLGLATLCI | GGQGIAAMVI | ERLN       |            |             | 394 |

SEQ ID NO: 85                      moltype = AA   length = 282  
 FEATURE                           Location/Qualifiers  
 source                            1..282  
                                  mol\_type = protein  
                                  organism = Clostridium acetobutylicum

SEQUENCE: 85

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MKKVCVIGAG | TMGSGIAQAF | AAKGFEVVL  | DIKDEFVDRG | LDFINKNLSK | LVKKGKIEEA | 60  |
| TKVEILTRIS | GTVDLNMAD  | CDLVIEAAVE | RMDIKKQIFA | DLDNICKPET | ILASNTSSLS | 120 |
| ITEVASATKR | PDKVIGHHFF | NPAPVMKLVE | VIRGIATSQE | TFDAVKETSI | AIGKDPVEVA | 180 |
| EAPGFVVNRI | LIPMINEAVG | ILAEGLASVE | DIDKAMKLGA | NHPMGPLELG | DFIGLDICLA | 240 |
| IMDVLYSETG | DSKYRPHITL | KKYVRAGWLG | RKSGKGFYDY | SK         |            | 282 |

SEQ ID NO: 86                      moltype = AA   length = 261  
 FEATURE                           Location/Qualifiers  
 source                            1..261  
                                  mol\_type = protein  
                                  organism = Clostridium acetobutylicum

SEQUENCE: 86

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MELNNVILEK | EGKVAVVTIN | RPKALNALNS | DTLKEMDYVI | GEIENDSEVL | AVILTGAGEK | 60  |
| SFVAGADISE | MKEMNTIEGR | KPGILGNKVF | RRLLELEKPV | IAAVNGFALG | GGCEIAMSCD | 120 |
| IRIASSNARF | GQPEVGLGIT | PGGGGTQRLS | RLVGMGMKQ  | LIFTAQNIKA | DEALRIGLVN | 180 |
| KVVEPSELNM | TAETIANKIV | SNAPVAVKLS | KQAINRGMQC | DIDTALAFES | EAFGECFSTE | 240 |
| DQKDAMTAFI | EKRKIEGPKN | R          |            |            |            | 261 |

SEQ ID NO: 87                      moltype = AA   length = 394  
 FEATURE                           Location/Qualifiers  
 source                            1..394  
                                  mol\_type = protein  
                                  organism = Cupriavidus necator

SEQUENCE: 87

|            |            |             |            |            |            |     |
|------------|------------|-------------|------------|------------|------------|-----|
| MTREVVVVSG | VRTAIGTFGG | SLKDVAPAE   | GALVVREALA | RAQVSGDDVG | HVVFGNVIQT | 60  |
| EPRDMLGRV  | AAVNGGVITN | APALTVMRLC  | GSGLQAIUSA | AQTILLGDTD | VAIGGGAESM | 120 |
| SRAPYLAPAA | RWGARMGDAG | LVDMMGLALH  | DPFHRIHMGV | TAENVAKEYD | ISRAQQDEAA | 180 |
| LESHRRASAA | IKAGYFKDQI | VPVVSXGRKG  | DVTFDTDEHV | RHDATIDMT  | KLRPVFVKEN | 240 |
| GTVTAGNASG | LNDAAAIVVM | MERAEAEERRG | LKPLARLVSY | GHAGVDPKAM | GIGPVPATKI | 300 |
| ALERAGLQVS | DLVDIEANEA | FAAQACAVTK  | ALGLDPAKVN | PNGSGISLGH | PIGATGALIT | 360 |
| VKALHELNRV | QGRYALVTMC | IGGGQGIAAI  | FERI       |            |            | 394 |

SEQ ID NO: 88                      moltype = AA   length = 397  
 FEATURE                           Location/Qualifiers  
 source                            1..397  
                                  mol\_type = protein  
                                  organism = Treponema denticola

SEQUENCE: 88

|            |             |             |             |             |             |     |
|------------|-------------|-------------|-------------|-------------|-------------|-----|
| MIVKPMVRIN | ICLNAHPQGC  | KKGVEDQIEY  | TKKRITAENV  | AGAKAPKNVL  | VLGCSNGYGL  | 60  |
| ASRITAAGFY | GAATIGVSFE  | KAGSETKYGT  | PGWYNNLAFD  | EAAKREGLYS  | VTIDGDAFSD  | 120 |
| EIKAQVIEEA | KKKGKIKFDLI | VYSLASPVRT  | DPDTGIMHKS  | VLKPFPGKTFT | GKTVDPTFTGE | 180 |
| LKEISAEPAN | DEEAAATVKV  | MGGEDWERWI  | KQLSKEGLE   | EGCITLAYS   | IGPEATQALY  | 240 |
| RKGTIGKAKE | HLEATAHRLN  | KENPSIRAFV  | SVNKGVLVTRA | SAVIPVIPLY  | LASLFKVMKE  | 300 |
| KGNHEGCIEQ | ITRLYAEERLY | RKDGITIPVDE | ENRIRIDDWE  | LEEDVQKAVS  | ALMEKVTGEN  | 360 |
| AESLTDLAGY | RHDFLASNGF  | DVEGINYEAE  | VERFDRI     |             |             | 397 |

SEQ ID NO: 89                      moltype = AA   length = 720  
 FEATURE                           Location/Qualifiers  
 source                            1..720  
                                  mol\_type = protein  
                                  organism = Cannabis sativa

SEQUENCE: 89

|            |            |            |             |            |            |     |
|------------|------------|------------|-------------|------------|------------|-----|
| MGKNYKSLDS | VVASDFIALG | ITSEVAETLH | GRLAEIVCNY  | GAATPQTWIN | IANHILSPDL | 60  |
| PFSLHQMIFY | GCYKDFGPAP | PAWIPDPEKV | KSTNLGALLE  | KRGKEFLGVK | YKDPISFSFH | 120 |
| FQEFVSRNPE | VYWRTVLMDE | MKISFSKDPE | CILRRDDINN  | PGGSEWLPGG | YLNSAKNCLN | 180 |
| VNSNKKLNDT | MIWVRDEGND | DLPLNKLTL  | QLRKRVLVVG  | YALEEMGLEK | GCAIADMPM  | 240 |
| HVDVAVIYLA | IVLAGYVVVS | IADSFSAPEI | STRRLRLSKAK | AIFTQDHIIR | GKKRIPLYSR | 300 |
| VVEAKSPMAI | VIPCSGSNIG | AELRDGDISW | DYFLERAKEF  | KNCEFTAREQ | PVDAYTNILF | 360 |
| SSGTTGEPKA | IPWTQATPLK | AAADGWSHLD | IRKGDVIVWP  | TNLGWMMPGW | LVYASLLNGA | 420 |
| SIALYNGSPL | VSGFAKFVQD | AKVTMLGVVP | SIVRSWKSTN  | CVSGYDWSTI | RCFSSSGEAS | 480 |
| NVDEYLWLMG | RANYKPVIM  | CGGTEIGGAF | SAGSFLQAQS  | LSSFSSQCMG | CTLYILDKN  | 540 |
| YMPKNKPGI  | GELALGPVFM | GASKTLLNGN | HHDVYFKGMP  | TLNGEVLRRH | GDIFELTSNG | 600 |
| YYHAHGRADD | TMNIGGIKIS | SIEIERVCNE | VDDRVFETTA  | IGVPLPGGGP | EQLVIFVFLK | 660 |
| DSNDTTIDLN | QLRLSFLNLG | QKCLNPLFKV | TRVPLSSLP   | RTATNKIMRR | VLRQQFSHF  | 720 |

SEQ ID NO: 90                      moltype = AA   length = 491  
 FEATURE                           Location/Qualifiers



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source          1..491
                 mol_type = protein
                 organism = Saccharomyces cerevisiae

SEQUENCE: 90
MKLSTKLCWC  GIKGRLRPQK  QQQLHNTNLQ  MTELKKQKTA  EQKTRPQNVG  IKGIQIYIPT  60
QCVNQSELEK  FDGVSQGYT  IGLGQTNMSF  VNDREDIYSM  SLTVLSKLIK  SYNIDTNKIG  120
RLEVGTETLI  DKSXSVKSVL  MQLFGENTDV  EGIDTLNACY  GGTNALFNSL  NWIESNAWDG  180
RDAIVVCGDI  AIYDKGAARP  TGGAGTVAMW  IGPDAPIVPD  SVRASymeHA  YDFYKPDFTS  240
EYPYVDGHFS  LTCYVKALDQ  VYKSYSKKAI  SKGLVSDPAG  SDALNVLYKF  DYNVPHVPTC  300
KLVTKSYGRL  LYNDFRANPQ  LPFEVDAELA  TRDYDESLTD  KNIEKTFVNV  AKPFHKERVA  360
QSLIVPTNTG  NMYTASVYAA  FASLLNVVGS  DDLQGKRVGL  FSYGSGLAAS  LYSCKIVGDV  420
QHIIKELDIT  NKLAKRITET  PKDYEAALIE  RENAHLKKNF  KPQGSIEHLQ  SGVYYLTNID  480
DKFRRSYDVK  K                                     491

SEQ ID NO: 91      moltype = AA  length = 381
FEATURE            Location/Qualifiers
source             1..381
                   mol_type = protein
                   organism = Saccharomyces cerevisiae

SEQUENCE: 91
MVAVRRKALS  ILAEPVLAS  DRLPYKNDY  DRVFGACCE  VIGYMLPVG  VIGPLVIDGT  60
SVHIPMATTE  GCLVASMRG  CKAINAGGGA  TTVLTKDGMT  RGPVVRPPTL  KRSGACKIWL  120
DSEEGQNAIK  KAFNSTSRFA  RLQHIQTCLA  GDLLFMRPRT  TTGDAMGMNM  ISKGVEYSLK  180
QMVVEYGWED  MEVSVSGNY  CTDKKPAAIN  WIEGRKSVV  AEATIPGDVV  RKVLKSDVSA  240
LVELNIAKNL  VGSAMAGSVG  GFNAHAANLV  TAVFLALGQD  PAQNVESNCS  ITLMKEVDGD  300
LRISVSMPSI  EVGTIGGGTV  LEPQGAAML  LGVRGPHATA  PGTNARQLAR  IVACAVLAGE  360
LSLCAALAA  HLVQSHMTHN  R                                     381

SEQ ID NO: 92      moltype = AA  length = 443
FEATURE            Location/Qualifiers
source             1..443
                   mol_type = protein
                   organism = Saccharomyces cerevisiae

SEQUENCE: 92
MSLPFLTSAP  GKVIIFGEHS  AVYNKPAVAA  SVSALRYYL  ISESSAPDTI  ELDFPDISFN  60
HKWSINDFNA  ITEDQVNSQK  LAKAQQATDG  LSQELVSLLD  PLLAQLSESF  HYHAAFCFLY  120
MPVCLCPHAK  NIKFSLKSTL  PIGAGLGSSA  SISVSLALAM  AYLGLIGSN  DLEKLSNDK  180
HIVNQWAFIG  ECKIHGTPSG  IDNAVATYGN  ALLFEKDSHN  GTINTNNFKF  LDDFPAIPMI  240
LTYTRIPRST  KDLVARVRVL  VTEKFEVVMK  PILDAMGECA  LQGLEIMTKL  SKCKGTDDEA  300
VTNNELVEQ  LLELIRINH  LVSIGVSH  GLELIKNSLD  DLRIKSTKLT  GAGGGGCSLT  360
LLRRDITQEQ  IDSPKKKLD  DFSYETFETD  LGGTGCCLLS  AKNLNKDLKI  KSLVPQLFEN  420
KTTTKQQIDD  LLLPGNTNLP  WTS                                     443

SEQ ID NO: 93      moltype = AA  length = 452
FEATURE            Location/Qualifiers
source             1..452
                   mol_type = protein
                   organism = Saccharomyces cerevisiae

SEQUENCE: 93
MSELRAFSAP  GKALLAGGYL  VLDTKYEA  VGLSARMHAV  AHPYGSLOGS  DKFEVRVSKS  60
QPKDGEWL  YH  ISPKSGFIPV  SIGGSKNPF  EKVIANVFSY  FKPNMDDYCN  RNLFFVIDIFS  120
DDAYHSQEDS  VTEHRGNRR  SFHSHRIEEV  PKTGLGSSAG  GLVTVLTAL  ASFFVSDLEN  180
NVDKYREVIH  NLAQVAHCQA  QGKIGSGFDV  AAAAYGSIRY  RRFPPALISN  LPDIGSATY  240
SKLAHLVDEE  DWNITIKSNH  LPSGLTLWMG  DIKNGSETVK  LVQVKVKNWD  SHMPESLKIY  300
TELDHANSR  F  MDGLSKLDRL  HETHDDYSQ  IFESLERND  C  TCQKYPEITE  VRDAVATIRR  360
SFRKITKESG  ADIEPPVQTS  LLDDCQTLKG  VLTCLIPGAG  GYDAIAVITK  QDVLRAQTA  420
NDKRFKSKVQ  W  LDVTQADWGV  RKEKDPETYL  DK                                     452

SEQ ID NO: 94      moltype = AA  length = 396
FEATURE            Location/Qualifiers
source             1..396
                   mol_type = protein
                   organism = Saccharomyces cerevisiae

SEQUENCE: 94
MTVYTASVTA  PVNIATLKYW  GKRDTKLNLP  TNSSISVTL  S  QDDLRLTSA  ATAPEFERDT  60
LWLNGEPHSI  DNERTQNC  LR  DLRLRKEME  SKDASLPTLS  QWKLHIVSEN  NFPTAAGLAS  120
SAAGFAALVS  AIAKLYQLPQ  STSEISRIR  KSGSACRSL  FGGYVAVEMG  KAEDGHDSMA  180
VQIADSSDWP  QMKACVLVVS  DIKDVSSSTQ  GMQLTVATSE  LFKERIEHV  PKRFEVMRKA  240
IVEKDFATFA  KETMMDNSNF  HATCLDSFPP  IFYMNDTSKR  IISWCHTINQ  FYGETIVAYT  300
FDAGPNAVLY  YLAENESKLF  AFYIKLFGSV  PGWDKKFTTE  QLEAFNHQFE  SSNFTARELD  360
LELQKDVAVR  ILTQVSGSPQ  ETNESLIDAK  TGLPKE                                     396

SEQ ID NO: 95      moltype = AA  length = 288
FEATURE            Location/Qualifiers
source             1..288
                   mol_type = protein
                   organism = Saccharomyces cerevisiae

SEQUENCE: 95
MTADNNSMPH  GAVSSYAKLV  QNQTPEIDLE  EFPEIPLQQ  RPNTRSSETS  NDESGETCFS  60

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|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| GHDEEQIKLM | NENCIVLDWD | DNAIGAGTKK | VCHLMENIEK | GLLHRAFSVF | IFNEQGELL  | 120 |
| QQRATEKITF | PDLWTNTCCS | HPLCIDDELG | LKGKLDKIK  | GAITAAVRKL | DHELGPED   | 180 |
| TKTRGKFHFL | NRIHYMAPSN | EPWGEHEIDY | ILFYKINAKE | NLTVPNPVNE | VRDFKWWSPN | 240 |
| DLKTMFADPS | YKFTPWFKII | CENYLFNWWE | QLDDLSEVEN | DRQIHRML   |            | 288 |

SEQ ID NO: 96                      moltype = AA   length = 335  
 FEATURE                           Location/Qualifiers  
 source                            1..335  
                                  mol\_type = protein  
                                  organism = *Saccharomyces cerevisiae*

SEQUENCE: 96

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MEAKIDELIN | NDPVWSSQNE | SLISKPYNHI | LLKPGKNFRL | NLIVQINRVM | NLPKDQLAIV | 60  |
| SQIVELLHNS | SLIIDIEDN  | APLRRGQTTS | HLIWGVPSTI | NTANYMYFRA | MLVSVQLTTK | 120 |
| EPLYHHLITI | FNEELINLHR | GQGLDIYWRD | FLPEIIPQOE | MYLNMVMNKT | GGLFRLTLRL | 180 |
| MEALSPSSH  | GHSLVPPINL | LGIYQIRDD  | YLNKDFQMS  | SEKGFADIT  | EGKLSFPIVH | 240 |
| ALNFTKTKGQ | TEQHNEILRI | LLRRTSDKDI | KLKLIQILEF | DTNSLAYTKN | FINQLVNMKI | 300 |
| NDNENKYLDP | LASHSDTATN | LHDELLYIID | HLSEL      |            |            | 335 |

SEQ ID NO: 97                      moltype = AA   length = 2233  
 FEATURE                           Location/Qualifiers  
 source                            1..2233  
                                  mol\_type = protein  
                                  organism = *Saccharomyces cerevisiae*

SEQUENCE: 97

|             |             |            |            |            |             |      |
|-------------|-------------|------------|------------|------------|-------------|------|
| MSEELFESS   | PQMEYEITN   | YSERHTELP  | HFIGLNTVDK | LEESPLRDFV | KSHGGHTVIS  | 60   |
| KILIANNGIA  | AVKEIRSVRK  | WAYETFGDDR | TVQFVAMATP | EDLEANAIEY | RMADQYIEVP  | 120  |
| GGTNMNNYAN  | VDLIVDIAER  | ADVDAVWAGW | GHASENPLP  | EKLSQSKRKV | IFIGPPGNAM  | 180  |
| RSLGDKISST  | IQAQSAKVP   | IPWSGTGVD  | VHVEKTKGLV | SVDDDIYQKG | CCTSPEDGLQ  | 240  |
| KAKRIGFPVM  | IKASEGGGK   | GIRQVEREED | FIALYHQAAN | EIPGSPIFIM | KLARARHLE   | 300  |
| QQLADQYGT   | NISLFGDSC   | VQRHOKIIE  | EAPVTIAKAE | TFHEMEKAAV | RLGKLVGYVS  | 360  |
| AGTVEYLYSH  | DDGKFYFLEL  | NPRQLVEHPT | TEMVSGVNL  | AAQLQIANGI | PMHRISDIRT  | 420  |
| LYGMNPHSAS  | EIDFEFKTD   | ATKKQRRPIP | KGHCTACRIT | SEDPNDGFKP | SGGTLHELNF  | 480  |
| RSSSNVWGYF  | SVGNNGNIHS  | FSDSQGHIF  | AFGENRQASR | KHMVVALKEL | SIRGDFRTTV  | 540  |
| EYLKLLLETE  | DFEDNTITTG  | WLDDLITHKM | TAEKPDPTLA | VICGAATKAF | LASEEARHXY  | 600  |
| IESLQKQVQL  | SKDLLQTMFP  | VDFIHEGKRY | KFTVAKSGND | RYTLFINGSK | CDIILRQLSD  | 660  |
| GGLLIAIGGK  | SHTIYWKEEV  | AATRLSVDSM | TTLLEVENDP | TQLRTPSPGK | LVKFLVENGE  | 720  |
| HIIKGQPYAE  | IEVMKMQMPL  | VSQENGIVQL | LKQPGSTIVA | GDIMAIMTLD | DPSKVKHALP  | 780  |
| FEGMLPDPFGS | PVIEGTPKPY  | KFKSLVSTLE | NILKGYDNQV | IMNASLQQLI | EVLNPNKLPY  | 840  |
| SEWKLHISAL  | HSRLPAKLDE  | QMEELVARSL | RRGAVFPARQ | LSKLIDMAVK | NPEYNPDKLL  | 900  |
| GAVVEPLADI  | AHKYSNGLEA  | HEHSIFVHFL | EYYEVEKLF  | NGPNVRENI  | ILKLRDENPK  | 960  |
| DLDKVALTVL  | SHSKVSANNN  | LILAILKHYQ | PLCKLSSKVS | AIFSTPLQHI | VELESKATAK  | 1020 |
| VALQAREILI  | QCALPSVKER  | TEQIEHILKS | SVVKVAYGSS | NPKRSEPDN  | ILKDLIDSNY  | 1080 |
| VVFDVLLQFL  | THQDPVVTAA  | AAQVYIRRAY | RAYTIGDIRV | HEGVTVPIVE | WKFQLPSAAF  | 1140 |
| STFPTVSKSM  | GMNRAVSVSD  | LSYVANSQSS | PLREGILMAV | DHLDDVDEIL | SQSLEVIPIRH | 1200 |
| QSSSNGPAPD  | RSQSSASLNS  | VANVCVASTE | GFSEEEIILV | RLREILDNLK | QELINASIRR  | 1260 |
| ITFMFGFKDG  | SYPKYTFNG   | PVYNENETIR | HIEPALAFOL | ELGRLSNFNI | KPIFTDNRI   | 1320 |
| HVYEAVSKTS  | PLDKRFFTRG  | IIRTGHIRDD | ISIQEYLTSE | ANRLMSDILD | NLEVTDTSNS  | 1380 |
| DLNHIFINFI  | AVFDISPEDV  | EAAFGGFLER | FGKRLRLRV  | SSAIIIRIIR | DPQTGAPVPL  | 1440 |
| RALINNVSQY  | VIKTEMYTEV  | KNAKGEWVFK | SLGKPGSMHL | RPIATPYPVK | EWLQPKRYKA  | 1500 |
| HLMGTTYVYD  | FPPELFRQASS | SQWKNFSADV | KLTDFFISN  | ELIEDENGEL | TEVEREPGAN  | 1560 |
| ATGMVAFKIT  | VKTPEYPRGR  | QFVVVANDIT | FKIGSGFQOE | DEFFNKVTEY | ARKRGIPRIY  | 1620 |
| LAANSGARIG  | MAEBIVPLFQ  | VANWDAANPD | KGFQYLYLTS | EGMETLKKFD | KENSVLTERT  | 1680 |
| VINGEERFVI  | KTIIGSEDGL  | GVECLRGSGL | IAGATSRAYH | DIFTITLVTC | RSVGIGAYLV  | 1740 |
| RLGQRAIQVE  | GQPIILTGAP  | AINKMLGREV | YTSNLQLGGT | QIMYNNGVSH | LTAVDLAGV   | 1800 |
| EKIVEWMSV   | PAKRNMPVPI  | LETKDTWDRP | VDFTPTNDET | YDVRWMIER  | ETESGFYGL   | 1860 |
| FDKGSFFETL  | SGWAKGVVVG  | RARLGGIPLG | VIGVETRTVE | NLIPADPANP | NSAETLIQEP  | 1920 |
| GQVWHPNSAF  | KTAQAINDFN  | NGEQLPMMIL | ANWRGFSGGQ | RDMFNEVLKY | GSFIVDALVD  | 1980 |
| YKQPIIIYIP  | PTGELRGGSW  | VVVDPTINAD | QMEMYADVNA | RAGVLEPOGM | VGIKFRREKL  | 2040 |
| LDTMNRLLDK  | YRELRSQLSN  | KSLAPEVHQQ | ISKQLADRER | ELLPIYGQIS | LQFADLHDS   | 2100 |
| SRMVAKGVIS  | KELEWTEARR  | FFFWRLRRRL | NEEYLIKRLS | HQVGEASRL  | KIARIRSWYP  | 2160 |
| ASVDHEDDRQ  | VATIEENYK   | TLDDKLKGLK | LESFAQDLAK | KIRSDHDNAI | DGLSEVIKML  | 2220 |
| STDDKEKLLK  | TLK         |            |            |            |             | 2233 |

SEQ ID NO: 98                      moltype = AA   length = 385  
 FEATURE                           Location/Qualifiers  
 source                            1..385  
                                  mol\_type = protein  
                                  organism = *Cannabis sativa*

SEQUENCE: 98

|            |            |            |            |            |             |     |
|------------|------------|------------|------------|------------|-------------|-----|
| MNHLRAEGPA | SVLAIGTANP | ENILLQDEFP | DYYFRVTKE  | HMTQLKEKFR | KICDKSMIRK  | 60  |
| RNCFLNEEHL | KQNPRLVEHE | MQTLDAQDM  | LVVEVPKLGK | DACAKAIKEW | GQPKSKIITHL | 120 |
| IFTSASTTDM | PGADYHCAKL | LGLSPSVKRV | MMYQLGCGY  | GTVLRIAKDI | AENNKGARVL  | 180 |
| AVCCDIMACL | FRGPSSEDL  | LLVGQAIFGD | GAAAVIVGAE | PDESVERGPI | FELVSTGQTI  | 240 |
| LPNSEGTIGG | HIREAGLIFD | LHKDVPMLIS | NNIEKCLIEA | FTPIGISDWN | SIFWITHPGG  | 300 |
| KAILDKVEEK | LHLKSDKFPD | SRHVLSEHGN | MSSSTVLFVM | DELKRSLLEE | GKSTTGDGFE  | 360 |
| WGVLFPGFPG | LTVERVVVRS | VPIKY      |            |            |             | 385 |

SEQ ID NO: 99                      moltype = AA   length = 101  
 FEATURE                           Location/Qualifiers

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source                1..101
                      mol_type = protein
                      organism = Cannabis sativa

SEQUENCE: 99
MAVKHLIVLK FKDEITEAQK EEFKTYVNL VNIIPAMKDV YWGKDVTKN KEEGYTHIVE 60
VTFESVETIQ DYIIHPAHVG FGDVYRSFWE KLLIFDYTPR K 101

SEQ ID NO: 100        moltype = AA length = 395
FEATURE              Location/Qualifiers
source                1..395
                      mol_type = protein
                      organism = Cannabis sativa

SEQUENCE: 100
MGLSSVCTFS FQNTYHTLLN PHNNNPKTSI LCYRHPKTP I KSYNNFPFSK HCSTKSFHLQ 60
NKCSESLSLA KNSIRAATTN QTEPPESDNH SVATKILNFG KACWKLQRPY TIIAFTSCAC 120
GLFGKELLHN TNLISWSLMF KAFFFLVAIL CIASFPTTIN QIYDLHIDRI NKPDLPLASG 180
EISVNTAWIM SIIVALFGLI ITIKMKGGPL YIFGYCFGIF GGIVYSVPPF RWKQNPSTAF 240
LLNFLAHII T NFTPYYASRA ALGLPFELRP SFTFLAPFMK SMGSALALIK DASDVEGDTK 300
FGISTLASKY GSRNLTLFCS GIVLLSYVAA ILAGIWPQA FNSNVMLLSH AILAPWLILQ 360
TRDFALTNVD PEAGRREYEF MWKLYYAEYL VYVFI 395

SEQ ID NO: 101        moltype = AA length = 544
FEATURE              Location/Qualifiers
source                1..544
                      mol_type = protein
                      organism = Cannabis sativa

SEQUENCE: 101
MKCSTFSFWF VCKIIFFFFS FNIQTSIANP RENFLKCFSQ YIPNNATNLK LVYTQNNPLY 60
MSVLNSTIHN LRFTSDTTPK PLVIVTPSHV SHIQGTILCS KKVGLQIRTR SGGHDSEGMS 120
YISQVPFVIV DLRNMRSIKI DVHSQTAWVE AGATLGEVYY WVNENKNNLS LAAGYCPTVC 180
AGGHFGGGGY GPLMRNYGLA ADNIIDAHV NVHGKVLDRK SMGEDLFWAL RGGGAESFGI 240
IVAWKIRLVA VPKSTMFSVK KIMEIHELK LVNKWQNIAY KYDKDLLMT HFITRNTDN 300
QKKNKTAIHT YFSSVFLGGV DSLVDLMNKS FPGLGIKKT CRQLSWIDTI IFYSGVVNYD 360
TDFNKEILL DRSAGQNGAF KIKLDYVKKP IPESVFVQIL EKLYEEDIGA GMYALYPYGG 420
IMDEISESAI PFPHRAGILY ELWYICSWEK QEDNEKHLNW IRNIYNFMTP YVSKNPRLAY 480
LNYRDLDIGI NDPKNPNNYT QARIWGEKYF GKNFDRLVKV KTLVDPNNFF RNEQSIPPLP 540
RHRH 544

SEQ ID NO: 102        moltype = AA length = 545
FEATURE              Location/Qualifiers
source                1..545
                      mol_type = protein
                      organism = Cannabis sativa

SEQUENCE: 102
MNCSTFSFWF VCKIIFFFLS FNIQISIANP QENFLKCFSE YIPNNPANPK FIYTQHDQLY 60
MSVLNSTIQN LRFTSDTTPK PLVIVTPSNV SHIQASILCS KKVGLQIRTR SGGHDAEGLS 120
YISQVPFAIV DLRNMHTVKV DIHSQTAWVE AGATLGEVYY WINEMNNFS FPGGYCPTVG 180
VGGHFGGGGY GALMRNYGLA ADNIIDAHV NVDGKVLDRK SMGEDLFWAI RGGGGENFGI 240
IAACKIKLVV VPSKATIFSV KKNMEIHGLV KLFNKWQNI YKYDKDLMLT THFRTRNITD 300
NHGKNKTTVH GYFSSIFLGG VDSLVDLMNK SPPELGIKKT DCKELSWIDT TIFYSGVVNY 360
NTANFKKEIL LDRSAGKTA FSKLDYVKK LIPETAMVKI LEKLYEEVVG VGMVLYPYG 420
GIMDEISESA IPFPHRAGIM YELWYTATWE KQEDNEKHIN WVRSVYNFTT PYVSQNPRLA 480
YLNRYDLDLG KTNPESPNNY TQARIWGEKY FGKNFNRLVK VTKADPMNF FRNEQSIPPL 540
PPRH 545

SEQ ID NO: 103        moltype = AA length = 1529
FEATURE              Location/Qualifiers
source                1..1529
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 103
MSAKAISEQT GKELLYKFIC TTSAIQNRFK YARVTPD TDW ARLLQDHPWL LSQNLVVKPD 60
QLIKRRGKLG LVGVNLTLDG VKSWLKPRLG QEATVGKATG FLKNFLIEPF VPHSQAEFPY 120
VCIYATREGD YVLFPHHEGGV DVGDDVAKAQ KLLVGVD EKL NPEDIKKHLL VHAPEDKKEI 180
LASFISGLFN FYEDLYFTYL EINPLVVT KD GYVLDLAAK VDATADYICK VKWGDIEFPP 240
PFGREAYPEE AYIADLDAKS GASLKLTLN PKGRIWTMVA GGGASVYSD TICDLGGVNE 300
LANYGEGYSGA PSEQQTYDYA KTILSLMTRE KHPDGKILII GGSIANFTNV AATFKGIVRA 360
IRDYQQPLKE HNVITIFVRG GPNYQEGLRV MGEVGKTTGI PIHVFGTETH MTAIVGMALG 420
HRPIPNQPT AAHTANFLLN ASGSTSTPAP SRTASFSESR ADEVAPAKKA KPAMPQDSVP 480
SPRSLQGKST TLFSRHTKAI VWGMQTRAVQ GMLDFDYVCS RDEPSVAAMV YPFTGDHKQK 540
FYWGHEKILI PVFKNMADAM RKHPEVDVLI NFASLRSAID STMETMNYAQ IRTIAIIAEG 600
IPEALTRKLI KKADQKGVTI IGPATVGGIK PGCFKIGNTG GMLDNILASK LYRPGSVAYV 660
SRSGGMSNEL NNISRTTDP VYEGVAIGGD RYPGSTFMDH VLRVQDTPGV KMIIVLGEIG 720
GTEEYKICRG IKEGRLTKPI VCWICIGCAT MFSSEVQFGH AGACANQASE TAVAKNQALK 780
EAGVFVPRSF DELGEIIQSV YEDLVANGVI VPAQEVPPPT VPMDYSWARE LGLIRKPASF 840
MTSICDERGQ ELIYAGMPIT EVFKEEMGIG GVLGLLWFQK RLPKYSCQFI EMCLMVTADH 900
GPAVSGAHNT IICARAGKDL VSSLTSGLLT IGRDFGGALD AAAKMFSKAF DSGIIPMEFV 960
NMKKEGKLI MGIGHRVKSI NNPDMRVQIL KDYVRQHPPA TPLDYALEV EKITTSKKPN 1020
LILNVDGLIG VAFVDMLRNC GSFTREEADE YIDIGALNGI FVLGRSMGFI GHYLDQKRLK 1080

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|            |            |            |            |            |            |      |
|------------|------------|------------|------------|------------|------------|------|
| QGLYRHPWDD | ISYVLPEHMS | MKLSGGGGSG | GGSGGGGSA  | EAWYNLGNAY | YKQGDYQKAI | 1140 |
| EYYQKALELD | PNNAEAWYNL | GNAYYKQGDY | QKAIEYYQKA | LELDPNNAEA | WYNLGNAYYK | 1200 |
| QGDYQKAIED | YQKALELDPN | NLQAEAWKNL | GNAYYKQGDY | QKAIEYYQKA | LELDPNNASA | 1260 |
| WYNLGNAYYK | QGDYQKAIEY | YQKALELDPN | NAKAWYRRGN | AYYKQGDYQK | AIEDYQKALE | 1320 |
| LDPNNRSRSA | GGGGSGGGGS | GGGGASSYYH | HHHHHLESTS | LYKKAGSGSN | LVAQLENEVA | 1380 |
| SLENENETLK | KKNLHKKDLI | AYLEKEIANL | RKKIEEGSAG | SAAGSGEFGS | AEAAAEKAAA | 1440 |
| KAGSAGSAAG | SSEFGSSYYH | HHHHHLESTS | LYKKAGSGSA | RNAYLRKKIA | RLKKDNLQLE | 1500 |
| RDEQNLEKII | ANLRDEIARL | ENEVASHEQ  |            |            |            | 1529 |

SEQ ID NO: 104                      moltype = AA   length = 821  
 FEATURE                              Location/Qualifiers  
 source                                1..821  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 104

|            |             |            |            |            |            |     |
|------------|-------------|------------|------------|------------|------------|-----|
| MKNCVIVSAV | RTAIGSFNGS  | LASTSAIDL  | ATVIKAAIER | AKIDSQHVDE | VIMGNVLQAG | 60  |
| LGQNPARQAL | LKSGLAETVC  | GFTVNVKCGS | GLKSVALAAQ | AIQAGQAQSI | VAGGMENMSL | 120 |
| APYLLDAKAR | SGYRLGDGV   | YDVILRDGLM | CATHGYHMG  | TAENVKEYG  | ITREMQDELA | 180 |
| LHSQRKAAAA | IESGAFTAEI  | VPVNVVTRKK | TFVFSQDEFP | KANSTAEALG | ALRPAPDKAG | 240 |
| TVTAGNASGI | NDGAAALVIM  | EESAALAAGL | TPLARIKSYA | SGGVPPALMG | MGPVPATQKA | 300 |
| LQLAGLQLAD | LDLIEANFAF  | AAQFLAVGKN | LGFDSEKVN  | NGGAIALGHP | IGASGARILV | 360 |
| TLLHAMQARD | KTLLGLATLCI | GGGQGIAMVI | ERLNKLSGGG | SGSGGGSGGG | GSAAEAWYNL | 420 |
| NAYYKQGDYQ | KAIEYYQKAL  | ELDPNNAEAW | YNLGNAYYKQ | GDYQKAIEYY | QKALELDPNN | 480 |
| ABAWYNLGN  | YYKQGDYQKA  | IEDYQKALEL | DPNNLQAEAW | KNLGNAYYKQ | GDYQKAIEYY | 540 |
| QKALELDPNN | ASAWYNLGN   | YYKQGDYQKA | IEYYQKALEL | DPNNAKAWYR | RGNAYYKQGD | 600 |
| YQKAIEDYQK | ALELDPNNRS  | RSAGGGGSGG | GGSGGGGASS | YHHHHHHHLE | STSLYKKAGS | 660 |
| GSNEVTTLN  | DAAFIENENA  | YLEKEIARLR | KEKAALNRNL | AHKKGSAGSA | AGSGEFGSAE | 720 |
| AAAKEAAAAK | GSAGSAAGSG  | EPGSSYYHHH | HHHLESTSLY | KKAGSGSQKV | AELKNRVAVK | 780 |
| LNRNEQLKNK | VEELKNRNAY  | LKNELATLEN | EVARLENDVA | E          |            | 821 |

SEQ ID NO: 105                      moltype = AA   length = 776  
 FEATURE                              Location/Qualifiers  
 source                                1..776  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 105

|            |            |            |             |             |             |     |
|------------|------------|------------|-------------|-------------|-------------|-----|
| MKKVCVIGAG | TMGSGIAQAF | AAKGFEVVL  | DIKDEFVDRG  | LDFINKNL    | LVKKGKIEEA  | 60  |
| TKVEILTRIS | GTVDLNMAD  | CDLVIEAAVE | RMDIKKQIFA  | DLDNICKPET  | ILASNTSSLS  | 120 |
| ITEVASATKR | PDKVGIMHFF | NPAPVMKLVE | VIRGIATSQE  | TFDAVKETS   | AIGKDPVEVA  | 180 |
| EAPGFVVNRI | LIPMINEAVG | ILAEIASVE  | DIDKAMKLGA  | NHPMGPLELG  | DFIGLDICLA  | 240 |
| IMDVLYSETG | DSKYRPHILL | KKYVRAGWLG | RKSGKGFYDY  | SKKLSGGGGS  | GGGGSGGGGS  | 300 |
| ABAWYNLGN  | YYKQGDYQKA | IEYYQKALEL | DPNNAEAWYN  | LGNAAYYKQGD | YQKAIEYYQK  | 360 |
| ALELDPNNAE | AWYNLGNAYY | KQGDYQKAIE | DYQKALELDP  | NNLQAEAWKN  | LGNAAYYKQGD | 420 |
| YQKAIEYYQK | ALELDPNNAS | AWYNLGNAYY | KQGDYQKAIE  | YYQKALELDP  | NNAKAWYRRG  | 480 |
| NAYYKQGDYQ | KAIEDYQKAL | ELDPNNRSRS | AGGGSGGGGG  | SGGGGASENL  | YPQGENLYFP  | 540 |
| GDSSSESCWC | GRKASETCSG | CNTARYCGSF | CQHKDWEKHH  | HICGQTLQAQ  | QGSAGSAAGS  | 600 |
| GEFGSAEAAA | KEAAAKAGSA | GSAAGSGEFG | SMAVSESQLK  | KMVSQYKYRD  | LTVRETNNVI  | 660 |
| TLYKDLKPV  | DSYVFNDGSS | RELMLTGTI  | PVPYRGNTYN  | IPICLWLLDT  | YPYNPPICFV  | 720 |
| KPTSSMTIKT | GKHVDANGKI | YLPYLHEWKH | PQSDLLGLLIQ | VMIVVFGDEP  | PVFSRP      | 776 |

SEQ ID NO: 106                      moltype = AA   length = 741  
 FEATURE                              Location/Qualifiers  
 source                                1..741  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 106

|            |            |            |            |            |             |     |
|------------|------------|------------|------------|------------|-------------|-----|
| MELNNVILEK | EGKVAVVTIN | RPKALNALNS | DTLKEMDYVI | GEIENDSEVL | AVILTGAGEK  | 60  |
| SFVAGADISE | MKEMNTIEGR | KFGILGNKVF | RRLELLEKPV | IAAVNGFALG | GGCEIAMSCD  | 120 |
| IRIASSNARF | GQPEVGLGIT | PGFGGTQRLS | RLVGMGMMAQ | LIFTAQNIKA | DEALRIGLVN  | 180 |
| KVVEPSELNM | TAKEIANKIV | SNAPVAVKLS | KQAINRGMQC | DIDTALAFES | EAFGEFCFSTE | 240 |
| DQKDAMTAFI | EKRKIEGFKN | RKLSGGGGSG | GGSGGGGSA  | EAWYNLGNAY | YKQGDYQKAI  | 300 |
| EYYQKALELD | PNNAEAWYNL | GNAYYKQGDY | QKAIEYYQKA | LELDPNNAEA | WYNLGNAYYK  | 360 |
| QGDYQKAIED | YQKALELDPN | NLQAEAWKNL | GNAYYKQGDY | QKAIEYYQKA | LELDPNNASA  | 420 |
| WYNLGNAYYK | QGDYQKAIEY | YQKALELDPN | NAKAWYRRGN | AYYKQGDYQK | AIEDYQKALE  | 480 |
| LDPNNRSRSA | GGGGSGGGGS | GGGGASGPLG | SPLTASMLAS | APPQEQKQML | GERLFPLIQA  | 540 |
| MHPTLAGKIT | GMLLEIDNSE | LLHMLESPES | LSRKVDEAVA | VLQAHQAKEA | AQKAGSAGSA  | 600 |
| AGSGEFGSAE | AAAKEAAAAK | GSAGSAAGSG | EPGSSNTMSV | PTDGAVTTSQ | IPASEQETLV  | 660 |
| RPKPLLLKLL | KSVGAQKDTY | TMKEVLFYLG | QYIMTKRLYD | EKQQHIYVCS | NDLLGDLFGV  | 720 |
| PSFSVKEHRK | IYTMIRNLV  | V          |            |            |             | 741 |

SEQ ID NO: 107                      moltype = AA   length = 820  
 FEATURE                              Location/Qualifiers  
 source                                1..820  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 107

|            |             |            |             |            |            |     |
|------------|-------------|------------|-------------|------------|------------|-----|
| MIVKPMVRNN | ICLNAHPQGC  | KKGVEDQIEY | TKKRITAEVK  | AGAKAPKNVL | VLGCSNGYGL | 60  |
| ASRITAAGFY | GAAITIGVSFE | KAGSETKYGT | PGWYNLAFD   | EAAKREGLYS | VTIDGDAFSD | 120 |
| EIKAQVIEEA | KKKGKIFDLI  | VYSLASPVRT | DPDGTGIMHKS | VLKPFKKTFT | GKTVDPFTGE | 180 |

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|            |            |            |            |             |            |     |
|------------|------------|------------|------------|-------------|------------|-----|
| LKEISAEPAN | DEEAAATVKV | MGGEDWERWI | KQLSKEGLLE | EGCITLAYS   | IGPEATQALY | 240 |
| RKGTIGKAKE | HLEATAHRLN | KENPSIRAFV | SVNKGVLTRA | SAVIPVPIPLY | LASLFKVMKE | 300 |
| KGNHEGCTEQ | ITRLYAERLY | RKDGITPVDE | ENRIRIDDWE | LEEDVQKAVS  | ALMEKVTGEN | 360 |
| AESLTDLAGY | RHDPLASNGF | DVEGINYEAE | VERFDRIKLS | GGGSGGGGS   | GGGSAEAWY  | 420 |
| NLGNAYYKQG | DYQKAIYYQ  | KALELDPNNA | EAWYNLGNAY | YKQGDYQKAI  | EYYQKALELD | 480 |
| PNNAEAWYNL | GNAYYKQGDY | QKAIEDYQKA | LELDPNNLQA | EAWKNLGNAY  | YKQGDYQKAI | 540 |
| EYYQKALELD | PNNASAWYNL | GNAYYKQGDY | QKAIYYQKA  | LELDPNNAKA  | WYRRGNAYYK | 600 |
| QGDYQKAIED | YQKALELDPN | NRSRSAGGGG | SGGGSGGGG  | ASSYHHHHH   | HLESTSLYKK | 660 |
| AGSGSNLLAT | LRSTAAVLEN | ENHVLEKEKE | KLRKEKEQLL | NKLEAYKGS   | GSAAGSGEFG | 720 |
| SAEAAAKEAA | AKAGSAGSAA | GSGEFGSSYY | HHHHHHLEST | SLYKKAGSGS  | KRIAYLRKKI | 780 |
| AALKKDNANL | EKDIANLENE | IERLIKEIKT | LENEVASHEQ |             |            | 820 |

SEQ ID NO: 108                      moltype = AA   length = 826  
 FEATURE                              Location/Qualifiers  
 source                                1..826  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 108

|            |            |            |            |            |             |     |
|------------|------------|------------|------------|------------|-------------|-----|
| MTREVVVVSG | VRTAIGTFGG | SLKDVAPAE  | GALVVREALA | RAQVSGDDVG | HVVFGNVIQT  | 60  |
| EPRDMLGRV  | AAVNGGVITN | APALTVMRLC | GSGLQAIVSA | AQTILLGDTD | VAIGGGAESM  | 120 |
| SRAPYLAPAA | RWGARMGDAG | LVDMMGLALH | DPFHRIHMGV | TAENVAKEDY | ISRAQQDEAA  | 180 |
| LESHRRASAA | IKAGYFKDQI | VPVVSCKGRK | DVTFDTDEHV | RHDATIDMT  | KLRPVFVKEN  | 240 |
| GTVTAGNASG | LNDAAAIVVM | MERAEAEERR | LKPLARLVSY | GHAGVDPKAM | GIGPVPA TKI | 300 |
| ALERAGLQVS | DLDVIEANEA | FAAQACAVTK | ALGLDPKAVN | PNGSGISLGH | PIGATGALIT  | 360 |
| VKALHELNRV | QGRYALVTMC | IYGGQGIAAI | FERIKLSGGG | SGGGSGGGG  | GSAAWYNL    | 420 |
| NAYYKQGDYQ | KAIYYQKAL  | ELDPNNAEAW | YNLGNAYYKQ | GDYQKAIYY  | QKALELDPNN  | 480 |
| AEAWYNLGN  | YKQGDYQKA  | IEDYQKALEL | DPNNLQAEAW | KNLGNAYYKQ | GDYQKAIYY   | 540 |
| QKALELDPNN | ASAWYNLGN  | YKQGDYQKA  | IEYYQKALEL | DPNNAKAWYR | RGNAYYKQGD  | 600 |
| YQKAIEDYQK | ALELDPNNRS | RSAGGGSGG  | GGSGGGGASD | VMWEYKWE   | GDAELYGPFT  | 660 |
| SAQMOTWVSE | GYFPDGVYCR | KLDPPGGQFY | NSKRIDFDLY | TGSAGSAGS  | GEFGSAEAAA  | 720 |
| KEAAAKAGSA | GSAAGSGEFG | SESDSVEFNN | ASIVYVNIKT | RFLDHPEIYR | SFLEILHTYQ  | 780 |
| KEQLHTKGRP | PRGMSSEEVF | TEVANLFRGQ | EDLLSEFGQF | LPEAKR     |             | 826 |

SEQ ID NO: 109                      moltype = AA   length = 849  
 FEATURE                              Location/Qualifiers  
 source                                1..849  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 109

|            |            |            |            |            |             |     |
|------------|------------|------------|------------|------------|-------------|-----|
| MKLSTKLCWC | GIKGRLRPQK | QQQLHNTNLQ | MTELKKQKTA | EQKTRPQNVG | IKGIQIYIPT  | 60  |
| QCVNQSELEK | PDGVSQGYT  | IGLGQTNMSF | VNDREDIYSM | SLTVLSKLIK | SYNIDTNKIG  | 120 |
| RLEVGTETLI | DKSKSVKSVL | MQLFGENTDV | EGIDTLNACY | GGTNALFNSL | NWIESNAWDG  | 180 |
| RDAIVVCGDI | AIYDKGAARP | TGGAGTVAMW | IGPDAPIVFD | SVRASymeHA | YDFYKPDFTS  | 240 |
| EYPYVDGHFS | LTCYVKALDQ | VYKSYSKKAI | SKGLVSDPAG | SDALNVLYKF | DYNVVFHVPTC | 300 |
| KLVTKSYGRL | LYNDFRANPQ | LFPEVDAELA | TRDYDESLTD | KNIKTPVNV  | AKPFHKERVA  | 360 |
| QSLIVPTNTG | NMYTASVYAA | FASLLNVVGS | DDLQGKRVGL | FSYSGSLAAS | LYSCKIVGDV  | 420 |
| QHIKELDIT  | NKLAKRITET | PKDYEAAL   | RENAHLKKNF | KPQGSIEHLQ | SGVYYLTNID  | 480 |
| DKFRRSYDVK | KKLSGGGGSG | GGSGGGGSA  | EAWYNLGNAY | YKQGDYQKAI | EYYQKALELD  | 540 |
| PNNAEAWYNL | GNAYYKQGDY | QKAIYYQKA  | LELDPNNAEA | WYNLGNAYYK | QGDYQKAIED  | 600 |
| YQKALELDPN | NLQAEAWKNL | GNAYYKQGDY | QKAIYYQKA  | LELDPNNASA | WYNLGNAYYK  | 660 |
| QGDYQKAIY  | YQKALELDPN | NAKAWYRRGN | AYYKQGDYQK | AIEDYQKALE | LDPNNRSRSA  | 720 |
| GGSGGGGGGS | GGSGGGSLGL | PPGWEVRSTV | SGRIYFVDHN | NRTTQFTDPR | LHGSAGSAG   | 780 |
| SGEFGSAEAA | AKAAAKAGS  | AGSAGSGSEF | SGSAMGPLPP | GWEKRTDSNG | RVYFVNHNTR  | 840 |
| ITQWEDPRS  |            |            |            |            |             | 849 |

SEQ ID NO: 110                      moltype = AA   length = 826  
 FEATURE                              Location/Qualifiers  
 source                                1..826  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 110

|            |            |            |            |             |            |     |
|------------|------------|------------|------------|-------------|------------|-----|
| MVAVRRKALS | ILAEAPVLAS | DRLPYKNYDY | DRVFGACCEN | VIGYMLPLVG  | VIGPLVIDGT | 60  |
| SYHIPMATTE | GCLVASAMRG | CKAINAGGGA | TTVLTKDGMT | RGPVVRFPPTL | KRSGACKIWL | 120 |
| DSEEGQNAIK | KAFNSTSRFA | RLQHIQTCLA | GDLLFMRFT  | TTGDAMGMNM  | ISKGVESL   | 180 |
| QMVEEYGWED | MEVSVSGSNY | CTDKKPAAIN | WIEGRKSVV  | AEATIPGDV   | RKVLKSDVSA | 240 |
| LVELNIAKNL | VGSAMAGSVG | GPNHAANLV  | TAVFLALGQD | PAQNVESNC   | ITLMKEVDGD | 300 |
| LRISVSMPSI | EVGTIGGGTV | LEPQGAAML  | LGVRGPHTA  | PGTNARQLAR  | IVACAVLAGE | 360 |
| LSLCAALAG  | HLVQSHMTHN | RKLSGGGGSG | GGSGGGGSA  | EAWYNLGNAY  | YKQGDYQKAI | 420 |
| EYYQKALELD | PNNAEAWYNL | GNAYYKQGDY | QKAIYYQKA  | LELDPNNAEA  | WYNLGNAYYK | 480 |
| QGDYQKAIED | YQKALELDPN | NLQAEAWKNL | GNAYYKQGDY | QKAIYYQKA   | LELDPNNASA | 540 |
| WYNLGNAYYK | QGDYQKAIY  | YQKALELDPN | NAKAWYRRGN | AYYKQGDYQK  | AIEDYQKALE | 600 |
| LDPNNRSRSA | GGSGGGGGGS | GGGASSYYH  | HHHHHLEST  | LYKKAGSEFF  | RRERNKMAA  | 660 |
| KCRNRRREL  | DTLQAEETDQ | EDEKSAEQTE | IANLLKEKEK | LEFILAARHP  | ACKIPDDLGF | 720 |
| PEEMSLEGS  | GSAAGSGEFG | SAEAAAKEAA | AKAGSAGSAA | GSGEFGSSYY  | HHHHHHLEST | 780 |
| SLYKKAGSGS | QKVESLKQKI | EELKQKRAQL | KNDIANLEKE | IAYAET      |            | 826 |

SEQ ID NO: 111                      moltype = AA   length = 1046  
 FEATURE                              Location/Qualifiers  
 source                                1..1046

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mol_type = protein
organism = synthetic construct

SEQUENCE: 111
MSLPFLTSAP GKVIIFGEHS AVYNKPAVAA SVSALRITYLL ISESSAPDTI ELDFFDISFN 60
HKWSINDPNA ITEDQVNSQK LAKAQQATDG LSQELVSLLD PLLAQLSESF HYHAAFCFLY 120
MFVCLCPHAK NIKFSLKSTL PIGAGLGSSA SISVSLALAM AYLGGILGNS DLEKLSSENDK 180
HIVNQWAFIG EKCIHGTSPG IDNAVATYGN ALLFEKDSHN GTINTNNFKF LDDFPAIPMI 240
LTYTRIPRST KDLVARVRVL VTEKFPEVMK PILDAMGECA LQGLEIMTKL SKCKGTDDEA 300
VETNNELYEQ LLELIRINHG LLVSIQVSHG GLELIKNLSD DLRIGSTKLT GAGGGGCSLT 360
LLRRDITQEQ IDSFKKKLQD DFSYETFETD LGGTGCCLLS AKNLNKDLKI KSLVPQLFEN 420
KTTTKQQIDD LLLPGNTNLP WTSKLSGGGG SGGGGSGGGG SAEAWYNLGN AYYKQGDYQK 480
AIEYYQKALE LDPNNAEAWY NLGNAYYKQG DYQKAIEYYQ KALELDPNNA EAWYNLGNAY 540
YKQGDYQKAI EDYQKALELD PNNLQAEAWK NLGNAYYKQG DYQKAIEYYQ KALELDPNNA 600
SAWYNLGNAY YKQGDYQKAI EYYQKALELD PNNAKAWYRR GNAYYKQGDY QKAIEDYQKA 660
LELDPNNRSE SAGGGSGGGG GSGGGGASME PAMEPETLEA RINRATNPLN KELDWASING 720
FCEQLNEDFE MDGLSKLDRL HKIQSPQEW EAIQALTVLET CMKSCGRFH DEVGKFRFLN 780
ELIKVVSPLY LGSRTSEKVK NKILELLYSW TVGLPEEVKI AEAYQMLKKQ GIVKSGSAGS 840
AAGSGEFGSA EAAAKEAAAK AGSAGSAGS GEGSGGAMG MAEAEGESLE SWLNKATNPS 900
NRQEDWEYII EGCDQINKEL EGCQIAVRLL AHKIQSPQEW EALQALTVE ACMKNCGRF 960
HNEVGKFRFL NELIKVVSPLY YLGDVSEKV KTKVIELLYS WTMALPEEAK IKDAYHMLKR 1020
QGIVQSDPFI PVDRTLIPSP PPRPKN 1046

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SEQ ID NO: 112      moltype = AA length = 868
FEATURE            Location/Qualifiers
source             1..868
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 112
MSELRAFSAP GKALLAGGYL VLDTKYEAFF VGLSARMHAV AHPYGSLLQS DKFEVRVSKS 60
QPKDGEWLH ISPKSGTIPV SIGGSKNPF EKVIANVFSY FKPNMDDYCN RNLFVIDIFS 120
DDAYHSQEDS VTEHRGNRRL SFHSHRIEEV PKTGLGSSAG GLVTVLTTAL ASFFVSDLEN 180
NVDKYREVIH NLAQVAHCQA QGKIGSGFDV AAAAYGSIRY RRPFPALISN LPDIGSATY 240
SKLAHLVDEE DWNITIKSNH LPSGLTLWMG DIKNGSETVK LVQKVKWYD SHMPESLKIY 300
TELDHANSRF MDGLSKLDRL HETHDDYSDQ IFESLERND CQKYPEITE VRDAVATIRR 360
SPRKITKESG ADIEPPVQTS LLDCCQTLKG VLTCLIPGAG GYDAIAVITK QDVLRAQTA 420
NDKRFYSKVQW LDVTQABWGV RKEKDPETYL DKKLSGGGGS GGGSGGGGS AEAWYNLGN 480
YYKQGDYQKA IEYQKALELD DPNNAEAWYN LGNAYYKQGD YQKAIEYYQK ALELDPNNAE 540
AWYNLGNAYY KQGDYQKAIE DYQKALELDP NNLQAEAWKN LGNAYYKQGD YQKAIEYYQK 600
ALELDPNNAS AWYNLGNAYY KQGDYQKAIE YYQKALELDP NNAKAWYRRG NAYYKQGDYQ 660
KAIEDYQKAL ELDPPNNRSE AGGGSGGGG SGGGSGSSY HHHHHHLEST SLYKAGSGS 720
QKVEELKNKI AELENRNVK KNRVAHLKQE IAYLKDELA HEPFGSAGSA AGSGEFGSAE 780
AAAKEAAAKA GSAGSAGSG EFGSSYYHH HHHLESTSLY KKAGSGSFEN VTHEFILATL 840
ENENAKLRRL EAKLERELAR LRNEVAWL 868

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SEQ ID NO: 113      moltype = AA length = 1008
FEATURE            Location/Qualifiers
source             1..1008
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 113
MTVYTASVTA PVNIATLKYY GKRDTKLNLP TNSSISVTL S QDDLRLTSTA ATAPEFERDT 60
LWLNGEPHSI DNERTQNCRL DLRQLRKEME SKDASLPTLS QWKLHIVSEN NFPTAAGLAS 120
SAAGFAALVS AIAKLYQLPQ STSEISRIR KSGGSACRSL FGGYVAVEMG KAEDGHDSMA 180
VQIADSSDWP QMKACVLVVS DIKDVSTSQ GMQLTVATSE LFKERIEHV PKRFEVMRKA 240
IVEKDFATFA KETMMDSNSF HATCLDSFPP IFYMNDTSKR IISWCHTINQ FYGETIVAYT 300
FDAGPNVAVLY LAENESKLF AFYIKLFGSV PGWDKKFTTE QLEAFNHQFE SSNFTARELD 360
LELQKDVAVR ILTQVSGSPQ ETNESLIDAK TGLPKEKLSG GGGSGGGGS GGGSAEAWYN 420
LGNAYYKQGD YQKAIEYYQK ALELDPNNAE AWYNLGNAYY KQGDYQKAIE YYQKALELDP 480
NNAEAWYNLG NAYYKQGDYQ KAIEDYQKAL ELDPPNNLQAE AWKNLGNAYY KQGDYQKAIE 540
YYQKALELDP NNASAWYNLG NAYYKQGDYQ KAIEYYQKAL ELDPPNNAKAW YRRGNAYYKQ 600
GDYQKAIEDY QKALELDPNN RSRAGGGGS GGGSGGGGA SAMADLEQKV LEMEASTYDG 660
VFIWKISDFP RKRQEAAGR IPAFISPAFY TSRYGYKMCL RIYLNQDGTG RGTLSLFFV 720
VMKGPNDALL RWPFPNQVTL MLLDQNNREH VIDAPRPDV SSSFPQRPVD MNIASGCPLF 780
CPVSKMEAKN SYVRDIAFI KAIVDLTGLG SAGSAGSGE FGSAAEAAKE AAKAGSAGS 840
AAGSGEFGSA SIKLQSSDGE IFEVDVEIAK QSVTIKTMLE DLGMDDEGDD DPVPLPNVNA 900
AILKKVIQWC THHKDDPPPP EDDENKEKRT DDIPVWDQEF LKVDQGTLE LILAANYLDI 960
KGLLDVTCCT VANMIKGTPT EIRKTFNIK NDFTEEEAQ VRKENQWC 1008

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SEQ ID NO: 114      moltype = AA length = 717
FEATURE            Location/Qualifiers
source             1..717
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 114
MTADNNSMPH GAVSSYAKLV QNQTPEIDLE EFPEIPLQQ RPNTRSSSETS NDESGETCFS 60
GHDEEQIKLM NENCIVLDWD DNAIGAGTKK VCHLMENIEK GLLHRAFSVF IFNEQGELL 120
QQRATEKITF PDLWTNTCCS HPLCIDDELG LKGLDDKIK GAITAAVRKL DHELGIPEDE 180
TKTRGKFHFL NRIHYMAPSN EPWGEHEIDY ILFYKINAKE NLTVNPNVNE VRDFKWPSPN 240
DLKTMFADPS YKFTFPWKII CENYLFNWWE QLDDLSEVEN DRQIHRMLKL SGGGSGGGG 300

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|            |             |            |             |             |             |     |
|------------|-------------|------------|-------------|-------------|-------------|-----|
| SGGGGSAEAW | YNLGNAYYKQ  | GDYQKAI EY | QKALELDPNN  | AEAWYNL GNA | YYKQGDYQKA  | 360 |
| IEYYQKALEL | DPNNAEAWYN  | LGNAYYKQGD | YQKAI EDYQK | ALELDPNNLQ  | AEAWKNL GNA | 420 |
| YYKQGDYQKA | IEYYQKALEL  | DPNNAEAWYN | LGNAYYKQGD  | YQKAI EYQK  | ALELDPNNAK  | 480 |
| AWYRRGNAYY | KQGDYQKAI E | DYQKALELDP | NNRSRSAGGG  | SGSGGGSGGG  | GASSYVHHHH  | 540 |
| HHLESTSLYK | KAGSGSNVTK  | ELKNYIQELE | ERNALKNLK   | EHLKFAKAE   | EFELAHHKFE  | 600 |
| GSAGSAAGSG | EFGSAEAAAK  | EAAAKAGSAG | SAAGSGEFGS  | SYHHHHHHHL  | ESTSLYKKAG  | 660 |
| SGSQKVAQLK | NRVAYKLKEN  | AKLENIVARL | ENDNANLEKD  | IANLEKDIAN  | LERDVAR     | 717 |

SEQ ID NO: 115                      moltype = AA    length = 798

FEATURE                              Location/Qualifiers

source                                1..798

mol\_type = protein

organism = synthetic construct

SEQUENCE: 115

|            |             |             |             |             |             |     |
|------------|-------------|-------------|-------------|-------------|-------------|-----|
| MEAKIDELIN | NDPVWSSQNE  | SLISKPNYHI  | LLKPGKNFRL  | NLIVQINRVM  | NLPKDQLAIV  | 60  |
| SQIVELLHNS | SLLEDDIEDN  | APLRRGQTTS  | HLIWGVPSTI  | NTANYMYFRA  | MQLVSQLTTK  | 120 |
| EPLYHHLITI | FNEELINLHR  | GQGLDIYWRD  | FLPEIPTQEQ  | MYLNMVMNKT  | GGLFRLTLRL  | 180 |
| MEALSPSSHH | GHSLVPPFINL | LGIIYQIRDD  | YLNLLKDFQMS | SEKGF AEDIT | EGKLSFPIVH  | 240 |
| ALNFTKTKGQ | TEQHNEILRI  | LLLRSTDKDI  | KLKLIQILEF  | DTNSLAYTKN  | FINQLVNMKI  | 300 |
| NDNENKYLDP | LASHSDTATN  | LHDELLYIID  | HLSELKLSGG  | GGSGGGSGGG  | GGSAEAWYNL  | 360 |
| GNAYYKQGDY | QKAI EYQKA  | LELDPNNAEA  | WYNLGNAYYK  | QGDYQKAI EY | YQKALELDPN  | 420 |
| NAEAWYNLGN | AYYKQGDYQK  | AI EDYQKALE | LDPNNLQAEA  | WKNLGNAYYK  | QGDYQKAI EY | 480 |
| YQKALELDPN | NASAWYNLGN  | AYYKQGDYQK  | AI EYQKALE  | LDPNNAKAWY  | RRGNAYYKQG  | 540 |
| YQKAI EDYQ | KALELDPNNR  | SRSAGGGSGG  | GGSGGGGGAS  | LCTMKKGPSG  | YGFNLHSDKS  | 600 |
| KPGQFIRSDV | PDSPAEASGL  | RAQDRIVEVN  | GVCMEGKQHG  | DVVS AIRAGG | DETKLLVVD   | 660 |
| EGSAGSAAGS | GEFGSAEAAA  | KEAAAKAGSA  | GSAAGSGEFG  | SSSGAI IYTV | ELKRYGGPLG  | 720 |
| ITISGTEEPF | DPIIISLSTK  | GGLAERTGAI  | HIGDRILAIN  | SSSLKGKPLS  | EAIHLLQ MAG | 780 |
| ETVTLKIKKQ | TDAQPASS    |             |             |             |             | 798 |

SEQ ID NO: 116                      moltype = AA    length = 883

FEATURE                              Location/Qualifiers

source                                1..883

mol\_type = protein

organism = synthetic construct

SEQUENCE: 116

|            |             |             |             |             |             |     |
|------------|-------------|-------------|-------------|-------------|-------------|-----|
| MNHLRAEGPA | SVLAIGTANP  | ENILLQDEFP  | DYYFRVTKSE  | HMTQLKEKFR  | KICDKSMIRK  | 60  |
| RNCFLENEHL | KQNPRLVEHE  | MQTL DARQDM | LVVEVPKL GK | DACAKAI KEW | GQPKSKITHL  | 120 |
| IFTSASTTDM | PGADYHCAKL  | LGLSPSVKRV  | MMYQLGCGYG  | GTVLRIAKDI  | AENNKGARVL  | 180 |
| AVCCDIMACL | FRGPSESLE   | LLVGQAIFGD  | GAAAVIVGAE  | PDES VGERPI | FELVSTGQTI  | 240 |
| LPNSEGTIGG | HIREAGLIFD  | LHKDVPMLIS  | NNIEKCLIEA  | FTPIGISDWN  | SIFWTHPGG   | 300 |
| KAILDKVEEK | LHLKSDKFPD  | SRHVLSEHGN  | MSSSTVLFVM  | DELKRKSL EE | GKSTTGDGFE  | 360 |
| WGVLPFGFPG | LTVERVVVRS  | VPIKYKLSGG  | GGSGGGGGSG  | GGSAEAWYNL  | GNAYYKQGDY  | 420 |
| QKAI EYQKA | LELDPNNAEA  | WYNLGNAYYK  | QGDYQKAI EY | YQKALELDPN  | NAEAWYNLGN  | 480 |
| AYYKQGDYQK | AI EDYQKALE | LDPNNLQAEA  | WKNLGNAYYK  | QGDYQKAI EY | YQKALELDPN  | 540 |
| NASAWYNLGN | AYYKQGDYQK  | AI EYQKALE  | LDPNNAKAWY  | RRGNAYYKQG  | DYQKAI EDYQ | 600 |
| KALELDPNNR | SRSAGGGSGG  | GGSGGGGGAS  | GNNLETYEWY  | NKSI SRDKAE | KLLLD TGKEG | 660 |
| AFMVRDSRTP | GTYYTVSVFTK | AIISENPCIK  | HYHIKETNDS  | PKRYVVAEKY  | VFDSIPLLIQ  | 720 |
| YHQYNGGGLV | TRLRYPVCGG  | SAGSAAGSGE  | FGSAEAAAKE  | AAAKAGSAGS  | AAGSGEFGSG  | 780 |
| SHPWFFGKIP | RAKAEEMLSK  | QRHDGAFLIR  | ESESAPGDFS  | LSVKFGNDVQ  | HFVKVLRDGAG | 840 |
| KYFLVWVKFN | SLNELVDYHR  | STSVSRNQOI  | FLRDIEQVPQ  | QPT         |             | 883 |

SEQ ID NO: 117                      moltype = AA    length = 665

FEATURE                              Location/Qualifiers

source                                1..665

mol\_type = protein

organism = synthetic construct

SEQUENCE: 117

|            |             |             |             |            |             |     |
|------------|-------------|-------------|-------------|------------|-------------|-----|
| MAVKHLIVLK | FKDEITEAQK  | EEFFKTYVNL  | VNIIPAMKDV  | YWGKDVTQKN | KEEGYTHIVE  | 60  |
| VTFESVETIQ | DYIIHPAHVG  | FGDVYRSFWE  | KLLIFDYTPR  | KKLSGGGGSG | GGSGGGGGSA  | 120 |
| EAWYNLGNAY | YKQGDYQKAI  | EYYQKALELD  | PNNAEAWYNL  | GNAYYKQGDY | QKAI EYQKA  | 180 |
| LELDPNNAEA | WYNLGNAYYK  | QGDYQKAI ED | YQKALELDPN  | NLQAEAWKNL | GNAYYKQGDY  | 240 |
| QKAI EYQKA | LELDPNNASA  | WYNLGNAYYK  | QGDYQKAI EY | YQKALELDPN | NAKAWYRRGN  | 300 |
| AYYKQGDYQK | AI EDYQKALE | LDPNNRSRSA  | GGSGGGGGGS  | GGGGASGQDR | SEATLIKRFK  | 360 |
| GEGVRYKAKL | IGIDEVSAAR  | GDKLCQDSMM  | KLKGVVAGAR  | SKGEHKQKIF | LTISFGGIKI  | 420 |
| FDEKTGALQH | HHAVHEISYI  | AKDITDHRAP  | GYVCGKEGNH  | RFVAIKTAQA | AEPVILDLRD  | 480 |
| LPQLIYELKQ | REELEKKAGS  | AGSAAGSGEF  | GSAEAAAKEA  | AAKAGSAGS  | AGSGEFGSGS  | 540 |
| HMSGQFWVTS | QKTEASERCQ  | LQGSYILRVE  | AEKLTLLTLG  | AQSQILEPLL | FWPYTLRLRY  | 600 |
| GRDKVMFSFE | AGRCRPSGPG  | TFTFQTSQGN  | DIFQAVEAAI  | QQQKAQKQVG | QAQDILRL EH | 660 |
| HHHHH      |             |             |             |            |             | 665 |

SEQ ID NO: 118                      moltype = AA    length = 797

FEATURE                              Location/Qualifiers

source                                1..797

mol\_type = protein

organism = synthetic construct

SEQUENCE: 118

|            |             |            |            |            |             |     |
|------------|-------------|------------|------------|------------|-------------|-----|
| MGLSSVCTFS | FQTNHYHTLLN | PHNNNPKTSL | LCYRHPKTPI | KYSYNNFPSK | HCSTKSFHLQ  | 60  |
| NKCSLSLSTA | KNSIRAATTN  | QTEPPESDNH | SVATKILNFG | KACWKLQRPY | TIIAFTSCAC  | 120 |
| GLFGKELLHN | TNLISWSLMF  | KAPFFLVAIL | CIASFTTTIN | QIYDLHIDRI | NKPDLLPLASG | 180 |

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|             |            |            |             |             |             |     |
|-------------|------------|------------|-------------|-------------|-------------|-----|
| EISVNTAWIM  | SIIVALFGLI | ITIKMKGGPL | YIFGYCFGIF  | GGIVYSVPPF  | RWKQNPSTAF  | 240 |
| LLNFLAHIIT  | NFTFYASRA  | ALGLPFELRP | SFTFLAPFMK  | SMGSALALIK  | DASDVEGDTK  | 300 |
| FGISTLASKY  | GSRNLTFLCS | GIVLLSYVAA | ILAGIWPQA   | FNSNVMLLSH  | AILAFWLILQ  | 360 |
| TRDFALTNYD  | PEAGRRFYEF | MWKLYYAEVL | VYVFIKLSGG  | GGSGGGSGSG  | GGSAEAWYNL  | 420 |
| GNAYYKQGDY  | QKAIEYYQKA | LELDPNNAEA | WYNLGNAYYK  | QGDYQKAIIEY | YQKALELDPN  | 480 |
| NAAEAWYNLGN | AYYKQGDYQK | AIEDYQKALE | LDPNNLQAEA  | WKNLGNAYYK  | QGDYQKAIIEY | 540 |
| YQKALELDPN  | NASAWYNLGN | AYYKQGDYQK | AIIEYYQKALE | LDPNNAKAWY  | RRGNAYYKQG  | 600 |
| DYQKAIEDYQ  | KALELDPNNR | SRSAGGGGSG | GGSGGGGGAS  | AEYVRALPFD  | NGNDEEDLPF  | 660 |
| KKGDILIRID  | KPEEQWNAE  | DSEKGRGMIP | VPYVEKYGSA  | GSAAGSGEFG  | SAEAAAKEAA  | 720 |
| AKAGSAGSAA  | SGSEFGSLIK | HMAEALPFD  | TGNSKLELNF  | KAGDVIFLLS  | RINKDWLEGT  | 780 |
| VRGATGIFPL  | SFVKILK    |            |             |             |             | 797 |

SEQ ID NO: 119                      moltype = AA    length = 3620  
 FEATURE                              Location/Qualifiers  
 source                                1..3620  
                                       mol\_type = protein  
                                       organism = synthetic    construct

SEQUENCE: 119

|             |             |             |             |             |             |      |
|-------------|-------------|-------------|-------------|-------------|-------------|------|
| MGSAGSAGS   | GEFGSAGSAA  | GSSEFGSAGS  | AAGSGEFSYY  | HHHHHLEST   | SLYKAGSGS   | 60   |
| ARNAYLRKKI  | ARLKKDNLQL  | ERDEQNLEKI  | IANLRDEIAR  | LENEVASHEQ  | GSAGSAGSG   | 120  |
| EPFAEAAKEA  | AAKAGSAGSA  | AGSGEFSYYH  | HHHHHLEST   | LYKKAGSGSN  | LVAQLENEVA  | 180  |
| SLNENETLK   | KKNLHKKDLI  | AYLEKEIANL  | RKKIEGSGAG  | SAAGSGEFGS  | AEAAAKEAAA  | 240  |
| KEAAAKEAAA  | KAGSAGSAG   | SGEFGSSYYH  | HHHHHLEST   | LYKKAGSGSQ  | KVAELKNRVA  | 300  |
| VKLNRLNEQL  | NKVLEELKNR  | AYLKNELATL  | ENEVARLEND  | VAEGSAGSAA  | GSSEFGAEAAA | 360  |
| KEAAAKAGSA  | GSAGSGEFGS  | YHHHHHLE    | STSLYKAGS   | GSNEVTTLN   | DAAFIENENA  | 420  |
| YLEKEIARLR  | KEKAALRNRL  | AHKKGSAGSA  | AGSGEFGSAE  | AAAKEAAAKE  | AAAKEAAAKA  | 480  |
| GSAGSAGSG   | EFGRSPPTIS  | NPPPLISSAK  | HPSVGSAGSA  | AGSGEFAEAA  | AKEAAAKAGS  | 540  |
| AGSAGSGEFG  | NFLQSRPEPT  | APPEESFRSG  | GSAGSAGSG   | EFGSAAEAAK  | EAAAKEAAAK  | 600  |
| EAAAKAGSAG  | SAAGSGEFGS  | SKGTGLNPNA  | KVWQEIAPGN  | GSAGSAGSG   | EPFAEAAKEA  | 660  |
| AAKAGSAGSA  | AGSGEFPDGG  | TTFEHLWSSL  | EPDSTYGSAG  | SAAGSGEFGS  | AEAAAKEAAA  | 720  |
| KEAAAKEAAA  | KAGSAGSAG   | SGEFGSSYYH  | HHHHHLEST   | LYKKAGSGSK  | RIAYLRKKIA  | 780  |
| ALKKDNANLE  | KDIANLENEI  | ERLIKEIKTL  | ENEVASHEQG  | SAGSAGSGE   | FAEAAAKEAA  | 840  |
| AKAGSAGSAA  | GSSEFSYYHH  | HHHHLESTSL  | YKAGSGSNL   | LATLRSTA    | LENENHVLEK  | 900  |
| EKEKLRKEKE  | QLLNKLEAYK  | GSAGSAGSG   | EFGSAAEAAK  | EAAAKEAAAK  | EAAAKAGSAG  | 960  |
| SAAGSGEFGS  | PATSQHPPPP  | PGRHSQAPSH  | GSAGSAGSG   | EPFAEAAKEA  | AAKAGSAGSA  | 1020 |
| AGSGEFELNS  | LLIILEAAEY  | LERRDRGSAG  | SAAGSGEFGS  | AEAAAKEAAA  | KEAAAKEAAA  | 1080 |
| KAGSAGSAG   | SGEFGSRPEPT | ISNPPPLISS  | AKHPSVGSAG  | SAAGSGEFAE  | AAAKEAAAKA  | 1140 |
| GSAGSAGSG   | EFNLFQSRPE  | PTAPPEESFR  | SGGSAGSAG   | SGEFGSAEAA  | AKEAAAKEAA  | 1200 |
| AKEAAAKAGS  | AGSAGSGEFG  | SSKGTGLNP   | NAKVVQEIAP  | GNGSAGSAG   | SGEFAEAAAK  | 1260 |
| EAAAKAGSAG  | SAAGSGEFPD  | GGTTFEHLWS  | SLEPDSTYGS  | AGSAGSGEFG  | GSAAAKEAA   | 1320 |
| AAAKEAAKEA  | AAKAGSAGSA  | AGSGEFGSSY  | YHHHHHLES   | TSLYKAGSG   | SKRIAYLRKK  | 1380 |
| IAALKKDNAN  | LEKDIAANLE  | EIERLIKEIK  | TLNEVASHE   | QGSAGSAGS   | GEFAEAAAKE  | 1440 |
| AAAKAGSAGS  | AGSGEFSYYH  | HHHHHLEST   | SLYKAGSGS   | NLLATLRSTA  | AVLENENHVL  | 1500 |
| EKEKEKLRKE  | KEQLLNKLEA  | YKGSAGSAG   | SGEFGSAEAA  | AKEAAAKEAA  | AKEAAAKAGS  | 1560 |
| AGSAGSGEFG  | GSALVDDAAD  | YEPPPSNNEE  | ALGSAGSAG   | SGEFAEAAAK  | EAAAKAGSAG  | 1620 |
| SAAGSGEFP   | LFDDPSYVNV  | QNLDKARQGS  | AGSAGSGEFG  | GSAAAKEAA   | AAKEAAAKEA  | 1680 |
| AAKAGSAGSA  | AGSGEFGSKN  | TKSMNFDNPV  | YRKTTEEGS   | AGSAGSGEFG  | AEAAAKEAAA  | 1740 |
| KAGSAGSAG   | SGEFPRLPST  | WIENKLYGMS  | DPNWSAGSA   | AGSGEFGSAE  | AAAKEAAAKE  | 1800 |
| AAAKEAAAKA  | GSAGSAGSG   | EPFGSVVDNSP | PPALPPKKRQ  | SAPSGSAGSA  | AGSGEFAEAA  | 1860 |
| AKEAAAKAGS  | AGSAGSGEFG  | TQRSKPQPAV  | PPRPSADLIL  | GSAGSAGSG   | EFGSAAEAAK  | 1920 |
| EAAAKEAAAK  | EAAAKAGSAG  | SAAGSGEFGS  | TDEEREETEE  | EVYLLNSTTL  | GSAGSAGSG   | 1980 |
| EPFAEAAKEA  | AAKAGSAGSA  | AGSGEFGDNV  | SGTQRLDSAT  | VRTYSCGSAG  | SAAGSGEFGS  | 2040 |
| AEAAAKEAAA  | KEAAAKEAAA  | KAGSAGSAG   | SGEFGSSYYH  | HHHHHLEST   | LYKKAGSGSQ  | 2100 |
| KVAQLKNRVA  | YKLENAKLE   | NIVARLENDN  | ANLEKDIAANL | EKDIAANLERD | VARGSAGSAA  | 2160 |
| GSSEFGAEAAA | KEAAAKAGSA  | GSAGSGEFGS  | YHHHHHLE    | STSLYKAGS   | GSNTVKELKN  | 2220 |
| YIQELEERNA  | ELKNLKEHLK  | FAKAELEFEL  | AAHKFEFGSAG | SAAGSGEFGS  | AEAAAKEAAA  | 2280 |
| KEAAAKEAAA  | KAGSAGSAG   | SGEFGSHDDS  | LPHPPQATDD  | SGHESDGSAG  | SAAGSGEFAE  | 2340 |
| AAAKEAAAKA  | GSAGSAGSG   | EPFGSPNAGSV | EQTTPKKPGLR | RRGSAGSAG   | SGEFGSAEAA  | 2400 |
| AKEAAAKEAA  | AKEAAAKAGS  | AGSAGSGEFG  | GSSYYHHHHH  | HLESTSLYKK  | AGSGSPENVT  | 2460 |
| HEFILTATLEN | ENAKLRLRLA  | KLERELARLR  | NEVAVLGSAG  | SAAGSGEFAE  | AAAKEAAAKA  | 2520 |
| GSAGSAGSG   | EFSSYYHHHH  | HLESTSLYKK  | AGSGSQKVEE  | LKNKIAELEN  | RNAVKKNRVA  | 2580 |
| HLKQBIAYLK  | DELAHEFEG   | SAGSAGSGE   | FGSAEAAAKE  | AAAKEAAAKE  | AAKAGSAGS   | 2640 |
| AAGSGEFGSV  | SSTKLVSFHD  | DSDEDLLHIG  | SAGSAGSGE   | FAEAAAKEAA  | AKAGSAGSAA  | 2700 |
| GSSEFAAATP  | ISTFHDDSD   | DLHLVGSAGS  | AAGSGEFGSA  | EAAAKEAAAK  | EAAAKEAAAK  | 2760 |
| AGSAGSAGS   | GEFGSSYYHH  | HHHHLESTSL  | YKAGSGSQK   | VESLKQKIEE  | LKQKQQLKN   | 2820 |
| DIALEKEITA  | YAEFGSAGSA  | AGSGEFAEAA  | AKEAAAKAGS  | AGSAGSGEFG  | SYHHHHHHHL  | 2880 |
| ESTSLYKAG   | SEFRRERNK   | MAAAKCRNRR  | RELDTLQAE   | TDQLEDEKSA  | LQTEIANLLK  | 2940 |
| EKEKLEFILA  | AHRPACKIPD  | DLGFPPEMSL  | EGSAGSAGS   | GEFGSAEAAA  | KEAAAKEAAA  | 3000 |
| KEAAAKAGSA  | GSAGSGEFG   | SFQMPADTPP  | PAYLPPEDPM  | TGSAGSAGS   | GEFAEAAAKE  | 3060 |
| AAKAGSAGS   | AAGSGEFERE  | SNEEPPPPYE  | DPYWGNGGSA  | GSAGSGEFG   | SAEAAAKEAA  | 3120 |
| AKEAAAKEAA  | AKAGSAGSAA  | GSSEFGSSYY  | HHHHHLEST   | SLYKAGSGS   | QKVAELKNRV  | 3180 |
| AVKLNRLNEQL | KNKVEELKNR  | NAYLKNELAT  | LENEVARLEN  | DVAEGSAGSA  | AGSGEFAEAA  | 3240 |
| AKEAAAKAGS  | AGSAGSGEFG  | SYHHHHHHHL  | ESTSLYKAG   | SGSNEVTTL   | NDAAFIENEN  | 3300 |
| AYLEKEIARL  | RKEKAALRN   | LAHKSSYYHH  | HHHHLESTSL  | YKAGSGSAR   | NAYLRKKIAR  | 3360 |
| LKKDNLQLER  | DEQNLEKIIA  | NLRDEIARLE  | NEVASHEQGS  | AGSAGSGEFG  | AEAAAKEAAA  | 3420 |
| KAGSAGSAG   | SGEFSYYHHH  | HHHLESTSLY  | KKAGSGSNLV  | AQLENEVASL  | ENENETLKKK  | 3480 |
| NLHKKDLIAY  | LEKEIANLRK  | KIEEGSAGSA  | AGSGEFGSAE  | AAAKEAAAKE  | AAAKEAAAKA  | 3540 |
| GSAGSAGSG   | EFGESEQLIS  | EEDLEQKLIS  | EEDLEQKLIS  | EEDLGSAGSA  | AGSGEFGSAG  | 3600 |
| SAAGSGEFGS  | AGSAGSGEFG  |             |             |             |             | 3620 |



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SEQ ID NO: 120      moltype = AA length = 452
FEATURE            Location/Qualifiers
source             1..452
                   mol_type = protein
                   organism = synthetic construct

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ARNAYLRKKI  ARLKKNLQL  ERDEQNLEKI  IANLRDEIAR  LENEVASHEQ  GSAGSAGSG  120
EFAEAAAKEA  AAKAGSAGSA  AGSGEFSYYH  HHHHHLESTS  LYKKAGSGSN  LVAQLENEVA  180
SLENETETLK  KKNLHKKDLI  AYLEKEIANL  RKKIEEGSAG  SAAGSGEFGS  AEAKEAEAAA  240
KEAAAKEAAA  KAGSAGSAGS  SGSEFGSSATR  ELDELMASLS  DFKIQGSAG  SAAGSGEFAE  300
AAAKEAAAKA  GSAGSAGSG  EFDLALSENW  AQEFLAAGDA  VDGSAGSAG  SGSEFGSAEA  360
AKEAAAKEAA  AKEAAAKAGS  AGSAGSGSEF  GSDYKDDDDK  DYKDDDDKDY  KDDDDKGSAG  420
SAAGSGEFGS  AGSAGSGSEF  GSAGSAGSG  EF  452

SEQ ID NO: 121      moltype = DNA length = 3303
FEATURE            Location/Qualifiers
source             1..3303
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 121
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gctagattgt  tgcaagatca  tccatgggtg  ttgtctcaaa  atttgggtgt  taaaccagat  180
caattgatta  aaagaagagg  taaattgggt  ttggttggtg  ttaatttgac  ttggatgggt  240
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gtttgtattt  atgtactag  agaaggtgat  tatgtttgt  ttcacatga  agtggtgtgt  420
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gttgatgcta  ctgctgatta  tatttga  gttaaatggg  gtgatattga  atttccacca  720
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ttgttttgg  gtatagctat  ggtttttatt  ggtcattatt  tggatcaaaa  aagattgaaa  3240
caaggtttgt  atagacatcc  atgggatgat  atttcttatg  ttttgccaga  acatatgtct  3300
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SEQ ID NO: 122      moltype = DNA length = 1182
FEATURE            Location/Qualifiers
source             1..1182
                  mol_type = other DNA
                  organism = synthetic construct

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tctggtgggt ttccaccagc ttgatgggtg atgggtccag ttccagctac tcaaaaagct 900
ttgcaattgg ctggtttgca attggctgat attgatttga ttgaagctaa tgaagctttt 960
gctgctcaat ttttggctgt tggtaaaaaa ttgggttttg attctgaaaa agttaatggt 1020
aatggtgggt ctattgtctt gggtcaccca attgggtgctt ctggtgctag aattttgggt 1080
actttgttgc atgctatgca agctagagat aaaacttttg gtttggtctac tttgtgtatt 1140
gggtgtggtc aaggtattgc tatggttatt gaaagattga at 1182

SEQ ID NO: 123      moltype = DNA length = 846
FEATURE            Location/Qualifiers
source             1..846
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 123
atgaaaaaag tttgtgttat tgggtgctgt actatgggtt ctggtattgc tcaagctttt 60
gctgctaaag gttttgaagt tgttttgaga gatattaaag atgaatttgt tgatagaggt 120
ttggatttta ttaataaaaaa ttgtctctaaa ttggttaaaa aaggtaaaaa tgaagaagct 180
actaaagtgg aaatttttgac tagaatttct ggtactgttg atttgaatat ggctgctgat 240
tgtgatattg ttattgaaagc tgcgtgtgaa agaattggata ttaaaaaaca aatttttctg 300
gatttgagata atatttgttaa accagaaaact attttggctt ctaatacttc ttctttgtct 360
attactgaag ttgcttctgc tactaaaaaga ccagataaaag ttattggtat gcattttttt 420
aatccagctc cagttatgaa attggttgaa gttattagag gtattgctac ttctcaagaa 480
acttttgatg ctgttaaaaga aacttctatt gctatttgta aagatccagt tgaagtgtct 540
gaagctccag gttttgttgt taatagaatt ttgattccaa tgattaatga agctgttggt 600
attttggctg aaggtattgc ttctgttgaa gatattgata aagctatgaa attgggtgct 660
aatcatccaa tgggtccatt ggaattgggt gattttattg gtttggtatg ttgtttggct 720
attatggatg ttttgtattc tgaaactggg gattctaaat atagaccaca tactttgttg 780
aaaaaatatg ttagagctgg ttgggtgggt agaaaaactcg gtaaaagggt ttatgattat 840
tctaaa 846

SEQ ID NO: 124      moltype = DNA length = 783
FEATURE            Location/Qualifiers
source             1..783
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 124
atggaattga ataattgtat tttggaaaaa gaaggtaaaag ttgctgttgt tactattaat 60
agaccaaaaag ctttgaatgc ttgaattct gatactttga aagaaatgga ttatgttatt 120
ggtgaaaattg aaaatgatgc tgaagttttg gctgttattt tgactgggtg tggtgaaaaa 180
tcttttgttg ctggtgctga tatttctgaa atgaaagaaa tgaatactat tgaaggtaga 240
aaatttggtta ttttgggtgaa taaagttttt agaagattgg aattgttgga aaaaccagtt 300
attgctgctg ttaatgtgtt tgctttgggt ggtggttgtg aaattgctat gtcttgtgat 360
attagaattg cttcttctaa tgcctagattt ggtcaaccag aagttgggtt ggggtattact 420
ccaggttttg gtgtgactca aagattgtct agattgggtg gtatgggtat ggctaataca 480
ttgattttta ctgctcaaaa tattaaagct gatgaagctt tgagaattgg tttggttaat 540
aaagttgttg aacctatcga attgatgaat actgctaaag aaattgctaa taaaattggt 600
tctaattgctc cagttgtctg taaattgtct aaacaagcta ttaatataggg tatgcaatgt 660
gatattgata ctgctttggc ttttgaatct gaagcttttg gtgaatgttt ttctactgaa 720
gatcaaaaag atgctatgac tgcttttatt gaaaaaagaa aaattgaagg ttttaaaaat 780
aga 783

SEQ ID NO: 125      moltype = DNA length = 1191
FEATURE            Location/Qualifiers
source             1..1191
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 125
atgattgtta aaccaatggg tagaaataat atttgtttga atgctcatcc acaaggttgt 60
aaaaaagggt ttgaagatca aattgaatat actaaaaaaa gaattactgc tgaagttaaa 120
gctggtgcta aagctccaaa aatgtttttg gttttgggtt gttctaattg ttatggtttg 180

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gcttctagaa ttactgctgc ttttgggtat ggtgctgcta ctattggtgt tctttttgaa 240
aaagctgggt ctgaaactaa atatgggtact ccaggttgggt ataataatgt ggcttttgat 300
gaagctgcta aaagagaagg tttgtattct gttactattg atggtgatgc tttttctgat 360
gaaattaaag ctcaagttat tgaagaagct aaaaaaaaaa gtattaaatt tgatttgatt 420
gtttattctt tggcttctcc agttagaact gatccagata ctggtattat gcataaatct 480
gttttgaaac catttggttaa aacttttact ggtaaaactg ttgatccatt tactggtgaa 540
ttgaaagaaa tttctgctga accagctaatt gatgaagaag ctgctgctac tgttaaagtt 600
atgggtgggt aagattggga aagatggatt aaacaattgt ctaaagaagg tttgttgga 660
gaaggttgta ttactttggc ttattcttat attggtccag aagctactca agctttgat 720
agaaaaggta ctattggtta agctaaagaa catttggaag ctactgctca tagattgaat 780
aaagaaaatc catctattag agcttttgtt tctgttaata aaggtttgggt tactagagct 840
tctgctgtta ttccagttat tccattgtat ttggtctctt tgtttaaagt tatgaaagaa 900
aaaggtaatc atgaaggttg tattgaacaa attactagat tgtatgctga aagattgtat 960
agaaaagatg ctgattttcc agttgatgaa gaaaatagaa ttagaattga tgattgggaa 1020
ttggaagaag atgttcaaaa agctgtttct gctttgatgg aaaaagttac tggtgaaaat 1080
gctgaatcct tgactgattt ggctgggtat agacatgatt ttttggtctc taatggtttt 1140
gatgttggaag gtattaatta tgaagctgaa gttgaaagat ttgatagaat t 1191

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SEQ ID NO: 126      moltype = DNA length = 1182
FEATURE            Location/Qualifiers
source              1..1182
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 126
atgactagag aagtgttgtt tgtttctggt gttagaactg ctattggtac ttttgggtgt 60
tctttgaaag atgttgctcc agctgaattg ggtgctttgg ttggttagaga agctttggct 120
agagctcaag tttctggtga tgatgttggt catgttgttt ttggttaagt tattcaaact 180
gaaccaagag atatgtattt gggtagagtt gctgctgtta atggtgggtg tactattaat 240
gctccagctt tgactgttaa tagattgtgt ggttctggtt tgaagctatg tgtttctgct 300
gctcaaaacta ttttgttggg tgatactgat gttgctattg gtggtggtgc tgaatctatg 360
tctagagctc catatttggc tccagctgct agatgggggtg ctagaatggg tgatgctggt 420
ttgggtgata tgatgttggg tgccttgcac gatccatttc atagaattca tatgggtgtt 480
actgctgaaa atgttgctaa agaataatgat atttctagag ctcaacaaga tgaagctgct 540
ttggaatctc atagaagagc ttctgctgct attaaagctg gttattttta agatcaaatt 600
gttccagttg tttctaaagg tagaaaaggt gatgttactt ttgatactga tgaacatggt 660
agacatgatg ctactattga tgatatgact aaattgagac cagtttttgt taaagaaaat 720
ggtagctgta ctgttggtta tgcttctggt ttgaatgatg ctgctgctgc tgttgttatg 780
atggaagagag ctgaagctga aagaagaggt ttgaaacctg tggctagatt ggtttcttat 840
ggtagctgct gtgttgatcc aaaagctatg ggtatttggt cagttccagc tactaaaatt 900
gctttgaaaa gagctggttt gcaagtttct gatttggatg ttattgaagc taatgaagct 960
tttgcctgctc aaagcttggc tgttactaaa gctttgggtt tggatccagc taaagttaat 1020
ccaaatgggt ctggtatttc ttgggtcatc ccaatttggt ctactggtgc tttgattact 1080
gttaaagctt tgcattgaatt gaatagagtt caaggtatag atgctttggt tactatgtgt 1140
attggtgggt gtcagggtat tgctgctatt tttgaaagaa tt 1182

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SEQ ID NO: 127      moltype = DNA length = 1476
FEATURE            Location/Qualifiers
source              1..1476
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 127
atgaaactct caactaaact ttgttggtgt ggtattaaag gaagacttag gccgcaaaag 60
caacaacaat tacacaatac aaacttgcaa atgactgaac taaaaaaca aaagaccgct 120
gaacaaaaaa ccagacctca aaatgtcggt attaaaggtta tccaaattta catcccaact 180
caatgtgtca accaatctga gctagagaaa ttgatggcg tttctcaagg taaatacaca 240
attggtctgg gccaaaccaa catgtctttt gtcaatgaca gagaagatat ctactcgatg 300
tccctaactg ttttgtctaa gttgatcaag agttacaaca tcgacaccaa caaaattggt 360
agattagaag tcggtactga aactctgatt gacaagtcca agtctgtcaa gctgtctgtg 420
atgcaattgt ttggtgaaaa cactgacgtc gaaggtattg acacgcttaa tgctgtttac 480
ggtaggtacca acgctgtgtt caactctttg aactggattg aatctaagcg atgggtggt 540
agagacggca ttgtagtttg cggtagattt gccatctacg ataagggtgc cgcaagacca 600
accggtgggt ccggtactgt tgcattgtgg atcggtctct atgctccaat tgtatttgac 660
tctgtaagag cttcttacat ggaacacgcc tacgattttt acaagccaga ttccaccagc 720
gaatatcctt acgtcgatgg tcatttttca ttaacttgtt acgtcaaggc tcttgatcaa 780
gtttacaaga gttattccaa gaaggctatt tctaaagggt tggtagcga tcccgtgggt 840
tcggatgctt tgaacgtttt gaaatatttc gactacaacg ttttccatgt tccaacctgt 900
aaattggtca caaaatcata cggtagatta ctatataacg atttcagagc caatcctcaa 960
ttgtcccgag aagttgacgc cgaattagct actcgcgatt atgacgaatc ttaaacgat 1020
aagaacattg aaaaaacttt tgttaatggt gctaagccat tccacaaaga gagagttgoc 1080
caatctttga ttgttccaac aaacacaggt aacatgtaca ccgcatctgt ttatgcccgc 1140
ttgcatctc tattaactca ttttgatct gacgacttac aaggcaagcg tgttggttta 1200
ttttcttacg gttccggttt agctgcatct ctatattctt gcaaaattgt tggtagcgtc 1260
caacatatta tcaaggaaat agatattact aacaaattag ccaagagaaat caccgaaact 1320
ccaaaggatt ccgaagctgc catcgaaattg agagaaaatg cccatttgaa gaagaacttc 1380
aaacctcaag gttccattga gcatttgcaa agtgggtgtt actacttgac caacatcgat 1440
gacaaattta gaagatctta cgatgttaaa aaataa 1476

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SEQ ID NO: 128      moltype = DNA length = 1143
FEATURE            Location/Qualifiers

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source          1..1143
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 128
atggttgccg tacgtaggaa ggctctttca attttggcag aagctcctgt attagcatct 60
gatcgtttac catataaaaa ttatgactac gaccgcgtat ttggcgcttg ttgtgaaaat 120
gttatagggt acatgccttt gcccggttgg gtatataggcc ccttggttat cgatggtaca 180
tcttatcata taccaatggc aactacagag ggttggttgg tagcttctgc catgcgtggc 240
tgtaaggcaa tcaatgctgg cgggtgtgca acaactgttt taactaagga tggtagtaca 300
agaggcccaa tagtcgcttt cccaactttg aaaagatctg gtgcctgtaa gatatggtta 360
gactcagaag agggacaaaa cgcaattaaa aaagctttta actctacatc aagatttgca 420
cgtctgcaac atattcaaac ttgtctagca ggagatttac tottcatgag atttagaaca 480
actactggty acgcaatggg tatgaatatg atttctaaag gtgtcgaata ctcatataag 540
caaatggtag aagagtatgg ctgggaagat atggaggttg tctccgtttc tggtaactac 600
tgtaccgaca aaaaaccagc tgcctcaaac tggatcgaag gtctgtgtaa gagtgtcgtc 660
gcagaagcta ctattcctgg tgatgttgtc agaaaagtgt taaaaagtga tgttccgcga 720
ttggttgtag tgaacatttg taagaatttg gttggatctg caatggctgg gtctgttggt 780
ggatttaacg cacatgcagc taatttagtg acagctgttt tcttgccatt aggacaagat 840
cctgcacaaa atgttgaaag ttccaactgt ataacttga tgaagaagt ggacgggtgat 900
ttgagaattt ccgtatccat gccatccatc gaagttagta ccacggctgg tggtagtctt 960
ctagaaccac aagggtccat gttggactta ttagggttaa gaggcccgca tgcataccgt 1020
cctggtacca acgcacgtca attagcaaga atagttgcct gtgcctgctt ggcaggtgaa 1080
ttatccttat gtgtgcctc agcagccggc catttggttc aaagtcatat gaccacaaac 1140
agg                                                1143

SEQ ID NO: 129      moltype = DNA length = 1332
FEATURE            Location/Qualifiers
source              1..1332
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 129
atgtcattac cgtttettaac ttctgcaccg ggaaaggtta ttatttttgg tgaacactct 60
gctgtgtaca acaagcctgc cgtcgtgctc agtgtgtctg cgttgagaac ctacctgcta 120
ataagcgagt catctgcacc agatactatt gaattggact tccggacat tagctttaat 180
cataagtggc ccatcaatga tttcaatgcc atcacggagg atcaagtaaa ctcccaaaaa 240
ttggccaagg ctcaacaagc caccgatggc ttgtctcagg aactcgttag tcttttggat 300
ccgttgtagt ctcaactatc ggaatccttc cactaccatg cagcgttttg ttctctgtat 360
atgtttgttt gcctatgccc ccatgccaa gaaatattaagt tttctttaa gtctacttta 420
cccatcggtg ctgggttggg ctcaagcgcc tctatttctg tatcactggc cttagctatg 480
gcctacttgg ggggggttaat aggtactaat gacttggaag agctgtcaga aaacgataag 540
catatagtga atcaatgggc cttcataggt gaaaagtgtg ttacgggtac cccttcaggga 600
atagataacg ctgtggccac ttatgttaat gccctgctat ttgaaaaaga ctacataaat 660
ggacaataaa acacaaaaca ttttaagtgc ttatagattt tccagcccat tccaatgatc 720
ctaacctata ctagaatttc aagggtctca aaagatcttg ttgctcgcgt tctgtgtgtg 780
gtcaccgaga aatttctcga agttatgaag ccaattctag atgccaatgg tgaatgtgcc 840
ctacaaggct taagatttca gactaagtta agtaaatgta aaggcaccga tgacgaggct 900
gtagaaacta ataataaact gtagaactaa ctattgggat tgataagaat aaatcatgga 960
ctgcttctgc caatcggtgt ttctctcctc ggattagaac ttattaaaaa tctgagcgat 1020
gatttgagaa ttggctccac aaaaacttacc ggtgctggtg gcggcggttg ctctttgact 1080
ttgttacgaa gagacattac tcaagagcaa attgacagct tcaaaaagaa attgcaagat 1140
gattttagtt acgagacatt tgaacacagac ttgggtggga ctggctgctg tttgttaagc 1200
gcacaaaatt tgaataaaga tcttaaaatc aaatccctag tattccaatt atttgaaat 1260
aaaactacca caaagcaaca aattgacgat ctattattgc caggaaacac gaatttacca 1320
tggacttcat aa                                                1332

SEQ ID NO: 130      moltype = DNA length = 1356
FEATURE            Location/Qualifiers
source              1..1356
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 130
atgtcagagt tgagagcctt cagtgcacca gggaaagcgt tactagctgg tggatattta 60
gttttagata caaaatatga agcattttgt gtcggattat cggcaagaat gcatgctgta 120
gccatcctt acggttcatt gcaaggtctc gataagtttg aagtcgctgt gaaaagtaaa 180
caatttaaag atggggagtg gctgtacctt ataagtccta aaagtggctt cattcctgtt 240
tcgataggcg gatctaagaa ccttttcatt gaaaaagtta tgcgtaaact atttagctac 300
tttaaaccta acatggacga ctactgcaat agaaacttgt tcgttattga tattttctct 360
gatgatgcct accatttctc ggaggatagc gttacggaac atcgtggcaa cagaagattg 420
agttttcatt cgcacagaat tgaagaagtt cccaaaacag ggctgggctc ctccggcagg 480
ttagtccagc ttttaactac agctttggcc tcttttttgg tatcggaact ggaaaaataat 540
gtagacaaat atagagaagt tattcataat ttgacacaag ttgctcattg tcaagctcag 600
ggtaaaattg gaagcgggtt tgatgtagcg gcggcagcat atggatctat cagatataga 660
gattccccac ccgcattaat ctctaatttg ccagatattg gaagtgtac ttacggcagt 720
aaactggcgc atttggttga tgaagaagac tggaaatatta cgattaaaag taaccattta 780
ccttcgggat taactttatg gatggcgat attaagaatg gttcagaaac agtaaaaactg 840
gtccagaagg taaaaaattg gtatgattcg catatgccag aaagcttgaa aatatatata 900
gaactcgatc atgcacaaatc tagatttatg gatggactat ctaaaactaga tgccttacac 960
gagactcatg acgattacag cgatcagata tttgagctc ttgagaggaa tgactgtacc 1020
tgtcaaaagt atcctgaatc cacagaagtt agagatgcag ttgccacaat tagacgttcc 1080

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|             |             |             |             |             |             |      |
|-------------|-------------|-------------|-------------|-------------|-------------|------|
| tttagaaaaa  | taactaaaga  | atctgggtgcc | gatatcgaac  | ctccccgtaca | aactagctta  | 1140 |
| ttggatgatt  | gccagacott  | aaaaggagtt  | cttactttgct | taatacctgg  | tgtgtggtggt | 1200 |
| tatgacgccca | ttgcagtgat  | tactaagcaa  | gatgttgatc  | ttaggggtcca | aaccgctaata | 1260 |
| gacaaaaagat | tttctaaggt  | tcaatggctg  | gatgtaactc  | aggctgactg  | gggtgttagg  | 1320 |
| aaagaaaaag  | atccgggaaac | ttatcttgat  | aaataa      |             |             | 1356 |

SEQ ID NO: 131      moltype = DNA   length = 1191  
 FEATURE            Location/Qualifiers  
 source              1..1191  
                      mol\_type = other DNA  
                      organism = synthetic   construct

SEQUENCE: 131

|             |            |             |            |             |            |      |
|-------------|------------|-------------|------------|-------------|------------|------|
| atgaccggttt | acacagcatc | cgttaccgca  | cccgtcaaca | tcgcaaccct  | taagtattgg | 60   |
| gggaaaaagg  | acacgaagtt | gaatctgccc  | accaattcgt | ccatatacgt  | gactttatcg | 120  |
| caagatgacc  | tcagaacggt | gacctctgcy  | gctactgcac | ctgagtttga  | acgcgacact | 180  |
| ttgtgggttaa | atggagaacc | acacagcatc  | gacaatgaaa | gaactcaaaa  | ttgtctgcgc | 240  |
| gacctacgcc  | aattaagaaa | ggaaatggaa  | tcgaaggacg | cctcattgcc  | cacattatct | 300  |
| caatgggaaac | tcacatttgt | ctccgaaaat  | aactttccta | cagcagctgg  | tttagcttcc | 360  |
| tcctgtgctg  | gctttgctgc | attggtctct  | gcaattgcta | agttatacca  | attaccacag | 420  |
| tcaacttcag  | aaatatctag | aatagcaaga  | aaggggtctg | gttcagcttg  | tagatcgttg | 480  |
| tttggcggat  | acgtggcctg | ggaaatggga  | aaagctgaag | atggctcatga | ttccatggca | 540  |
| gtacaaatcg  | cagacagctc | tgactggcct  | cagatgaaag | cttgtgtcct  | agttgtcagc | 600  |
| gatattaaaa  | aggatgtgag | ttccactcag  | ggtatgcaat | tgaccgtggc  | aacctccgaa | 660  |
| ctattttaaag | aaagaattga | acatgtcgta  | ccaaagagat | ttgaagtcac  | gcgtaaaagc | 720  |
| attgttgaaa  | aagatttcgc | cacctttgca  | aaggaaacaa | tgatggattc  | caactcttcc | 780  |
| catgccacat  | gtttggactc | tttccctcca  | atattctaca | tgaatgacac  | ttccaagcgt | 840  |
| atcatcagtt  | gggtccacac | cattaatcag  | ttttacggag | aaacaatcgt  | tgcatacacg | 900  |
| tttgatgcag  | gtccaaatgc | tgtgtgtgac  | tacttagctg | aaaatgagtc  | gaaactcttt | 960  |
| gcattttatct | ataaatttgt | tggctctggt  | cctggatggg | acaagaaatt  | tactactgag | 1020 |
| cagcttgagg  | ctttcaacca | tcaatttgaa  | tcatctaact | ttactgcacg  | tgaattggat | 1080 |
| cttgagttgc  | aaaaggatgt | tgccagagtg  | attttaactc | aagtcggttc  | aggcccaaaa | 1140 |
| gaaaacaaag  | aattcttgat | tgacgcaaaag | actggtctac | caaaggaata  | a          | 1191 |

SEQ ID NO: 132      moltype = DNA   length = 867  
 FEATURE            Location/Qualifiers  
 source              1..867  
                      mol\_type = other DNA  
                      organism = synthetic   construct

SEQUENCE: 132

|            |            |             |            |             |             |     |
|------------|------------|-------------|------------|-------------|-------------|-----|
| atgactgccg | acaacaatag | tatgccccat  | ggtgcagtat | ctagttacgc  | caaattagtg  | 60  |
| caaaacccaa | cacctgaaga | catttttgaa  | gagtttctcg | aaattattcc  | attacaacaa  | 120 |
| agacctaata | cccgatctag | tgagacgtca  | aatgacgaaa | gcggagaaaac | atgtttttct  | 180 |
| ggctcatgat | aggagcaaat | taagttaatg  | aatgaaaatt | gtattgtttt  | ggattgggac  | 240 |
| gataatgcta | ttggtgcogg | taccaagaaa  | gtttgtcatt | taatggaaaa  | tattgaaaag  | 300 |
| ggtttactac | atcgtgcatt | ctccgtcttt  | attttcaatg | aacaaggtga  | attactttta  | 360 |
| caacaaagag | ccactgaaaa | aataactttc  | cctgatcttt | ggactaacac  | atgctgctct  | 420 |
| catccactat | gtattgatga | cgaattaggt  | ttgaagggtg | agctagacga  | taagattaag  | 480 |
| ggcgctatta | ctgcggcggg | gagaaaacta  | gatcatgaat | taggtattcc  | agaagatgaa  | 540 |
| actaagacaa | ggggtaagtt | tcacttttta  | aacagaatcc | attacatggc  | accaagcaat  | 600 |
| gaaccatggg | gtgaacatga | aattgattac  | atcctatttt | ataagatcaa  | cgctaagaaa  | 660 |
| aacttgactg | tcaacccaaa | cgtcaatgaa  | gttagagact | tcaaatgggt  | ttcaccaaaat | 720 |
| gatttgaaaa | ctatgtttgc | tgacccaagt  | tacaagttta | cgcttgggtt  | taagattatt  | 780 |
| tgcgagaatt | acttattcaa | ctgggtgggag | caattagatg | acctttctga  | agtggaaaaat | 840 |
| gacaggcaaa | ttcatagaat | gctataaa    |            |             |             | 867 |

SEQ ID NO: 133      moltype = DNA   length = 1059  
 FEATURE            Location/Qualifiers  
 source              1..1059  
                      mol\_type = other DNA  
                      organism = synthetic   construct

SEQUENCE: 133

|             |            |            |             |             |             |      |
|-------------|------------|------------|-------------|-------------|-------------|------|
| atggcttcag  | aaaaggaaat | aagaagagaa | agattcttga  | acgtattccc  | aaagttagtt  | 60   |
| gaagaattga  | acgctagttt | gttagcttat | ggtatgccta  | aagaagcctg  | cgattgggtat | 120  |
| gctcactctt  | taaacataca | tactccaggt | ggtaaattga  | atagaggttt  | gagtgtagtt  | 180  |
| gatacttatg  | ctatcttctg | taacaaaacc | gttgaacaat  | taggtcaaga  | agaatacgaa  | 240  |
| aaggctcgta  | tcttgggttg | gtgtattgaa | ttgttgcaag  | catacttttt  | ggttgccgat  | 300  |
| gacatgatgg  | ataagtctat | aacaagaaga | ggtcaaccat  | gctgggtacaa | agttccagaa  | 360  |
| gttggtgaaa  | tagccataaa | tgatgctttt | atgttgggaag | ccgctatcta  | taaatgtgtg  | 420  |
| aagtacacatt | tcagaaacga | aaagtactac | atcgatatta  | ccgaattatt  | ccacgaagtt  | 480  |
| actttccaaa  | cagaattggg | tcaattgatg | gatttgataa  | ctgcacctga  | agataaagtt  | 540  |
| gacttgtcaa  | agttttcctt | gaagaaacat | tcatctcatg  | tcacotttga  | aactgcttat  | 600  |
| tactccttct  | atttgccagt | cgcttgggt  | atgtacgtag  | ctggatttac  | tgatgaaaaa  | 660  |
| gacttgaagc  | aagcaagaga | tgttttgata | ccttgggggt  | aatacttcca  | aatccaagat  | 720  |
| gactacttag  | actgtttcgg | tactccagaa | caaataaggta | aaatcggtac  | agatattcaa  | 780  |
| gacaataagt  | gcagttgggt | tattaacaag | gctttggaat  | tagcatctgc  | cgaacaaaaga | 840  |
| aagacttttg  | atgaaaacta | cggtaaaaag | gactcagttg  | ctgaagcaaa  | gtgtaagaaa  | 900  |
| attttttaag  | atttgaagat | tgaacaattg | taccatgaat  | acgaagaatc  | catcgctaaa  | 960  |
| gacttaaaag  | caaagattag | tcaagttgat | gaatcaagag  | gttttaaaag  | cgacgttttg  | 1020 |
| acagctttct  | tgaataaggt | ctacaagaga | tcaaagtag   |             |             | 1059 |

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SEQ ID NO: 134      moltype = DNA length = 1155
FEATURE            Location/Qualifiers
source             1..1155
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 134
atgaatcatt tgagagctga aggtccagct tctgttttgg ctattggtac tgctaatacca 60
gaaaatattt tgttgcaaga tgaatttcca gattattatt ttagagttac taaatctgaa 120
catatgactc aattgaaaga aaaatttaga aaaatttgtg ataaatctat gattagaaaa 180
agaaattgtt ttttgaatga agaacatttg aaacaaaatc caagattggg tgaacatgaa 240
atgcaaaactt tggatgctag acaagatatg ttggttgttg aagttccaaa attgggtaaa 300
gatgcttgtg ctaaaagctat taaagaatgg ggtcaaccaa aatctaaaat tactcatttg 360
atttttactt ctgctttctac tactgatatg ccaggtgctg attatcattg tgctaaattg 420
ttgggtttgt ctccatctgt taaaagagtt atgatgtatc aattgggttg ttatgggtgt 480
gggtactgtt tgagaattgc taaagatatt gctgaaaata ataaagggtc tagagttttg 540
gctgttttgt ttgatattat ggcttgtttg tttagaggtc catctgaatc tgatttggaa 600
ttgttgggtg gtcaagctat ttttgggtgat ggtgctgctg ctgttatgtt tgggtgctgaa 660
ccagatgaat ctgttgggtg aagaccaatt ttgaattgg tttctactgg tcaaaactatt 720
ttgccaaaat ctgaagggtac tatttgggtt catattagag aagctgtgtt gatttttggat 780
ttgcataaag atgttccaat gttgatattt aataatattg aaaaatgttt gattgaagct 840
tttactccaa ttggtatttc tgattggaat tctatttttt ggattactca tccaggtggg 900
aaagctattt tggataaaagt tgaagaaaaa ttgcatttga aatctgataa atttgttgat 960
tctagacatg ttttgtctga acatggtaat atgtcttctt ctactgtttt gtttgttatg 1020
gatgaattga gaaaaagatc ttggaagaa ggtaaatcta ctactggtga tggttttgaa 1080
tgggtgtttt tgtttgtgtt tgggtccaggt ttgactgttg aaagagttgt tgttagatct 1140
gttccaatta aatat                                     1155

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SEQ ID NO: 135      moltype = DNA length = 303
FEATURE            Location/Qualifiers
source             1..303
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 135
atggctgtta aacatttgat tgttttgaat tttaaagatg aaattactga agctcaaaaa 60
gaagaatttt ttaaaactta tgttaatttg gttaatatata ttccagctat gaaagatgtt 120
tattgggtga aagatgtttac tcaaaaaaat aaagaagaag gttatactca tattgttgaa 180
gttacttttg aatctgttga aactattcaa gattatata ttcactcagc tcatgttggg 240
tttggtagtg tttatagatc tttttgggaa aaattgttga tttttgatta tactccaaga 300
aaa

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SEQ ID NO: 136      moltype = DNA length = 1185
FEATURE            Location/Qualifiers
source             1..1185
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 136
atgggtttgt cttctgtttg tactttttct tttcaacta attatcatatc ttgttgaat 60
ccacataata ataataccaa aacttctttg ttgtgttata gacatccaaa aactccaatt 120
aaatattctt ataataattt tccatctaaa cattgttcta ctaaatcttt tcatttgcaa 180
aataaatggt ctgactcttt gtctattgct aaaaattcta ttagagctgc tactactaat 240
caaaactgaac caccagaatc tgataatcat tctgttgcta ctaaaatttt gaattttggg 300
aaagcttgtt ggaattgca aagaccatat actattattg cttttacttc ttgtgcttgg 360
ggtttgtttg gtaagaatt gtgtcataat actaatttga tttcttgggc ttgatgtttt 420
aaagcttttt ttttttgggt tgcatttttg tgtattgctt cttttactac tactattaat 480
caaatattat atttgcata tgcataaatt aataaaccag atttgccatt ggcttctggg 540
gaaatttctg ttaatactgc ttggattatg tctattattg ttgctttggt tggtttgatt 600
attactatta aaatgaaaag tgggtccattg tatatttttg gttattgttt tggatttttt 660
gggtggtattg tttattctgt tccaccattt agatggaac aaaaatccatc tactgctttt 720
ttgttgaatt ttttgggtca tattattact aattttactt tttattatgc ttctagagct 780
gctttgggtt tgccatttga attgagacca tcttttactt ttttgggtgc ttttatgaaa 840
tctatgggtt ctgctttggc ttgtattaaa gatgcttctg atgttgaagg tgatactaaa 900
tttggatttt ctactttggc tttcaaatat ggttctagaa atttgacttt gttttgttct 960
ggatttgggt tgttgcctta tgggtgctgt attttggctg gtattatttg gccacaagct 1020
tttaattcta atgttatgtt gttgtctcat gctatttttg ctttttgggt gattttgcaa 1080
actagagatt ttgctttgac taattatgat ccagaagctg gtagaagatt ttatgaattt 1140
atgtggaaat tgtattatgc tgaatatttg gtttatgttt ttatt                                     1185

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SEQ ID NO: 137      moltype = DNA length = 6702
FEATURE            Location/Qualifiers
source             1..6702
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 137
atgagcgaag aaagcttatt cgagctcttct ccacagaaga tggagtaaga aattacaaa 60
tactcagaaa gacatacaga acttccaggt catttcattg gcctcaatac agtagataaa 120
ctagaggagt ccccgtaaag ggactttgtt aagagtcacg gtggtcacac ggtcatatcc 180
aagatcctga tagcaataaa tggatttgcc gccgtgaaag aaattagatc cgctcagaaaa 240
tgggcatacg agacgttcgg cgatgacaga accgtccaat tcgtcgccat ggccacccca 300

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|            |             |             |             |             |             |      |
|------------|-------------|-------------|-------------|-------------|-------------|------|
| gaagatctgg | aggccaacgc  | agaatatatc  | cgtatggccg  | atcaatacat  | tgaagtgcc   | 360  |
| ggtggtacta | ataataacaa  | ctacgctaac  | gtagacttga  | tcgtagacat  | cgccgaaaga  | 420  |
| gcagacgtag | acgcgcgatg  | ggctggctgg  | ggtcacgcct  | ccgagaatcc  | actattgcct  | 480  |
| gaaaaattgt | cccagctctaa | gaggaaagtc  | atctttattg  | ggcctccagg  | taacgccaatg | 540  |
| aggctctttg | gtgataaaat  | ctcctctacc  | attgtcgctc  | aaagtgcata  | agtcoccatgt | 600  |
| attccatggt | ctggtagccg  | tgttgacacc  | gttcacgtgg  | acgagaaaac  | cggtctggctc | 660  |
| tctgtcgacg | atgacatcta  | tcaaaagggt  | tgttgtacct  | ctcctgaaga  | tggtttaca   | 720  |
| aaggccaagc | gtattgggtt  | tectgtcatg  | attaaaggcat | ccgaagggtg  | tgggtgtaaa  | 780  |
| ggtatcagac | aagttgaacg  | tgaagaagat  | ttcatcgctt  | tataccacca  | ggcagccaac  | 840  |
| gaaattccag | gctcccccat  | tttcatcatg  | aagttggccg  | gtagagcgcg  | tcacttgga   | 900  |
| gttcaactgc | tagacagatc  | gtacgggtaca | aatatttccct | tgttcggtag  | agactgttcc  | 960  |
| gttcagagac | gtcatcaaaa  | aattatcgaa  | gaagcaccag  | ttacaattgc  | caaggctgaa  | 1020 |
| acatttcacg | agatggaaaa  | ggctgcccgc  | agactgggga  | aactagtcgg  | ttatgtctct  | 1080 |
| gcgggtaccg | tggagtatct  | atattctcat  | gatgatggaa  | aattctactt  | tttagaattg  | 1140 |
| aaccaagat  | tacaagtcga  | gcattccaaca | acggaaatgg  | tctccggtgt  | taacttacct  | 1200 |
| gcagctcaat | tacaaatcgc  | tatgggtatc  | cctatgcata  | gaataagtga  | cattagaact  | 1260 |
| ttatatggta | tgaattcctca | ttctgcctca  | gaaatcgatt  | tcgaattcaa  | aactcaagat  | 1320 |
| gccaccaaga | aacaaagaag  | acattattcca | aagggtcatt  | gtacgcgttg  | tcgtatcaca  | 1380 |
| tcagaagatc | caaacgatgg  | attcaagcca  | tcgggtggta  | ctttgcatga  | actaaacttc  | 1440 |
| cgttcttctc | ctaattgtttg | gggttacttc  | tcctgtgggt  | acaatggtaa  | tattcactcc  | 1500 |
| ttttcggact | ctcagttcgg  | ccattattttt | gcttttgggt  | aaaaagaca   | agcttccagg  | 1560 |
| aaacacatgg | tgttggccct  | gaaggaattg  | tcctattagg  | gtgatttcag  | aactactgtg  | 1620 |
| gaataactga | tcaaaactttt | ggaaactgaa  | gatttcgagg  | ataacactat  | taccacgggt  | 1680 |
| tgggtggacg | atttgattac  | tcataaaatg  | accgctgaaa  | agcctgatcc  | aactcttgcc  | 1740 |
| gtcatttgcg | gtgcgcgtac  | aaaggctttc  | ttagcatctg  | aagaagcccg  | ccacaagtat  | 1800 |
| atcgaatcct | tcaaaaaggg  | acaagttcta  | tctaagacc   | tactgcgaac  | tatgttccct  | 1860 |
| gtagatttta | tccatgaggg  | taaaagatac  | aagttcaccg  | tagctaaatc  | cggtaatgac  | 1920 |
| cggtacacat | tatttatcaa  | gggttctaaa  | tgtgatatac  | tactgcgtca  | actatctgat  | 1980 |
| gggtgtcttt | tgattgcca   | aggcggtaaa  | tcgcatacca  | tctattggaa  | agaagaagtt  | 2040 |
| gctgtacaaa | gattatccgt  | tgactctatg  | actactttgt  | tggaaagtga  | aaacgatcca  | 2100 |
| acccagttgc | gtactccatc  | cctgggtaaa  | ttgggttaaa  | tcttgggtga  | aaatgggtga  | 2160 |
| cacattatca | agggccaacc  | atatgcagaa  | attgaagtta  | tgaaaatgca  | aatgcctttg  | 2220 |
| gtttctcaag | aaaatggtat  | cgtccagtta  | ttaaagcaac  | ctggttctac  | cattgttgca  | 2280 |
| gggtgatata | tggcttttat  | gactcttgac  | gatccatcca  | aggtaacgca  | cgctctacca  | 2340 |
| tttgaaggt  | tgtgcccaga  | ttttgtttct  | ccagttatcg  | aaggaaacaa  | acctgcctat  | 2400 |
| aaattcaagt | cattagtgct  | tactttggaa  | aacattttga  | aggtttatga  | caaccaagtt  | 2460 |
| attatgaagt | ttccttgcca  | acaattgata  | gaggttttga  | gaaatccaaa  | actgccttac  | 2520 |
| tcagaatgga | aactacacat  | ctctgcttta  | cattcaagat  | tgccctgctaa | gctagatgaa  | 2580 |
| caaatggaag | agttagttgc  | acgtttctttg | agacgttggt  | ctgttttccc  | agctagacaa  | 2640 |
| ttaagttaat | tgattgatat  | ggcgtggaag  | aatcctgaat  | acaaccccca  | caaattgtctg | 2700 |
| ggcgccgtcg | tggaaaccatt | ggcggtatatt | gctcataagt  | actctaaccg  | gtagaagcc   | 2760 |
| catgaacatt | ctatatattgt | ccatttcttg  | gaagaatatt  | acgaagtga   | aaagtatttc  | 2820 |
| aatggtccaa | atggttcgtga | ggaaaatattc | attctgaaat  | tgctgtgatga | aaaccctaaa  | 2880 |
| gatctagata | aagttgcgct  | aactgttttg  | tctcattcga  | aagtttccagc | gaagaataac  | 2940 |
| ctgatccatg | ctatcttgaa  | acattatcaa  | ccattgtgca  | agttatcttc  | ttaaagttct  | 3000 |
| gccattttct | ctactcctct  | acaacatatt  | gttgaactag  | aatctaaggc  | tacgcctaag  | 3060 |
| gtcgtcttac | aagcaagaga  | aattttgatt  | caaggcgctt  | taccttcggt  | caaggaaaga  | 3120 |
| actgaacaaa | ttgaacatatt | cttaaaatcc  | tctgtttgta  | aggttgccata | tggctcatcc  | 3180 |
| aatccaaagc | gctctgaaac  | agatttgat   | atcttgagg   | acttgatcga  | ttctaattac  | 3240 |
| ggtgtgttcg | atgttttact  | tcaattccta  | acccatcaag  | accagttgt   | gactgctgca  | 3300 |
| gctgctcaag | tctatattcg  | tcgtgcttat  | cgtgcttaca  | ccataggaga  | tattagagtt  | 3360 |
| cacgaaggtg | tcacagttcc  | aattgttgaa  | tggaaattcc  | aactaccttc  | agctgcgttc  | 3420 |
| tcacaccttc | caactgttaa  | atctaaaaatg | gggtatgaaca | gggtctgttc  | tgtttcagat  | 3480 |
| tgtcatatg  | ttgcaaacag  | tcagtcattct | ccgttaagag  | aaggtatttt  | gatggctgtg  | 3540 |
| gatcatttag | atgatgttga  | tgaatttttg  | tcacaaagtt  | tggaaagttat | tcctcgtcac  | 3600 |
| caatcttctt | ctaaccgacc  | tgtctctgat  | cgttctgtga  | gctccgcac   | gtagagtaac  | 3660 |
| gttgctaatg | tttgtgttgc  | ttctacagaa  | gggttcgaat  | ctgaagagga  | aattttggt   | 3720 |
| agggtgagag | aaattttgga  | tttgaataag  | caggaaattaa | tcaatgcttc  | tatccgtcgt  | 3780 |
| atcacattta | tgttcgggtt  | ttaaagatggg | tcttatccaa  | agttattatc  | ttttaacggg  | 3840 |
| ccaaattata | acgaaaatga  | aacaattcgt  | cacattgagc  | cggtcttggc  | cttccaactg  | 3900 |
| gaattagga  | gattgtccaa  | ctcaacatt   | aaaccaattt  | tcaactgata  | tagaaacatc  | 3960 |
| catgtctacg | aagctgttag  | taagacttct  | ccattggata  | agagattctt  | tacaagaggt  | 4020 |
| attattagaa | cggtgcatat  | ccgtgatgac  | atttctattc  | aagaatatct  | gacttctgaa  | 4080 |
| gctaacagat | tgatgagtg   | tattttgga   | aatttagaag  | tcaccgacac  | ttcaaatctt  | 4140 |
| gatttgaatc | atatcttcat  | caacttcatt  | gcggtgtttg  | atatctctcc  | agaagatgct  | 4200 |
| gaagccgctc | tcggtgtgtt  | cttagaaaaga | tttggttaaga | gattgttgag  | attgcgtgtt  | 4260 |
| tcttctgcg  | aaattagaat  | catcatcaaa  | gatcctcaaa  | caggtgcccc  | agtaccattg  | 4320 |
| cgtgccttga | tcaataacgt  | ttctgtttat  | gttatcaaaa  | cagaaatgta  | caccgaagtc  | 4380 |
| aagaacgcga | aaggtgaatg  | ggtatttaag  | tctttgggt   | aacctggatc  | catgcattta  | 4440 |
| agacattatg | ctactcctta  | ccctgttaag  | gaatgggtgc  | aacccaaaacg | ttataaggca  | 4500 |
| cacttgatgg | gtaccacata  | tgtctatgac  | ttccagaat   | tattccgcca  | agcatcgta   | 4560 |
| tcccaatgga | aaaatttctc  | tgcagatgtt  | aagttaacag  | atgatttctt  | tattttccaa  | 4620 |
| gagttgatgg | aagatgaaa   | cggcgaaat   | actgaggtgg  | aaagagaacc  | tgggtcccaac | 4680 |
| gctattggta | tgggtgtcct  | taagattact  | gtaaagactc  | ctgaatatcc  | aagaggccgt  | 4740 |
| caatttgggt | tgttggctaa  | cgaatccaca  | ttcaagatcg  | gttcccttgg  | tccacaagaa  | 4800 |
| gacgaattct | tcaataaggt  | tactgaatat  | gctagaagac  | gtggtatccc  | aagaatttac  | 4860 |
| ttggctgcaa | actcaggtgc  | cagaattggg  | atggctgaag  | agattgttcc  | actatttcaa  | 4920 |
| gttgcatgga | atgatgctgc  | caatccggac  | aagggtctcc  | aatacttata  | cttaacaagt  | 4980 |
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SEQ ID NO: 138      moltype = DNA length = 1632
FEATURE
source             Location/Qualifiers
                   1..1632
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 138
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SEQ ID NO: 139      moltype = DNA length = 1635
FEATURE
source             Location/Qualifiers
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                   mol_type = other DNA
                   organism = synthetic construct

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accacagcca ggttttgggg tgaaaaaatc ttccgcaaaa atttcaatag atttggttaag 1560
gtaaaaacta aggcggatcc aaacaatttt tttagaaatg agcagagtat tccgcccctg 1620
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SEQ ID NO: 140      moltype = DNA length = 2160
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                    organism = synthetic construct

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SEQ ID NO: 141      moltype = DNA length = 4587
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source              Location/Qualifiers
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                    mol_type = other DNA
                    organism = synthetic construct

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caattgatta aaagaagagg taaattgggt ttgggttggtg ttaatttgac tttggatgg 240
gttaaatctt ggttgaaacc aagattgggt caagaagcta ctgttggtta agctactggt 300
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FEATURE             Location/Qualifiers
source              1..2463
                    mol_type = other DNA
                    organism = synthetic construct

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aatgcttatt ataaacaagg tgattatcaa aaagctattg aagattatca aaaagctttg 1500
gaattggatc caaataatag atctagatct gctggtgggt gtggttctgg tgggtggtgg 1560
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tgtaatactg ctagatattg tgggtctttt tgtcaacata aagattggga aaaacatcat 1740
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ggttctgctg ctggttctgg tgaatttggg tctatggctg tttctgaatc tcaattgaaa 1920
aaaatggttt ctaaatataa atatagagat ttgactgtta gagaaactgt taatgttatt 1980
actttgtata aagatttgaa accagttttg gattcttatg tttttaatga tgggttctct 2040
agagaattga tgaatttgac tgggtactatt ccagttccat atagaggtaa tacttataat 2100

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attccaattt gtttggtggtt gttggatact tatccatata atccaccaat ttgttttgtt 2160
aaaccaactt cttctatgac tattaaaaact ggtaaacatg ttgatgctaa tggtaaaatt 2220
tatttgccat atttgcatga atggaacat ccacaatctg atttggtggg ttgtattcaa 2280
gttatgattg ttgttttttg tgatgaacca ccagtttttt ctagacca 2328

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SEQ ID NO: 144      moltype = DNA length = 2223
FEATURE             Location/Qualifiers
source               1..2223
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 144
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ggtgaaattg aaaatgattc tgaagttttg gctgttattt tgactggtgc tggtgaaaaa 180
tcttttgttg ctggtgctga tatttctgaa atgaaagaaa tgaatactat tgaaggtaga 240
aaatttggtt ttttgggtta taaagttttt agaagattgg aattgttgga aaaaccagtt 300
attgctgctg ttaattggtt tgctttgggt ggtggttgtg aaattgctat gctttgtgat 360
attagaattg cttcttctaa tgctagattt ggtcaaccag aagttggttt ggggtattact 420
ccaggttttg gtggtactca aagattgtct agattggttg gtatgggtat ggctaaacaa 480
ttgattttta ctgctcaaaa tattaaagct gatgaagctt tgagaattgg tttggttaat 540
aaagttgttg aaccatctga attgatgaat actgctaaag aaattgctaa taaaattggt 600
tctaattgctc cagttgctgt taaattgtct aaacaagcta ttaatagagg tatgcaatgt 660
gatattgata ctgctttggc ttttgaatct gaagcttttg gtgaatgttt ttctactgaa 720
gatcaaaaag atgctatgac tgcttttatt gaaaaaagaa aaattgaagg ttttaaaaat 780
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caaaaagcta ttgaatatta tcaaaaagct ttggaattgg atccaaaata tgcttctgct 1260
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gggtggtggt gtgcttctgg tccattgggt tctccattga ctgcttctat gttggcttct 1560
gctccaccac aagaaacaaa acaaatgttg ggtgaaagat tgtttccatt gattcaagct 1620
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gaaaaaacac aacatattgt ttattgttct aatgatttgt tgggtgattt gtttggttgt 2160
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gtt 2223

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SEQ ID NO: 145      moltype = DNA length = 2460
FEATURE             Location/Qualifiers
source               1..2460
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 145
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gctggtgcta aagctccaaa aaatgttttg gttttgggtt gttctaattg ttatggtttg 180
gcttctagaa ttactgctgc ttttggttat ggtgctgcta ctattggtgt ttcttttgaa 240
aaagctgggt ctgaaactaa atatggtact ccaggttggt ataataattt ggcttttgat 300
gaagctgcta aaagagaagg tttgtattct gttactattg atggtgatgc tttttctgat 360
gaaattaaag ctcaagttat tgaagaagct aaaaaaaaag gtattaaatt tgatttgatt 420
gtttattctt tggcttctcc agttagaact gatccagata ctggtattat gcataaatct 480
gttttgaaac catttggtta aacttttact ggtaaaaactg ttgatccatt tactggtgaa 540
ttgaaagaaa tttctgctga accagcta atgatgaaga ctgctgctac tgttaaaagt 600
atgggtgggt aagattggga aagatggatt aaacaattgt ctaaagaagg tttgttgga 660
gaaggttgta ttactttggc ttattcttat attggtccag aagctactca agctttgat 720
agaaaaggta ctattggtta agctaaagaa catttggaag ctactgctca tagattgaat 780
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tctgctgcta ttccagttat tccattgtat ttggttctt tgtttaaagt tatgaaagaa 900
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gctgaatctt tgactgattt ggtggtttat agacatgatt ttttggcttc taatggtttt 1140
gatgttgaag gtatttaata tgaagctgaa gttgaaagat ttgatagaat taaattgtct 1200
gggtggtggt gttctggttg tgggtggtct ggtggtggtg gttctgctga agcttggtat 1260
aatttgggta atgcttatta taacaagggt gattatcaaa aagctattga atattatcaa 1320
aaagcttttg aattggatcc aaataatgct gaagcttggt ataatttggg taatgcttat 1380

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|------------|-------------|------------|-------------|-------------|------------|------|
| tataaacaag | gtgattatca  | aaaagctatt | gaatattatc  | aaaaagcttt  | ggaattggat | 1440 |
| cacaataatg | ctgaagcttg  | gtataatttg | ggtaatgctt  | attataaaca  | aggtgattat | 1500 |
| caaaaagcta | ttgaagatta  | tcaaaaagct | ttggaattgg  | atccaaataa  | tttgcaagct | 1560 |
| gaagcttggg | aaaatttggg  | taatgcttat | tataaacaag  | gtgattatca  | aaaagctatt | 1620 |
| gaatattatc | aaaaagcttt  | ggaattggat | cacaataatg  | cttctgcttg  | gtataatttg | 1680 |
| ggtaatgctt | attataaaca  | aggtgattat | caaaaagcta  | ttgaatatta  | tcaaaaagct | 1740 |
| ttggaattgg | atccaaataa  | tgctaaagct | tggtatagaa  | gaggtaatgc  | ttattataaa | 1800 |
| caaggtgatt | atcaaaaagc  | tattgaagat | tatcaaaaag  | ctttggaatt  | ggatccaaat | 1860 |
| aatagatcta | gatctgctgg  | tggttggtgg | tctggtggtg  | gtggttctgg  | tggttggtgg | 1920 |
| gcttcttctt | attatcatca  | tcatcatcat | catttggaat  | ctacttcttt  | gtataaaaaa | 1980 |
| gctggttctg | gttctaattt  | gttggtactt | ttgagatcta  | ctgctgctgt  | tttggaatat | 2040 |
| gaaaatcatg | ttttgaaaaa  | agaaaaagaa | aaattgagaa  | aagaaaaaga  | acaattgttg | 2100 |
| aataaattgg | aagcttataa  | aggttctgct | ggttctgctg  | ctggttctgg  | tgaatttggt | 2160 |
| ctgctgtaag | ctgctgctaa  | agaagctgct | gctaaagctg  | gttctgctgg  | ttctgctgct | 2220 |
| ggttctggtg | aatttggttc  | ttcttattat | catcatcatc  | atcatcattt  | ggaatctact | 2280 |
| tctttgtata | aaaaagctgg  | ttctggttct | aaaagaattg  | cttattttgag | aaaaaaaatt | 2340 |
| gctgctttga | aaaaagataa  | tgctaatttg | gaaaaagata  | ttgctaattt  | ggaaaatgaa | 2400 |
| attgaaagat | tgattaaaaga | aattaaaact | ttggaataatg | aagttgcttc  | tcatgaacaa | 2460 |

SEQ ID NO: 146                   moltype = DNA   length = 2478  
 FEATURE                        Location/Qualifiers  
 source                         1..2478  
                               mol\_type = other DNA  
                               organism = synthetic construct

SEQUENCE: 146

|             |             |             |             |             |            |      |
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| atgactagag  | aagttgttgt  | tgtttctggt  | gttagaactg  | ctattggtac  | ttttggtggt | 60   |
| tctttgaaag  | atgttgctcc  | agctgaattg  | ggtgctttgg  | ttgttagaga  | agctttggct | 120  |
| agagctcaag  | ttctggttga  | tgatgttggt  | catgttggtt  | ttggtaatgt  | tattcaaact | 180  |
| gaaccaagag  | atatgtatgt  | gggtagagtt  | gctgctgtta  | atggtggtgt  | tactattaat | 240  |
| gctccagctt  | tgactgttaa  | tagatttgtt  | ggttctggtt  | tgcaagctat  | tgtttctgct | 300  |
| gctcaaaacta | tttgttggg   | tgatactgat  | gttgctattg  | gtggtggtgc  | tgaatctatg | 360  |
| tctagagctc  | catatttggt  | tcagctgctg  | agatgggggtg | ctagaatggg  | tgaattgctg | 420  |
| ttggttgata  | tgatgttggg  | tgctttgcat  | gatccatttc  | atagaattca  | tatgggtggt | 480  |
| actgctgaaa  | atgttgctaa  | agaatatgat  | atctctagag  | ctcaacaaga  | tgaagctgct | 540  |
| ttggaatctc  | atagaagagc  | ttctgctgct  | attaaagctg  | gttattttta  | agatcaaat  | 600  |
| gttccagttg  | tttctaagg   | tgaaaaaggt  | gatgttactt  | ttgatactga  | tgaacatggt | 660  |
| agacatgatg  | ctactattga  | tgatatgact  | aaattgagac  | cagtttttgt  | taaagaaaat | 720  |
| ggtactgtta  | ctgctggtta  | tgttctggtt  | ttgaatgatg  | ctgctgctgc  | tgtttgttat | 780  |
| atggaagag   | ctgaagctga  | aagaagaggt  | ttgaaacctt  | tggttagatt  | ggtttcttat | 840  |
| ggtcatgctg  | gtgttgatcc  | aaaagctatg  | ggtattggtc  | cagtttccagc | tactaaaatt | 900  |
| gctttgaaaa  | gagcttggtt  | gcaagtttct  | gatttggatg  | ttattgaagc  | taatgaagct | 960  |
| tttctgctgc  | aagcttgctg  | tgttactaaa  | gctttgggtt  | ttggtccagc  | taaatgtta  | 1020 |
| ccaaatgggt  | ctggttattt  | ttgggtctat  | ccaattgggtg | ctactggtgc  | tttgattact | 1080 |
| gttaaaagctt | tgcatgaatt  | gaatagagtt  | caaggtagat  | atgctttggt  | tactatgtgt | 1140 |
| attggtggtg  | gtcaaggtat  | tgctgctatt  | tttgaagaaa  | ttaaattgtc  | tggttggtgt | 1200 |
| ggttctggtg  | gtggtggttc  | tggttggtgt  | ggttctgctg  | aagcttggtg  | taatttggtt | 1260 |
| aatgcttatt  | ataaacaagg  | tgattatcaa  | aaagctattg  | aatattatca  | aaaagctttg | 1320 |
| gaattggatc  | caaataatgc  | tgaagcttgg  | tataatttgg  | gtaatgctta  | ttataaacaa | 1380 |
| gggtgattatc | aaaaagctat  | tgaattattt  | caaaaagctt  | tggaattgga  | tccaaataat | 1440 |
| gctgaagctt  | ggtataattt  | gggtaatgct  | tattataaac  | aaggtgatta  | tcaaaaagct | 1500 |
| attgaagatt  | atcaaaaagc  | tttggaattg  | gatccaaata  | atttgcaagc  | tgaagcttgg | 1560 |
| aaaaatttgg  | tgaaatgctta | ttataaacaa  | ggtgattatc  | aaaaagctat  | tgaattattt | 1620 |
| caaaaagctt  | tgaatttggg  | tccaaataat  | gcttctgctt  | ggtataattt  | gggtaatgct | 1680 |
| tattataaac  | aaggtgatta  | tcaaaaagct  | attgaattat  | atcaaaaagc  | tttggaattg | 1740 |
| gatccaaata  | atgctaaaagc | ttggtataga  | agaggtaatg  | cttattataa  | acaaggtgat | 1800 |
| tatcaaaaag  | ctattgaaga  | ttatcaaaaa  | gctttggaat  | tggtccaaa   | taatagatct | 1860 |
| agatctgctg  | gtggttggtg  | ttctggtggt  | ggtggttctg  | gtggtggtg   | tgcttctgat | 1920 |
| ggtatgtggg  | aatataaatg  | ggaaaatact  | ggtgatgctg  | aattgatggt  | tccatttact | 1980 |
| tctgctcaaa  | tgcaaaactg  | ggtttctgaa  | ggttattttc  | cagatggtgt  | ttattgtaga | 2040 |
| aaattggatc  | caccaggtgg  | tcaattttat  | aattctaaaa  | gaattgattt  | tgatttggat | 2100 |
| actggttctg  | ctggttctgc  | tgctggttct  | ggtgaatttg  | gttctgctga  | agctgctgct | 2160 |
| aaagaagctg  | ctgctaaagc  | tggttctgct  | ggttctgctg  | ctggttctgg  | tgaatttggt | 2220 |
| tctgaatctg  | attctgttga  | atttaataat  | gctatttctt  | atgttaataa  | aattaaaact | 2280 |
| agatttttgg  | atcatccaga  | aatttataga  | tcttttttgg  | aaattttgca  | tacttatcaa | 2340 |
| aaagaacaat  | tgcatactaa  | aggttagacca | tttagaggta  | tgtctgaaga  | agaagttttt | 2400 |
| actgaagttg  | ctaatttggt  | tagaggtcaa  | gaagatttgt  | tgtctgaatt  | tggtcaattt | 2460 |
| ttgccagaag  | ctaaaaga    |             |             |             |            | 2478 |

SEQ ID NO: 147                   moltype = DNA   length = 2550  
 FEATURE                        Location/Qualifiers  
 source                         1..2550  
                               mol\_type = other DNA  
                               organism = synthetic construct

SEQUENCE: 147

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| caacaacaat | tacacaatac | aaacttgcaa | atgactgaac | taaaaaaaca | aaagaccgct  | 120 |
| gaacaaaaaa | ccagacctca | aaatgtcggg | attaaaggta | tccaaattta | catcccaact  | 180 |
| caatgtgtca | ccaactctga | gctagagaaa | tttgatggcg | tttctcaagg | taaatcacaca | 240 |
| attggtctcg | gccaaaccaa | catgtctttt | gtcaatgaca | gagaagatat | ctactcgatg  | 300 |

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atgcaattgt ttggtgaaaa cactgacgtc gaaggatttg acacgcttaa tgctgtttac 480
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tctgctggtt ctgctgctgg ttctggtgaa ttggttcttg gtgctatggg tccattgcca 2460
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agaattactc aattgggaaga tccaagatct 2550

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SEQ ID NO: 148      moltype = DNA length = 2478
FEATURE             Location/Qualifiers
source              1..2478
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 148
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gttataggtt acatgccttt gcccggttgt gttataggcc ccttggttat cgatggtaca 180
tcttatcata taccaatggc aactacagag ggttggtttg tagcttctgc catgctgggc 240
tgtaaggcaa tcaatgctgg ccgtggtgca acaactgttt taactaagga tggtagtaca 300
agaggcccag tagtccgttt cccaactttg aaaagatctg gtgcctgtaa gatattggtta 360
gactcagaag agggacaaa cgaatttaaa aaagctttta actctacatc aagatttgca 420
cgtctgcaac atattcaaac ttgtctagca ggagatttac tcttcatgag atttagaaca 480
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caaatggtag aagagtatgg ctgggaagat atggagggtt tctccgtttc tggtaactac 600
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SEQ ID NO: 149      moltype = DNA length = 3141
FEATURE             Location/Qualifiers
source              1..3141
                    mol_type = other DNA
                    organism = synthetic construct

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cataagtggt ccatcaatga ttccaatgcc atcacccagg atcaagtaaa ctcccaaaaa 240
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SEQ ID NO: 150      moltype = DNA length = 2604
FEATURE             Location/Qualifiers
source              1..2604
                    mol_type = other DNA
                    organism = synthetic construct

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gccatccctt acggttcatt gcaagggtct gataagtttg aagtgctgtg gaaaagtaaa 180
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SEQ ID NO: 151      moltype = DNA length = 3027
FEATURE             Location/Qualifiers
source               1..3027
                    mol_type = other DNA
                    organism = synthetic construct

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caagatgacc tcagaacggt gacctctgcg gctactgcac ctgagtttga acgcgacact 180
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SEQ ID NO: 152      moltype = DNA length = 2154
FEATURE             Location/Qualifiers
source               1..2154
                     mol_type = other DNA
                     organism = synthetic construct

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SEQ ID NO: 153      moltype = DNA length = 2448
FEATURE             Location/Qualifiers
source               1..2448
                     mol_type = other DNA
                     organism = synthetic construct

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SEQ ID NO: 154      moltype = DNA length = 2649
FEATURE             Location/Qualifiers
source               1..2649
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 154
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catatgactc aattgaaaga aaaatttaga aaaatttgtg ataaatctat gattagaaaa 180
agaaatttgt ttttgaatga agaacttttg aaacaaaatc caagattggt tgaacatgaa 240
atgcaaaact tggatgctag acaagatatg ttggttgttg aagttccaaa attgggtaaa 300
gatgcttgtg tcaaaagctat taaagaatgg ggtcaaccaa aatctaaaaa tactcatttg 360
atTTTTactt ctgcttctac tactgatatg ccaggtgctg attatcattg tgctaataatg 420
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gctgctgcta aagctggttc tgcgtggtct gctgctggtt ctggtgaatt tgggtctggt 2340
tctcatccat ggttttttgg taaaattcca agagctaaag ctgaagaaat gttgtctaaa 2400
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|             |            |            |            |            |            |      |
|-------------|------------|------------|------------|------------|------------|------|
| ttgtctgtta  | aatttggtta | tgtgtgtcaa | cattttaaag | ttttgagaga | tggtgtcgtg | 2520 |
| aaatattttt  | tgtgggttgt | taaatttaat | tctttgaatg | aattgggtga | ttatcataga | 2580 |
| tctactttctg | tttctagaaa | tcaacaaatt | tttttgagag | atattgaaca | agttccacaa | 2640 |
| caaccaact   |            |            |            |            |            | 2649 |

SEQ ID NO: 155                      moltype = DNA   length = 1995  
 FEATURE                              Location/Qualifiers  
 source                                1..1995  
                                       mol\_type = other DNA  
                                       organism = synthetic construct

SEQUENCE: 155

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| atggctgtta  | aacatttgat  | tgttttga   | aaatgaatg   | aaattactga | agctcaaaaa  | 60   |
| gaagaatttt  | ttaaaactta  | gttaattttg | gttaattatta | ttccagctat | gaaagatggt  | 120  |
| tattggggta  | aagatgttac  | tcaaaaaaat | aaagaagaag  | gttatactca | tattgttgaa  | 180  |
| gttacttttg  | aatctgttga  | aactattcaa | gattatatta  | ttcatccagc | tcatgttggt  | 240  |
| tttgggtgatg | tttatagatc  | tttttgggaa | aaattgttga  | tttttgatta | tactccaaga  | 300  |
| aaaaaattgt  | ctgggtgttg  | tggttctggt | gggtgtgtgt  | ctgggtgttg | tggttctgct  | 360  |
| gaagcttggt  | ataatttggg  | taatgcttat | tataaacaag  | gtgattatca | aaaagctatt  | 420  |
| gaatattatc  | aaaaagcttt  | ggaattggat | ccaataatg   | ctgaagcttg | gtataatttg  | 480  |
| ggtaatgctt  | attataaaca  | aggtgattat | caaaaagcta  | ttgaatatta | tcaaaaagct  | 540  |
| ttggaatttg  | atccaaataa  | tgctgaagct | tggtataatt  | tggttaattg | ttattataaa  | 600  |
| caaggtgatt  | atcaaaaagc  | tattgaagat | tatcaaaaag  | ctttggaatt | ggatccaaat  | 660  |
| aatttgcaag  | ctgaagcttg  | gaaaaatttg | ggtaatgctt  | attataaaca | aggtgattat  | 720  |
| caaaaagcta  | ttgaatatta  | tcaaaaagct | ttggaatttg  | atccaaataa | tgcttctgct  | 780  |
| tggtataatt  | tggtgaatgc  | ttattataaa | caaggtgatt  | atcaaaaagc | tattgaatat  | 840  |
| tatcaaaaag  | ctttggaatt  | ggatccaaat | aatgctaaag  | cttgggtatg | aagaggtaat  | 900  |
| gcttattata  | aacaaggtga  | ttatcaaaaa | gctattgaag  | attatcaaaa | agctttggaa  | 960  |
| ttggatccaa  | ataatagatc  | tagatctgct | gggtgtgtgt  | gttctgtgtg | tggtgtgtct  | 1020 |
| gggtgtgtgt  | gtgttctctg  | tcaagataga | tctgaagcta  | ctttgattaa | aagattttaa  | 1080 |
| ggtagaagtg  | ttagatataa  | agctaaattg | attggtattg  | atgaagtttc | tgctgctaga  | 1140 |
| gggtataaat  | tgtgtcaaga  | ttctatgatg | aaattgaaag  | gtgtgtgtgc | tggtgtctaga | 1200 |
| tctaaaggtg  | aacataaaca  | aaaaaatttt | ttgactattt  | cttttgggtg | tattaaaaat  | 1260 |
| tttgatga    | aaactgtgtc  | tttgcaacat | catcatgctg  | ttcatgaaat | ttcttatatt  | 1320 |
| gctaaagata  | tactgatca   | tagagctttt | ggttatgttt  | tggttaaaag | aggtaatcat  | 1380 |
| agatttgttg  | ctattaaaac  | tgctcaagct | gctgaaccag  | ttattttgga | tttgagagat  | 1440 |
| ttgtttcaat  | tgattttatga | attgaaacaa | agagaagaat  | tggaaaaaaa | agctgtgtct  | 1500 |
| gtcgttctcg  | ctgctgtgtg  | tggtgaattt | ggttctgctg  | aagctgctgc | taaagaagct  | 1560 |
| gctgctaag   | ctggttctgc  | tggttctgct | gctggttctg  | gtgaatttgg | ttctgtgtct  | 1620 |
| catatgggtt  | ctcaattttg  | ggttacttct | caaaaaactg  | aagcttctga | aagatgtggt  | 1680 |
| ttgcaaggtt  | cttatatttt  | gagagttgaa | gctgaaaaat  | tgactttgtt | gactttgggt  | 1740 |
| gctcaatctc  | aaatttttga  | accattgttg | ttttggccat  | atactttgtt | gagaagatat  | 1800 |
| ggtagagata  | aagttatgtt  | ttcttttgaa | gctggtagaa  | gatgtccatc | tggtccaggt  | 1860 |
| actttttact  | ttcaaaacttc | tcaaggtaat | gatatttttc  | aagctgttga | agctgctatt  | 1920 |
| caacaacaaa  | aagctcaagg  | taaagttggt | caagctcaag  | atattttgag | attggaacat  | 1980 |
| catcatcatc  | atcat       |            |             |            |             | 1995 |

SEQ ID NO: 156                      moltype = DNA   length = 2391  
 FEATURE                              Location/Qualifiers  
 source                                1..2391  
                                       mol\_type = other DNA  
                                       organism = synthetic construct

SEQUENCE: 156

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| atgggtttgt | cttctgtttg | tactttttct  | tttcaaaacta | attatcatatc | tttgttgaat | 60   |
| ccacataata | ataatccaaa | aacttctttg  | ttgtgttata  | gacatccaaa  | aactccaat  | 120  |
| aaatattctt | ataataattt | tccatctaaa  | cattgttcta  | ctaaatcttt  | tcattgtcaa | 180  |
| aataaatggt | ctgaatcttt | gtctattgct  | aaaaattcta  | ttagagctgc  | tactactaat | 240  |
| caactgaac  | caccagaatc | tgataatcat  | tctgttgcta  | ctaaaaattt  | gaattttggt | 300  |
| aaagcttggt | ggaatgtgca | aagaccatat  | actattattg  | cttttacttc  | ttgtgcttgt | 360  |
| ggtttgtttg | gtaagaattt | gttgcataat  | actaatttga  | tttcttggtc  | tttgatgttt | 420  |
| aaagcttttt | tttttttggt | tgctattttg  | tgtattgctt  | cttttactac  | tactattaat | 480  |
| caaatttatg | atttgcatat | tgatagaatt  | aataaaccag  | atttgccatt  | ggcttctggt | 540  |
| gaaatttctg | ttataactgc | ttggattatg  | tctattattg  | ttgctttggt  | tggtttgatt | 600  |
| attactatta | aaatgaaagg | tggtccattg  | tattattttg  | gttattgttt  | tggtattttt | 660  |
| gggtgtattg | tttattctgt | tccaccattt  | agatggaaac  | aaaatccatc  | tactgctttt | 720  |
| ttgttgaatt | ttttggctca | tattattact  | aattttactt  | tttattatgc  | ttctagagct | 780  |
| gctttgggtt | tgccatttga | attgagacca  | tcttttactt  | ttttgttggt  | ttttatgaaa | 840  |
| tctatgggtt | ctgctttggc | tttgattaaa  | gatgcttctg  | atgttgaagg  | tgatactaaa | 900  |
| tttggtattt | ctactttggc | ttctaaatat  | ggttctagaa  | atttgacttt  | gttttgttct | 960  |
| ggatttgttt | tgttgtctta | tgttgctgct  | attttggctg  | gtattatttg  | gccacaagct | 1020 |
| tttaattcta | atgttatgtt | gttgctctcat | gctatttttg  | ctttttgggt  | gatttttgca | 1080 |
| actagagatt | ttgctttgac | taattatgat  | ccagaagctg  | gtagaagatt  | ttatgaattt | 1140 |
| atgtggaaat | tgtattatgc | tgaatatttg  | gtttatgttt  | ttattaaatt  | gtctggtggt | 1200 |
| gggtgttctg | gtggtgtgtg | ttctgtgtgt  | gggtgttctg  | ctgaagcttg  | gtataatttg | 1260 |
| ggtaatgctt | attataaaca | aggtgattat  | caaaaagcta  | ttgaatatta  | tcaaaaagct | 1320 |
| ttggaatttg | atccaaataa | tgctgaagct  | tggtataatt  | tggttaattg  | ttattataaa | 1380 |
| caaggtgatt | atcaaaaagc | tattgaatat  | tatcaaaaag  | ctttggaatt  | ggatccaaat | 1440 |
| aatgctgaag | cttgggtata | tttgggtaat  | gcttattata  | aacaaggtga  | ttatcaaaaa | 1500 |
| gctattgaag | attatcaaaa | agctttggaa  | ttggatccaa  | ataatttgca  | agctgaagct | 1560 |
| tggaataatt | tggttaattg | ttattataaa  | caaggtgatt  | atcaaaaagc  | tattgaatat | 1620 |

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|             |             |             |             |             |            |      |
|-------------|-------------|-------------|-------------|-------------|------------|------|
| tatcaaaaag  | ctttggaatt  | ggatccaaat  | aatgcttctg  | cttgggtataa | tttgggtaat | 1680 |
| gcttattata  | aacaaggtga  | ttatcaaaaa  | gctattgaa   | attatcaaaa  | agctttggaa | 1740 |
| ttggatccaa  | ataatgctaa  | agcttgggtat | agaagaggt   | atgcttatta  | taacaaggt  | 1800 |
| gattatcaaa  | aagctattga  | agattatcaa  | aaagctttgg  | aattggatcc  | aaataataga | 1860 |
| tctagatctg  | ctgggtgggtg | tggttctgg   | ggtgggtggt  | ctgggtgggtg | tggtgcttct | 1920 |
| gctgaatatg  | ttagagcttt  | gtttgatttt  | aatggtaatg  | atgaagaaga  | tttgccattt | 1980 |
| aaaaaagggtg | atattttgag  | aattagagat  | aaaccagaag  | aacaatgggtg | gaatgctgaa | 2040 |
| gattctgaag  | gtaaaagagg  | tatgattcca  | gttccatatg  | ttgaaaaata  | tggttctgct | 2100 |
| ggttctgctg  | ctggttctgg  | tgaatttgg   | tctgctgaag  | ctgctgctaa  | agaagctgct | 2160 |
| gctaaagctg  | gttctgctgg  | ttctgctgct  | ggttctgggtg | aatttgggtc  | tttgattaaa | 2220 |
| catatgagag  | ctgaagcttt  | gtttgatttt  | actggtaatt  | ctaaattgga  | attgaatttt | 2280 |
| aaagctgggtg | atgttatttt  | tttggtgtct  | agaattaata  | aagattgggt  | ggaaggtact | 2340 |
| gttagagggtg | ctactgggtat | ttttccattg  | tcttttgta   | aaattttgaa  | a          | 2391 |

SEQ ID NO: 157      moltype = DNA    length = 8226  
 FEATURE            Location/Qualifiers  
 source              1..8226  
                     mol\_type = other DNA  
                     organism = synthetic construct

SEQUENCE: 157

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| atgagcgaag  | aaagcttatt  | cgagcttctt  | ccacagaaga  | tggagtagca  | aattacaaac  | 60   |
| tactcagaaa  | gacatacaga  | acttccaggt  | catttccattg | gcctcaatac  | agtagataaa  | 120  |
| ctagaggagt  | cccggtaag   | ggactttgtt  | aagagtcacg  | gtgggtcacac | ggctcatatcc | 180  |
| aagatcctga  | tagcaataaa  | tggtattgct  | gccgtgaaag  | aaatttagatc | cgtcagaaaaa | 240  |
| tgggcatacg  | agacgttctg  | cgatgacaga  | accgtccaat  | tcgtcgccat  | ggccacccca  | 300  |
| gaagatctgg  | aggccaacgc  | agaatatatc  | cgtatggcgc  | atcaatacat  | tgaagtgcga  | 360  |
| gggtggtacta | ataataacaa  | ctacgctaac  | gtagacttga  | tcgtagacat  | cgcggaaga   | 420  |
| gcagacgtag  | agcccgtag   | ggctgggtgg  | ggtcacgcct  | cagagaatcc  | actattgcct  | 480  |
| gaaaaattgt  | cccagcttaa  | gaggaaagtc  | atctttattg  | ggcctccagg  | taacgcccag  | 540  |
| aggctcttag  | gtgataaaat  | ctcctctacc  | attgtcgcct  | aaagtgtcaa  | agtcoccatgt | 600  |
| attccatgg   | ctggtaccgg  | tgttgacacc  | gttcacgtgg  | acgagaaaac  | cggctcggct  | 660  |
| tctgtcgacg  | atgacatcta  | tcaaaagggt  | tgttgtacct  | ctcctgaaga  | tggtttacaa  | 720  |
| aaggccaacg  | gtatttggtt  | tctgtctatg  | attaaggcat  | cgaaggtgg   | tggttggtaaa | 780  |
| ggatctagac  | aagttgaacg  | tgaagaagat  | tctatcgctt  | tataccacca  | ggcagccaac  | 840  |
| gaaattccag  | gctcccccat  | tttcatcatg  | aagttggcgc  | gtagagcgcg  | tcacttggaa  | 900  |
| gttcaactgc  | tagcagatca  | gtacggtaca  | aatatttcc   | tggtcggtag  | agactgttcc  | 960  |
| gttcagagag  | gtcatcaaaa  | aattatcgaa  | gaagcaccag  | ttacaattgc  | caaggctgaa  | 1020 |
| acatttcacg  | agatggaaaa  | ggctgcgcgc  | agactgggga  | aactagtcgg  | ttatgtctct  | 1080 |
| gcccgtaccg  | tggagtagct  | atattctcat  | gatgatggaa  | aattctactt  | tttagaattg  | 1140 |
| aaaccaagat  | tacaagtcga  | gcatacaaca  | acggaaatgg  | tctccggtgt  | taacttacct  | 1200 |
| gcagctcaat  | tacaaatcgc  | tatgggtatc  | cctatgcata  | gaataagtga  | cattagaact  | 1260 |
| ttataggtta  | tgaatcctca  | ttctgcctca  | gaaatcgatt  | tcgaattcaa  | aaactcaagat | 1320 |
| gccaccaaga  | acaaagaag   | acctattcca  | aagggtcatt  | gtaccgcctt  | tcgtatcaca  | 1380 |
| tcagaagatc  | aaaacgtagg  | attcaagcca  | tcgggtggta  | ctttgcatga  | actaaacttc  | 1440 |
| cgttcttctc  | ctaattgttg  | gggttaacttc | tccgtgggta  | acaatggtaa  | tattcactcc  | 1500 |
| ttttcggact  | ctcagttcgg  | ccatattttt  | gcttttggtg  | aaaatagaca  | agcttccagg  | 1560 |
| aaacacatgg  | ttgttgccct  | gaagggaattg | tccattaggg  | gtgatttcag  | aaactactgtg | 1620 |
| gaatacttga  | tcaaaacttt  | ggaaactgaa  | gatttcgagg  | ataacactat  | taccaccgg   | 1680 |
| tggttgacg   | atttgattac  | tcataaaatg  | accgctgaaa  | agcctgatcc  | aactcttgcc  | 1740 |
| gtcattttgc  | gtgcgcgtac  | aaaggtcttc  | ttagcatctg  | aagaagcccg  | ccacaagtat  | 1800 |
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| aaagctgggt  | ctgctgggttc  | tgtgctgggt  | tctggtgaat  | ttggttcttc   | ttattatcat  | 6240 |
| catcatcatc  | atcatttggga  | atctacttct  | ttgtataaaa  | aagctgggttc  | tggttctcaa  | 6300 |
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| gggttctggtg | aaatttgcgc   | tgctactcca  | atttctaact  | ttcatgatga   | ttctgatgaa  | 8160 |
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| gatattgcta  | atttggaaaa   | agaaattgct  | tatgctgaaa  | ctgggttctgc  | tggttctgct  | 8520 |
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| gaatctactt  | ctttgtataa   | aaaagctggt  | tctgaatttt  | ttagaagaga   | aagaaataaa  | 8700 |
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| actgatcaat  | tgggaagatga  | aaaatctgct  | ttgcaaaactg | aaatttctaa   | tttgggtgaa  | 8820 |
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| actgggttctg | ctgggttctgc | tgtgtgttct  | ggtgaatttg   | ctgaagctgc  | tgctaaagaa | 9180  |
| gctgctgcta  | aagctggttc  | tgtgtgttct  | gctgctggtt   | ctggtgaatt  | tgaagagaa  | 9240  |
| tctaatagaag | aaccaccacc  | accatatgaa  | gatccatatt   | gggtgaatgg  | tggttctgct | 9300  |
| ggttctgctg  | ctgggttctgg | tgaatttgg   | tctgctgaag   | ctgctgctaa  | agaagctgct | 9360  |
| gctaagaag   | ctgctgctaa  | agaagctgct  | gctaagaagctg | gttctgctgg  | ttctgctgct | 9420  |
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| gctgttaaat  | tgaatagaaa  | tgaacaattg  | aaaaataaag   | ttgaagaatt  | gaaaaataga | 9600  |
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| tcttattatc  | atcatcatca  | tcatcatttg  | gaatctactt   | ctttgtataa  | aaaagctggt | 9840  |
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| gcttatttgg  | aaaaagaaat  | tgctagattg  | agaaaagaaa   | aagctgcttt  | gagaaataga | 9960  |
| ttggctcata  | aaaaatctta  | tatcatcatc  | catcatcatc   | atttggaaatc | tacttctttg | 10020 |
| tataaaaaag  | ctgggttctgg | ttctgctaga  | aatgcttatt   | tgagaaaaaa  | aatgtctaga | 10080 |
| ttgaaaaaag  | ataatttgca  | attggaaga   | gatgaacaaa   | atttggaaaa  | aattatttgc | 10140 |
| aatttgagag  | atgaaattgc  | tagattggaa  | aatgaagtgg   | cttctcatga  | acaaggttct | 10200 |
| ctggttctg   | ctgctggttc  | tggtgaattt  | gctgaagctg   | ctgctaaaga  | agctgctgct | 10260 |
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| catcatcatt  | tggaatctac  | ttcttctgat  | aaaaaagctg   | gttctggttc  | taatttgggt | 10380 |
| gctcaattgg  | aaaatgaagt  | tgcttctttg  | gaaaatgaaa   | atgaaacttt  | gaaaaaaaaa | 10440 |
| aatgtgcata  | aaaaagattt  | gattgcttat  | ttggaaaaaag  | aaattgctaa  | tttgagaaaa | 10500 |
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SEQUENCE: 159

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| ggttctggtg  | aatttgggttc | tgtgtgttct  | gctgctggtt | ctggtgaatt | ttcttattat  | 120  |
| catcatcatc  | atcatcattt  | ggaatctact  | tctttgtata | aaaaagctgg | ttctggttct  | 180  |
| gctagaaatg  | cttatttgag  | aaaaaaaatt  | gctagattga | aaaaagataa | tttgcaattg  | 240  |
| gaaagagatg  | aacaaaattt  | ggaaaaaatt  | attgctaatt | tgagagatga | aattgtctaga | 300  |
| ttggaaaaatg | aagttgcttc  | tcatgaacaa  | ggttctgctg | gttctgctgc | tggttctggt  | 360  |
| gaatttctg   | aagctgctgc  | taaagaagct  | gctgctaaag | ctggttctgc | tggttctgct  | 420  |
| ctggttctg   | gtgaatttct  | ttattatcat  | catcatcatc | atcatttgg  | atcatttct   | 480  |
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| gattttaaaa  | ttcaagggtg  | ttctgctggt  | tctgctgctg | gttctggtga | atttctgtaa  | 900  |
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| gaatttgatt  | tggttctgct  | tgaatttgg   | gctcaagaat | tttctgctgc | tggtgatgct  | 1020 |
| gttgatggtt  | tgtctggttc  | tgtgctggtt  | ttggttctgc | tgaagctgct |             | 1080 |
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| gctggttctg  | ctgctggttc  | tggtgaattt  | ggttctgatt | ataaagatga | tgatgataaa  | 1200 |
| gattataaag  | atgatgatga  | taagatttat  | aaagatgatg | atgataaagg | ttctgctggt  | 1260 |
| tctgctgctg  | gttctggtga  | atttgggttct | gctggttctg | ctgctggttc | tggtgaattt  | 1320 |
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SEQUENCE: 160

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| QLIKRRKGLG  | LVGVNLTLDG  | VKSWLKPRLG  | QEATVGKATG | FLKNFLIEPF | VPHSQAEIFY | 120 |
| VCIYATREGD  | YVLFPHHEGGV | DVGVDVDAKAQ | KLLVGVDEKL | NPEDIKKHLL | VHAPEDKKEI | 180 |
| LASFISGLFNF | FYEDLYPTYL  | EINPLVVTKD  | GVYVLDLAAK | VDATADYICK | VKWDIEFFPP | 240 |
| PFGREAYPEE  | AYIADLDAKS  | GASLKLTLN   | PKGRIWTMVA | GGGASVVYS  | TICDLGGVNE | 300 |
| LANYGEYSGA  | PSEQQTYDYA  | KTILSLMTRE  | KHPDGKILII | GGSIANFTNV | AATFKGIVRA | 360 |
| IRDYQGPLKE  | HEVTFIVRRG  | GPNYQEGLRV  | MGEVGKTTGI | PIHVFGETH  | MTAIVGMALG | 420 |
| HRPIPNQPT   | AAHTANFLN   | ASGSTSTPAP  | SRTASFSESR | ADEVAPAKKA | KPAMPQDSVP | 480 |
| SPRSLQGKST  | TLFSRHFKAI  | VWGMQTRAVQ  | GMLDFDYVCS | RDEPSVAAMV | YPFTGDHKQK | 540 |
| FYWGHEKILI  | PVKFNMDAM   | RKHPEVDVLI  | NFASLSAYD  | STMETMNYAQ | IRTIAIIAEG | 600 |
| IPEALTRKLI  | KKADQKGVTI  | IGPATVGGIK  | PGCFKIGNTG | GMLDNILASK | LYRPGSVAYV | 660 |
| SRSGMSNEL   | NNIISRTTDG  | VYEGVAIGGD  | RYPGSTFMDH | VLRYQDTPGV | KMIVVLGEIG | 720 |



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| EAGVFPVRSF | DELGEIIQSV | YEDLVANGVI  | VPAQEVPPPT | VPMDYSHARE | LGLIRKPASF | 840  |
| MTSICDERGQ | ELIYAGMPIT | EVFKKEEMGIG | GVLGLLWFQK | RLPKYSCQFI | EMCLMVTADH | 900  |
| GPAVSGAHNT | IICARAGKDL | VSSLTSGLLT  | IGDRFGGALD | AAAKMFSAF  | DSGIIPMEFV | 960  |
| NMKKKEGLI  | MGIGHRVKSI | NNPDMRVQIL  | KDYVRQHPPA | TPLLDYALEV | EKITTSKKPN | 1020 |
| LILNVDGLIG | VAFVDMRLNC | GSFTREEADE  | YIDIGALNGI | FVLGRSMGFI | GHYLDQKRLK | 1080 |
| QGLYRHPWDD | ISYVLPEHMS | MKLSGGGGSG  | GGSGGGGSA  | EAWYNLGNAY | YKQGDYQKAI | 1140 |
| EYYQKALELD | PNNAEAWYNL | GNAYYKQGDY  | QKAIEYYQKA | LELDPNNAEA | WYNLGNAYYK | 1200 |
| QGDYQKAIED | YQKALELDPN | NLQAEAWKNL  | GNAYYKQGDY | QKAIEYYQKA | LELDPNNASA | 1260 |
| WYNLGNAYYK | QGDYQKAIEY | YQKALELDPN  | NAKAWYRRGN | AYYKQGDYQK | AIEDYQKALE | 1320 |
| LDPNNRSRSA | GGGGSGGGGS | GGGGASSYYH  | HHHHHLESTS | LYKKAGSGSN | LVAQLENEVA | 1380 |
| SLENENETLK | KKNLHKKDLI | AYLEKETIANL | RKKIEEGSAG | SAAGSGEFGS | AEAAAKEAAA | 1440 |
| KAGSAGSAAG | SGEFGSSYYH | HHHHHLESTS  | LYKKAGSGSA | RNAYLRKKIA | RLKKNLQLE  | 1500 |
| RDEQNLEKII | ANLRDEIARL | ENEVASHEQG  | SG         |            |            | 1532 |

SEQ ID NO: 161                      moltype = AA   length = 824  
 FEATURE                              Location/Qualifiers  
 source                                1..824  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 161

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MKNCVIVSAV | RTAIGSFNGS | LASTSAIDL  | ATVIKAAIER | AKIDSQHVE  | VIMGNVLQAG | 60  |
| LGQNPARQAL | LKSGLAETVC | GFTVNKVC   | GLKSVALAAQ | AIQAGQAQSI | VAGGMENMSL | 120 |
| APYLLDAKAR | SGYRLGDQV  | YDVILRDGLM | CATHGYHMG  | TAENVAKYEG | ITREMQDELA | 180 |
| LHSQRKAAAA | IESGAFTAEI | VPVNVVTRKK | TFVFSQDEFP | KANSTAEALG | ALRPAFDKAG | 240 |
| TVTAGNASGI | NDGAAALVIM | EESAALAAGL | TPLARIKSYA | SGGVPPALMG | MGPVPATQKA | 300 |
| LQLAGLQLAD | IDLIEANFAF | AAQFLAVGKN | LGFDSEKVIN | NGGAIALGHP | IGASGARILV | 360 |
| TLLHAMQARD | ITLGLATLCI | GGGQGIAMVI | ERLNKLSGGG | GGGGGGSGGG | GSAAWYNL   | 420 |
| NAYYKQGDYQ | KAIEYYQKAL | ELDPNNAEAW | YNLGNAYYKQ | GDYQKAIEYY | QKALELDPNN | 480 |
| AEAWYNLGN  | AYYKQDYQKA | IEDYQKALEL | DPNNLQAEAW | KNLGNAYYKQ | GDYQKAIEYY | 540 |
| QKALELDPNN | AWYNLGNAYY | YKQGDYQKA  | IEYYQKALEL | DPNNAKAWYR | RGNAYYKQGD | 600 |
| YQKAIEDYQK | ALELDPNNRS | RSAGGGSGGG | GGSGGGGASS | YHHHHHHHLE | STSLYKKAGS | 660 |
| GSNEVTTLN  | DAAFIENENA | YLEKEIARLR | KEKAALNRRL | AHKKGSAGSA | AGSGEFGSAE | 720 |
| AAAKEAAAAA | GSAGSAAGSG | EFSSSYHHH  | HHHLESTSLY | KKAGSGSQKV | AELKNRVAVK | 780 |
| LNRNEQLKNK | VEELKNRNAY | LKNELATLEN | EVARLENDVA | EGSG       |            | 824 |

SEQ ID NO: 162                      moltype = AA   length = 779  
 FEATURE                              Location/Qualifiers  
 source                                1..779  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 162

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MKKVCVIGAG | TMGSGIAQAF | AAKGFEVVL  | DIKDEFVDRG | LDFINKNLSK | LVKKGKIEEA | 60  |
| TKVEILTRIS | GTVDLNMAD  | CDLVIEAAVE | RMDIKKQIFA | DLDNICKPET | ILASNTSSLS | 120 |
| ITEVASATKR | PDKVIHMHF  | NPAPVMKLVE | VIRGIATSQE | TFDAVKETS  | AIGKDPVEVA | 180 |
| EAPGFVVNRI | LIPMINEAVG | ILAEIASVE  | DIDKAMKLGA | NHPMGPLELG | DFGLDLCILA | 240 |
| IMDVLYSETG | DSKYRPHILL | KKYVRAGWLG | RKSGKGFYDY | SKKLSGGGGS | GGGGSGGGGS | 300 |
| AEAWYNLGN  | AYYKQDYQKA | IEYYQKALEL | DPNNAEAWYN | LGNAYYKQGD | YQKAIEYYQK | 360 |
| ALELDPNNAE | AWYNLGNAYY | KQGDYQKAIE | DYQKALELDP | NNLQAEAWKN | LGNAYYKQGD | 420 |
| YQKAIEYYQK | ALELDPNNAS | AWYNLGNAYY | KQGDYQKAIE | YYQKALELDP | NNAKAWYRRG | 480 |
| NAYYKQGDYQ | KAIEDYQKAL | ELDPNNRSRS | AGGGSGGGGG | SGGGGASENL | YPQGENLYFQ | 540 |
| GDSSESCWNC | GRKASETCGS | CNTARYCGSF | CQHKDWEKHH | HICGQTLQAQ | QGSAGSAAGS | 600 |
| GEFGSAEAAA | KEAAAKAGSA | GSAAGSGEFG | SMAVSESQK  | KMVSKYKYRD | LTVRETNNVI | 660 |
| TLYKDLKPV  | DSYVFNDGSS | RELMLTGTI  | PVPYRGNTYN | IPICLWLLDT | YPYNPPICFV | 720 |
| KPTSSMTIKT | GKHVDANGKI | YLPYLHEWKH | PQSDLLGLLI | VMIVVFGDEP | PVFSRPGSG  | 779 |

SEQ ID NO: 163                      moltype = AA   length = 744  
 FEATURE                              Location/Qualifiers  
 source                                1..744  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 163

|            |             |            |            |            |             |     |
|------------|-------------|------------|------------|------------|-------------|-----|
| MELNNVILEK | EGKVAVVTIN  | RPKALNALNS | DTLKEMDYVI | GEIENDSEVL | AVILTGAGEK  | 60  |
| SFVAGADISE | MKEMNTIEGR  | KFGILGNKVF | RRLELLEKPV | IAAVNGFALG | GGCEIAMSCD  | 120 |
| IRIASSNARF | GQPEVGLGIT  | PGFGGTQRLS | RLVGMGMAKQ | LIFTAQNIKA | DEALRIGLVN  | 180 |
| KVVEPSELMN | TAKETIANKIV | SNAPVAVKLS | KQAINRGMQC | DIDTALAFES | EAFGEFCFSTE | 240 |
| DQKDAMTAFI | EKRKIEGFKN  | RKLSSGGGSG | GGSGGGGSA  | EAWYNLGNAY | YKQGDYQKAI  | 300 |
| EYYQKALELD | PNNAEAWYNL  | GNAYYKQGDY | QKAIEYYQKA | LELDPNNAEA | WYNLGNAYYK  | 360 |
| QGDYQKAIED | YQKALELDPN  | NLQAEAWKNL | GNAYYKQGDY | QKAIEYYQKA | LELDPNNASA  | 420 |
| WYNLGNAYYK | QGDYQKAIEY  | YQKALELDPN | NAKAWYRRGN | AYYKQGDYQK | AIEDYQKALE  | 480 |
| LDPNNRSRSA | GGGGSGGGGS  | GGGGASGPLG | SPLTASMLAS | APPQEQKQML | GERLFPPLIQA | 540 |
| MHPTLAGKIT | GMLLEIDNSE  | LLHMLSPES  | LRSKVDEAVA | VLQAHQAKEA | AQKAGSAGSA  | 600 |
| AGSGEFGSAE | AAAKEAAAAA  | GSAGSAAGSG | EPGNTNMSV  | PTDGAVTTSQ | IPASEQETLV  | 660 |
| RPKPLLLKLL | KSVGAQKDTY  | TMKEVLFYLG | QYIMTKRLYD | EKQQHIVYCS | NDLLGDLFGV  | 720 |
| PSFSVKEHRK | IYTMIRYNLV  | VGSG       |            |            |             | 744 |

SEQ ID NO: 164                      moltype = AA   length = 823  
 FEATURE                              Location/Qualifiers  
 source                                1..823

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mol_type = protein
organism = synthetic construct

SEQUENCE: 164
MIVKPMVRNM ICLNAHPQGC KKGVEDQIEY TKKRITAEVK AGAKAPKNVL VLGCNSGYGL 60
ASRITAAPGY GAATIGVSFE KAGSETKYGT PGWYNNLAFD EAAKREGLYS VTIDGDAFSD 120
EIKAQVIEEA KKKGIKFDLI VYSLASPVRT DPDTGIMHKS VLKPFGKTFT GKTVDPFTGE 180
LKEISAEPAN DEEAATYKV MGGEDWERWI KQLSKEGLE EGCITLAYS IGPEATQALY 240
RKGITIGKAKE HLEATAHRLN KENPSIRAFV SVNKGLVTRA SAVIPVIPLY LASLFKVMKE 300
KGNHEGCIQ ITRLYAERLY RKDGTIPVDE ENRIRIDDWE LEEDVQKAVS ALMEKVTGEN 360
AESLTDLAGY RHDPLASNGF DVEGINYEA VERFDRIKLS GGGSGGGGS GGGSGAEAWY 420
NLGNAYYKQG DYQKAEIYYQ KALELDPNNA EAWYNLGNAY YKQGDYQKAI EYYQKALELD 480
PNNAEAWYNL GNAYYKQGDY QKAIEDYQKA LELDPNNLQA EAWKNLGNAY YKQGDYQKAI 540
EYYQKALELD PNNASAWYNL GNAYYKQGDY QKAEIYYQKA LELDPNNAKA WYRRGNAYYK 600
QGDYQKAIED YQKALELDPN NRSRSAGGGG SGGSGGGGGG ASSYYHHHHH HLESTSLYKK 660
AGSGSNLLAT LRSTAAVLEN ENHVLKEKE KLKKEKEQLL NKLEAYKGS GSAAGSGEFG 720
SAEAAAKEAA AKAGSAGSAA GSGEFGSSYY HHHHHHLEST SLYKKAGSGS KRIAYLRKKI 780
AALKKDNANL EKDIANLENE IERLIKEIKT LENEVASHEQ GSG 823

SEQ ID NO: 165      moltype = AA length = 829
FEATURE            Location/Qualifiers
source              1..829
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 165
MTREVVVVSG VRTAIGTFGG SLKDVAPAEI GALVVREALA RAQVSGDDVG HVVFGNVIQT 60
EPRDMYLGRV AAVNGGVITN APALTVNRLC GSGLQAIVSA AQTILLGDTD VAIGGGAESM 120
SRAPYLAPAA RWGARMGDAG LVDMMLGALH DPFHRIHMGV TAENVAKEYD ISRAQQDEAA 180
LESHRRASAA IKAGYFKDQI VPVVSKGRKG DVTFTDEHV RHDATIDMT KLRPVFVKEN 240
GTVTAGNASG LNDAAAVVUM MERAEEARRG LKPLARLVSY GHAGVDPKAM GIGPVPATKI 300
ALERAGLQVS DLDVIEANEA FAQAQCAVTK ALGLDPAKVN PNGSGISLGH PIGATGALIT 360
VKALHELNRV QGRYALVTMC IGGQGIIAAI FERIKLSGGG GSGGGGGGGG GSAAEWYNLG 420
NAYYKQGDYQ KAIEYYQKAL ELDPNNAEAW YNLGNAYYKQ GDYQKAEIYY QKALELDPNN 480
AEAWYNLGN AYYKQGDYQKA IEDYQKALEL DPNNLQAEAW KNLGNAYYKQ GDYQKAEIYY 540
QKALELDPNN ASAWYNLGN AYYKQGDYQKA IEYYQKALEL DPNNAKAWYR RGNAYYKQGD 600
YQKAIEDYQK ALELDPNNRS RSAGGGGSGG GSGGGGGASD VMWEYKWEKT GDAELYGPFT 660
SAQMOTWVSE GYFPDGVYCR KLDPPGGQFY NSKRIDFDLY TGSAGSAGS GEFGSAAAA 720
KEAAAKAGSA GSAAGSGEFG SEDSVFEFNN AISYVNKIKT RFLDHPEIYR SFLEILHTYQ 780
KEQLHTKGRP PRGMSSEEVF TEVANLFRGQ EDLLSEFGQF LPEAKRGSG 829

SEQ ID NO: 166      moltype = AA length = 852
FEATURE            Location/Qualifiers
source              1..852
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 166
MKLSTKLCWC GIKGLRLPQK QQQLHNTNLQ MTELKKQKTA EQKTRPQNVG IKGIQIYIPT 60
QCVNQSELEK PDGVSQGYT IGLGQTNMSF VNDREDIYSM SLTVLSKLIK SYNIDTNKIG 120
RLEVGTETLI DKSXSVKSVL MQLFGENTDV EGIDTLNACY GGTNALFNSL NWIESNAWDG 180
RDAIVVCGDI AIYDKGAARP TGGAGTVAMW IGPDAPIVFD SVRASMEHA YDFYKPDFTS 240
EYPYVDGHFS LTCYVKALDQ VYKSYSKKAI SKGLVSDPAG SDALNVLYKF DYNVVFHVPTC 300
KLVTXSYGRL LNYDFRANPQ LPFEVDAELA TRDYDES LTD KNIKTFVNV AKPFHKERVA 360
QSLIVPTNTG NMYTASVYAA FASLLNVVGS DDLQGKRVGL FSYGSLAAS LYSCKIVGDV 420
QHIKELDIT NKLAKRITET PKDYEAALIEL RENAHLLKNF KPQGSIEHLQ SGVYYLTNID 480
DKFRRSYDVY KKLSGGGGSG GGGSGGGGSA EAWYNLGNAY YKQGDYQKAI EYYQKALELD 540
PNNAEAWYNL GNAYYKQGDY QKAEIYYQKA LELDPNNAEA WYNLGNAYYK QGDYQKAIED 600
YQKALELDPN NLQAEAWKNL GNAYYKQGDY QKAEIYYQKA LELDPNNAEA WYNLGNAYYK 660
QGDYQKAEIY YQKALELDPN NAKAWYRRGN AYYKQGDYQK AIEDYQKALE LDPNNRSRSA 720
GGGSGGGGSG GGGGASLGPL PPGWEVRSV SGRIYFVDHN NRTTQFTDPR LHGSAGSAG 780
SGEFGSAEAA AKEAAAKAGS AGSAGSGSEF GSGAMGPLPP GWKRTDSNG RYVFVNHNTR 840
ITQWEDPRSG SG 852

SEQ ID NO: 167      moltype = AA length = 829
FEATURE            Location/Qualifiers
source              1..829
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 167
MVAVRRKALS ILAEAPVLAS DRLPYKNYDY DRVFGACCEN VIGYMLPVG VIGPLVIDGT 60
SYHIPMATTE GCLVASAMRG CKAINAGGGA TTVLTGDMGT RGPVVRFPPTL KRSGACKIWL 120
DSEEGQNAIK KAFNSTSRFA RLQHIQTCLA GDLLFMRPRT TTGDAMGMNM ISKGVEYSLK 180
QMVEEYGWED MEVSVSGNY CTDKKPAAIN WIEGRKSVV AEATIPGDV RVKVLKSDVSA 240
LVELNIAKNL VGSAMAGSVG GFNAHAANLV TAVFLALGQD PAQNVESNCS ITLMKEVDGD 300
LRISVMPSTI EVGTIGGTIV LEPQGAMLDL LGVRGPHATA PGTNARQLAR IVACAVLAGE 360
LGLCAALAG HLVSHTMTHN RKLSGGGGSG GGGSGGGGSA EAWYNLGNAY YKQGDYQKAI 420
EYYQKALELD PNNAEAWYNL GNAYYKQGDY QKAEIYYQKA LELDPNNAEA WYNLGNAYYK 480
QGDYQKAIED YQKALELDPN NLQAEAWKNL GNAYYKQGDY QKAEIYYQKA LELDPNNAEA 540
WYNLGNAYYK QGDYQKAEIY YQKALELDPN NAKAWYRRGN AYYKQGDYQK AIEDYQKALE 600
LDPNNRSRSA GGGSGGGGSG GGGGASSYYH HHHHHLEST LYKKAGSEFF RRERNKMAA 660
KCRNRRRELDT LQAETDQL EDEKSALQTE IANLLKEKEK LEFILAHRP ACKIPDDLGF 720

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|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| PEEMSLEGSA | GSAAGSGEFG | SAEAAAKEAA | AKAGSAGSAA | GSGEFGSSYY | HHHHHHLEST | 780 |
| SLYKKAGSGS | QKVESLQKI  | EELKQRKAQL | KNDIANLEKE | IAYAETGSG  |            | 829 |

SEQ ID NO: 168                      moltype = AA   length = 1049  
 FEATURE                              Location/Qualifiers  
 REGION                                1..1049  
                                       note = chimeric enzyme  
 source                                1..1049  
                                       mol\_type = protein  
                                       organism = synthetic construct

SEQUENCE: 168

|             |             |            |            |            |             |      |
|-------------|-------------|------------|------------|------------|-------------|------|
| MSLPPLTSAP  | GKVIIFGEHS  | AVYNKPAVAA | SVSALRTYLL | ISESSAPDTI | ELDFPDISFN  | 60   |
| HKWSINDFNA  | ITEDQVNSQK  | LAKAQQATDG | LSQELVSLLD | PLLAQLSESF | HYHAAPCFLY  | 120  |
| MFVCLCPHAK  | NIKFSCLKSTL | PIGAGLGSSA | SISVSLALAM | AYLGGLIGSN | DLEKLESENDK | 180  |
| HIVNQWAFIG  | EKCIHGTPSG  | IDNAVATYGN | ALLFEKDSHN | GTINTNNFKF | LDDFPAIPMI  | 240  |
| LYTTRIPRST  | KDLVARVRVL  | VTEKFPEVMK | PILDAMGECA | LQGLEIMTKL | SKCKGTDDEA  | 300  |
| VETNNELYEQ  | LLELIRINHG  | LLVSGVSHP  | GLELIKNLSD | DLRIGSTKLT | GAGGGGCSLT  | 360  |
| LLRRDITQEQ  | IDSFKKKLQD  | DFSJETFTD  | LGGTGCCLLS | AKNLNKDLKI | KSLVPQLFEN  | 420  |
| KTTTKQQIDD  | LLLPGNNTLP  | WTSKLSGGGG | SGGGSGGGG  | SAEAWYNLGN | AYYKQGDYQK  | 480  |
| AIEYYQKALE  | LDPNNAEAWY  | NLGNAYYKQG | DYQKAIEYYQ | KALELDPNNA | EAWYNLGNAY  | 540  |
| YKQGDYQKAI  | EDYQKALELD  | PNNLQAEAWK | NLGNAYYKQG | DYQKAIEYYQ | KALELDPNNA  | 600  |
| SAWYNLGNAY  | YKQGDYQKAI  | EYYQKALELD | PNNAKAWYRR | GNAYYKQGDY | QKAIEDYQKA  | 660  |
| LELDPNNRSR  | SAGGGSGGGG  | GSGGGGASME | PAMEPETLEA | RINRATNPLN | KELDWASING  | 720  |
| FCEQLNEDFE  | GPPLATRLLA  | HKIQSPQEW  | AIQALTVLET | CMKSCGKRPH | DEVGKFRFLN  | 780  |
| ELIKVVPSPKY | LGSRTSEKVK  | NKILELLYSW | TVGLPEEVKI | AEAYQMLKKQ | GIVKSGSAGS  | 840  |
| AAGSGEFGSA  | EAAAKEAAAK  | AGSAGSAAGS | GEFGSGAMGS | MAEAEGESLE | SWLNKATNPS  | 900  |
| NRQEDWEYII  | GFCDQINKEL  | EGPQIAVRLL | AHKIQSPQEW | EALQALTVLE | ACMKNCGRRF  | 960  |
| HNEVGKFRFL  | NELIKVSPK   | YLGDRVSEKV | KTKVIELLYS | WTMALPEEAK | IKDAYHMLKR  | 1020 |
| QGIVQSDPPI  | PVDRTLIPSP  | PPRPKNKSG  |            |            |             | 1049 |

SEQ ID NO: 169                      moltype = AA   length = 871  
 FEATURE                              Location/Qualifiers  
 source                                1..871  
                                       mol\_type = protein  
                                       organism = synthetic construct

SEQUENCE: 169

|            |            |            |            |             |            |     |
|------------|------------|------------|------------|-------------|------------|-----|
| MSELRAFSAP | GKALLAGGYL | VLDTKYEAFF | VGLSARMHAV | AHPYGSLSQGS | DKFEVRVKSK | 60  |
| QPKDGEWLYH | ISPKSGFIPV | SIGGSKNPFI | EKVIANVFSY | FKPNMDDYCN  | RNLFVIDIFS | 120 |
| DDAYHSQEDS | VTEHRGNRRL | SPHSHRIEEV | PKTGLGSSAG | GLVTVLTTAL  | ASFFVSDLEN | 180 |
| NVDKYREVIH | NLAQVAHQCA | QKIGSGFDV  | AAAAYGSIKY | RRFPALISN   | LPDIGSATYQ | 240 |
| SKLAHLVDEE | DWNITIKSNH | LPSGLTLWMG | DIKNGSETVK | LVQKVKNWYD  | SHMPESLKIY | 300 |
| TELDANSRF  | MDGLSKLDR  | HETHDDYSDQ | IFESLERND  | TCQKYPEITE  | VRDAVATIRR | 360 |
| SPRKITKESG | ADIEPPVQTS | LLDDCQTLKG | VLTCLIPGAG | GYDAIAVITK  | QDVLRAQTA  | 420 |
| NDKRFSPKQV | LDVTQADWGV | RKEKDPETYL | DKKLSGGGGS | GGGGSGGGGS  | AEAWYNLGN  | 480 |
| YYKQGDYQKA | IEYYQKALEL | DPNNAEAWYN | LGNAAYYKQG | YQKAIEYYQK  | ALELDPNNAE | 540 |
| AWYNLGNAYY | KQGDYQKAIE | DYQKALELDP | NNLQAEAWKN | LGNAAYYKQG  | YQKAIEYYQK | 600 |
| ALELDPNNAS | AWYNLGNAYY | KQGDYQKAIE | YYQKALELDP | NNAKAWYRRG  | NAYYKQGDYQ | 660 |
| KAIEDYQKAL | ELDPNNRSRS | AGGGSGGGGG | SGGGGASSYY | HHHHHHLEST  | SLYKKAGSGS | 720 |
| QKVEELKNKI | AELENRNAV  | KNRVAHLKQE | IAYLKDELAA | HEFEGSAGSA  | AGSGEFGSAE | 780 |
| AAAKEAAAKA | GSAGSAAGSG | EPGSSYYHHH | HHHLESTSLY | KKAGSGSFEN  | VTHEFILATL | 840 |
| ENENAKLRRL | EAKLERELAR | LRNEVAWLGS | G          |             |            | 871 |

SEQ ID NO: 170                      moltype = AA   length = 1011  
 FEATURE                              Location/Qualifiers  
 source                                1..1011  
                                       mol\_type = protein  
                                       organism = synthetic construct

SEQUENCE: 170

|             |            |            |            |            |             |      |
|-------------|------------|------------|------------|------------|-------------|------|
| MTVYTASVTA  | PVNIATLKYW | GKRDTKLNLP | TNSSISVTL  | QDDLRLTSTA | ATAPEFERDT  | 60   |
| LWLNGEPHSI  | DNERTQNCRL | DLRQLRKEME | SKDASLPTLS | QWKLHIVSEN | NFPTAAGLAS  | 120  |
| SAAGFAALVS  | AIKLYQLPQ  | STSEISRIAR | KSGGSACRSL | FGGYVAVEMG | KAEDGHDSMA  | 180  |
| VQIADSSDWP  | QMKACVLVVS | DIKKDVSSTQ | GMQLTVATSE | LFKERIEHVV | PKRFEVVRKA  | 240  |
| IVEKDFATFA  | KETMMDNSF  | HATCLDSFPP | IFYMNDTSKR | IISWCHTINQ | FYGETIVAYT  | 300  |
| FDAGPNVAVLY | YLAENESKLF | AFIYKLFSGV | PGWDKKFTTE | QLEAFNHQFE | SSNFTARELD  | 360  |
| LELQKDVAVR  | ILTQVGSGPQ | ETNESLIDAK | TGLPKEKLSG | GGGSGGGGSG | GGGSAEAWYN  | 420  |
| LGNAAYYKQG  | YQKAIEYYQK | ALELDPNNAE | AWYNLGNAYY | KQGDYQKAIE | YYQKALELDP  | 480  |
| NNAEAWYNLG  | NAYYKQGDYQ | KAIEDYQKAL | ELDPNNLQAE | AWKNLGNAYY | KQGDYQKAIE  | 540  |
| YYQKALELDP  | NNASAWYNLG | NAYYKQGDYQ | KAIEDYQKAL | ELDPNNAKAW | YRRGNAYYKQ  | 600  |
| GDYQKAIEDY  | QKALELDPNN | RSRSAGGGGS | GGGGSGGGGA | SAMADLEQKV | LEMEASTYDG  | 660  |
| VFIWKISDFP  | RKRQEAIVAG | IPAIFSPAFY | TSRYGYKMCL | RIYLNQDGTG | RGTHLSLFFV  | 720  |
| VMKGPNDALL  | RWPNQKVTL  | MLLDQNNREH | VIDAFRPDVT | SSSFQRPVND | MNIASGCPLF  | 780  |
| CPVSKMEAKN  | SYVRDDAIFI | KAIVDLTGLG | SAGSAAGSGE | FGSAEAAAKE | AAAKAGSAGS  | 840  |
| AAGSGEFGSA  | SIKLQSDGGE | IFEVDVEIAK | QSVTIKTMLE | DLGMDDEGDD | DPVPLPNVNA  | 900  |
| AILKKVIQWC  | THHKDDPPPP | EDDENKEKRT | DDIPVWDQEF | LKVDQGTLFE | LILAAANYLDI | 960  |
| KGLLDVVTCKT | VANMIKGTPT | EEIRKTFNIK | NDFTEEEEAQ | VRKENQWCGS | G           | 1011 |

SEQ ID NO: 171                      moltype = AA   length = 720  
 FEATURE                              Location/Qualifiers  
 source                                1..720

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mol_type = protein
organism = synthetic construct

SEQUENCE: 171
MTADNNSMPH GAVSSYAKLV QNQTPEIDLE EPPEIPLQQ RPNTRSSETS NDESGETCFS 60
GHDEEQIKLM NENCIVLDWD DNAIGAGTKK VCHLMENIEK GLLHRAFSVF IFNEQGELLL 120
QQRATEKITS PDLWNTTCCS HPLCIDDELG LKGKLDDEKIK GAITAAVRKL DHELGIPED 180
TKTRGKFHFL NRIHYMAPSN EPWGEHEIDY ILFYKINAKE NLTVPNPVNE VRDFKWVSPN 240
DLKTMFADPS YKFTPWFKII CENYLFNWWE QLDDLSEVEN DRQIHRMLKL SGGGSGGGG 300
SGGGGSAAEW YNLGNAYYKQ GDYQKAIEYY QKALELDPNN AEAWYNLGN YKQGDYQKA 360
IEYYQKALEL DPNNAAEWYN LGNAYYKQGD YQKAIEDYQK ALELDPNNLQ AEAWNLGN 420
YYKQGDYQKA IEYYQKALEL DPNNASAWYN LGNAYYKQGD YQKAIEYYQK ALELDPNNAK 480
AWYRRGNAYY KQGDYQKAIE DYQKALELDP NNRSRASAGG GSGGGSGGG GASSYHHHH 540
HHLESTSLYK KAGSGSNTVK ELKNYIQELE ERNAELKNL EHLKFAKAE EFELAHHKE 600
GSAGSAAGSG EFGSAEAAK EAAAKAGSAG SAAGSGEFGS SYHHHHHHHL ESTSLYKKAG 660
SGSQVAKLQK NRVAYKLKEN AKLENIVARL ENDNANLEKD IANLEKDIAN LERDVARGSG 720

SEQ ID NO: 172      moltype = AA length = 801
FEATURE            Location/Qualifiers
source              1..801
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 172
MEAKIDELIN NDPVSSQNE SLISKPYNHI LLKPGKNFRL NLIVQINRVM NLPKQQLAIV 60
SQIVELLHNS SLLIDDIEDN APLRRGQTTS HLIWGVPTSI NTANYMYFRA MQLVSQLTTK 120
EPLYHWLITI FNEELINLHR GQGLDIYWRD FLPEIITQEQ MYLNMVMNKT GGLFRLTLRL 180
MEALSPSSHG GHSLVPFINL LGIYYQIRDD YLNLKDFQMS SEKGFAEDIT EGKLSFPIVH 240
ALNFTKTKQG TEQHNELIRI LLLRTSDKDI KLKLIQILEF DTNSLAYTKN FINQLVNMIM 300
NDNENKYLDP LASHSDPATN LHDELLYIID HLSELKLSGG GSGGGSGGG GSAEAWYNL 360
GNAYYKQGDY QKAIEYYQKA LELEDPNNAEA WYNLGNAYYK QGDYQKAIEY YQKALELDPN 420
NAEAWYNLGN AYYKQGDYQK AIEDYQKALE LDPNNLQAEA WKNLGNAYYK QGDYQKAIEY 480
YQKALELDPN NASAWYNLGN AYYKQGDYQK AIEYYQKALE LDPNNAKAWY RRGNAYYKQG 540
DYQKAIEDYQ KALELDPNNR SRSAGGGGSG GGGSGGGGAS LCTMKKGPSG YGFNLHSDKS 600
KPGQPIRSVD PDSPAEASGL RAQDRIVEVN GVCMEGKHG DVVSAIRAG DETKLLVDR 660
EGSAGSAAGS GEFGSAAEAA KEAAAKAGSA GSAAGSGEFG SSSGAIITYV ELKRYGGPLG 720
ITISGTEEPF DPTIISLTK GGLAERTGAI HIGDRILAIN SSSLKGKPLS EAIHLLQMG 780
ETVTLKIKKQ TDAQPASSGS G 801

SEQ ID NO: 173      moltype = AA length = 547
FEATURE            Location/Qualifiers
source              1..547
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 173
MKCSTFSFWF VCKIIFFFFS FNIQTSIANP RENFLKCFSQ YIPNNATNLK LVYTQNNPLY 60
MSVLNSTIHN LRFTSDTPK PLVIVTPSHV SHIQGTILCS KKVGLQIRTR SGGHDSGMS 120
YISQVPFVIV DLRNMRSIKI DVHSQTAWVE AGATLGEVY WVNENENLS LAAGYCPTVC 180
AGGHFPGGGY GPLMRNYGLA ADNIIDAHV NVHGKVLDRK SMGEDLFWAL RGGGAESFGI 240
IVAWKIRLVA VPKSTMFSVK KIMEIHELK LVNKWQNIAY KYDKDLLMT HFITRITDN 300
QKKNKTAIHT YFSSVFLGGV DSLVDLMNKS FPELGIKKT CRQLSWIDTI IFYSGVNYD 360
TDNFKKEILL DRSAGQNGAF KIKLDYVKK IPESVFVQIL EKLYEEDIGA GMYALYPYG 420
IMDEISESAI PPHRAGILY ELWYICSWEK QEDNEKHLNW IRNIYNFMTP YVSKNPRLAY 480
LNRYDLDIGI NDPKNPNNTY QARIWGEKYF GKNFDRLVKV KTLVDPNNF RNEQSIPPL 540
RHRHSGG 547

SEQ ID NO: 174      moltype = AA length = 548
FEATURE            Location/Qualifiers
source              1..548
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 174
MNCSTFSFWF VCKIIFFFLS FNIQISIANP QENFLKCFSE YIPNNPANPK FIYTQHDQLY 60
MSVLNSTIQN LRFTSDTPK PLVIVTPSNV SHIQASILCS KKVGLQIRTR SGGHDAEGLS 120
YISQVPFAIV DLRNMHTVKV DIHSQTAWVE AGATLGEVY WINEMNENFS FPGGYCPTVG 180
VGGHFGGGY GALMRNYGLA ADNIIDAHV NVDGKVLDRK SMGEDLFWAI RGGGGENFGI 240
IAACKIKLVV VPSKATIFSV KKNMEIHGLV KLFNKWQNI YKYDKDLMLT THFRTRNITD 300
NHGKNKTTVH GYFSSIPLGG VDSLVDLMNK SFPELGIKT DCKELSWIDT TIFYSGVNY 360
NTANFKKEIL LDRSAGKKA FSKLDYVKK LIPETAMVKI LEKLYEEVGV GMYVLYPYG 420
GIMDEISESA IPFHRAGIM YELWYTATWE QEDNEKHIN WVRSVYNFTT PYVSQNPRLA 480
YLNRYDLDLG KTNPESPNY TQARIWGEKY FGKNFNRLVK VKTKADPNF FRNEQSIPPL 540
PPRHHGSG 548

SEQ ID NO: 175      moltype = AA length = 886
FEATURE            Location/Qualifiers
source              1..886
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 175
MNHLRAEGPA SVLAIGTANP ENILLQDEFP DYYFRVTKSE HMTQLKEKFR KICDKSMIRK 60
RNCFLNEEHL KQNPRLVEHE MQTLARQDM LVVEVPKLGK DACAKAIKEW GQPKSKITHL 120

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-continued

|            |             |             |            |             |             |     |
|------------|-------------|-------------|------------|-------------|-------------|-----|
| IPTSASTTDM | PGADYHCAKL  | LGLSPSVKRV  | MMYQLGCVGG | GTVLRIAKDI  | AENNKGARVL  | 180 |
| AVCCDIMACL | FRGPSEDLE   | LLVGQAIFGD  | GAAAVIVGAE | PDESVMGERPI | FELVSTGQTI  | 240 |
| LPNSEGTIGG | HIREAGLIFD  | LHKDVPMLIS  | NNIEKCLIEA | FTPIGISDWN  | SIFWITHPGG  | 300 |
| KAILDKVEEK | LHLKSDKFVD  | SRHVLSEHGN  | MSSSTVLFVM | DELKRKSLLE  | GKSTTGDGFE  | 360 |
| WGVLPFGPGP | LTVERVVRS   | VPIKYKLSGG  | GGSGGGGGSG | GGSAEAWYNL  | GNAYYKQGDY  | 420 |
| QKAIEYYQKA | LELDPNNAEA  | WYNLGNAYYK  | QGDYQKAIEY | YQKALELDPN  | NAEAWYNLGN  | 480 |
| AYYKQGDYQK | AIEDYQKALE  | LDPNNLQAEA  | WKNLGNAYYK | QGDYQKAIEY  | YQKALELDPN  | 540 |
| NASAWYNLGN | AYYKQGDYQK  | AIIEYYQKALE | LDPNNAKAWY | RRGNAYYKQG  | DYQKAIEDYQ  | 600 |
| KALELDPNNR | SRSAGGGGSG  | GGSGGGGGAS  | GNNLETYEWY | NKSISRDKAE  | KLLLDTGKEG  | 660 |
| AFMVRDSRTP | GTYYTVSVFTK | AIISENPCIK  | HYHIKETNDS | PKRYVVAEKY  | VFDSIPLLIQ  | 720 |
| YHQYNGGGLV | TRLRYPVCGG  | SAGSAAGSGE  | FGSAEAAAKE | AAAKAGSAGS  | AAGSGEGFGSG | 780 |
| SHPWFFGKIP | RAKAEEMLSK  | QRHGDGAFLIR | ESESAPGDFS | LSVKFGNDVQ  | HFKVLRDAG   | 840 |
| KYFLWVVKFN | SLNELVDYHR  | STSVSRNQOI  | FLRDIEQVPQ | QPTGSG      |             | 886 |

SEQ ID NO: 176                      moltype = AA   length = 668  
 FEATURE                              Location/Qualifiers  
 source                                1..668  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 176

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MAVKHLIVLK | FKDEITEAQK | EEFFKTYVNL | VNIIPAMKDV | YWGKDVTKQN | KEEGYTHIVE | 60  |
| VTFESVETIQ | DYIIHPAHVG | FGDVYRSFWE | KLLIPDYTPR | KKLSGGGGSG | GGSGGGGSA  | 120 |
| EAWYNLGNAY | YKQGDYQKAI | EYYQKALELD | PNNAEAWYNL | GNAYYKQGDY | QKAIEYYQKA | 180 |
| LELDPNNAEA | WYNLGNAYYK | QGDYQKAIED | YQKALELDPN | NLQAEAWKNL | GNAYYKQGDY | 240 |
| QKAIEYYQKA | LELDPNNAEA | WYNLGNAYYK | QGDYQKAIEY | YQKALELDPN | NAKAWYRRGN | 300 |
| AYYKQGDYQK | AIEDYQKALE | LDPNNRSRSA | GGSGGGGGSG | GGGGASGQDR | SEATLIKRFK | 360 |
| GBGVRYKAKL | IGIDEVSAAR | GDKLCQDSMM | KLKGVVAGAR | SKGEHKQKIF | LTISFGGIKI | 420 |
| FDEKTGALQH | HHAVHEISYI | AKDITDHRAF | GYVCGKEGNH | RFVAIKTAQA | AEPVILDLRD | 480 |
| LPQLIYELKQ | REELEKKAGS | AGSAAGSGEF | GSAAEAAAKE | AAKAGSAGSA | AGSGEFGSGS | 540 |
| HMSGQFWVTS | QKTEASERCG | LQGSYILRVE | AEKLTLLTLG | AQSQILEPLL | FWPYTLRLRY | 600 |
| GRDKVMFSFE | AGRRCPSGPG | TFTFQTSQGN | DIFQAVEAAI | QQQKAQKQVG | QAQDILRLEH | 660 |
| HHHHHSGS   |            |            |            |            |            | 668 |

SEQ ID NO: 177                      moltype = AA   length = 800  
 FEATURE                              Location/Qualifiers  
 source                                1..800  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 177

|            |             |            |             |            |            |     |
|------------|-------------|------------|-------------|------------|------------|-----|
| MGLSVCTFS  | FQTNHYHTLLN | PHNNNPKTSL | LCYRHPKTP   | KYSYNNFPSK | HCSTKSFHLQ | 60  |
| NKCSSELSIA | KNSIRAATTN  | QTEPPESDNH | SVATKILNFG  | KACWKLQRPY | TIIAFTSCAC | 120 |
| GLFGKELLHN | TNLISWSLMF  | KAFFFLVAIL | CIASFTTTIN  | QIYDLHIDRI | NKPDPLASG  | 180 |
| ETSVNTAWIM | SIIVALFGLI  | ITIKMKGGPL | YIFGYCFGIF  | GGIVYSVPPF | RWKQNPSTAF | 240 |
| LLNFLAHIIT | NFTFYASRA   | ALGLPFELRP | SFTFLLAPFM  | SMGSALALIK | DASDVEGDTK | 300 |
| FGISTLASKY | GSRNLTFLCS  | GIVLLSYVAA | ILAGIWPQA   | FNSNVMLLSH | AILAFWLILQ | 360 |
| TRDFALTNYD | PEAGRRFYEF  | MWKLYYAEVL | VYVFIKLSGG  | GGSGGGGGSG | GGSAEAWYNL | 420 |
| GNAYYKQGDY | QKAIEYYQKA  | LELDPNNAEA | WYNLGNAYYK  | QGDYQKAIEY | YQKALELDPN | 480 |
| NAEAWYNLGN | AYYKQGDYQK  | AIEDYQKALE | LDPNNLQAEA  | WKNLGNAYYK | QGDYQKAIEY | 540 |
| YQKALELDPN | NASAWYNLGN  | AYYKQGDYQK | AIIEYYQKALE | LDPNNAKAWY | RRGNAYYKQG | 600 |
| DYQKAIEDYQ | KALELDPNNR  | SRSAGGGGSG | GGSGGGGGAS  | AEYVRALPDF | NGNDEEDLFP | 660 |
| KKGDILIRID | KPEEQWNAE   | DSGKRGKMP  | VPYVEKYGSA  | GSAAGSGEFG | SABAAAKEAA | 720 |
| AKAGSAGSAA | GSGEFGSLIK  | HMRAEALPDF | TGNSKLELNF  | KAGDVIFLLS | RINKDWLEGT | 780 |
| VRGATGIFPL | SFVKILKSGS  |            |             |            |            | 800 |

SEQ ID NO: 178                      moltype = AA   length = 2744  
 FEATURE                              Location/Qualifiers  
 source                                1..2744  
                                       mol\_type = protein  
                                       organism = synthetic   construct

SEQUENCE: 178

|            |            |            |            |            |             |      |
|------------|------------|------------|------------|------------|-------------|------|
| MSEESLFESS | PQKMEYEITN | YSERHTELP  | HFIGLNTVDK | LEESPLRDFV | KSHGGHTVIS  | 60   |
| KILIANNGIA | AVKEIRSVRK | WAYETFGDDR | TVQFVAMATP | EDLEANAIEY | RMADQYIEVP  | 120  |
| GGTNMNNYAN | VDLIVDIAER | ADVDVWAGW  | GHASENPLLP | EKLSQSKRKV | IFIGPPGNAM  | 180  |
| RSLGDKISST | IQAQSAKVP  | IPWSGTGVD  | VHVDEKTLGV | SVDDDIYQKG | CCTSPEDGLQ  | 240  |
| KAKRIGFPVM | IKASEGGGK  | GIRQVEREED | FIALYHQAAN | EIPGSPIFIM | KLARARHLE   | 300  |
| VQLLADQYGT | NISLFGDRCS | VQRRHQKIE  | EAPVTIAKAE | TFHEMEKAAV | RLGKLVGYVS  | 360  |
| AGTVEYLYSH | DDGKFYFLEL | NPRLQVEHPT | TEMVSGVNL  | AAQLQIAMGI | PMHRISDIRT  | 420  |
| LYGMNPHSAS | EIDFEFTQD  | ATKKQRRPIP | KGHCTACRIT | SEDPNDGFKP | SGGTLHELNF  | 480  |
| RSSSNVWGYF | SVGNNGNIHS | FSDSQFGHIF | AFGENRQASR | KHMVVALKEL | SIRGDFRTTV  | 540  |
| EYLIKLETE  | DFEDNTITTG | WLDDLITHKM | TAEKPDPTLA | VICGAATKAF | LASEEARHKY  | 600  |
| IESLQKGQVL | SKDLLQTMFP | VDFIHEGKRY | KFTVAKSGND | RYTLFINGSK | CDIILRQLSD  | 660  |
| GGLLIAIGGK | SHTIYWKEEV | AATRLSVDSM | TTLLEVENDP | TQLRTPSPGK | LVKFLVENGE  | 720  |
| HIKQGPYAE  | IEVMMQMPL  | VSQENGIVQL | LKQPGSTIVA | GDIMAIMTLD | DPSKVKHALP  | 780  |
| FEGMLPDFGS | PVIEGTKPAY | FKKSLVSTLE | NILKGYDNQV | IMNASLQQLI | EVLARNPKLPY | 840  |
| SEWKLHISAL | HSRLPAKLDE | QMEELVARSL | RRGAVFPARQ | LsklidMAVK | NPEYNPKLL   | 900  |
| GAVVEPLADI | AHKYSNGLEA | HEHSIFVHFL | EYYEVEKLF  | NGPNVREENI | ILKLRLDENPK | 960  |
| DLDKVALTVL | SHSKVSANNN | LILAILKHYQ | PLCKLSSKVS | AIFSTPLQHI | VELESKATAK  | 1020 |
| VALQAREILI | QGALPSVKER | TEQIEHILKS | SVVKVAYGSS | NPKRSEPDNL | ILKDLIDSNY  | 1080 |
| VVFDVLLQFL | THQDPVVTAA | AAQVYIRRAY | RAYTIGDIRV | HEGVTVPIVE | WKFQLPSAAF  | 1140 |

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|            |             |            |            |            |             |      |
|------------|-------------|------------|------------|------------|-------------|------|
| STFPTVKSKM | GMNRAVSVD   | LSYVANSQSS | PLREGILMAV | DHLDDVDEIL | SQSLEVIPRH  | 1200 |
| QSSSNGPAPD | RSGSSASLSN  | VANVCVASTE | GFSEEEIILV | RLREILDLNK | QELINASIRR  | 1260 |
| ITFMFGFKDG | SYPKYYTFNG  | PNYNENETIR | HIEPALAQOL | ELGRLSNFNI | KPIFTDNRNI  | 1320 |
| HVYEAVSKTS | PLDKRFFTRG  | IIRTGHIRDD | ISIQEYLTS  | ANRLMSDILD | NLEVTDTNS   | 1380 |
| DLNHIFINFI | AVFDISPEDV  | EAAFGGFLER | FGKRLRLRLV | SSAEIRIIK  | DPQTGAPVPL  | 1440 |
| RALINNVSGY | VIKTEMYTEV  | KNAKGWVFK  | SLGKPGSMHL | RPIATPYPVK | EWLQPKRYKA  | 1500 |
| HLMGTTYVVD | PELFRQASS   | SQWKNFSADV | KLTDFFISN  | ELIEDENGEL | TEVEREFGAN  | 1560 |
| AIGMVAFKIT | VKTPEYPRGR  | QFVVVANDIT | FKIGSFGPQE | DEFFNKVTEY | ARKRGIPRIY  | 1620 |
| LAANSGARIG | MAEEIVPLFQ  | VAWDAANPD  | KGFQYLYLTS | EGMETLKKFD | KENSVLTERT  | 1680 |
| VINGEERFVI | KTIIGSEGDG  | GVECLRGSGL | IAGATSRAYH | DIFTITLVTC | RSVGIGAYLV  | 1740 |
| RLGQRAIQVE | GQPIILTAP   | AINKMLGREV | YTSNLQLGQT | QIMYNNGVSH | LTAVDLAGV   | 1800 |
| EKIVEWMSYV | PAKRNMPVPI  | LETQDTPDRP | VDFTPTNDET | YDVRWMEGR  | ETESGFPEYGL | 1860 |
| FDKGSFFETL | SGWAKGVVVG  | RARLGGIPLG | VIGVETRTVE | NLIPADPAMP | NSAETLIQEP  | 1920 |
| GQVWHPNSAF | KTAQAINDFN  | NGEQLPMMIL | ANWRGFSGGQ | RDMFNEVLKY | GSFIVDALVD  | 1980 |
| YKQPIIYIP  | PTGBELRGSSW | VVVDPTINAD | QMEMYADVNA | RAGVLEPQGM | VGKFRPREKL  | 2040 |
| LDTMNRLLDK | YRELRSQSLN  | KSLAPEVHQQ | ISKQLADRER | ELLPIYQGIS | LQFADLHDS   | 2100 |
| SRMVAKGVIS | KELEWTARR   | FFFWRLLRRL | NEEYLIKRLS | HQVGEASRL  | KIARIRSWYP  | 2160 |
| ASVDHEDDRQ | VATWIEENYK  | TLDDKLKGLK | LESFAQDLAK | KIRSDHDNAI | DGLSEVIKML  | 2220 |
| STDDKEKLLK | YRELRSQSLN  | SGGGSGGGG  | SABAWNLGN  | AYYKQGDYQK | AIYYQKALE   | 2280 |
| LDPNNAEAWY | NLGNAYYKQG  | DYQKAIEYYQ | KALELDPNNA | EAWYNLGNAY | YKQGDYQKAI  | 2340 |
| EDYQKALELD | PNNLQAEAWK  | NLGNAYYKQG | DYQKAIEYYQ | KALELDPNNA | SAWYNLGNAY  | 2400 |
| YKQGDYQKAI | EYYQKALELD  | PNNAKAWYRR | GNAYYKQGDY | QKAIEDYQKA | LELDPNNSR   | 2460 |
| SAGGGSGGGG | GSGGGGASGS  | HMLRGAQSIQ | PTANLDRD   | LVLVNLVMEV | RAVLELKNEL  | 2520 |
| AQLPPEGVYV | VVKNVGLTLR  | KLIGSVDDLL | PSLPSSSRTE | IEGTQKLLNK | DLAELINKMR  | 2580 |
| LAQONAVTSL | SEECRQMLT   | ASHTLAVDAK | NLLDAVDQAK | VLANLAHPA  | EGSAGSAGS   | 2640 |
| GEFGSAEAAA | KEAAAKAGSA  | GSAAGSGEFG | SGAMATPGSE | NVLPREPLIA | TAVKFLQNSR  | 2700 |
| VRQSPLATRR | AFLKKKGLTD  | EETDMAFQQS | GTAADPESSL | WGS        |             | 2744 |

SEQ ID NO: 179      moltype = AA    length = 3623  
 FEATURE            Location/Qualifiers  
 source              1..3623  
                      mol\_type = protein  
                      organism = synthetic    construct

SEQUENCE: 179

|             |            |            |            |            |            |      |
|-------------|------------|------------|------------|------------|------------|------|
| MGSAGSAGS   | GEFGSAGSAA | GSGEFGSAGS | AAGSGEFSYY | HHHHHHLEST | SLYKKAGSGS | 60   |
| ARNAYLRKKI  | ARLKKDNLQL | ERDEQNLEKI | IANLRDEIAR | LENEVASHEQ | GSAGSAGSG  | 120  |
| EFAEAAAKEA  | AAKAGSAGSA | AGSGEFSYYH | HHHHHLESTS | LYKKAGSGSN | LVAQLENEVA | 180  |
| SLNENETLK   | KKNLHKDLI  | AYLEKEIANL | RKKIEEGSAG | SAAGSGEFGS | AEAAAKEAAA | 240  |
| KEAAAKEAAA  | KAGSAGSAG  | SGEFGSSYYH | HHHHHLESTS | LYKKAGSGSQ | KVAELKNRVA | 300  |
| VKLNREQLK   | NKVBELKNRN | AYLKNELATL | ENEVARLEND | VAEGSAGSAA | GSGEFAEAAA | 360  |
| KEAAAKAGSA  | GSAAGSGEFS | YHHHHHHHLE | STSLYKKAGS | GSNEVTTLN  | DAAFIENENA | 420  |
| YLEKETIARL  | KEKAALRNRL | AHKKSAGSA  | AGSGEFGSAE | AAAKEAAAKE | AAAKEAAAKA | 480  |
| GSAGSAGSG   | EFGSRPTIS  | NPPPLISSAK | HPSVGSAGSA | AGSGEFAEAA | AKEAAAKAGS | 540  |
| AGSAAGSGEF  | NFLQSRPEPT | APPEESFRSG | GSAGSAGSG  | EFGSAEAAAK | EAAAKEAAAK | 600  |
| EAAAKAGSAG  | SAAGSGEFGS | SKGTGLNPNA | KVWQEIAPGN | GSAGSAGSG  | EFAEAAAKEA | 660  |
| AAKAGSAGSA  | AGSGEFPDGG | TTFEHLWSSL | EPDSTYGSAG | SAAGSGEFGS | AEAAAKEAAA | 720  |
| KEAAAKEAAA  | KAGSAGSAG  | SGEFGSSYYH | HHHHHLESTS | LYKKAGSGSK | RIAYLRKKIA | 780  |
| ALKKDNANLE  | KDIANLENEI | ERLIKEIKTL | ENEVASHEQG | SAGSAGSGE  | FAEAAAKEAA | 840  |
| AKAGSAGSAA  | GSGEFSYYHH | HHHHLESTSL | YKKAGSGSNL | LATLRSTA   | LENENHVLEK | 900  |
| EKEKLRKEKE  | QLLNKLEAYK | GSAGSAGSG  | EFGSAEAAAK | EAAAKEAAAK | EAAAKAGSAG | 960  |
| SAAGSGEFGS  | PATSQHPPPP | PGHRSQAPSH | GSAGSAGSG  | EFAEAAAKEA | AAKAGSAGSA | 1020 |
| AGSGEFELNS  | LILLEAAEY  | LERDRGSAG  | SAAGSGEFGS | AEAAAKEAAA | KEAAAKEAAA | 1080 |
| KAGSAGSAG   | SGEFGSRPPT | ISNPPPLISS | AKHPSVGSAG | SAAGSGEFAE | AAAKEAAAKA | 1140 |
| GSAGSAGSG   | EFNPLQSRPE | PTAPPEESFR | SGGSAGSAG  | SGEFGSAEAA | AKEAAAKEAA | 1200 |
| AKEAAAKAGS  | AGSGEFGSAG | GSSKGTGLNP | NAKVWQEIAP | GNGSAGSAG  | SGEFAEAAAK | 1260 |
| EAAAKAGSAG  | SAAGSGEFPD | GGTTFEHLWS | SLEPDSTYGS | AGSAGSGEFG | GSAAAKEAAE | 1320 |
| AAKEAAAKEA  | AAKAGSAGSA | AGSGEFGSSY | YHHHHHHLES | TSLYKKAGSG | SKRIAYLRKK | 1380 |
| IAALKKDNAN  | LEKDIANLEN | EIERLIKEIK | TLNEVASHE  | QGSAGSAGS  | GEFAEAAAKE | 1440 |
| AAAKAGSAGS  | AAGSGEFSYY | HHHHHHLEST | SLYKKAGSGS | NLLATLRSTA | AVLENENHVL | 1500 |
| EKEKLRKEKE  | KEQLLNKLEA | YKGSAGSAG  | SGEFGSAEAA | AKEAAAKEAA | AKEAAAKAGS | 1560 |
| AGSAGSGEFG  | GSALVDDAAD | YEPSPNNNEE | ALGSAGSAG  | SGEFAEAAAK | EAAAKAGSAG | 1620 |
| SAAGSGEFPRE | LFDDPSYVNV | QNLDKARQGS | AGSAGSGEFG | GSAAAKEAAE | AAKEAAAKEA | 1680 |
| AAKAGSAGSA  | AGSGEFGSKN | TKSMNFDNPV | YRKTTEEEGS | AGSAGSGEFG | AEAAAKEAAA | 1740 |
| KAGSAGSAG   | SGEFRSLPST | WIENKLYGMS | DPNWSAGSAG | AGSGEFGSAE | AAAKEAAAKE | 1800 |
| AAAKEAAAKA  | GSAGSAGSG  | EFGSVVDNSP | PPALPPKKRQ | SAPSGSAGSA | AGSGEFAEAA | 1860 |
| AKEAAAKAGS  | AGSAGSGEFG | TQRSKPQPAV | PPRPSADLIL | GSAGSAGSG  | EFGSAEAAAK | 1920 |
| EAAAKEAAAK  | AAAKAGSAG  | SAAGSGEFGS | TDEEREETEE | EVYLLNSTTL | GSAGSAGSG  | 1980 |
| EFAEAAAKEA  | AAKAGSAGSA | AGSGEFGDNV | SGTQRLDSAT | VRTYSCGSAG | SAAGSGEFGS | 2040 |
| AEAAAKEAAA  | KEAAAKEAAA | KAGSAGSAG  | SGEFGSSYYH | HHHHHLESTS | LYKKAGSGSQ | 2100 |
| KVAQLKNRVA  | YKLKENAKLE | NIVARLENDN | ANLEKDIANL | EKDIANLERD | VARGSAGSAA | 2160 |
| GSGEFAEAAA  | KEAAAKAGS  | SAAGSGEFGS | YHHHHHHHLE | STSLYKKAGS | GSNTVKELKN | 2220 |
| YIQELEERNA  | ELKNLKEHLK | FAKAELEFEL | AAHKFEFSAG | SAAGSGEFGS | AEAAAKEAAA | 2280 |
| KEAAAKEAAA  | KAGSAGSAG  | SGEFGSHDDS | LPHPQQAATD | SGHESDGSAG | SAAGSGEFAE | 2340 |
| AAAKEAAAKA  | AGSAGSAGSV | EFGSPNAGSV | EQTPKKPLGR | RRGSAGSAG  | SGEFGSAEAA | 2400 |
| AKEAAAKEAA  | AKEAAAKAGS | AGSAGSGEFG | GSSYYHHHHH | HLESTSLYKK | AGSGSFENV  | 2460 |
| HEFILATLEN  | ENAKLRLREA | KLERELARLR | NEVAWLGSAG | SAAGSGEFAE | AAAKEAAAKA | 2520 |
| GSAGSAGSG   | EFSYYHHHHH | HLESTSLYKK | AGSGSQKVEE | LKNKIAELEN | RNAVKKNRVA | 2580 |
| HLKQETIAYL  | DELAHAFEG  | SAGSAGSGE  | FGSAEAAAKE | AAAKEAAAKE | AAAKAGSAGS | 2640 |
| AAGSGEFGSV  | SSTKLVSFHD | DSDEDLLHIG | SAGSAGSGE  | FAEAAAKEAA | AKAGSAGSAA | 2700 |

-continued

|            |            |            |            |            |            |      |
|------------|------------|------------|------------|------------|------------|------|
| GSGEFAAATP | ISTFHDDSD  | DLHLVGSAGS | AAGSGEFGSA | EAAAKEAAAK | EAAAKEAAAK | 2760 |
| AGSAGSAAGS | GEFGSSYYHH | HHHHLESTSL | YKKAGSGSQK | VESLKQKIEE | LKQRKAQLKN | 2820 |
| DIANLEKEIA | YAETGSAGSA | AGSGEFAEAA | AKBAKAGS   | AGSAAGSGEF | SYHHHHHHHL | 2880 |
| ESTSLYKKAG | SEFFRRERNK | MAAAKCRNRR | RELDTLQAE  | TDQLEDEKSA | LQTEIANLLK | 2940 |
| EKEKLEFILA | AHRPACKIPD | DLGFPEEMSL | EGSAGSAAGS | GEFGSAEAAA | KEAAAKEAAA | 3000 |
| KEAAAKAGSA | GSAAGSGEFG | SFQMPADTPP | PAYLPPEDPM | TGSAGSAAGS | GEFAEAAAKE | 3060 |
| AAAKAGSAGS | AAGSGEFERE | SNEEPPPPYE | DPYWGNGGSA | GSAAGSGEFG | SAEAAAKEAA | 3120 |
| AKEAAAKEAA | AKAGSAGSAA | GSGEFGSSYY | HHHHHHLEST | SLYKKAGSGS | QKVAELKNRV | 3180 |
| AVKLNREQL  | KNKVEELKNR | NAYLKNELAT | LENEVARLEN | DVAEGSAGSA | AGSGEFAEAA | 3240 |
| AKEAAAKAGS | AGSAAGSGEF | SYHHHHHHHL | ESTSLYKKAG | SGSNEVTTL  | NDAAFIENEN | 3300 |
| AYLEKEIARL | RKEKAALNRN | LAHKSSYYHH | HHHHLESTSL | YKKAGSGSAR | NAYLRKKIAR | 3360 |
| LKKDNLQLER | DEQNLEKIIA | NLRDEIARLE | NEVASHEQGS | AGSAAGSGEF | AEAAAKEAAA | 3420 |
| KAGSAGSAAG | SGEFSYYHHH | HHHLESTSLY | KKAGSGSNLV | AQLENEVASL | ENENETLKKK | 3480 |
| NLHKKDLIAY | LEKEIANLRK | KIEEGSAGSA | AGSGEFGSAE | AAAKEAAAKE | AAAKEAAAKA | 3540 |
| GSAGSAAGSG | EFGSEQKLIS | EEDLEQKLIS | EEDLEQKLIS | EEDLGSAGSA | AGSGEFGSAG | 3600 |
| SAAGSGEFGS | AGSAAGSGEF | GSG        |            |            |            | 3623 |

SEQ ID NO: 180                   moltype = AA   length = 455  
 FEATURE                       Location/Qualifiers  
 source                         1..455  
                               mol\_type = protein  
                               organism = synthetic construct

SEQUENCE: 180

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MGSAGSAAGS | GEFGSAGSAA | GSGEFGSAGS | AAGSGEFSYY | HHHHHHLEST | SLYKKAGSGS | 60  |
| ARNAYLRKKI | ARLKKDNLQL | ERDEQNLEKI | IANLRDEIAR | LENEVASHEQ | GSAGSAAGSG | 120 |
| EFAEAAAKEA | AAKAGSAGSA | AGSGEFSYYH | HHHHHHLEST | LYKKAGSGSN | LVAQLENEVA | 180 |
| SEENENETLK | KKNLHKDLI  | AYLEKEIANL | RKKIEEGSAG | SAAGSGEFGS | AEAAAKEAAA | 240 |
| KEAAAKEAAA | KAGSAGSAAG | SGEFGSSATR | ELDELMAALS | DFKIQGGSAG | SAAGSGEFAE | 300 |
| AAAKEAAAKA | GSAGSAAGSG | EPDLALSENW | AQEFLAAGDA | VDGSAGSAAG | SGEFGSAEAA | 360 |
| AKEAAAKEAA | AKEAAAKAGS | AGSAAGSGEF | GSDYKDDDDK | DYKDDDDKDY | KDDDDKGSAG | 420 |
| SAAGSGEFGS | AGSAAGSGEF | GSAGSAAGSG | EFGSG      |            |            | 455 |

SEQ ID NO: 181                   moltype = DNA   length = 4596  
 FEATURE                       Location/Qualifiers  
 source                         1..4596  
                               mol\_type = other DNA  
                               organism = synthetic construct

SEQUENCE: 181

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| atgagtgcta  | aggcaatttc  | tgaacaaact  | ggtaagaat   | tggtgtacaa | gtttatttgt  | 60   |
| actacatcag  | ccatccaaaa  | tagattcaaa  | tacgctagag  | ttacccacga | tactgactgg  | 120  |
| gctagattgt  | tacaagatca  | tccatgggtg  | ttatctcaaa  | acttggttgt | caaacctgac  | 180  |
| caattaatta  | agagaagagg  | taaatggggt  | ttagtagggt  | ttaatttgac | attggatggt  | 240  |
| gtaaagtctt  | gggtgaaacc  | aagattaggt  | caagaagcca  | cagttggtta | agctaccggt  | 300  |
| ttcttgaaaa  | attctctgat  | cgaaccattt  | gtccctcatt  | cacaagccga | agaattctat  | 360  |
| gtatgtatct  | acgtactact  | agaggggtgac | tatgttttat  | ttcatcacga | aggtggtgtc  | 420  |
| gacgtagggt  | acgttgacgc  | caaggctcaa  | aagttgttgg  | ttggtgtcga | tgaagaattg  | 480  |
| aaccacgaag  | acattaaaaa  | gcatttgttg  | gttcacgcac  | ctgaagataa | aaaggaaata  | 540  |
| ttggcctcct  | ttataagtg   | ttgtttta    | ttctacgaag  | atttgtactt | cacctacttg  | 600  |
| gaaatttaacc | cattagtagt  | tactaaggat  | ggtgtatatg  | ttttggactt | agctgcacaa  | 660  |
| gttgatgcaa  | cagccgacta  | catttgttaag | gtcaaatggg  | gtgacatcga | atttccacct  | 720  |
| ccattcgcta  | gagaagctta  | tcagaagaa   | gcctacattg  | ctgatttgga | cgctaagtct  | 780  |
| gggtgatcat  | tgaagttgac  | attgttgaac  | cctaagagta  | gaatttgga  | catggttgct  | 840  |
| gggtggtggt  | ctagtgtcgt  | atattctgat  | actatatgag  | acttggttgg | tggttaacga  | 900  |
| ttggcaaac   | tcgttgtaata | ctcaggtgcc  | ccatccgaac  | aacaacata  | cgattacgct  | 960  |
| aagaccatct  | tgctcctaat  | gactagagaa  | aagcatcctg  | atggtaaaat | cttgatcatc  | 1020 |
| gggtggtagta | tcgcacaaatt | tactaacgtt  | gccgctacat  | tcaagggtat | cgctcagagct | 1080 |
| ataagagatt  | accaaggtcc  | attgaaggaa  | cacgaagtaa  | caatattcgt | tagaagaggt  | 1140 |
| ggctcctaact | accaagaagg  | ttgagagtc   | atgggtgaa   | taggttaaac | cactggtata  | 1200 |
| ccaatccatg  | tccttggtac  | agaaacccac  | atgactgcaa  | tagttggtat | ggccttaggt  | 1260 |
| catagaccac  | tcctcaatca  | acctccaacc  | gcagccacac  | ctgcaaat   | ctgttgtaac  | 1320 |
| gcctctggtt  | caacttccac  | accagctcct  | tctagaacag  | caagtttctc | tgaatcaaga  | 1380 |
| gctgatgaag  | tcgctccagc  | taagaagaca  | aaaccagcca  | tgctcaaga  | ctccgttcca  | 1440 |
| agtcctagat  | ctttgcaggg  | taaatctact  | actttgtttt  | ctagacatac | taaggctata  | 1500 |
| gtatggggtta | tgcaaaccaag | agcagttcaa  | ggcatgttgg  | atttcgacta | tggtttgtagt | 1560 |
| agagatgaac  | catctgttgc  | tgcaatgggtc | tatcctttta  | ctggtgacca | taagcaaaaa  | 1620 |
| ttctactggg  | gtcacaaagga | aatattgatc  | ccagttttta  | agaacatggc | cgatgctatg  | 1680 |
| agaaaacatc  | ctgaagtcga  | cgtattgatt  | aaactcgctc  | cattaagatc | cgcttaacgat | 1740 |
| ttcacaattg  | aaacccatgaa | ctacgctcaa  | ataagaacca  | tcgctatcat | tcgagaaggt  | 1800 |
| attccagaag  | ccttgactag  | aaagttgatt  | aagaagagctg | atcaaaaagg | tgccacaata  | 1860 |
| atcggtccag  | ctacocgtagg | tggtatttaag | cctggttgtt  | tcaagatcgg | taacactggt  | 1920 |
| ggtagtgttg  | ataacatatt  | ggcatctaag  | ttgtatagac  | caggttcagt | cgcttacgta  | 1980 |
| tcacgaagtg  | gtggtatgtc  | caacgaattg  | aacaacatca  | tcagtagaac | tacagatggt  | 2040 |
| gtatacgaag  | gtgtgtctat  | tggtggtgac  | agataccag   | gttctacttt | tatggatcat  | 2100 |
| gtattgagat  | atacgaacac  | acctggtggt  | aaaatgattg  | ttgtcttggg | tgaatagggt  | 2160 |
| ggtactgaag  | aatacaagat  | atgcagaggt  | atcaagaag   | gtagattgac | aaagccaatc  | 2220 |
| gtttgtgtgt  | gcattgtgtac | ttgtgcaaca  | atgttttctt  | cagaagtcca | attcggtcat  | 2280 |
| gcaggtgcct  | gcgctaatac  | agcttcagaa  | acagcagttg  | ccaagaacca | agcattaaaa  | 2340 |
| gaagccggtg  | ttttgtctcc  | tagatctttc  | gatgaattag  | gtgaatcat  | tcaatcagtc  | 2400 |
| tatgaagact  | tggtagctaa  | tggtgtaatt  | gttccagcac  | aagaagtccc | tcacactact  | 2460 |

-continued

|             |             |            |            |            |            |      |
|-------------|-------------|------------|------------|------------|------------|------|
| gtccctatg   | attactcttg  | ggcaagagaa | ttgggtttaa | ttagaaagcc | agctagtttt | 2520 |
| atgacctcta  | tatgtgatga  | aagaggtcaa | gaattgatct | atgctggtat | gcctattact | 2580 |
| gaagtattca  | aagaagaaat  | gggtatcggt | gggtgtttta | gtttgtgtg  | gttccaaaag | 2640 |
| agattgccaa  | agtactcttg  | tcaattcatt | gaaatgtgct | taatggttac | agctgatcat | 2700 |
| ggctctgctg  | tctcaggtgc  | acacaatacc | ataatctgcg | ctagagctgg | taaagatttg | 2760 |
| gtttcttctt  | tgacctcagg  | tttgtaaact | attggtgaca | gatttggtgg | tgcattagac | 2820 |
| gccgctgcaa  | agatgttttc  | aaaagctttc | gattccggta | taatcccaat | ggaatcgtt  | 2880 |
| aataagatga  | aaaaggagg   | taaattgata | atgggtatcg | gtcatcgtgt | taagtctatc | 2940 |
| aataaccctg  | atatgagagt  | acaaatcttg | aaggactatg | ttagacaaca | ctttccagcc | 3000 |
| acacctttgt  | tagattacgc  | tttggaaagt | gaaaagatta | ccacttctaa | aaagccaaat | 3060 |
| ttgatcttga  | acgttgatgg  | tttaattggt | gttgcttttg | tcgacatgtt | gagaaactgt | 3120 |
| ggttccttca  | ctagagaaga  | agctgatgaa | tatatcgaca | ttggtgcatt | gaatggtatc | 3180 |
| ttgttttag   | gtagatctat  | gggtttcatt | ggtcattact | tggatcaaaa | gagattaaag | 3240 |
| caaggtttgt  | acagacatcc  | atgggatgac | atttcttacc | ttttacctga | acacatgtca | 3300 |
| atgaaattgt  | ctggtggtgg  | tggttctggt | ggtggtggtt | ctggtggtgg | tggtagtgcc | 3360 |
| gaagcttggt  | acaatttggg  | taacgcatac | tacaagcagg | gtgactacca | aaaggcaatt | 3420 |
| gaattattac  | aaaaggcctt  | ggaattagac | ccaaataacg | cagaagcctg | gtataatttg | 3480 |
| ggtaattgct  | attataaaca  | gggtgactat | caaaaggcta | tcgaatacta | ccaaaaggca | 3540 |
| ttggaattag  | accctaataa  | cgctgaagca | tggtataatt | tgggtaacgc | ttattataag | 3600 |
| cagggtgact  | atcaaaaagc  | catcgaaagc | tacaaaagg  | ctttggaatt | agatccaaat | 3660 |
| aacttacaag  | ccgaagcttg  | gaagaatttg | ggtaacgctt | actataaaca | gggtgactac | 3720 |
| caaaaagcaa  | tgaaatacta  | tcaaaaagct | ttagaattgg | accctaataa | cgcatacagc | 3780 |
| tggtaacaatt | tgggttaatgc | ttactataag | cagggtgact | atcagaagcc | cattgaatac | 3840 |
| tatcaaaaagg | ctttagaatt  | ggatccaaat | aacgctaaag | catggtacag | acgttgtaac | 3900 |
| gcttattaca  | aacagggtga  | ctaccagaaa | gccattgaag | attatcaaaa | ggctttggaa | 3960 |
| ttggtacctta | acaacagatc  | tagatcagct | gggtggtggt | gttctggtgg | tgggtgttct | 4020 |
| gggtggtggt  | gtgcttcttc  | atattaccat | caccatcacc | atcacttgga | atccacaagt | 4080 |
| ttatacaaaa  | aggctggttc  | tggttcaaat | ttggtcgcac | aattgaaaaa | cgaagtagcc | 4140 |
| tctttagaaa  | atgaaaacga  | aaccttgaaa | aagaaaaact | tacataagaa | agatttgatc | 4200 |
| gcttatttgg  | aaaaggaaat  | cgcataattg | agaaaagaaa | ttgaagaagg | tagtgacagt | 4260 |
| tctgcccgtg  | gttctggtga  | atttggttca | gctgaagcag | ccgctaagga | agcagccgct | 4320 |
| aaagccggtt  | cagctggttc  | cgcagccggt | tctggtgaat | tcggttccag | ttactatcac | 4380 |
| catcaccatc  | atcacttgga  | atccactagt | ttatataaga | aagcaggttc | tgttccagca | 4440 |
| agaaatgcct  | acttgagaaa  | gaaaaatagc | agattaaaga | aagataaact | gcaattggaa | 4500 |
| agagatgaac  | aaaatttggg  | aaagattatc | gccaaactaa | gagatgaaat | cgctagattg | 4560 |
| gaaaatgaag  | ttgcatccca  | tgaacaaggt | agtggt     |            |            | 4596 |

SEQ ID NO: 182      moltype = DNA    length = 2472  
 FEATURE            Location/Qualifiers  
 source              1..2472  
                      mol\_type = other DNA  
                      organism = synthetic    construct

SEQUENCE: 182

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| atgaaaaact  | gtgtaatcgt  | ttctgctggt  | agaactgcaa  | ttggttcctt  | taatggtagt  | 60   |
| ttggcctcta  | catcagctat  | tgtattgggt  | gctaccgtca  | tcaaagctgc  | aattgaaaga  | 120  |
| gcaaaagattg | attctcaaca  | tgtcgacgaa  | gtaataatgg  | gtaacgtttt  | gcaagctggt  | 180  |
| ttagggtcaaa | atccagcaag  | acaagccttg  | ttaaaaatctg | gttttagcaga | aactgtatgt  | 240  |
| ggtttcacag  | ttaataaagg  | ctcgcggttc  | ggtttgaagt  | cagttgcttt  | agccgctcaa  | 300  |
| gctatacaag  | cagggtcaagc | ccaatctatc  | gtcgctgggt  | gtatggaaaa  | tatgtcattg  | 360  |
| gcaccttatt  | tgtagatgc   | aaaagccaga  | tcagggttata | gattaggtga  | cgggtcaagt  | 420  |
| tacgacgtta  | ttttgagaga  | tggttttaatg | tgcgctactc  | atggttatca  | catgggtatt  | 480  |
| acagcagaaa  | atgttgccaa  | agaatacggg  | ataaccagag  | aatgcaaga   | tgaattggca  | 540  |
| ttacattccc  | aaagaaagcc  | agccgctgca  | atcgaaaagt  | gtgcttttac  | tcagaaaaat  | 600  |
| gtcccagtaa  | acgttgtcac  | aagaaagaaa  | acttctggtt  | tctcccaaga  | tgaattccca  | 660  |
| aaagctaata  | gtaccgctga  | agcattgggt  | gctttaagac  | ctgcattcga  | caaggccggt  | 720  |
| accgtaactg  | ccggttaatgc | ttctggtata  | aacgatgggt  | ccgctgcatt  | ggttatcatg  | 780  |
| gaagaatcag  | ccgcttttagc | agccggtttg  | acaccttag   | ctagaattaa  | atcttatgca  | 840  |
| tcagggtggtg | ttccacctgc  | tttgatgggt  | atgggtccag  | tccctgctac  | ccaaaaggca  | 900  |
| ttgcaattag  | ccggttttgc  | attggtctgat | atcgacttaa  | tcgaagcaaa  | cgaagccttt  | 960  |
| gctgcacaat  | tcttggcagt  | tggtaaaaat  | ttgggtttcg  | actccgaaaa  | ggttaatgtc  | 1020 |
| aacggtggtg  | ccatttgctt  | gggtcatcca  | ataggtgctt  | cagggtgcaag | aatcttggtt  | 1080 |
| acattgttgc  | atgccatgca  | agctagagat  | aaaaccttgg  | gtttagctac  | tttgtgtatc  | 1140 |
| gggtggtggtc | aaggtatcgc  | aatggttatc  | gaaagattga  | ataagttgtc  | tgggtggtggt | 1200 |
| ggttctggtg  | gtggtggttc  | tggttggtggt | ggtagtgcag  | aagcctggta  | caatttgggt  | 1260 |
| aacgcttact  | acaagcaggg  | tgactaccaa  | aaggcaatcg  | aatactacca  | aaaggccttg  | 1320 |
| gaattagatc  | caataaacgc  | tgaagcatgg  | tataatttgg  | gtaatgccta  | ttataaacag  | 1380 |
| ggtgactatc  | aaaaagctat  | tgaatattac  | caaaaggcat  | tggaaattaga | tcctaataac  | 1440 |
| gccgaagctt  | ggataatttt  | gggtaacgcc  | tattataagc  | aggggtgacta | tcaaaaggcc  | 1500 |
| atcgaagatt  | acaaaaaggc  | tttggaattg  | gatccaaaca  | acttgcaagc  | agaagcctgg  | 1560 |
| aagaatttgg  | gtaacgctta  | ttacaacacg  | gggtgactacc | aaaaagctat  | tgaatactat  | 1620 |
| caaaaagcct  | tagaatttga  | tcctaataac  | gcttctgcat  | ggtaacaatt  | gggtaatgcc  | 1680 |
| tactataaac  | agggtgacta  | ccagaaggct  | attgaatact  | acaaaaaagc  | attagaattg  | 1740 |
| gatccaaata  | acgccaaggc  | ttggtacaga  | cgtggttaatg | cctattacaa  | gcagggtgac  | 1800 |
| taccagaag   | ccatagaaga  | ctatcaaaaa  | gccttggaaat | tggatccctaa | caacagatcc  | 1860 |
| agaagtgcgtg | gtggtggtgg  | ttctggtggt  | gggtggttctg | gtggtggtgg  | tgttcttcca  | 1920 |
| tattaccatc  | accatcacca  | tcacttgtaa  | tctacatcat  | tatacaaaaa  | ggctggttcc  | 1980 |
| ggtagtaaatg | aagttactac  | attgaaaaac  | gatgccgctt  | ttatcgaaaa  | cgaaaacgca  | 2040 |
| tacttgaaaa  | aggaattcgc  | cagattaaga  | aaggaaaaag  | cagccttgag  | aaatagatta  | 2100 |
| gcccataaaa  | agggttccgc  | tggtagtgtc  | gcaggttctg  | gtgaatttgg  | ttcagctgaa  | 2160 |



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gaaggttctg gt 2472

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source              1..2337
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gatttggaac acatctgcaa acctgaaaca atcttagcct caaacacctc ttcattgtcc 360
attactgaag tcgctagtgc aacaaaaaga ccagataagg taataggcat gcatttcttt 420
aatccagctc ctgttatgaa gttggttagaa gttatttagag gtatagcaac atctcaagaa 480
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aagcagggtg actatcaaaa agctatcgaa gattacaaa aggcattgga attggatcct 1200
aacaacttac aagcagaagc ctggaagaat ttgggtaacg catattacaa acaggggtgac 1260
taccaaaaag ccattgaata ttatcaaaaa gctttggaat tggatccaaa taacgcttca 1320
gcatggtaga atttgggtaa tgctattac aagcagggtg actatcagaa agctattgaa 1380
tattatcaaa aggccttgga attagatcct aataacgcca aggccttgga cagacgtggt 1440
aatgcctatt acaagcaggg tgactaccag aaggccattg aagactatca aaaagccttg 1500
gaattggatc caaacaacag atctagatca gctggtggtg gtggttctgg tgggtggtggt 1560
tctggtggtg gtggtgcttc cgaaaatttg tacttccaa gtgaaaactt gtacttccag 1620
ggtagactca gtgaaagttg ttggaattgc ggtagaaaaa cctccgaaac ctgtagtgggt 1680
tgcaacactg ctagatatgt tggttctttt tgccaacaca aagattggga aaagcatcac 1740
catatttggt gtc aaacatc acaagcacaa caaggttctg ccggttcagc tgcaggttct 1800
ggtagaattg ttcccgctga agccgctgca aaagaagccg ctgcaaaagg aggttccgcc 1860
ggtagtgccg ctggtagtgg tgaattcggg tctatggcag ttccgaaag tcaattgaag 1920
aaaaatgggt ctaagtacaa gtacagagat ttgactgtta gagaaacagt taacgtcatc 1980
actttgtaca aggtattgaa gccagtcctg gactcatagc tttttaatga tggttcttca 2040
agagaattga tgaacttaac ttgtacaata ccagttcctt accgtggtaa cacttacaac 2100
atccaatctg gtttggtggt gttagatata tatccttaca atccacctat ctgcttcgtc 2160
aaaccaacat ccagtatgac cattaaaaact ggtaaacatg ttgatgctaa cggtaaaata 2220
tatttgccat acttacacga atggaagcat cctcaatcag acttggtggg tttaatccaa 2280
gtaatgatcg tcgtatttgg tgacgaacca cctgttttct ctgaccagg ttcaggt 2337

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SEQ ID NO: 184      moltype = DNA length = 2232
FEATURE             Location/Qualifiers
source              1..2232
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 184
atggaattga acaacgttat attgaaaaag gagggtaaaag tcgctgttgt cactataaat 60
agaccaaaag cattgaacgc cttgaaactc gatacattga aggaaatgga ctacgttatc 120
ggtagaaattg aaaacgatcc agaagtctta gcagtaattt tgaccggtgc cggtagaaaa 180
tcctttgttg ccggtgctga tatcagtga atgaaggaaa tgaacactat cgaaggtaga 240
aagttcggtt tcttgggttaa caaggttttc agaagattgg aattggtgga aaagcctggt 300
atagctgcag tcaatgggtt cgctttgggt ggtggtgttg aaatcgcaat gtcctgcgat 360
attagaatag cttcttcaaa cgcaagattt ggtcaaccag aagtcgggtt aggtattaca 420
cctggttttg gtggtaccac aagattatct agattggttg gtatgggtat ggccaagcaa 480
ttgattttta ctgctcaaaa catcaaggct gatgaagcat tgagaatcgg ttggttaat 540
aaggtagttg aaccatctga attgatgaac accgccaagg aaatcgctaa taagattggt 600
tctaattgct cagttgtgtg caagttgagt aagcaagcta taaatcgtgg tatgcaatgt 660
gatatcgaca ctgcatggc cttcgaatct gaagcatttg gtgaatgctt ctcaacagaa 720
gatcaaaaag acgcaatgac cgcctttatc gaaaagagaa agatagaagg ttcaaaaac 780
agaaagttat ctggtggtgg ttggtctggt ggtggtggtt ctggtggtgg tggtagtgct 840
gaagcatggg caaatttggg taacgcttac tacaagcagg gtgactacca aaaggcaatc 900
gaatactacc aaaaggcctt ggaattggac ccaataacgc ccgaagcttg gtataattg 960
ggtaatgcct attataaaca ggtgactat caaaaagcta tagaatacta ccaaaaggca 1020
ttggaattgg accctaataa cgcagaagcc ttgtataatt tgggtaacgc ctattataag 1080
cagggtgact atcaaaaggc catagaagac taccaaaagg ctttggaaat ggatccaaac 1140
aacttacaag ctgaagcatg gaagaatttg ggtaacgctt attacaaaca ggtgactac 1200

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caaaaagcta ttgaatatta tcaaaaagct ttagaattag accctaataa cgcctctgct 1260
tggtagcaatt tgggtaatgc ctactataaa cagggtgact accagaaggc tattgaaat 1320
taccaaaaag ctttagaatt ggatccaaat aacgcaaagg cctggtagac acgtggtaat 1380
gcctattaca agcagggtga ctaccagaaa gccattgaag attatcaaaa agctttggaa 1440
ttggatcccta acaacagatc cagaagtgtc ggtgggtggg gttctgggtg tgggtgttct 1500
ggtggtgggtg gtgcttctgg tccattgggt tccctttga ctgcatcaat gtttagcttc 1560
gcaccacctc aagaacaaaa gcaaatgttg ggtgaaagat tatcccat gatacaagct 1620
atgcatcccta ctttagcagg taaaatcaca ggcattgtgt tggaaatcga taactctgaa 1680
ttgttacaca tgttagaatc ccagaaaagt ttgagatcta aagttagcga agccgtagct 1740
gttttgcaag ctcatcaagc aaaagaagcc gctcaaaagg ccggttcagc tggttccgca 1800
gccggtagtg gtgaatttgg ttctgtgaa gctgcagcca aagaagctgc agccaaggca 1860
ggtagtgccg gttctgtgct aggttctggt gaattcgggt ccaataccaa catgagtgct 1920
ccaactgatg gtgctgtaac tacatctcaa attcctgcat cagaacaaga aactttagtt 1980
agaccaaagc ctttgtgtgt gaagtgtgtg gtagctcaaaa agatacctac 2040
actatgaagg aagttttatt ttatttgggt caatacatca tgacaaagag attatacgt 2100
gaaaagcaac cacatatcgt ttactgttca aacgatttgt tgggtgactt gtttgggtga 2160
ccatctttct cagttaagga acacagaaag atctatacaa tgatatacag aaatttgggtc 2220
gtaggttctg gt 2232

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SEQ ID NO: 185      moltype = DNA length = 2469
FEATURE            Location/Qualifiers
source              1..2469
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 185
atgactcgtaa agccaatggt tagaaacaac atctgtttga acgctcatcc tcaaggttgc 60
aaaaagggtg tagaagatca aatcgaatac accaaaaaga gaatcactgc agaagttaaa 120
gccggtgcta aagcacctaa gaatgttttg gtcttaggtt gttccaacgg ttatggtttg 180
gctagtagaa taacagctgc atttggttac ggtgccgcta ccacgggtgt ttctctcgaa 240
aaggctggtg gtgaacccaa atatgttact ccaggttgggt acaataaact ggcatattgat 300
gaagcagcca agagagaagg ttatactct gtcactatag atggtgacgc ttctcagat 360
gaaatcaagg cacaagttat tgagaagcc aaaaagaagc gtataaaat cgatttgatc 420
gtttactcct tagcaagtcc agtcagaaca gatcctgaca ccggtataat gcataagtct 480
gttttgaagc catcgggtta aactttcaca ggtaaaacag tcatccttt caccgggtgaa 540
ttgaaagaaa tatctgctga accagcaaat gatgaagaag ctgcagccac agtaaaagt 600
atgggtgggtg aagactggga aagatggatc aagcaattgt ccaagaagg tttgttgaa 660
gaaggttgta tcaccttagc ttattcatac attggtcctg aagccactca agctttgtat 720
agaaaaaggt caatcggtaa agctaaagaa catttggaa ccaccgctca cagattaaat 780
aaggaaccac catctatcag agcatttgtt tctgtaataa agggtttagt tactagagca 840
tccgccgtta tccagtgatc tcccttgat ttggctagtt tgtttaaggt tatgaaggaa 900
aagggttaacc atgaaggttg catagaacaa atcactagat tgtacgcaga aagattatac 960
agaaaggatg gtacaattcc agttgacgaa gaaaacagaa tcagaatcga tgactgggaa 1020
ttggaagaag atgtccaaaaa ggcagtatct gccttaattg aaaaagttac cggtgaaaac 1080
gctgaatcat tgactgattt ggcaggttat agacacgact ttttagcctc taatggtttc 1140
gatgtcgaag gtattaaacta cgaagcagaa gtagaagaat tcgacagaat taaattgtct 1200
ggtggtgggtg gttctggttg tgggtgttct ggtggtgggt gtagtgctga agcatgggtat 1260
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aaggctttgg aattggaccc taataacgcc gaagcttgggt acaatttggg taatgcctac 1380
tataaacagg gtgactatca aaaaagcaatt gaattattac aaaaaggcctt ggaattagac 1440
ccaaataacg cagaagcctg gtacaatttg ggtaacgcct actataagca gggtgactat 1500
aaaaaggcta ttgaagacta ccaaaaggca ttggaattag atcctaataa cttgcaagct 1560
gaagcatgga aaaatttggg taatgcctat tataaacagg gtgactacca aaaagctatt 1620
gaatactatc aaaaagcctt ggaaattggac ccaaataacg cctcagcttg gtataatttg 1680
ggtaatgcat actacaacaa gggtagactat cagaaggcaa tagaactata tcaaaaagcc 1740
ttagaatttg atcctaataa cgcaaaaagc tggtagagac gtggaatgct ctactacaag 1800
cagggtgact atcagaagcc gatagaagat tatcaaaagg cattggaatt ggatccaaac 1860
aacagatcta gatcagctgg tgggtggtgt tctggtggtg gtggttcttg tgggtggtgt 1920
gcttcttcat attaccatca ccatcaccat cacttggaa ccacaagttt atataagaaa 1980
gcaggttctg gttcaaatth gttagccact ttgagatcaa cagctgcagt attggaaaac 2040
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aataagtttg aagcctacaa aggttctgct ggttcagccg ctggttccgg tgaattcgg 2160
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ggttctgggtg aatttgggtc cagttactat caccatcacc atcatcactt ggaatccact 2280
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gctgcattaa agaaagataa cgccaacttg gaaaaggaca tcgctaattt ggaaaacgaa 2400
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ggttcagggt 2469

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SEQ ID NO: 186      moltype = DNA length = 2487
FEATURE            Location/Qualifiers
source              1..2487
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 186
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tctttaaaag atgttgaccc agccgaattg ggtgcattag tcgtaagaga agctttggca 120
agagcccaag tttcagggtg cgatgtcgggt catgttgtct tcggtaacgt tatccaaaca 180
gaaccaagag atatgtatth gggtagagta gctgcagtta atggtgggtg taccataaac 240
gctctgcat taactgtcaa cagattgtgt ggtagtgtgt tacaagctat tgtttctgcc 300

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gctcaaacaa tattgttagg tgacaccgac gttgctatcg gtgggtgtgc tgaatctatg 360
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ttgggttgaca tgatgttggg tgcctttgcat gatccattcc atagaatcca catgggtgta 480
actgcagaaa acgttgccaa ggaatacgc atctcaagag cacaacaaga cgaagctgca 540
ttagaatcac acagaagagc atccgcccgt attaaagccg gttactttaa ggatcaataa 600
gttccagtag tttctaaagg tagaaagggt gacgttacct tcgatactga cgaacatggt 660
agacacgacg ctactattga tgacatgaca aagttaagac ctgttttcgt caaggaaaat 720
ggtagctgta cagctggtaa tgcactcgtt ttgaacgatg cagccgctgc agtcgtaatg 780
atggaaagag ccgaagctga aagaagaggt ttgaaacat tagctagatt ggtttcttat 840
ggtagctgct gtgctgatcc taaagcaatg ggtagaggtc cagttcctgc tactaagatc 900
gcattggaaa gagccgggtt acaagtcctc gatttggacg taattgaagc caatgaagct 960
tttgcgctc aagcatgtgc cgttacaaaa gccctgggtt tagatccagc taaggccaat 1020
cctaaccgta gtggtatctc ttaggtgcat ccaattggtg caaccgggtc cttgataact 1080
gttaaggctt tgacggaatt gaacagagta caaggtagat atgcattagt tacaatgtgc 1140
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ggttctgtgt gtggtggttc tgggtggtgt ggtagtgctg aagcatggta caatttgggt 1260
aacgcttact acaagcaggg tgactaccaa aaggcaatcg aatattacca aaaagccttg 1320
gaattagacc caaataacgc cgaagcttgg tataatttgg gtaatgccta ttataaacag 1380
ggtagactac aaaaagctat agaatactac caaaaggcat tggaaattga ccctaataac 1440
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ttgccagaag caaaaagagg ttccgggt

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SEQ ID NO: 187      moltype = DNA length = 2169
FEATURE             Location/Qualifiers
source               1..2169
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 187
atgggtaaaa attacaagtc attggattcc gttgtcgcaa gtgactttat tgccttgggt 60
ataactctcg aagtcgcaga aacattgcat ggtagattag ccgaaattgt atgtaactac 120
ggtagctgcaa ccccaacaac ttggatcaac atagcaaaac atatctgtgc accagatttg 180
cctttctcct tgcaccaaatt gttgttttat ggttgcatac aggatttcgg tccgtgctca 240
cctgcatgga tccagacccc tcaaaaagggt aagtcaacta atttgggtgc tttgttagaa 300
aagagaggta aagaattctt ggggtttaag tacaaggatc caatctcttc tttttctcac 360
ttccaaagat tttctgtcag aaacctgtaa gtatactgga gaacagtttt gatggatgaa 420
atgaaaaataa gttctctctaa ggaaccagaa tgtatcttga gaagagatga catcaacaac 480
ccagggtggt tgaatgggtt gccagggtgt tatttgaact cagctaaaaa ttgcttgaac 540
gttaactcca ataagaattt gaatgatact atgattgtct ggagagatga aggcaacgat 600
gaactgtccat tgaataagtt gacattggat caattgagaa agagagattg gttggtcggt 660
tacgatttag aagaaatggg ttgggaaaaa ggttgtgcca tagctatcga tatgctatg 720
catgtagacg ctgtagttat ctatttgggt attgttttag caggttacgt cgtagtctct 780
atagctgatt cattttccgc accagaaatc tcaactagat tgagattatc caaagcaaa 840
gccatattca cacaagatca catcatcaga ggtaaaaaga gaatcccttt atactcaaga 900
gtcgtagaag ccaaatcccc aatggctata gttatccctt gtagtgggtc taacattggt 960
gcagaattaa gagatggtga catatcttgg gattactttt tggaaagagc caaagaattc 1020
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tccagtggta ctacagggtg accaaaagca ataccttggg cacaagccac ccctttaaa 1140
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tgtgtctcag gttatgattg gtccaccatc agatgctttt cttcatccgg tgaagcctct 1440
aatgtcgacg aatatttgtg gttaatgggt agagctaact acaagccagt tatcgaaatg 1500
tgtggtggta ccgaaatttg ttgtgcattc tcagccgggt cctttttaca agctcaatca 1560
ttgagttctt tttcatccca atgtatgggt tgcacattgt acatcttggg taagaacggt 1620
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gattctaacg acacaaccat agatttgaac caattgagat tatcttttaa cttgggttta 2040
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agaactgcta caaacaagat tatgagaaga gtcttgagac aacaattcag tcattttgaa 2160
ggttctggtg                                     2169

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SEQ ID NO: 188      moltype = DNA length = 2556
FEATURE            Location/Qualifiers
source              1..2556
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 188
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caacaacaat tgcataacac aaacttgcaa atgacogaat tgaagaaaca aaagactgct 120
gaacaaaaga ctagaccaca aaacgttggt attaaaggta tccaaatcta tatccctaca 180
caatgtgtca atcaatctga attggaagaa ttgatgggtg tatcacaggg taaatacact 240
atcgggttag gtcaaacaaa catgtctttc gtaaacgata gagaagacat ctattctatg 300
tcattgactg tttgtgccaa gttgataaaa agttacaaca tcgatacaaa caagattggt 360
agattggaag ttggtaccga aactttgatc gataagtcca agagtgtcaa gtctgtattg 420
atgcaattgt tcggtgaaaa taaccgatgt gaaggtatcg acactttaaa tgcttggtat 480
ggtgtgacta acgcattatt caattcattg aactggatcg aatccaatgc ctgggatggt 540
agagatgcta ttgttgtctg cggtgacatc gctatctatg acaaagggtc tgcaagacca 600
accggtggtg caggtactgt tgccatgtgg ataggtccag atgcacctat cgtttttgac 660
tctgtcagag catcatacat ggaacatgcc tacgatttct acaaacccaga ctccacctcc 720
gaatacctct acggttgatg tcacttttct ttgactttgt acgtcaaggc tttggacca 780
gtatacaagt cttactctaa gaaagcaata tctaagggtt tggtttcaga tccagctggt 840
tcgcagcgat taaacgctct gaagtacttc gattacaacg ttttccatgt cctcatatgc 900
aagttggtta ccaagtccta cggtagattg ttgtacaacg atttcagagc taaccacaaa 960
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aagaacatcg aaaagacttt cgtaaacggt gcaaaagccat tccacaaaaga aagagttgcc 1080
caatcattaa ttgtccctac aaataccggt aacatgtata cagcctcagt ttacgcgcgt 1140
tttgttctct tgttaaatga tgtaggtagt gatgacttgc aaggtaaaaag agttggttta 1200
ttctcctatg gtagtggttt agcagcctct ttgtactctt gtaagattgt aggtgacgtt 1260
caacacatta ttaaggaatt ggacatcact aataagttgg ctaagagaat cactgaaaca 1320
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gacaaattca gaagatcata cgatgttaaa aagaaattgt ctggtggtgg tggttctggt 1500
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tacaagcagg gtgactacca aaaagcaatc gaattattacc aaaagccctt ggaattagac 1620
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caaaaggcta tcgaatacta ccaaaaggca ttggaattag accctaataa cgcagaagcc 1740
tggtacaatt tgggtaacgc ctactataag cagggtgact atcaaaaagc catagaagac 1800
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ggtaatgcct actacaaaca ggggtgactac caaaaggcaa ttgaattata tcaaaaagcc 1920
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cagggtgact ccagaaaagc cattgaatat tatcaaaagg ctttagaatt ggatccaaat 2040
aacgcaaaag cctggtatag acgttggtat gcctactaca agcaggggtg ctatcagaag 2100
gctattgaag attatcaaaa agcttttgaa ttggatccaa acaacagatc cagaagtgtc 2160
ggtggtggtg gttctggtgg ttgtggttct ggtggtggtg gtgcttcttt gggctccttg 2220
ccacctggtt gggaagtaag atccacagtt agtggtagaa tctatttcgt tgatcataac 2280
aacagaacta cacaattcac cgacccaaga ttgcacggtt ctgctggttc agccgctggt 2340
tctggtgaat ttggttccgc agaagcagcc gctaaggaaag cagccgctaa agccggttcc 2400
gctggtatg cagccggtg ttggtgaattt ggttctggtg ctatgggttc attaccacct 2460
ggttgggaaa agagaacaga ttctaaccgt agagtctact tcgtaaacca taataccaga 2520
attactcaat gggaagatcc tagatctggt tcaggt                                     2556

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SEQ ID NO: 189      moltype = DNA length = 2487
FEATURE            Location/Qualifiers
source              1..2487
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 189
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gatagattac cttacaagaa ctacgattac gacagagtat ttggtgcttg ttgcgaaaac 120
gttattggtt atatgccatt gcctgtcggg gtaatcggtc cattagttat tgatggtaca 180
tcttaccata tccctatggc aactacagaa ggttgtttgg ttgcatcagc catgagaggt 240
tgcaaggcaa ttaatgctgg ttggtgtgct accactgttt taaccaaaga ttgtagtact 300
agagggtccg ttgtcagatt tctacttttg aagagatccg gtgcttgtaa aatatgggta 360
gatagtgaag aaggtcaaaa tgccatcaaa aaggctttta actccactag tagattcgca 420
agattgcaac atattcaaac atgcttagcc ggtgacttgt tgtttatgag attcagaaca 480
accactggtg acgctatggg tatgaatatg atatctaagg gtgtogaata ctcatggaag 540
caaatggtag aagaatacgg ttgggaagat atggaagtag tttctgttcc aggcaactac 600
tgtactgaca aaaagccagg tgaacttaac tggatagaag gtctgtgtaa atctgtcgtg 660
gctgaagcaa caatacctgg tgacgttggt agaaagggtt tgaactctga cgtatcagct 720
ttggttgaat tgaacatcgc taaaaatttg gttggttccg ccatggctgg tagtgctggt 780
ggttttaatg cacatgccgc taacttagtt acagcagctt tcttgccctt aggtcaagat 840
ccagctcaaa acgtgaatat tccaactgt atcaccttga tgaagaaagt tgatggtgac 900
ttaagaatat ccgttagtat gccatcaata gaagtgggta caatcggtgg ttggtaccgtc 960
ttggaacctc aaggtgcaat gttagatttg ttagggtgta gaggtccaca tgcaactgcc 1020
cctggtacaa atgctagaca attggcaaga attgtcgctt gtgcagatt agctggtgaa 1080
ttgtccttat gcgcagcctt ggctgcaggt cacttagttc aaagtcatat gacacacaac 1140
agaaagttgt ctggtggtgg ttggtctggt ggtggtggtt ctggtggtgg ttgtagtgcc 1200

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|              |             |            |             |             |             |      |
|--------------|-------------|------------|-------------|-------------|-------------|------|
| gaagcttggt   | ataatttggg  | taacgcata  | tacaagcagg  | gtgactacca  | aaaggccatc  | 1260 |
| gaatactacc   | aaaaggcttt  | ggaattggac | ccaaataacg  | cagaagcctg  | gtacaatttg  | 1320 |
| ggtaattgctt  | actataaaca  | gggtgactat | caaaaggcaa  | ttgaatatta  | ccaaaaggcc  | 1380 |
| ttggaattag   | accctaataa  | cgctgaagca | tggtacaatt  | tggtgaacgc  | ctactataag  | 1440 |
| caggggtgact  | atcaaaaagc  | tattgaagac | tacccaaaag  | cattgggaatt | agatccaaat  | 1500 |
| aaacttgcaag  | ccgaagcttg  | gaaaaatttg | ggtaacgcctt | actacaaaca  | gggtgactac  | 1560 |
| caaaaagcta   | ttgaatacta  | tcaaaaagct | ttggaattgg  | accctaataa  | cgcactgtgc  | 1620 |
| ttggtataaatt | tggttaatgc  | ttattataaa | cagggtgact  | accagaaggc  | aatagaatac  | 1680 |
| tatcaaaaag   | ccttggaatt  | agaccctaat | aacgcctaaag | catggtatag  | acgttggaat  | 1740 |
| gcttactata   | agcagggtga  | ctaccagaaa | gctatagaag  | attatcaaaa  | ggcattggaa  | 1800 |
| ttggatccta   | acaacagatc  | tagatcagct | gggtgtggtg  | gttctggtgg  | tggtggttct  | 1860 |
| gggtgtggtg   | gtgcttccag  | ttattaccat | caccatcacc  | atcacttgga  | atccactagt  | 1920 |
| ttatacaaaa   | aggcagggttc | agaatttttc | agaagagaaa  | gaaataagat  | ggccgctgca  | 1980 |
| aaatgtagaa   | acagaagaag  | agaattgaca | gataccttac  | aagctgaaac  | cgatcaattg  | 2040 |
| gaagacgaaa   | agctctgatt  | gcaaaactga | atagccaatt  | tggtgaagga  | aaagggaaaag | 2100 |
| ttggaattca   | ttttagccgc  | tcatagacca | gcttgcaaaa  | ttctgatga   | cttgggtttc  | 2160 |
| ccagaagaaa   | tggtctttaga | aggttccgca | ggtagtgacg  | ccggttccgg  | tgaatttggt  | 2220 |
| agtgctgaag   | ctgcagccaa  | ggaagctgca | gccaaagctg  | gttctgcagg  | ttcagctgca  | 2280 |
| ggttccggtg   | aaatcgggttc | ttcatactat | caccatcacc  | atcatcactt  | ggaatctacc  | 2340 |
| tcattataca   | aaaaggctgg  | ttccggtagt | caaaagggtg  | aatctttgaa  | gcaaaagatt  | 2400 |
| gaagaattga   | agcaaaagaaa | agcccaattg | aagaatgata  | tcgctaactt  | agaaaaggaa  | 2460 |
| atcgccctacg  | ctgaactggt  | ttctggt    |             |             |             | 2487 |

SEQ ID NO: 190                      moltype = DNA   length = 3147  
 FEATURE                              Location/Qualifiers  
 source                                1..3147  
                                       mol\_type = other DNA  
                                       organism = synthetic construct

SEQUENCE: 190

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| atgagtttac  | catttttgac | atctgctcct  | ggtaaagtta  | ttatattcgg  | tgaacatagt  | 60   |
| gcgctctata  | ataagccagc | tgctgctgca  | tctgtatcag  | ctttgagaac  | atacttgttg  | 120  |
| atctctgaat  | cttcagcacc | tgataccatc  | gaattggatt  | tcccagacat  | ctcatcctaat | 180  |
| cacaagtgtg  | ccattaatga | tttcaacgct  | atcacccgaag | accaagtaaa  | ctcacaaaag  | 240  |
| ttggccaaaag | ctcaacaagc | aactgatggt  | ttgtcacaaag | aattagtttc  | cttgttagac  | 300  |
| ccattgttgg  | ctcaattgtc | cgaaaagtttc | cattaccacg  | ccgctttctg  | tttctgtgac  | 360  |
| atgttctgtt  | gtttatgccc | tcattgctaag | aatatcaaat  | tttctttgaa  | gtctactttg  | 420  |
| ccaatttggt  | caggttttag | ttccagtgcc  | tctatatcag  | tttcttagc   | attggccatg  | 480  |
| gcttatattgg | gtggtttgat | aggttagtaac | gatttggaaa  | agttgtctga  | aaacgacaa   | 540  |
| catatcgta   | accaatgggc | atccatcggt  | gaaaaatgca  | ttccaggtac  | ttctagtgtg  | 600  |
| atagataatg  | agcttgccac | atatggtaac  | gctttgttat  | tcgaaaagga  | ctctcataac  | 660  |
| ggtagcatca  | acactaaca  | cttcaagttc  | ttggatgact  | ttcttgcaat  | accaatgatc  | 720  |
| ttgacttaca  | caagaattcc | aagatctact  | aaagatttgg  | tagctagagt  | cagagtattg  | 780  |
| gttacagaaa  | agttccctga | agttatgaag  | ccaatcttgg  | atgcaatggg  | tgaatgtgac  | 840  |
| ttgcaaggtt  | tggaaatcat | gacaaaagtg  | tcaaaagtga  | agggtagtga  | tgacgaagct  | 900  |
| gttgaaacaa  | ataacgaatt | gtacgaacaa  | ttgttggaat  | tgatcagaat  | caatcatggt  | 960  |
| ttgttagttt  | caatttggtg | ctccaccaca  | ggtttagaat  | tgataaagaa  | cttgtcagat  | 1020 |
| gacttaagaa  | tcggtttcac | aaaattgacc  | gggtgctggt  | gtggtggttg  | ttctttgacc  | 1080 |
| ttgttaagaa  | gagatatcac | tcagaaccaa  | atcgacagtt  | ttaaaaagaa  | attgcaagat  | 1140 |
| gacttctctt  | acgaaccttt | cgaacacagat | ttgggtggta  | ctggttggtg  | cttgtgtgca  | 1200 |
| gctaagaatt  | tgaacaaaga | tttgaagatt  | aaatccttgg  | ttttccaat   | gttcgaaaat  | 1260 |
| aagactacaa  | ccaagcaaca | aatcgatgac  | ttgtgtgtgc  | ctggttaatac | aaacttgcca  | 1320 |
| tggacctcaa  | aattatctgg | tggttggtggt | tctggtggtg  | gtggttctg   | tggttggtggt | 1380 |
| agtgctgaag  | catggtataa | tttgggtaac  | gcataattaca | agcaggttga  | ctacccaaaag | 1440 |
| gctatcgatt  | actacaaaaa | ggcatttgaa  | ttggacccta  | ataacgcgga  | agcttggtac  | 1500 |
| aaatttgggt  | atgcttacta | taaacagggt  | gactatcaaa  | aggccattga  | atattaccaa  | 1560 |
| aaggctttgg  | aattggaccc | aaataacgca  | gaagcctggt  | acaatttggg  | taacgcttac  | 1620 |
| tataagcagg  | gtgactatca | aaaagcaatt  | gaagactacc  | aaaaggcctt  | agaattggat  | 1680 |
| cctaataact  | tgcaagctga | agcatggaaa  | aatttgggta  | acgcttatta  | taaacagggt  | 1740 |
| gactacaaaa  | aagccattga | atactatcaa  | aaagcattgg  | aattggatcc  | aaataacgcc  | 1800 |
| tctgcttggt  | ataatttggg | taatgcttat  | tataagcagg  | gtgactacca  | gaaagccata  | 1860 |
| gaataactatc | aaaaagcttt | ggaattagac  | cctaataacg  | caaaagcctg  | gtatagacgt  | 1920 |
| ggtaattgctt | actacaaaca | gggtgactat  | cagaaggcaa  | tagaagatta  | tcaaaaagct  | 1980 |
| ttagaattag  | accocaaata | cagaagtaga  | tctgctggtg  | gtggtggttc  | tggttggtggt | 2040 |
| ggttctggtg  | gtggtggtgc | ttctatggaa  | cctgcaatgg  | aaccagaaac  | attggaaagc  | 2100 |
| agaatcaata  | gagctaccaa | tcctttgaac  | aaggaattgg  | attgggcttc  | tattaatggt  | 2160 |
| ttctgtgaac  | aattgaacga | agacttcgaa  | ggtccacctt  | tagcaacaag  | attattggcc  | 2220 |
| caataaaattc | aatcaccaca | agaatgggaa  | gcaatacaag  | ccttaaccgt  | cttggaaact  | 2280 |
| tgtatgaagt  | cctcggttaa | aagattccac  | gatgaagttg  | gtaaatcag   | atttttgaa   | 2340 |
| gaattgatca  | aggttgtctc | acctaagtat  | ttgggtagta  | gaacatctga  | aaagggttaa  | 2400 |
| aacaagatct  | tggaaattgt | gtactcctgg  | accgtaggtt  | taccagaaga  | agtttaagatc | 2460 |
| gctgaagcat  | accaaatggt | gaagaacaaa  | ggtattgtta  | agtcaggttc  | cgccggtagt  | 2520 |
| gcagccggtt  | ctggtgaatt | cggttctgca  | gaagctgcag  | ccaagggaagc | tcagcccaaa  | 2580 |
| gctggttcag  | caggttccgc | tgacagttct  | ggtgaatttg  | gttcagggtc  | aatgggttcc  | 2640 |
| atggccgaag  | ctgaaggtga | aagtttgtaa  | tccttggtta  | ataaggctac  | aaatccatca  | 2700 |
| aacagacaag  | aagatttgga | atatatcatt  | ggtttctgtg  | accaaatcaa  | taaggaattg  | 2760 |
| gaaggtcttc  | aaatagctgt | tagattattg  | gcacataaga  | tccaatctcc  | acaagaatgg  | 2820 |
| gaagccttac  | aagctttgac | tgttttagaa  | gcttgtagta  | agaattgcgg  | tagaagattt  | 2880 |
| cacaacgaag  | tcggtataat | cagatttttg  | aatgaattaa  | ttaaggtagt  | tagtccaaaa  | 2940 |
| tacttaggtg  | acagagtttc | tgaagggtt   | aagaccaaag  | tcatagaatt  | gttgtagtct  | 3000 |

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|            |             |            |            |            |            |      |
|------------|-------------|------------|------------|------------|------------|------|
| tggactatgg | ccttgccctga | agaagctaag | atcaaagatg | cataccatat | gttgaagaga | 3060 |
| caaggtatag | tccaatcaga  | tcacacctac | ccagtagaca | gaactttgat | tccatctcca | 3120 |
| ccaccaagac | ctaaaaatgg  | ttccgggt   |            |            |            | 3147 |

SEQ ID NO: 191                   moltype = DNA   length = 2610  
 FEATURE                       Location/Qualifiers  
 source                         1..2610  
                               mol\_type = other DNA  
                               organism = synthetic construct

SEQUENCE: 191

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| atgtccgaat  | taagagcttt  | tagtgcacct  | ggtaaagcct  | tgtagctgg  | tggttatttg  | 60   |
| gttttgata   | caaagtacga  | agcattcggt  | gtcggtttgt  | cagccagaat | gcatgcagtc  | 120  |
| gcccaccctt  | acggttcttt  | acaaggttct  | gataagttcg  | aagtaagagt | caagtctaag  | 180  |
| caattcaagg  | acgggtgaatg | gttatacctat | atatctccaa  | agtcaggttt | tattcctgtt  | 240  |
| tccataggtg  | gtagtataaaa | tccattctac  | gaaaagggtta | ttgcaaacgt | cttttcttac  | 300  |
| ttcaagccta  | acatggatga  | ctactgtaac  | agaaacttgt  | tcgtcatcga | tatatctct   | 360  |
| gatgacgctt  | atcattctca  | agaagactca  | gtaactgaac  | acagaggtaa | tagaagattg  | 420  |
| tcctttcata  | gtcacagaat  | tgaagaagtt  | ccaaaaaccg  | gtttagggtc | ttcagctggt  | 480  |
| ggtttagtca  | ctgtattgac  | tacagcttta  | gcctcctttt  | tcgttagtga | tttgaaaaac  | 540  |
| aacgtagaca  | agtacagaga  | agttattcat  | aatttggcac  | aagtagccca | ctgccaaaga  | 600  |
| caaggtaaaa  | tcgggttcggg | ttttgatggt  | gctgcagccg  | cttatgggtc | aattagatac  | 660  |
| agaagattcc  | cacctgcttt  | gatatcta    | ttgcccagata | tcggttctgc | tacatatggt  | 720  |
| tcaaagttgg  | cacatttggg  | tgatgaagaa  | gactggaaca  | tcacaattaa | atccaacccat | 780  |
| ttgcttagtg  | gtttgacctt  | atggatgggt  | gacattaaga  | atggttctga | aactgttaag  | 840  |
| ttggtccaaa  | aagtaaaaga  | ctggtagcat  | tctcatatgc  | cagaatcatt | gaagatctac  | 900  |
| acagaattag  | acatgcttaa  | ttccagattc  | atggatgggt  | tgagtaaat  | agacagattg  | 960  |
| catacccacg  | atgactactc  | tgaatcaaat  | ttcgaatcat  | tggaagaaa  | cgactgtact  | 1020 |
| tgccaaaaat  | accagaaata  | ccagaaagta  | agagatgccg  | ttgctaccat | aagaagatct  | 1080 |
| tttagaaaaga | tcactaaagga | atcaggtgca  | gatatcgaa   | cacctgttca | aacatctttg  | 1140 |
| ttagatgact  | gtcaaacctt  | gaaggggtgc  | ttaacttgct  | tgattccagg | tgctgggtgt  | 1200 |
| tatgatgcaa  | tgcogtcat   | cactaaacaa  | gatgtagact  | tgagagctca | aacagcaaac  | 1260 |
| gataagagat  | tttcaaaagt  | ccaatgggtta | gatgtaaccc  | aagctgactg | gggtgttaga  | 1320 |
| aaagaaaagg  | atcctgaaac  | ttacttggac  | aaaaagttat  | ctggtgggtg | tggttctggt  | 1380 |
| gggtggtggt  | ctggtggtg   | ggtagtgct   | gaagcatggt  | acaatttggg | taacgcatac  | 1440 |
| tacaagcagg  | gtgactacca  | aaaggccata  | gaatactacc  | aaaaggcttt | ggaattggac  | 1500 |
| ccaaataaac  | ccgaagcttg  | gtataatttg  | ggtaagtctt  | attataaaca | gggtgactat  | 1560 |
| caaaaggcaa  | tcgaataacta | ccaaaaggcc  | ttggaattag  | accctaataa | cgcagaagcc  | 1620 |
| tggtataatt  | tggttaacgc  | ttattataag  | caggggtgact | atcaaaaagc | tatcgaagac  | 1680 |
| taccaaaagg  | catctggaatt | agatccaaat  | aacttgcaag  | ctgaagcatg | gaagaatttg  | 1740 |
| ggtaacgctt  | actataaaaca | gggtgactac  | caaaaagcca  | ttgaatatta | tcaaaaagct  | 1800 |
| ttggaatttg  | atcctaataa  | cgctctgct   | tggtacaatt  | tggttaatgc | ttactataag  | 1860 |
| caggggtgact | atcagaaggc  | tattgaatat  | tatcaaaagg  | ctttagaatt | ggacccta    | 1920 |
| aacgcaaaag  | cttggtacag  | acgtggtaac  | gcttattaca  | aacagggtga | ctaccagaaa  | 1980 |
| gctattgaag  | attatcaaaa  | ggcattggaa  | ttggatccta  | acaacagatc | cagaagtgtc  | 2040 |
| gggtggtggt  | gttctggtg   | tggtggttct  | gggtggtggt  | gtgtctccag | ttattaccat  | 2100 |
| caccatcacc  | atcatttgga  | atcatcatca  | ttatacaaaa  | aggctgggtc | cggtagtcaa  | 2160 |
| aaggttgaag  | aattgaaaaa  | taagatagcc  | gaattggaaa  | acagaaaacg | tgtaaaaaag  | 2220 |
| aaacagagtc  | cacatttgaa  | acaagaataa  | gcctacttga  | aggatgaatt | agcagcccat  | 2280 |
| gaatttgaag  | gttctgccc   | ttcagctgca  | ggttctggtg  | aattcgggtc | agctgaagcc  | 2340 |
| gtgcgaaaag  | aagccgctgc  | aaaggccggt  | tcgctggtga  | gtgcccgtgg | ttctggtgaa  | 2400 |
| tttggttctt  | catactatca  | ccatcaccat  | catcacttgg  | aactctactc | attatataaa  | 2460 |
| aaaggccggt  | ccggttagtt  | cgaaaacggt  | acacatgaat  | tcattttggc | taccttgtaa  | 2520 |
| aacgaaaacg  | caaagttaag  | aagattggaa  | gccaaagttg  | aaagagaatt | agctagattg  | 2580 |
| agaaatgaag  | ttgcatggtt  | aggttctggt  |             |            |             | 2610 |

SEQ ID NO: 192                   moltype = DNA   length = 3033  
 FEATURE                       Location/Qualifiers  
 source                         1..3033  
                               mol\_type = other DNA  
                               organism = synthetic construct

SEQUENCE: 192

|            |            |            |             |             |             |      |
|------------|------------|------------|-------------|-------------|-------------|------|
| atgacagttt | ataccgcttc | tgtcaccgca | cctgtaaaata | ttgctacttt  | gaaatactgg  | 60   |
| ggtaaaagag | atactaagtt | gaatttgcca | acaaactctt  | caatctcagt  | tacattgtcc  | 120  |
| caagatgact | taagaacctt | gacttctgct | gcaactgctc  | ctgaattcga  | aagagataca  | 180  |
| ttgtgggtga | atggtgaacc | acattctatc | gacaacgaaa  | gaactcaaaa  | ctgtttgaga  | 240  |
| gatttgagac | aattgagaaa | ggaaatggag | agtaaggatg  | cttctttgcc  | tacattgagt  | 300  |
| caatggaagt | tgacacatag | ttctgaaaac | aacttcccaa  | ccgcccgtgg  | tttggcatcc  | 360  |
| agtgacgccc | gtttcgctgc | attagtctct | gcaatcgcca  | agttgtacca  | attgccacaa  | 420  |
| agtacatctg | aaatcagtag | aatcgctaga | aaaggttcag  | gttccgcatg  | tagatcttta  | 480  |
| tttggtggtt | acgtgcgatg | ggaaatgggt | aaagccgaag  | acggtcatga  | ttcaatggcc  | 540  |
| gtacaaatag | ctgactcttc | agattggcct | caaagtgaag  | cttgctgtct  | ggttgtctca  | 600  |
| gacatcaaaa | aggatgtatc | cagtacacaa | ggcatgcaat  | tgactgttgc  | aacatccgaa  | 660  |
| ttggttaaa  | aaagaatcga | acacgtagtt | ccaaaaagat  | tcgaagtcac  | gagaaaaggct | 720  |
| atcgtagaaa | aggattttgc | caccttcgct | aaggaaaacta | tgatggacag  | taactctttc  | 780  |
| catgcaactt | gtttggattc | atttccacct | attttctata  | tgaacgcacac | ctcaaagaga  | 840  |
| ataatctctc | gggtccacac | tatcaaccaa | ttctacggtg  | aaacaatcgt  | tgcttacacc  | 900  |
| ttcgatgcag | gtcctaagtc | cgtcttgat  | tacttagccg  | aaaacgaatc  | aaagttgttc  | 960  |
| gtctttatat | ataagttggt | tggttcoggt | ccaggttggg  | ataaaaagtt  | cactacagaa  | 1020 |
| caattggaag | cttttaatca | tcaattcgaa | tcttcaaac   | ttactgccag  | agaattggac  | 1080 |

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|            |             |             |            |             |             |      |
|------------|-------------|-------------|------------|-------------|-------------|------|
| ttagaattgc | aaaaggatgt  | agctagagtt  | atcttgaccc | aagttgggtc  | aggctctcaa  | 1140 |
| gaaactaacg | aatccttgat  | agatgctaag  | acaggtttgc | caaaagaaaa  | attgtctggt  | 1200 |
| ggtggtggtt | ctggtggtgg  | tggttctggt  | ggtggttgta | gtgctgaagc  | atggtataat  | 1260 |
| ttgggtaacg | cttattacaa  | gcagggtgac  | taccaaagg  | ccatcgataa  | ctaccaaag   | 1320 |
| gctttggaat | tggaccctaa  | taacgcgaa   | gcttggtaca | atttgggtaa  | tgctactat   | 1380 |
| aaacagggtg | actatcaaaa  | agcaattgaa  | tattaccaa  | aggccttgga  | attggaccca  | 1440 |
| aataacgcag | aagcctggta  | caatttgggt  | aacgcctact | ataagcagg   | tgactatcaa  | 1500 |
| aaggctatcg | aagattacca  | aaaggcatta  | gaattggatc | ctaataactt  | gcaagctgaa  | 1560 |
| gcatggaaaa | atttgggtaa  | tgctattat   | aaacagggtg | actacaaaa   | agctattgaa  | 1620 |
| tactatcaaa | aagctttaga  | attagaccca  | aataacgcct | cagcttggtg  | taatttgggt  | 1680 |
| aatgcatact | acaaacagg   | tgactatcag  | aaggcaattg | aatactatca  | aaaggcatta  | 1740 |
| gaattagatc | ctaataacgc  | aaaagcctgg  | tatagacgtg | gtaatgccta  | ctacaagcag  | 1800 |
| ggtgactatc | agaaggcgat  | tgaagactac  | caaaaggcat | tggaaattgga | tccaaacaac  | 1860 |
| agatcaagat | ccgctgggtg  | tggtggttct  | ggtggttggt | gttctggtgg  | tggtggtgct  | 1920 |
| tctgcaatgg | ccgatttgga  | acaaaaggta  | ttggaaatgg | aagctagtac  | atatgacggt  | 1980 |
| gtttttattt | ggaagatctc  | tgatttccca  | agaaaaagac | agaagctgt   | tgcaagtaga  | 2040 |
| atccttgcta | tttttagtcc  | agcattctac  | acctctagat | acggttacaa  | gatgtgtttg  | 2100 |
| agaatatatt | tgaatggtga  | cggtactggt  | agaggtaact | atttgccttt  | gtttttcgtc  | 2160 |
| gtaatgaagg | gtcctaata   | tgctttggtg  | agatggcctt | ttaatcaaaa  | ggttaccttg  | 2220 |
| atgtgtgttg | atcaaaaacg  | cagagaacac  | gttatcgacg | cttttagacc  | tgatgtcact  | 2280 |
| tcacgttctt | tcacaaagacc | agttaatgat  | atgaacattg | cttctggttg  | tcctttgttt  | 2340 |
| tgccagctct | caaagatgga  | agctaaaaat  | tcctatggtg | gagatgacgc  | catcttcatt  | 2400 |
| aaggctatcg | ttgatttgac  | tggttttaggt | tcagcaggtt | ccgcccgttg  | ttctggtgaa  | 2460 |
| tttggttcgc | ccgaagcagc  | cgctaaggaa  | gcagccgcta | aagcaggtag  | tgcccgttct  | 2520 |
| gcagccggtc | ctggcggaatt | tggtagtggc  | tctattaaat | tgcaatcatc  | cgacgggtgaa | 2580 |
| atcttcgaag | ttgatgtcga  | aatagcaaa   | caatctgtta | ccataaaaa   | tatgttgtaa  | 2640 |
| gatttggtga | tggatgacga  | agtgacgat   | gatccagttc | ctttgccaaa  | tgtaaacgct  | 2700 |
| gcaatattga | agaaagttat  | tcgaatgggc  | acacatcaca | aggacgatcc  | acctccacct  | 2760 |
| gaagacgatg | aaataaagga  | aaagagaact  | gacgatattc | cagtatggga  | ccaagaattc  | 2820 |
| ttgaaggttg | atcaaggtac  | attgttcgaa  | ttgatcttgg | ccgctaacta  | tttgacatc   | 2880 |
| aagggtttgt | tagatgtaac  | atgtaaaacc  | gttgctaaca | tgatcaagg   | taaaacacca  | 2940 |
| gaagaaatca | gaaagacctt  | taatatgaag  | aatgatttca | ctgaagaaga  | agaagcacaa  | 3000 |
| gttagaaagg | aaaaccaatg  | gtgcggttct  | ggt        |             |             | 3033 |

SEQ ID NO: 193                      moltype = DNA   length = 2160  
 FEATURE                              Location/Qualifiers  
 source                                1..2160  
                                          mol\_type = other DNA  
                                          organism = synthetic   construct

SEQUENCE: 193

|             |            |             |             |             |             |      |
|-------------|------------|-------------|-------------|-------------|-------------|------|
| atgactgctg  | ataataactc | tatgccacat  | ggtgcggtat  | cttcatacgc  | taagttggtt  | 60   |
| caaaacaaaa  | cacctgaaga | tatcttgtaa  | gaattccagc  | aaatcatccc  | tttgcaacaa  | 120  |
| agaccacaaa  | ctagatccag | tgaaacatcc  | aacgatgaaa  | gtggtgaaac  | ctgtttttca  | 180  |
| ggtcatgacg  | aagaacaaat | taaaattgatg | aacgaaaact  | gcacgtgatt  | ggattgggat  | 240  |
| gacaatgcaa  | taggtgccc  | tactaagaaa  | gtttgtcatt  | tgatggaaaa  | catagaaaa   | 300  |
| ggtttggtgc  | acagagcttt | ctccgttttt  | atattcaatg  | aacagggtga  | attgttattg  | 360  |
| caacaaagag  | caacagaaaa | gatcaccttt  | ccagatttgt  | ggactaatac  | atgttgtctc  | 420  |
| catcctttgt  | gcatttgatg | cgaattaggt  | ttgaagggtg  | aattggatga  | caaaattaa   | 480  |
| ggtgctataa  | ctgctgcagt | cagaaaaatta | gatcatgaat  | tgggtatacc  | agaagacgaa  | 540  |
| accaagactc  | gtggttaaat | ccatttctta  | aacagaatcc  | actatatggc  | tccatctaac  | 600  |
| gaaccttggg  | gtgaacatga | aatcgattac  | atcttatttt  | acaagattaa  | tgcaaaaggaa | 660  |
| aaacttgacg  | ttaaccacaa | cgtaaatgaa  | gtcagagatt  | tcaaatgggt  | ttctccta    | 720  |
| gatttgaaga  | ccatgtttgc | tgaccatca   | tataagtta   | ctccttggtt  | caagatcatc  | 780  |
| tgtgaaaact  | actgttttaa | ctggtgggaa  | caattagatg  | actgtctgga  | agttgaaaac  | 840  |
| gatagacaaa  | tccatagaat | gttgaaaattg | tctggtggtg  | gtggttctgg  | tggtggtggt  | 900  |
| tctggtggtg  | gtggtagtgc | cgaagcttgg  | tacaatttgg  | gtaacgctta  | ctacaagcag  | 960  |
| ggtgactacc  | aaaaggcaat | cgaatactac  | caaaaggcct  | tggaaattgga | cccaataaac  | 1020 |
| gcagaagcct  | ggtataattt | gggttaattg  | tattataaac  | agggtgacta  | tcaaaaggct  | 1080 |
| attgaatatt  | acaaaaaggc | attggaattg  | gaccctaata  | acgctgaagc  | atggtataat  | 1140 |
| ttgggtaacg  | cctattataa | gcagggtgac  | tatcaaaaag  | ccatcgaa    | ctaccaaag   | 1200 |
| gctttggaat  | tggatccaaa | caacttacaa  | gccgaagctt  | ggaagaattt  | gggtaacgct  | 1260 |
| tattacaaac  | agggtgacta | ccaaaaagct  | attgaatact  | atcaaaaagc  | cttagaatta  | 1320 |
| gaccctaata  | acgcatctgc | ctggtacaa   | ttgggttaatg | cctattacaa  | gcagggtgac  | 1380 |
| tatcagaagg  | ctattgataa | ctacaaaaaa  | gcattggaat  | tggatccaaa  | taacgcctaag | 1440 |
| gcatgggtaca | gacgtggtga | tgctattac   | aagcagggtg  | actatcaaaa  | ggcgattgaa  | 1500 |
| gattatcaaa  | aagctttgga | attggatcct  | aacaacagat  | ctagatcagc  | tggtggtggt  | 1560 |
| ggttctggtg  | gtggtggttc | tggtggtggt  | ggtgcttctt  | catattacca  | tcaccatcac  | 1620 |
| catcacttag  | aatccacaag | tttgtacaaa  | aaggctggtt  | ctggttcaaa  | cacogttaag  | 1680 |
| gaattaaaga  | actacatcca | agaattggaa  | gaaagaaacg  | cagaattgaa  | aaatttgaag  | 1740 |
| gaacatttga  | agtttgccaa | ggctgaatta  | gaattcgaat  | tgggcgcctc  | caaaatttga  | 1800 |
| ggttcgcgtg  | gtagtgcagc | cggttcggtg  | gaattcggtg  | gtgcagaagc  | tgacgcaaaa  | 1860 |
| gaagctgcag  | ccaagctggt | ttctgcaggt  | tcagctgcag  | gttctggtga  | atttgggttc  | 1920 |
| agttactatc  | accatcacca | tcatcactta  | gaatccacta  | gtttgtataa  | aaaggccggt  | 1980 |
| tctggttcac  | aaaaagtcgc | aatagagtag  | cctacaagtg  | gaaggaaaa   |             | 2040 |
| gctaagtttg  | aaaacattgt | cgcaagatta  | gaaaacgata  | atgccaaact  | ggaagaaagc  | 2100 |
| atcgctaatt  | tggaaaagg  | tattgcaaac  | ttgaaaagag  | atgttgccag  | aggttctggt  | 2160 |

SEQ ID NO: 194                      moltype = DNA   length = 2403  
 FEATURE                              Location/Qualifiers

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source          1..2403
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 194
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aatttgatcg tacaaatcaa cagagttatg aatttgccata aggatcaatt ggctatcggt 180
tctcaaatag tcgaattggt gcataactcc agtttggtga tcgatgacat cgaagataac 240
gcaccattga gaagagggtca aactacatcc cacttaattt ggggtgtccc tagtactatt 300
aataccgcaa actacatgta ctccagagcc atgcaattgg tatcacaatt gaccactaag 360
gaaccattgt accattgggt gatcacaaat tttaacgaag aattgattaa ttgacacaga 420
ggtcagggtt tggatatcta ttggagagac ttcttaccag aaattatacc taccacagaa 480
atgtacttga acatggtaat gaataagact ggtggtttgt ttagattgac cttgagatta 540
atggaagcct tgctctccat ttccatccac ggtcattcat tggttccttt cataaacttg 600
ttgggtatca tctatcaaat cagagatgac tacttgaatt tgaaggattt ccaaatgtcc 660
agtgaaaagg gtttcgcaga agacataact gagggtaaat tgcattccc aatcgtccat 720
gccttaaaat tcacaaaaac caagggtcaa accgaacaac acaatgaaat cttaagaatt 780
ttgttattga gaacttctga taaggacata aagttgaagt tgatccaaat cttggaattc 840
gataccaact cattggccta cactaagaac ttcatcaacc aattgggtta catgattaag 900
aatgataacg aaaataagta cttgccagat ttggcctccc atagtgcac tgctacaaat 960
ttgcacgatg aattgtgtga catcatcgac catttgtccg aattgaaatt atctggtggt 1020
ggtggttctg gtggtggtgg ttctggtggt ggtggttagt cagaagcctg gtacaaactg 1080
ggtaacgctt actacaagca ggggtgactac caaaaaggcta tcgaatacta ccaaaaggca 1140
ttggaattag acccaataa cgctgaagca tggtaacaat taggcaacgc atattataaa 1200
cagggtgact atcaaaaggc tactagaatac taccaaaagg ctttgaatt ggaccctaat 1260
aacgccgaag cttggtacaa cttgggtaat gcttattaca agcagggtga ctatcaaaaa 1320
gcaattgaag actacaaaa agccttgga ttagatccaa ataacttgca agcagaagcc 1380
tggagaagac taggcaacgc atactataaa cagggtgact accaaaagc cattgaatat 1440
tatcaaaaa ctttggaaat agaccctaat aacgcttctg cttggtataa cttaggcaat 1500
gcctattata agcagggtga ctatcagaaa gctattgaat attatcaaaa ggccttgga 1560
ttggacccaa ataacgccaa ggcttggtac agcgtggtga acgcatacta caaacagggt 1620
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agagctcaag atagaattgt cgaagtaaat ggtgtctgca tggaaaggta acaacacggt 1920
gacgttggtt ctgctattag agctggtggt gacgaaacta agttattggt agttgacaga 1980
gaaggttccg ccggtagtgc tgcaggttct ggtgaatttg gttcagctga agccgctgca 2040
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ggtggtttag ctgaaagaac aggtgcaatc catattgtgt acagaatatt ggctatcaat 2280
tcatccagtt tgaaggttaa accattgtca gaagctatcc acttattgca aatggcaggt 2340
gaaaccgtta ctttgaaaat caaaaagcaa acagatgcac aacctgctc ttcaggttct 2400
ggt 2403

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SEQ ID NO: 195      moltype = DNA length = 2658
FEATURE            Location/Qualifiers
source              1..2658
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 195
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catatgacgc agttgaagga aaagtttaga aagatctgtg ataagagtat gatcagaaag 180
aggaactgct tcttaaacga agagcatttg aagcaaaatc ctagattagt ggaacacgag 240
atgcaaacat tggatgctag gcaggacatg ttagttgtcg aagttcctaa attgggtaaa 300
gatgcatgtg ccaaaagctat taaggaaatgg ggtcaaccca agtctaagat aactcatttg 360
atttttacta gtgctagcac tacagatatg cctggtgcag actatcactg tgccaaacta 420
cttggtttat cgccctctgt gaagagagtt atgatgtatc aactaggttg ctacgggtgt 480
ggtagctgtc ttagaatcgc taagacatt gcagaaaata acaagggtgc caggggtctt 540
gctgtatggt gcgatattat ggcttgcttg tttagaggtc catcagaatc cgatttggag 600
ctgttggttg gtcaagctat ttccggtgac ggtgctgcag ctgttattgt tgggtgcagaa 660
cctgataggt cagtcogtga aagaccaatc tttgaattgg tttctacogg tcaaacgatt 720
ttaccaaata gtgaaggtag aataggtggt catatcagag aagctggttt gatattcgat 780
ttgcacaaag acgttccat gctaatatct aacaacatcg aaaagttct gatcagaggt 840
tttaccacca tcggtatttc cgattggaat agtatattct ggatcacgca tccaggtggt 900
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gatgaactta gaaagagatc tttggaagag ggtaaaagca ccacgggtga cggttttgaa 1080
tgggtgttcc tttttgatt cggtcccggt ttgaccgtcg aaagagtatg tgttagatct 1140
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ggtggtagtg cagaagcctg gtacaatttg ggtaacgctt actacaagca ggggtgactac 1260
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gcctattata agcagggtga ctatcagaag gcaatagagg actacaaaaa agccctagaa 1500
cttgatccaa ataatttgca ggcagaagcc tgggaagaat tgggtaatgc ttactataaa 1560
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aacgcttctg cttggtacaa tttgggcaac gcttactaca aacaggggtga ctacccaaaa 1680
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gggtggtggt ctggtggtgg tgggtgcttc ggtaacaact tagaaacata cgagtggtac 1920
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aaatactttt tgtgggtcgt aaagtcaat tccttgaacg aattagtga ttaccataga 2580
tcaacttccg ttgtaggaa ccaacagatt ttcttgagag atatcgaaac agttccacaa 2640
cagcctacag gttctgga 2658

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SEQ ID NO: 196      moltype = DNA length = 2004
FEATURE            Location/Qualifiers
source              1..2004
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 196
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tactggggtg aagacgtgac ccaaaagaat aaggaagagg gttataccca tatagtagaa 180
gtgacgttgc aatcagttga aactatccaa gattacatca tacacctgc tcatgttggc 240
tttggtgacg tctacagatc cttctgggaa aagttgctga tcttcgatta cactccaaga 300
aagaaattgt ctggtggtgg tggttctggt ggtggtggtt ctggtggtgg tggtagtgca 360
gaagcctggt ataatttggg aaacgcttat tacaacagg gtgactacca aaaggccatc 420
gagttattcc aaaaagctct tgaactggac ccaataaacc ctgagggcatg gtataatttg 480
ggtaacgcac actataagca aggtgactac caaaaggcaa ttgaatatta ccaaaaggcc 540
ttggagttag accctaataa cgccgaagct tgggtacaatt tgggtaatgc ctactataaa 600
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caaaaagcta ttgaatacta ccaaaaagct ctgggaattgg accctaataa cgcttctgct 780
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gcctactaca agcaggggtg ctacccaaaa gctattgagg actacccaaa agcacttgaa 960
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acttttacat tccagacatc acaaggcaat gatatcttcc aggcagttga agccgctatt 1920
caacagcaaa aagcccaggg taaagtcgga caggctcaag acattctaag attggaacac 1980
catcaccatc atcatggttc tgggt 2004

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SEQ ID NO: 197      moltype = DNA length = 2400
FEATURE            Location/Qualifiers
source              1..2400
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 197
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aagtatagct acaacaattt cccatcgaaa cattgtagta ctaagagctt ccatttgcaa 180
aataagtgtc ccgaatcttt gtctatcgct aagaactcaa ttagagctgc aactacaaat 240
cagacggaac cactgagtc ggataatcac tctgtagcca ccaaaatttt gaacttggt 300
aaagcttggt ggaagctgca aagaccatac acaataatag ccttcacctc ctgtgcttgc 360
ggttgtttg gtaagaact gttgcataac acaaaattga tttcgtggtc ttgatgttc 420
aaggcatttt tctttttggt tgcattcctt tgcatcgctt cttttaccac gactattaat 480
caaatctatg atttgcacat gcacagaatt aataagcccg atttgcact agcttcagg 540
gaaatctccg ttaatactgc atggattatg tcaatcattg tcgcttgggt cggtttaatc 600
atcacaaata aaatgaaagg tgggtccattg tacatcttgc gctactgttt cggtatatc 660
gggtggtatg tatattccgt tccacctttt agatggaaac aaaaccccag taccgctttc 720
ttactaaatt tcttggcaca tatcatcaca aacttcacct tctactacgc ttctagagct 780

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gctttgggtt  tgcattcgca  attaagacca  tcttttacat  ttttctgggc  ttttatgaaa  840
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ttcggcatca  gtaccttggc  tagcaagtac  ggttctagaa  atttgacttt  gttttgttca  960
ggtatcgat  tgttatccta  cgtggcagcc  attttagccg  gtatcatttg  gccacaagct  1020
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acgagagatt  ttgctttaac  taattatgac  cctgaggcag  gtagaagatt  ctacgaattc  1140
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ggtggttctg  gtggtggttg  ttctggtggt  ggtggttagt  ctgaagcatg  gtacaactta  1260
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ttagaactag  acccaaacaa  tgccgaggct  tggataaact  tgggcaatgc  ttattacaaa  1380
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| gcaggtctctg | ctgctggttc  | tgccgaattt  | tcctactatc  | atcatcatca  | tcactacttg  | 8640 |
| gaatccacaa  | gtttatacaa  | gaagcgggtt  | agtgaatttt  | tcagaagaga  | aagaaacaag  | 8700 |
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FEATURE             Location/Qualifiers
source              1..1365
                    mol_type = other DNA
                    organism = synthetic construct

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catcaccatc accatcactt ggaatccacc agtttataca aaaaggctgg ttctggttca 180
gctagaaaacg catatttgag aaagaaaatt gctagattga agaaagataa cttgcaattg 240
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gcttatttgc aaaaggaaat cgcaaaactg agaaagaaaa tagaagaagg ttcgctggt 660
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SEQ ID NO: 203      moltype = DNA length = 19104
FEATURE             Location/Qualifiers
source              1..19104
                    mol_type = other DNA
                    organism = synthetic construct

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gctagattgt tacaagatca tccatggttg ttatctcaaa acttggttgt caaacctgac 180
caattaatta agagaagagg taaattgggt ttagtagggt ttaatttgac attggtggt 240
gtaaagtctt ggttgaaacc aagattaggt caagaagcca cagttggtta agctaccggt 300
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| gacgtagggtg | acgtttgacgc | caaggctcaa  | aagttgttgg  | ttggtgtcga  | tgaaaagtgtg | 480  |
| aaccacagaag | acattaaaaa  | gcattttgttg | gttcacgcac  | ctgaagataa  | aaaggaaata  | 540  |
| ttggcctcct  | ttataagtgg  | tttgtttaat  | ttctacgaag  | atttgtactt  | cacctacttg  | 600  |
| gaaattaaac  | cattagtagt  | tactaaggat  | ggtgtatatg  | ttttggactt  | agctgcacaaa | 660  |
| ggtgatgcaa  | cagccgacta  | cattttgaag  | gtcaaatggg  | gtgacatcga  | atttccacct  | 720  |
| ccatcggtta  | gagaagctta  | tccagaagaa  | gcctacattg  | ctgatttggg  | cgctaagtct  | 780  |
| ggtgcatcat  | tgaagttgac  | attgttgaac  | cctaaaggta  | gaattttggg  | catgggttgc  | 840  |
| ggtggtgggtg | ctagtgtcgt  | atattctgat  | actatatcg   | acttgggtgg  | tgttaacgaa  | 900  |
| ttggcaaac   | acggtgaata  | ctcaggtgcc  | ccatccgaac  | aacaaacata  | cgattacgct  | 960  |
| aagaccatct  | tgtcctta    | gactagagaa  | aagcatcctg  | atggtaaaat  | cttgatcatc  | 1020 |
| ggtggttagta | tcgcaaat    | tactaacgtt  | gccgctacat  | tcaagggtat  | cgtcagagct  | 1080 |
| ataagagatt  | accaaggtcc  | attgaaggaa  | cacgaagtaa  | caatattcgt  | tagaagaggt  | 1140 |
| gggtcctaact | ccaagaagg   | tttgagagt   | atgggtgaag  | taggtaaaac  | cactggtata  | 1200 |
| ccaatccatg  | tctttgtgtac | agaaacccac  | atgactgcaa  | tagttggtat  | ggccttaggt  | 1260 |
| catagaccac  | tcctaatca   | acctccaacc  | gcagccacac  | ctgcaaat    | cttgttaaac  | 1320 |
| gcctctggtt  | caacttccac  | accagctcct  | tctagaacag  | caagtttctc  | tgaatcaaga  | 1380 |
| ggtgatgaag  | tcgctccagc  | taagaagca   | aaaccagcca  | tgctcaca    | ctcgttcca   | 1440 |
| agtctcatgat | acttgcagg   | taaatctact  | actttgtttt  | ctagacatac  | taaggctata  | 1500 |
| gtatgggtta  | tgcaaacaa   | agcagttcaa  | ggcatgttgg  | atttcgacta  | tgtttgtagt  | 1560 |
| agagatgaac  | catctgttgc  | tgcaatgggtc | tatcctttta  | ctggtgacca  | taagcaaaaa  | 1620 |
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| agaaaacatc  | ctgaagtcga  | cgtattgatt  | aaactcgctc  | cattaagatc  | cgcttacgat  | 1740 |
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| attccagaag  | ccttgactag  | aaagtgtgatt | aagaaagctg  | atcaaaaagg  | tgtcacataa  | 1860 |
| atcggtccag  | ccctcgtagg  | tggtatttaag | cctggttgtt  | tcaagatcgg  | taacactggt  | 1920 |
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| tcagaagtgt  | gtggtatgtc  | caacgaattg  | aacaacatca  | tcagtagaac  | tacagatggt  | 2040 |
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moltype = DNA length = 18738

FEATURE

Location/Qualifiers

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source          1..18738
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| ggtcctaact  | accaagaagg  | tttgagagtc  | atgggtgaag | taggtaaaa   | cactggtata  | 1200 |
| ccaatccatg  | tccttggtag  | agaaacccac  | atgactgcaa | tagttggtat  | ggccttaggt  | 1260 |
| catagaccaa  | tccttaatac  | acctccaacc  | gcagcccaca | ctgcaaaatt  | cttgttaaac  | 1320 |
| gcctctggtt  | caacttccac  | accagctcct  | tctagaacag | caagtttctc  | tgaatcaaga  | 1380 |
| gctgatgaag  | tcgctccagc  | taagaagaca  | aaaccagcca | tgccctcaaga | ctcgttcca   | 1440 |
| agtcctagat  | ctttgcaggg  | taaatctact  | actttgtttt | ctagacatac  | taaggctata  | 1500 |
| gtatggggtg  | tgcaaaacaag | agcagttcaa  | ggcatgttgg | atttcgacta  | tggttgtagt  | 1560 |
| agagatgaac  | catctgttgc  | tgcaatgggt  | tatcctttta | ctggtgacca  | taagcaaaaa  | 1620 |
| ttctactggg  | gtcacaaagg  | aataattgatc | ccagttttta | agaacatggc  | cgatgctatg  | 1680 |
| agaaaaacac  | ctgaagtcca  | cgtattgatt  | aacttcgcct | cattaagatc  | cgcttacgat  | 1740 |
| tctacaatgg  | aaaccatgaa  | ctacgctcaa  | ataagaacca | tcgctatcat  | tcgagaaggt  | 1800 |
| attccagaag  | ccttgactag  | aaagtgtgatt | aagaaagctg | atcaaaaagg  | tgccacaata  | 1860 |
| atcggtccag  | ctaccgtagg  | tggtatttaag | cctggttgtt | tcaagatcgg  | taaacctggt  | 1920 |
| ggtatgttgg  | ataacatatt  | ggcatctaag  | ttgtatagac | caggttcagt  | cgcttacgta  | 1980 |
| tcagaagagt  | gtggtatgtc  | caacgaattg  | aacaacatca | tcagtagaac  | tacagatggt  | 2040 |
| gtatacgaag  | gtgttgctat  | tggtggtgac  | agatacccag | gttctacttt  | tatggatcat  | 2100 |
| gtattgagat  | atcaagacac  | acctggtggt  | aaaatgattg | ttgtcttggg  | tgaatatagg  | 2160 |
| ggtactgaag  | aatacaagat  | atgcagaggt  | atcaaaagag | gtagattgac  | aaagccaatc  | 2220 |
| gtttgttggg  | gcatttggtac | ttgtgcaaca  | atgttttctt | cagaagttca  | attcggtcat  | 2280 |
| gcaggtgcct  | gcgctaatac  | agcttcagaa  | acagcagttg | ccaagaacca  | agcattaaaa  | 2340 |
| gaagccgggt  | tttttgtccc  | tagatctttc  | gatgaattag | gtgaaatcat  | tcaatcagtc  | 2400 |
| tatgaagact  | tggtagctaa  | tggtgtaatt  | gttccagcac | aagaagttcc  | tccacctact  | 2460 |
| gtccctatgg  | attactcttg  | ggcaagagaa  | ttgggtttta | ttagaaagcc  | agctagtgtt  | 2520 |
| atgacctcta  | tatgtgatga  | aagaggtcaa  | gaattgatct | atgctggtat  | gcctattact  | 2580 |
| gaagtattca  | aagaagaaat  | gggtatcggg  | ggtgttttag | ggttcttggg  | gttccaaaag  | 2640 |
| agattgccaa  | agtaactctg  | tcaattcatt  | gaaatgtgct | taattggttac | agctgatcat  | 2700 |
| ggtcctctgt  | tctcaggtgc  | acacaatacc  | ataatctcgc | ctagagctgg  | taaagatttg  | 2760 |
| gtttcttctt  | tgacctcagg  | ttgtttaact  | attggtgaca | gatttgggtg  | tgcattagac  | 2820 |
| gcgctgcgaa  | agatgttttc  | aaaaactttc  | gattccggta | taatcccaat  | ggaattcggt  | 2880 |
| aataagatga  | aaaaggaggg  | taaattgata  | atgggtatcg | gtcatcgtgt  | taagtctatc  | 2940 |



-continued

|             |             |             |             |             |             |      |
|-------------|-------------|-------------|-------------|-------------|-------------|------|
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| acacctttgt  | tagattacgc  | tttggaggtt  | gaaaagatta  | ccacttctaa  | aaagccaaat  | 3060 |
| ttgatcttga  | acgttgatgg  | tttaattggt  | gttgcttttg  | togacatggt  | gagaaactgt  | 3120 |
| ggttccctca  | ctagagaaga  | agctgatgaa  | tatatcgaca  | ttggtgcatt  | gaatggatc   | 3180 |
| tttgttttag  | gtagatctat  | gggtttcatt  | ggtcattact  | tggatcaaaa  | gagattaaag  | 3240 |
| caagggttgt  | acagacatcc  | atgggatgac  | atttcttacg  | ttttacctga  | acacatgtca  | 3300 |
| atgaaattgt  | ctggtgggtg  | tggttctggt  | ggtgggtggt  | ctggtgggtg  | tggtagtgc   | 3360 |
| gaagcttggt  | acaatttggg  | taacgcatac  | tacaagcagg  | gtgactacca  | aaaggcaatt  | 3420 |
| gaatattacc  | aaaaggcctt  | ggaattagac  | ccaaataacg  | cagaagcctg  | gtataatttg  | 3480 |
| ggtaattgct  | attataaaca  | gggtgactat  | caaaaggcta  | tcgaatacta  | ccaaaaggca  | 3540 |
| ttggaattag  | accctaataa  | cgctgaagca  | tgggtataatt | tgggtaacgc  | ttattataag  | 3600 |
| caggggtgact | atcaaaaagc  | catcgaagac  | tacaaaaagg  | ctttggaatt  | agatccaaat  | 3660 |
| aacttacaag  | ccgaagcttg  | gaagaatttg  | ggtaacgcct  | actataaaca  | gggtgactac  | 3720 |
| caaaaagcaa  | ttgaattacta | tcaaaaagct  | ttagaattgg  | accctaataa  | cgcatcagcc  | 3780 |
| tgggtacaatt | tgggtaatgc  | ttactataag  | caggggtgact | atcagaaggc  | cattgataac  | 3840 |
| tatcaaaaagg | ctttagaatt  | ggatccaaat  | aacgctaagg  | catgggtacag | acgtggtaac  | 3900 |
| gcttattaca  | aacaggggtga | ctaccagaaa  | gccattgaag  | attatcaaaa  | ggctttggaa  | 3960 |
| ttggatccta  | acaacagatc  | tagatcagct  | ggtgggtggt  | gttctggtg   | tggtggttct  | 4020 |
| ggtgggtggt  | gtgcttcttc  | atattaccat  | caccatcacc  | atcacttggg  | atccacaagt  | 4080 |
| ttatacaaaa  | aggctggttc  | tggttcaaat  | ttggtcgac   | aattggaaaa  | cgaagtagcc  | 4140 |
| tcttttagaaa | atgaaaacga  | aaccttgaaa  | aagaaaaact  | tacataagaa  | agatttgcac  | 4200 |
| gcttatttgg  | aaaaggaaat  | cgcaaatttg  | agaaagaaaa  | tgaaagaagg  | tagtgcaggt  | 4260 |
| tctgcgcgtg  | gttctggtga  | atttgggttc  | gctgaagcag  | ccgctaagga  | agcagccgct  | 4320 |
| aaagccggtt  | cagctggttc  | cgcagccggt  | tctggtgaat  | tcggttccag  | ttactatcac  | 4380 |
| catcacatcc  | atcacttgga  | atccacaagt  | ttataataga  | aagcagggtt  | tggttcagca  | 4440 |
| agaaatgcct  | acttgagaaa  | gaaaaatagct | agattaaaga  | aagataaact  | gcaattggaa  | 4500 |
| agagatgaac  | aaaatttggg  | aaagattatc  | gccaaactaa  | gagatgaaat  | cgctagattg  | 4560 |
| gaaaatgaag  | ttgcatacca  | tgaacaagggt | agtgggtgcta | ctaacttctc  | ttgttgaag   | 4620 |
| caagcaggtg  | acgttgaaga  | aaatccaggt  | ccaatgaaaa  | actgtgtaat  | cgtttctgct  | 4680 |
| gttagaactg  | caatttggttc | ctttaatggt  | agtttggcct  | ctacatcagc  | tattgatttg  | 4740 |
| ggtgctaccg  | tcatacaagg  | tgcatttgaa  | agagcaaaag  | ttgattctca  | acatgtcgac  | 4800 |
| gaagtaataa  | tgggttaacgt | tttgcaagct  | ggtttaggtc  | aaaatccagc  | aagacaagcc  | 4860 |
| ttgttaaaat  | ctggttttagc | agaaactgta  | tgtggtttca  | cagttaataa  | ggctcgcggt  | 4920 |
| tctggtttga  | agtcaagtgc  | tttagccgct  | caagctatac  | aagcagggtca | agcccaatct  | 4980 |
| atcgctcgctg | gtggatgga   | aaatatgtca  | ttggcacctt  | atttggtaga  | tgcaaaagcc  | 5040 |
| agatcaggtt  | atagattagg  | tgacggtcaa  | gtatacgacg  | ttattttgag  | agatggttta  | 5100 |
| atgtgcgctg  | ctcatgggtt  | ttacatgggt  | attacagcag  | aaaatgttgc  | caaagaatac  | 5160 |
| gggtataacca | gagaaatgca  | agatgaattg  | gcattacatt  | cccaagaaaa  | ggcagccgct  | 5220 |
| gcaatcgaaa  | gtggtgcttt  | tactgcagaa  | attgtccagc  | taaacgttgt  | cacaagaaag  | 5280 |
| aaaaactttcg | ttctctccca  | agatgaattc  | ccaaaagcta  | atagtaccgc  | tgaagcattg  | 5340 |
| gggtgctttaa | gacctgcatt  | cgacaaggcc  | ggtaccgtaa  | ctgcccgtta  | tgttctggtt  | 5400 |
| ataaacgatg  | gtgcccgtgc  | attggttatc  | atggaagaat  | cagccgcttt  | agcagccggt  | 5460 |
| ttgacacctt  | ttagtattat  | taaatcttat  | gcacaggtg   | gtgttccacc  | tgctttgatg  | 5520 |
| gggtatgggtc | cagtcctctg  | tacccaaaag  | gcattgcaat  | tagccggttt  | gcaattggct  | 5580 |
| gatatcgact  | taatcgaaag  | aaacgaagcc  | tttgcgtcac  | aattcttggtc | agttggttaa  | 5640 |
| aaatttgggtt | tcgactccga  | aaagggtta   | gtcaacggtg  | gtgccattgc  | tttgggtcat  | 5700 |
| ccaatagggtg | cttcaggtgc  | aagaatcttg  | gttacattgt  | tgcatgccat  | gcaagctaga  | 5760 |
| gataaaacct  | tgggttttagc | tactttgtgt  | atcggtggtg  | gtcaaggtat  | cgcaatgggt  | 5820 |
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| ggtggttagtg | cagaagcctg  | gtacaatttg  | ggtaacgctt  | actacaagca  | gggtgactac  | 5940 |
| caaaaaggcaa | tcgaataacta | ccaaaaggcc  | ttggaattag  | atccaaataa  | cgctgaagca  | 6000 |
| tgggtataatt | tgggttaatgc | ctattataaa  | caggggtgact | atcaaaaagc  | tattgaaat   | 6060 |
| tacaaaaagg  | cattgggaatt | agatccta    | aacgcgaag   | cttgggtata  | tttgggtaac  | 6120 |
| gcctattata  | agcaggggtga | ctatcaaaa   | gccatcgaa   | attaccaaaa  | ggctttggaa  | 6180 |
| ttggatccaa  | acaacttgcg  | agcagaagcc  | tgggaagaatt | tgggtaacgc  | ttattacaaa  | 6240 |
| caggggtgact | accaaaaagc  | tattgaaat   | tatcaaaaag  | ccttagaatt  | ggatccta    | 6300 |
| aacgcttctg  | catggtacaa  | tttgggtaat  | gcctactata  | aacaggggtga | ctaccagaag  | 6360 |
| gctattgaa   | actacaaaaa  | agcattagaa  | ttggatccaa  | ataacgccaa  | ggcttgggtac | 6420 |
| agacgtggta  | atgcctatta  | caagcaggtg  | gactaccaga  | aagccataga  | agactatcaa  | 6480 |
| aaagccttgg  | aattggatcc  | taacaacaga  | tccagaagtg  | ctggtgggtg  | tggttctggt  | 6540 |
| gggtggtggtt | ctggtgcttc  | tcatattacc  | atcaccatca  | ccatcacttg  | 6600        |      |
| gaatctacat  | cattatacaa  | aaaggctggt  | tccggtagta  | atgaagttac  | tacattggaa  | 6660 |
| aacgatgccg  | cttttatcga  | aaacgaaaa   | gcatacttgg  | aaaaggaaat  | cgccagatta  | 6720 |
| agaaaggaaa  | aggcagcctt  | gagaaataga  | ttagcccata  | aaaagggttc  | cgctggtagt  | 6780 |
| gctgcaggtt  | ctggtgaatt  | tggttcagct  | gaagccgctg  | caaaagaagc  | cgctgcgaag  | 6840 |
| gcaggttctg  | ccggttcagc  | cgtctggttc  | ggtgaattcg  | gttccagtta  | ctatcaccat  | 6900 |
| caccatcatc  | acttggaaat  | tacttcatta  | tataaaaaag  | ccggttccgg  | tagtcaaaaa  | 6960 |
| gtcgctgaat  | taaagaacag  | agtagctgtt  | aagttgaaca  | gaaacgaaca  | attgaaaaat  | 7020 |
| aaggtagaag  | aattgaaaaa  | tagaaacgcc  | tacttaaaga  | atgaattggc  | aacattggaa  | 7080 |
| aacgaagtgc  | ctagattgga  | aaatgatgta  | gcagaaggtt  | ctgggt      | 7125        |      |

SEQ ID NO: 208                      moltype = AA   length = 7  
 FEATURE                              Location/Qualifiers  
 source                                1..7  
                                       mol\_type = protein  
                                       organism = synthetic construct

SEQUENCE: 208  
 GVKESLV

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SEQ ID NO: 209      moltype = AA length = 723
FEATURE            Location/Qualifiers
source              1..723
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 209
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PFSLHQMLFY GCYKDFGPAP PAWIPDPEKV KSTNLGALLE KRGKEFLGVK YKDPISFSFH 120
FQEFVSRNPE VYWRTVLMDE MKISFSKDPE CILRRDDINN PGGSEWLPFG YLNSAKNCLN 180
VNSNKKLNDT MIVWRDEGND DLPLNKLTLQ QLRKRVVLVG YALEEMGLEK GCAIADMPM 240
HVDVAVIYLA IVLAGYVVVS IADSFSAPEI STRLRLSKAK AIFTQDHIIR GKRIPLYSR 300
VVEAKSPMAI VIPCSGSNIG AELRDGDISW DYFLERAKEF KNCEFTAREQ PVDAYTNILF 360
SSGTTGEPKA IPWTQATPLK AAADGWSHLD IRKGDVIVWP TNLGMMMGFW LVYASLLNGA 420
SIALYNGSPL VSGFAKFVDQ AKVTMLGVVP SIVRSWKSTN CVSGYDWSTI RCFSSSGEAS 480
NVDEYLWLMG RANYPKPIEM CGGTEIGGAF SAGSFLQAQS LSSFSSQCMG CTLYILDKNQ 540
YMPKKNKPGI GELALGPMVF GASKTLNNGN HHDVYFKGMP TLNGEVLRRH GDIFELTSNG 600
YYHAHGRADD TMNIGGIKIS SIEIERVCNE VDDRVFETTA IGVPPPLGGGP EQLVIFVFLK 660
DSNDTTIDLN QLRLSFLNGL QKKLNPLFKV TRVPLSSLP RTATNKIMRR VLRQQFSHF 720
GSG                                                    723

SEQ ID NO: 210      moltype = AA length = 127
FEATURE            Location/Qualifiers
source              1..127
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 210
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RYGRDKVMFS FEAGRRCPSG PGTFTFQTSQ GNDIQAVEA AIQQQKAQK VGQAQDILRL 120
EHHHHHHH                                                    127

SEQ ID NO: 211      moltype = AA length = 2741
FEATURE            Location/Qualifiers
source              1..2741
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 211
MSEESLFESS PQKMEYEITN YSERHTELPQ HFIGLNTVDK LEESPLRDFV KSHGGHTVIS 60
KILIANNGIA AVKEIRSVRK WAYETFGDDR TVQFVAMATP EDLEANA EYI RMADQYIEVP 120
GGTNMNNYAN VDLIVDIAER ADVDAVWAGW GHASENPLLP EKLSQSKRKV IFIGPPGNAM 180
RSLGDKISST IVAQSAKVPK IPWSGTGVDV VHVDEKTGLV SVDDDIYQKG CCTSPEDGLQ 240
KAKRIGFPVM IKASEGGGGK GIRQVEREED FIALYHQAAN EIPGSPIFIM KLAGRARHLE 300
VQLLADQYGT NISLFGSDCS VQRHQQKIE EAPVTIAKAE TFHEMEKAAV RLKGLVGYVS 360
AGTVEYLYSH DDGKFYFLEL NPRLQVEHPT TEMVSGVNLV AAQLQIANGI PMHRISDIRT 420
LYGMNPHSAS EIDFEFKTQD ATKKQRRPIP KGHCTACRIT SEDPNDGFKP SGGTLHELNF 480
RSSSNVWGYF SVGNNGNIHS FSDSQFGHIF AFGENRQASR KHMVVALKEL SIRGDFRTTV 540
EYLKLLLETE DFEDNTITG WLDLITHKM TAEKPDPTLA VICGAATKAF LASEEARHKY 600
IESLQKGQVL SKDLLQTMFP VDFIHEGKRY KFTVAKSGND RYTLFINGSK CDIILRLQSD 660
GGLLIAIGK SHTIYWKKEV AATRLSVDSM TTLLEVENDP TQLRTPSPGK LVKFLVENGE 720
HIIKGQPYAE IEVMMQMPLV VSQENGIVQL LKQPGSTIVA GDIMAIMTLD DPSKVKHALP 780
FEGMLPDPFGS PVIEGTPKPY KFKSLVSTLE NILKGYDNQV IMNASLQQLI EVLRNPKLPY 840
SEWLKHLISAL HSRLPAKLDE QMEELVARSL RRGAVFPARQ LSKLIDMAVK NPEYNPKLL 900
GAVVEPLADI AHKYSIVVHL EEYVEVEKLF NGPNVRENI ILKLRDENPK 960
DLDKVALTVL SHSKVSANNN LILAILKHYQ PLCKLSSKVS AIFSTPLQHI VELESKATAK 1020
VALQAREILI QGALPSVKER TEQIEHILKS SVVKVAYGSS NPKRSEPDN ILKDLIDSNY 1080
VVFDVLLQFL THQDPVPLTA AAQVYIRRAY RAYTIGDIRV HEGVTVPIVE WKFLPSAAF 1140
STFPTVKS KM GMNRAVSVSD LSYVANSQSS PLREGILMAV DHLDDVDEIL SQSLEVI PRH 1200
QSSSNGPAPD RSGSSASLSN VANVCVASTE GFSEEEILV RLREILDNLK QELINASIRR 1260
ITFMFGFKDG SYPKYTFNG PNYNETIR HIEPALAQL ELGRLSNFNI KPIFTNDRNI 1320
HVEAVSKTS PLDKRFFTRG IIRTGHIRDD ISIQEYLTSE ANRLMSDILD NLEVTDTSNS 1380
DLNHIFINFI AVFDISPEDV EAAFGGFLER FGKRLRLRV SAEIRIIK DPQTGAPVPL 1440
RALINNVSGY VIKTEMYTEV KNAKGEWVFK SLGKPGSMHL RPIATPYPVK EWLQPKRYKA 1500
HLMGTTYVYD PPELFRQASS SQWKNFSAV KLTDDFFISN ELIEDENGEL TEVEREPGAN 1560
ATGMVAFKIT VKTPEYPRGR QFVVVANDIT FKIGSFGPQE DEFFNKVTEY ARKRGPRIY 1620
LAANSGARIG MAEBIVPLFQ VAWNDAANPD KGFQYLYLTS EGMETLKKFD KENSVLTER 1680
VINGEERFVI KTIIGSEDGL GVECLRGSGL IAGATSRAYH DIFTITLVTC RSVGIGAYLV 1740
RLGQRAIQVE GQPIILTGAP AINKMLGREV YTSNLQGGT QIMYNNGVSH LTAVDLAGV 1800
EKIVEWMSYV PAKRNPVPVI LETKDTWDRP VDFTPTNDET YDVRWMI EGR ETESGFEYGL 1860
FDKGSFFETL SGWAKGVVVG RARLGGIPLG VIGVETRTVE NLIPADPANP NSAETLIQEP 1920
GQVWHPNSAF KTAQAINDFN NGEQLPMMIL ANWRGFSGGQ RDMFNEVLKY GSFIVDALVD 1980
YKQPIIIYIP PTGELRGGSW VVVDPTINAD QMEMYADVNA RAGVLEPQGM VGKIFRREKL 2040
LDTMNRLLDDK YRELRSLSN KSLAPEVHQI ISKQLADRER ELLPIYQIS LQFADLHDS 2100
SRMVAKGVIS KELEWTEARR FFFWRLRRL NEEYLIKRLS HQVGEASRL KIRIRSWYP 2160
ASVDHEDDRQ VATWIEENYK TLDDKLKGLK LESFAQDLAK KIRSDHDNAI DGLSEVIKML 2220
STDDKEKLLK TLKKLSGGGG SGGGGSGGGG SAEAWYNLGN AYYKQGDYQK AIEYYQKALE 2280
LDPNNAEAWY NLGNAYYKQG DYQKAIEYYQ KALELDPNNA EAWYNLGNAY YKQGDYQKAI 2340
EDYQKALELD PNNLQAEAWK NLGNAYYKQG DYQKAIEYYQ KALELDPNNA SAWYNLGNAY 2400
YKQGDYQKAI EYYQKALELD PNNAKAWYRR GNAYYKQGDY QKAIEDYQKA LELDPNNRSR 2460
SAGGGSGGGG SGGGGSGSGS HMRLGAQSIQ PTANLDRD TD LVYLVNMELV RAVLELKNEL 2520
AQLPPEGYVV VVKNVGLTLR KLIGSVDDL L PSLPSSRTE IEGTQKLNLK DLABELINKMR 2580

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|            |            |            |            |            |            |      |
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| GEFGSAEAAA | KEAAAKAGSA | GSAGSGEFG  | SGAMATPGSE | NVLPREPLIA | TAVKFLQNSR | 2700 |
| VRQSPLATRR | AFLKKKGLTD | EEIDMAFQQS | GTAADPESSL | W          |            | 2741 |

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SEQ ID NO: 212                   moltype = DNA   length = 57  
 FEATURE                       Location/Qualifiers  
 source                         1..57  
                               mol\_type = other DNA  
                               organism = synthetic construct

SEQUENCE: 212  
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What is claimed is:

1. A host cell capable of producing one or more cannabi- 15  
 noids, said host cell comprising:

- (a) a first exogenous nucleic acid encoding a first poly-  
 peptide having CBGA synthase activity and comprising  
 a first heterologous interaction domain,
- (b) a second exogenous nucleic acid encoding a second 20  
 polypeptide having olivetolic acid cyclase activity and  
 comprising a second heterologous interaction domain,
- (c) a third exogenous nucleic acid encoding a third  
 polypeptide having olivetol synthase activity and com-  
 prising a third heterologous interaction domain, 25
- (d) a fourth exogenous nucleic acid encoding a fourth  
 polypeptide having trans-2-enoyl-CoA reductase activ-  
 ity and comprising a fourth heterologous interaction  
 domain,
- (e) a fifth exogenous nucleic acid encoding a fifth poly- 30  
 peptide having enoyl-CoA hydratase activity and com-  
 prising a fifth heterologous interaction domain,
- (f) a sixth exogenous nucleic acid encoding a sixth  
 polypeptide having 3-hydroxybutyryl-CoA dehydroge- 35  
 nase activity and comprising a sixth heterologous inter-  
 action domain,
- (g) a seventh exogenous nucleic acid encoding a seventh  
 polypeptide having acetyl-CoA acetyltransferase activ-  
 ity and comprising a seventh heterologous interaction  
 domain, 40
- (h) an eighth exogenous nucleic acid encoding an eighth  
 polypeptide having ATP citrate lyase activity and com-  
 prising an eighth heterologous interaction domain,
- (i) a ninth exogenous nucleic acid encoding a ninth  
 polypeptide having geranyl pyrophosphate synthase 45  
 activity and comprising a ninth heterologous interac-  
 tion domain,
- (j) a tenth exogenous nucleic acid encoding a tenth  
 polypeptide having isopentyl-diphosphate isomerase  
 activity and comprising a tenth heterologous interac- 50  
 tion domain,
- (k) an eleventh exogenous nucleic acid encoding an  
 eleventh polypeptide having diphospho-mevalonate  
 decarboxylase activity and comprising an eleventh het- 55  
 erologous interaction domain,
- (l) a twelfth exogenous nucleic acid encoding a twelfth  
 polypeptide having phosphomevalonate kinase activity  
 and comprising a twelfth heterologous interaction  
 domain,
- (m) a thirteenth exogenous nucleic acid encoding a thir- 60  
 teenth polypeptide having mevalonate kinase activity  
 and comprising a thirteenth heterologous interaction  
 domain,
- (n) a fourteenth exogenous nucleic acid encoding a four- 65  
 teenth polypeptide having HMG-CoA reductase activ-  
 ity and comprising a fourteenth heterologous interac-  
 tion domain,

(o) a fifteenth exogenous nucleic acid encoding a fifteenth  
 polypeptide having HMG-CoA synthase activity and  
 comprising a fifteenth heterologous interaction domain,  
 and

(p) a sixteenth exogenous nucleic acid encoding a poly-  
 peptide scaffold comprising a peptide ligand for each of  
 said first to fifteenth heterologous interaction domains,  
 wherein each of said first to fifteenth heterologous inter-  
 action domains is different,

wherein each peptide ligand for each of said first to  
 fifteenth heterologous interaction domains is different,  
 wherein said polypeptide scaffold comprises, in an order  
 extending in a first direction away from said peptide  
 ligand for said first heterologous interaction domain,  
 (1) said peptide ligand for said second heterologous  
 interaction domain, (2) said peptide ligand for said  
 third heterologous interaction domain, (3) said peptide  
 ligand for said fourth heterologous interaction domain,  
 (4) said peptide ligand for said fifth heterologous  
 interaction domain, (5) said peptide ligand for said  
 sixth heterologous interaction domain, (6) said peptide  
 ligand for said seventh heterologous interaction  
 domain, and (7) said peptide ligand for said eighth  
 heterologous interaction domain, and

wherein said polypeptide scaffold comprises, in an order  
 extending in the other direction away from said peptide  
 ligand for said first heterologous interaction domain,  
 (1) said peptide ligand for said ninth heterologous  
 interaction domain, (2) said peptide ligand for said  
 tenth heterologous interaction domain, (3) said peptide  
 ligand for said eleventh heterologous interaction  
 domain, (4) said peptide ligand for said twelfth heter-  
 ologous interaction domain, (5) said peptide ligand for  
 said thirteenth heterologous interaction domain, (6)  
 said peptide ligand for said fourteenth heterologous  
 interaction domain, (7) said peptide ligand for said  
 fifteenth heterologous interaction domain, (8) said pep-  
 tide ligand for said seventh heterologous interaction  
 domain, and (9) said peptide ligand for said eighth  
 heterologous interaction domain.

2. The host cell of claim 1, further comprising (q) a  
 seventeenth exogenous nucleic acid encoding an acetyl-CoA  
 carboxylase and comprising a seventeenth heterologous  
 interaction domain, and (r) an eighteenth exogenous nucleic  
 acid encoding a polypeptide scaffold comprising a peptide  
 ligand for each of said eighth and seventeenth heterologous  
 interaction domains.

3. The host cell of claim 1, wherein said host cell further  
 comprises an exogenous nucleic acid encoding a cannabidi-  
 olic acid synthase and a cannabichromenic acid synthase.

4. The host cell of claim 1, wherein said host cell further  
 comprises an exogenous cannabidiolic acid synthase.

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5. The host cell of claim 1, wherein said host cell further comprises an exogenous cannabichromenic acid synthase.

6. The host cell of claim 1, wherein said host cell is a bacterial or a yeast host cell.

7. The host cell of claim 6, wherein said bacterial cell is selected from the group consisting of *Escherichia coli*, *Bacillus*, *Brevibacterium*, *Streptomyces*, and *Pseudomonas* cells.

8. The host cell of claim 6, wherein said yeast cell is selected from the group consisting of *Pichia pastoris*, *Saccharomyces cerevisiae*, *Yarrowia lipolytica*, *Kluyveromyces marxianus*, and *Komagataella phaffii* cells.

9. The host cell of claim 1, wherein said host cell is an algae or a plant cell.

10. The host cell of claim 9, wherein said algae is *Dunaliella* sp., *Chlorella variabilis*, *Euglena mutabilis*, or *Chlamydomonas reinhardtii* cells.

11. The host cell of claim 9, wherein said plant cell is a *Cannabis* or tobacco cell.

12. The host cell of claim 1, wherein each of said polypeptides is of the formula: enzyme-linker1-spacer-linker2-motif1-linker3-motif2, wherein linker1, linker2, and linker3 are the same or different, wherein motif1 and motif2 are the same or different, and wherein motif1 and motif2 form said heterologous interaction domain.

13. The host cell of claim 12, wherein said scaffold polypeptide comprises a linker between each adjacent peptide ligand.

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14. The host cell of claim 13, wherein said scaffold polypeptide is tagged with a MYC tag, FLAG tag, or HA tag.

15. The host cell of claim 12, wherein said linker is a flexible GS-rich sequence flanking a rigid  $\alpha$ -helical moiety.

16. The host cell of claim 12, wherein said spacer is the cTPR6 spacer.

17. The host cell of claim 1, wherein a constitutive promoter is operably linked to one or more of said exogenous nucleic acids encoding said polypeptides or to said sixteenth exogenous nucleic acid encoding said polypeptide scaffold.

18. The host cell of claim 1, wherein a first constitutive promoter is operably linked to one or more of said exogenous nucleic acids encoding said polypeptides and a second constitutive promoter is operably linked to said sixteenth exogenous nucleic acid encoding said polypeptide scaffold.

19. The host cell of claim 18, wherein said constitutive promoter used to express said polypeptide scaffold has weaker constitutive activity level than said constitutive promoter used to express said polypeptides.

20. The host cell of claim 1, wherein each said exogenous nucleic acid comprises an inducible promoter operably linked to the sequence encoding said polypeptide or said polypeptide scaffold.

21. The host cell of claim 20, wherein said promoter is the GAL1-10 promoter.

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